ST. VRAIN VALLEY SCHOOLS
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Request for Proposal
RFP # 2027-011
Boiler Replacement Projects – 2 Locations

SCHEDULE OF ACTIVITIES	DATES
Request for Proposal Posted on Bidnet/Rocky Mountain E- Purchasing	**November 9, 2026**
Mandatory Site Visit @ 2 Locations	**9:30a.m., November 17, 2026**
Written Questions Due Date and Time	**3:00p.m., November 24, 2026**
Addenda/Response to Questions	**December 1, 2026**
RFP Response Deadline	**2:00p.m., December 8, 2026**

THIS IS NOT A COMPETITIVE BID: The request for proposal process allows the District to determine the vendor that best meets the needs of the District in accordance with the evaluation criteria listed herein.

It shall be agreed and understood that Submissions are considered to be proprietary working documents offered by prospective firms while under evaluation by the District and, as such, are not subject to official opening or public inspection.

Once an award has been made and an agreeable contract executed, Submissions become subject to the Colorado Open Records Act (CORA). Vendors must proactively identify and clearly mark any specific data they contend meets the legal definition of a "trade secret" or confidential/proprietary information. Wholesale labeling of an entire proposal is not permitted, and the District reserves the right to evaluate the legal validity of any such claim.

Responses will be received until 2:00 pm, Tuesday, December 8, 2026.

Responses must be submitted in **electronic format only through the Bidnet/Rocky Mountain E-Purchasing website** (www.bidnetdirect.com/colorado). This site offers both "free" and "fee for service" registration options that allow for full access to the documents and for electronic submission of proposals. (Note: "free" registration may take up to 24 hours to process. Please plan accordingly.) All responses are time-stamped by the system upon receipt. **Late responses will not be accepted.**

RFP responses must be submitted and accepted before the response deadline, as indicated in the Schedule of Activities. It is the responsibility of the respondent to ensure that all RFP documents are uploaded and accepted prior to the designated date and time.

QUESTIONS
Repsondents may ***NOT*** contact ***any*** District officials or staff other than the Buyer listed herein prior to the RFP opening. ***All*** questions must be received by 3:00 PM, Tuesday,

November 24, 2026. All questions ***MUST*** be submitted in writing through the Bidnet/Rocky Mountain E-Purchasing website (www.bidnetdirect.com/colorado)

Any necessary Addendum will be posted on the Bidnet/Rocky Mountain E-Purchasing System website. Respondents are responsible for checking the Bidnet/Rocky Mountain Bid System to make sure that they are fully aware of all "Response to Questions" and Addenda which may be issued. *Free registration* ***will not*** *receive email notifications of Addenda*

RFP SUBMITTAL: RFP Submittal package need only contain the following as one single electronic document:

1. RFP Response Form with Signature, Exhibit 1
2. Proposal Response Requirements (Items 1-9, as defined)
3. Statement of Insurance, Exhibit 2
4. District Contract Acknowledgement Form, Exhibit 3
5. Addendum(s) if any
6. Any other pages or documents where signatures are required.

Enclosed are detailed specifications as determined by the District. Please note clearly on your RFP Responses any authorized alternatives or deviations from said specifications and include justification for acceptance of same. RFP Responses must include all requirements as defined. All responses are understood F.O.B. destination prepaid.

Before award of this RFP, the successful vendor may be required to satisfy the District as to their experience and competence to provide the service and product specified on the attached documents. The District may make such investigations as it deems necessary to determine the ability of the vendor to meet the required needs, and the vendor shall furnish to the District all such information and data for this purpose as the District may request. The District reserves the right to reject any responses if the evidence submitted by the vendor or the investigation of such vendor fails to satisfy the District that such vendor is properly qualified to carry out the obligations of the contract and to deliver the work or product contemplated therein.

The District reserves the right to reject any and all RFP Responses, waive irregularities, and accept the RFP Response or portion of any response which, in its opinion, best serves the needs and purposes of the District. The St. Vrain Valley School District RE-1J is an "Equal Opportunity Employer."

Sincerely,

Kim Myers

Kim Myers, NIGP-CPP, CPPB
Senior Procurement Specialist
ST. VRAIN VALLEY SCHOOL DISTRICT RE-1J

STANDARD TERMS AND CONDITIONS

1. RFP Responses must be signed by an authorized member of your company or will not be considered a valid response.

2. The District will receive electronic responses submitted through the Rocky Mountain E-Purchasing website (www.bidnetdirect.com/colorado). RFP responses must be submitted before the acceptance deadline as indicated in the Schedule of Activities. It is the responsibility of the respondent to ensure that all required documents are uploaded by the designated date and time.

3. All electronic files uploaded in the Rocky Mountain E-Purchasing System must be in a common format accessible by software programs utilized by the District. Those common formats are generally described as Microsoft Word (.doc or.docx), Microsoft Excel (.xls or .xlsx), Microsoft PowerPoint (.ppt or pptx), or Adobe Portable Document Format (.pdf.). Respondents will not secure, password protect, or lock uploaded files; the District must be able to open and view the contents of the files. Files that contain macros will not be accepted. Respondents will not disable or restrict the ability of the District to print the contents of an uploaded file. Scanned documents or images must be of sufficient quality, no less than 150 dpi, to allow for reading or interpreting the words, drawings, images, or sketches. The District may disqualify any response that does not meet the criteria stated in this paragraph.

   Any uploaded files that cannot be assessed or determined to be corrupt by the District will not be considered.

4. Only responses received from those Respondents who request a copy of the documents from Rocky Mountain E-Purchasing System will be accepted.

5. Prices stated must be in units specified. In case of a discrepancy between the unit price and the extension, the unit price shall prevail.

6. Prices shall be firm for the period indicated, but not less than ninety days following the RFP acceptance date.

7. Collusion. The Respondent, by affixing its signature, certifies that its Proposal is made without previous understanding, agreement, or connection either with any persons, firms, or corporations offering a Proposal for the same items or with the District. The Respondent also certifies that its Proposal is in all respects fair, without outside control, collusion, fraud, or otherwise illegal action. To ensure the integrity of the District's public procurement process, all Respondents are hereby placed on notice that any and all Respondents who falsify the certifications required in conjunction with this section shall be prosecuted to the fullest extent of the law.

8. The tradename of an item being proposed must be indicated by the Respondent when requested. The nature of some items dictates a sole source situation. Specifications are provided to identify the product/service required and establish an acceptable quality level.

9. Unless alternates are requested, the Respondent will not be allowed to offer more than one price on each item even though they may feel that they have two or more types or styles that will meet specifications. The Respondent must determine which to offer. If said Respondent should submit more than one price on any item, all prices will be rejected for that item. It is the Respondent's right to address alternatives in a separate document but not as a part of this RFP.

10. Samples of items, when required, must be furnished free of expense prior to the opening of Proposals and, if not destroyed, will be returned upon the Respondent's request at the Respondent's expense.

11. Special terms and conditions imposed by any Respondent on any item Proposed, other than those terms and conditions as set forth herein, shall result in rejection of said Proposal for those items or item as being non-responsive to this invitation. Examples: imposing "all or none" requirements, stipulating a minimum quantity over the quantity specified, proposing an alternate on a "No Substitute" item, etc.

12. Each Respondent shall agree to furnish the proposed product, services, equipment, or materials that are called for on the face of this proposal in strict accordance with the conditions, requirements, and specifications of this Request for Proposal. In the event the District includes trade names as a part of the work description of any item, any participating Respondent may submit quotations on other trade named products, providing they are equivalent unless the wording "no substitute" is indicated. When the description includes the wording "no substitute" in addition to the trade name, a quotation of the price will be accepted on the trade named product indicated. All disputes concerning grades and quality of merchandise or work shall be determined by the Purchasing Manager or authorized representative.

13. All materials, supplies, and equipment furnished or services performed under the terms of this purchase order or contractual agreement shall comply with the requirements and standards specified in the **Williams-Steiger Occupational Safety and Health Act of 1970** (Public Law 91-596), as well as with other applicable federal, state and local codes.

14. Time is of the essence. Time of delivery is a part of the consideration and the delivery date given by the Respondent in the RFP must be complied with unless otherwise instructed by the Buyer. This is very important and ***must*** be noted in the Proposal space provided. When a date is set for the delivery of merchandise or the performance of work, said merchandise must be delivered, or work performed, in accordance with the specifications or description herein contained on or before said date, or the order to the delinquent party may be canceled and awarded to the next lowest qualified responsible Proposal. In such case, the District will have the right to buy such articles at market price for immediate delivery, and an excess in cost of same over price named hereon is to be paid by the Respondent under this contract or deducted from any money due or hereafter coming to him. Failure to pay the said amount to the District upon demand will result in removal from the Districts' Bidder's List and/or legal action.

15. The District reserves the right to cancel, without penalty, any awards occurring as a result of this RFP. Written notice will be given to the awarded vendor thirty (30) days prior to contract cancellation advising reasons for the proposed cancellation, procedures to correct the problem if any, and the date the contract will be canceled. The District will be the sole judge on non-performance.

16. Barring extenuating circumstances, an award will be made based on the evaluation criteria as defined herein. The Buyer reserves the right to waive any technical or formal error or omissions and to reject any and all Responses, or to award a contract for the items hereon, either in part or whole, if deemed to be in the best interest of the District to do so.

17. Proposals may not be changed after the RFP acceptance time. The exception would be if there was a misinterpretation of the unit for which the Proposal was requested. No dollar amount change would be allowed, only a clarification as to the unit your RFP response represents. This must be done in writing 24 hours after notification to the Respondent from the Buyer.

18. The submittal of a Response shall constitute an irrevocable offer to contract with the District in accordance with the terms of said RFP, which offer may not be withdrawn until or unless rejected or not accepted by the District.

19. FREIGHT TERMS: F.O.B. Destination Freight Prepaid. All freight charges must be included in prices submitted in the RFP response. Prices quoted are to be firm for delivery to the District Warehouse or point designated in the District.

20. The District is exempt from State of Colorado Sales Tax and Federal Excise Tax. The contract amount shall not include the cost of any such taxes, including those on any material, supplies, or equipment used or installed in the work. The Respondent shall file an "Application for Exemption Certificate" with the Colorado Department of Revenue and submit copies of such certificate to the District upon award of the contract and prior to commencement of any work.

21. The Respondent agrees to indemnify and hold the District harmless from claims, suits, or action of every nature and description brought against it for or on account of any injuries or damages received or sustained by any party or parties, caused in whole or in part by or from the acts of the Respondents, his servants or agents. To this extent, the Respondent agrees to furnish adequate Public Liability and Property Damage Insurance, the amount of which will be determined by the District whenever such insurance, in the opinion of the District, is deemed necessary.

22. The issuance of a signed Purchase Order shall constitute a contract that must be fulfilled by the Respondent.

SPECIAL RFP TERMS AND CONDITIONS

Where there appears to be variances or conflicts between the Standard Terms and Conditions and the Special Terms and Conditions and/or Technical Specification outlined in this RFP Solicitation, the Special Conditions and/or the Technical Specifications shall prevail.

1. **PURPOSE OF RFP**: The purpose of this Request for Proposal is to obtain responses from qualified mechanical companies for Boiler Replacement Projects – 2 Locations (Black Rock Elementary and Eagle Crest Elementary) including but not limited to the removal, haul away and proper disposal of the existing boilers, purchase, installation and startup of new high efficiency Boilers.

2. **MANDATORY SITE VISIT: A Mandatory Site Visit is scheduled to begin at 9:30a.m. MT, Tuesday, November 17, 2026 at Black Rock Elementary School, 2000 Mountain View Blvd., Erie, CO 80516 and then traveling to Eagle Crest Elementary, 4444 Clover Basin Dr., Longmont, CO 80503.** The Respondent is required to attend this Mandatory Site Visit in order to submit a proposal. Respondents who fail to attend the mandatory site visit shall be considered non-responsive and ineligible for award. Additionally, all of the Respondent's Subcontractor(s) are required to attend this site visit in order to become familiar with the specifications and site conditions. It is recommended that attendees plan on approximately 1.5 hours to complete the site visits. No additional scheduled site visits will be available for this project.

3. **ELIGIBILITY OF RESPONDENTS**:
   The Respondent shall be able to provide evidence that they:
   1). Have an established satisfactory record of performance for a reasonable period of time.
   2) Have sufficient financial support, equipment and organization to ensure that they can satisfactorily provide the products and services if awarded a contract. The term "equipment and organization" as used herein shall be construed to mean a fully equipped and well-established company in line with the best business practices in the industry and as determined by the proper authorities of the District.

   The District reserves the right, before awarding the contract, to require a respondent to submit such evidence of its qualifications as it may deem necessary and may consider any evidence available to it (including, but not limited to, the financial, technical and other qualifications and abilities of the respondent, including past performance and experience with the District) in making the award in the best interest of the District.

4. **ADJUSTMENT OF QUANTITIES:** While the quantities stipulated in this Request for Proposal will be used by the District for purposes of response comparability, it is hereby agreed and understood that the District has the right to adjust/reduce the quantities ordered in conjunction with this RFP based on budgetary restrictions.

5. **CONSIGNMENT OF CONTRACT NOT ALLOWED**. Responses will only be considered from firms that are directly engaged in the business of manufacturing or distributing the goods and/or performing the services described in this Request for Proposal. Accordingly, no part of this Contract can be consigned to another vendor.

6. **LABOR, MATERIALS, AND EQUIPMENT TO BE SUPPLIED BY THE RESPONDENT**: Unless otherwise provided in this Request for Proposal, the respondent shall furnish all labor, materials, and equipment necessary to complete the project as specified. When not specifically identified in this RFP, such materials and equipment shall be of a suitable type and grade for the purpose. All materials, workmanship, and equipment shall be subject to the inspection and approval of the District's Project Manager.

7. **MATERIALS WILL BE SUPPLIED BY THE RESPONDENT**: The respondent hereby agrees that all materials supplied by the respondent in conjunction with this solicitation shall be new, warranted for their merchantability, and fit for a particular purpose. In the event any of the materials supplied by the respondent are found to be defective or do not conform to specifications, the District reserves the right to either cancel the order and return such materials to the respondent at the respondent's expense or require the respondent to replace the materials at the respondent's expense.

8. **METHOD OF AWARD IN THE BEST INTEREST OF THE DISTRICT:** All proposals submitted in response to this RFP will be reviewed for responsiveness prior to referral to the Evaluation Committee. The Evaluation Committee will score all responsive proposals in accordance with the evaluation criteria provided. The proposals will be evaluated to establish the award or award(s) of this solicitation. Multiple Awards may be designated, if determined to be in the best interest of the District. If the District determines that there was inadequate competition in any of the areas, the District may not issue any awards. Notwithstanding, the District reserves the right to alter this method of award if such alteration results in cost savings to the District. Awards for this proposal will be based on the highest overall evaluated score, conforming to the RFP requirements and determined to be in the best interest of the District.

   For this Request for Proposal, the Evaluation Committee will score Proposals based on the following criteria.

   - Company Information
   - Company Qualifications & Experience, and Project understanding
   - Staff/Team Qualifications and Experience
   - Proposed Solution, by Site
   - Project Schedule & Timeline
   - References
   - Warranty/Guarantee
   - Proposed Pricing, by Site
   - Completeness of the submittal

9. **PROTECTION OF PROPERTY**: The Respondent shall be held responsible for repairing or replacing any and all property which is damaged by reason of the respondent's operation on the property to the satisfaction of the District.

10. **ACCIDENT PREVENTION**: The Respondent shall be required to take safety precautions to protect persons and District property. All contractors and sub-contractors shall conform

to all OSHA, State, and County regulations while performing under the terms and conditions of this Contract. Any fines levied by the above-mentioned authorities because of inadequacies to comply with these requirements shall be borne solely by the Respondent which is responsible for same. Barricades shall be provided by the respondent when work is performed in areas traversed by persons or when deemed necessary by the District's Project Manager.

11. **CLEAN UP AFTER PROJECT IS COMPLETED**: All unusable materials and debris shall be removed from the District premises. At completion, the successful Respondent shall thoroughly clean up all areas where work has been involved as mutually agreed with the District's Project Manager. The successful respondent shall be responsible for disposing of environmentally hazardous waste materials in a manner that is consistent with regulations stipulated by the United States Environmental Protection Agency (EPA) as well as with any state or locally prescribed procedures.

12. **INDEMNIFICATION OF DISTRICT BY SUCCESSFUL RESPONDENT**: The successful Respondent shall indemnify and save the District harmless from any and all claims, liabilities, losses, and causes of action which may arise out of the fulfillment of the Respondent's contractual obligations as outlined in this RFP. The Respondent shall pay all claims and losses of any nature whatever in connection therewith, and shall defend all suits, in the name of the District when applicable, and shall pay all costs and judgments which may issue thereon.

13. **PRICES SHALL BE FIXED AND FIRM FOR THE TERM OF THE CONTRACT:** If awarded a contract under this solicitation, the prices proposed by the Respondent shall remain fixed and firm during the term of the contract; provided, however, that the respondent may offer incentive discounts from this fixed price to the District at any time during the contractual term.

14. **WARRANTY SHALL BE FOR ONE YEAR**: In addition to all other warranties that may be supplied by the Respondent, the Respondent shall warrant its product and/or service against faulty labor and/or defective material for a minimum period of one year after the date of acceptance of the labor, materials and/or equipment by the District. This warranty requirement shall remain in force for the full period identified above; regardless of whether the Respondent is under contract with the District at the time of the defect. Any payment by the District on behalf of goods and services received from the Respondent does not constitute waiver of these warranty provisions.

15. **DEFICIENCIES IN WORK TO BE CORRECTED BY RESPONDENT:** The successful Respondent shall promptly correct all deficiencies and/or defects in work and/or any work that fails to conform to the Contract Documents. All corrections shall be made within Thirty (30) calendar days after such rejected defects, deficiencies, and/or non-conformance verbally reported to the Respondent by the District's Project Manager. Respondent shall bear all costs of correcting such rejected work.

    If the Respondent fails to correct the work within the period specified in this RFP, the District reserves the right to place the Respondent in default of its contractual obligations, obtain the services of another vendor to correct the deficiencies, and charge the

Respondent for these costs; either through a deduction from the final payment over to the Respondent or through invoicing.

16. **METHOD OF PAYMENT AFTER WORK IS COMPLETED**: Upon completion and acceptance of the work required in conjunction with this RFP, the successful Respondent shall submit an invoice to the District's Financial Services Department. The invoice shall reflect the total value of the contract and shall reference the appropriate purchase order number, the address of the workplace, and all itemized material and labor costs; if applicable. The date of the invoice shall not exceed thirty (30) calendar days from the date that the work was completed and accepted. Under no circumstances shall the invoice be submitted to the District in advance of such completion and acceptance.

17. **TERM OF CONTRACT**: The Contract or Contracts shall commence on the individual Contract document date and shall remain in effect until such time as the product, commodities, equipment and/or services acquired in conjunction with this Request for Proposal have been completed and accepted by the District's authorized representative. Due to the funding source, separate (by location) contracts maybe required. Notwithstanding, it shall be understood and agreed that any required warranty period which exceeds this term shall remain in full force for the duration of the warranty period.

18. **PERFORMANCE OF CONTRACT:** In order to protect the vested interests of the District, and to ensure the efficient utilization of resources, the awarded vendor shall comply with all contractual obligations contained herein. With respect to these obligations, the District will report any non-compliance issues to the vendor for corrective action.

19. **PERFORMANCE AND PAYMENT BOND:** The awarded contractor will be required to furnish Performance and Payment bonds if the contract total amount is over $50,000. The Contractor shall furnish to St. Vrain Valley School District, a 100% Performance Bond and 100% Payment Bond within TEN (10) business days after notification of award of a contract. The cost of all bonds shall be included in Contractor's price offer. Industry Standard P&P Bonds or the AIA forms are acceptable. Bonds shall: 1) Be for the full amount of the contract price. 2) Guarantee the Contractor's faithful performance of the work under this contract, and the prompt and full payment for all labor and materials involved therein. 3) Guarantee protection to the District against liens of any kind. 4) Be, when a surety bond is furnished, from a surety company operating lawfully in the state of Colorado and shall be accompanied with an acceptable "Power-of-Attorney" form attached to each bond copy. All bonds required above shall be issued by companies authorized to do business under the laws of the State of Colorado. The Company must be rated no less than "A-“by the latest edition of Best's Insurance Guide which is published by A.M. Best Company, Oldwick, New Jersey. If the Respondent fails to deliver the specified performance bond within this specified time frame, including any extensions that may be granted by the District, the District shall declare the Respondent in default of the contractual terms and conditions and the Respondent and shall surrender its bid guaranty.

20. **HAZARDOUS WASTE DISPOSAL:** Successful respondent shall, at all times, fully comply with all state, federal and local, laws, rules, and regulations, including without limitation all applicable provisions of 6 C.C.R. § 1007-2.

21. **GOVERNMENT STANDARDS AND REQUIREMENTS:** All items or services to be purchased or supplied in conjunction with this RFP shall be in accordance with all federal, state, and local government standards to include, but not be limited to, those issued by the Office of Safety and Health Administration (OSHA), the National Institute of Occupational Safety Hazards (NIOSH) the National Fire Protection Association (NFPA), and the National Electric Code (NEC).

22. **PROJECT START AND COMPLETION DATES:** Project start and completion dates shall be coordinated with the District's Project Manager. All work shall be performed in accordance with good commercial practice and the work schedule and completion dates shall be adhered to by the successful Respondent; except in such cases where the completion date will be delayed due to acts of God, strikes, or other causes beyond the control of the Respondent. In these cases, the Respondent shall notify the District of the delays in advance of the original completion date so that a revised completion date can be negotiated.

    Should the respondent to whom the contract is awarded fail to complete the work within the time required, it is hereby agreed and understood that the District reserves the right to cancel the contract with the respondent and to secure the services of another vendor to complete the work. If the District exercises this right, the District shall be responsible for reimbursing the Respondent for work that was completed and found acceptable to the District in accordance with the RFP specifications. In addition, the District may, at its sole discretion, request payment from the Respondent, through an invoice or credit memo, for any additional costs over and beyond the original proposed price which were incurred by the District as a result of having to secure the services of another vendor.

23. **DISTRICT CONTRACT:** The parties agree and understand that District is a Colorado public school district and, as such, is subject to certain state and federal laws that cannot be waved. Attached here it is either the (a) Vendor Contract, (b) AIA Contract Document Supplemental Terms, or (c) SV A-001 Construction Service Agreement (collectively, the "District Contract"). The District Contract is attached hereto and incorporated herein by this reference. By submitting a response pursuant to this Request for Proposal, Respondent represents to the District that it has reviewed the District Contract and is prepared to sign the District Contract without modification, unless any such requested modifications are communicated to District at the same time as the response is being made.

SCOPE OF WORK

St. Vrain Valley School District is requesting separate proposed solutions and pricing for each of the 2 project locations, meeting the District and the proposed manufacture requirements & specifications. The Scope of Work is to include all permits, materials, labor, equipment, tools, lift, rentals, site protection, job cleanup, haul away of the existing unit, trash & hazardous material removal and proper disposal. All differences or exclusions must be identified in your proposal.

Requirements & Specifications

All proposed products, equipment, material and work is to follow the Scope of Work, Specifications and Requirements including the attached District Technical Guideline.

- Attached as, Technical Guideline, Edition 2026.

District Approved Boiler Manufacturers are as follows: *No Substitute

- Fulton Endura
- Reillo
- Raypak
- Lochinvar

Black Rock Elementary:
Replacement of existing Burnham Cast Iron Boiler Model GE615/11, 2.532 Mbh., Serial# 63130074-00-6101-0066. Replacement boiler must be high efficiency including required intake, exhaust, natural gas and hydronic piping changes.

- Unwire existing burner
- Removal and proper disposal of existing Burnham boiler
- Furnish and install high efficiency boiler (piping is primary/secondary configuration)
- Connect new boiler to existing piping
- Startup and Combustion tuning
- Re-insulate any disturbed insulations on new and existing piping connections
- Coordination and completion of state and city inspections
- State and County Permitting
- Complete replacement boiler documentation with the State
- Freight
- Provide all as-built documentation to owner
- Controls by SVVSD personnel.

Eagle Crest Elementary:
Replacement of existing Weil McClain boiler Model 688, 1.358 Mbh., No Serial# replacement boiler must be high efficiency including required intake, exhaust, natural gas and hydronic piping changes.

- Unwire existing burner
- Removal and proper disposal of existing Weil McClain boiler
- Furnish and install high efficiency boiler including piping as shown on the attached
  - Piping Drawing_10.16.26
- Connect new boiler to piping
- Startup and Combustion tuning

- Re-insulate any disturbed insulations on new and existing piping connections
- Coordination and completion of state and city inspections
- State and Boulder County Permitting
- Complete replacement boiler documentation with the State
- Freight
- Provide all as-built documentation to owner
- Controls by SVVSD personnel.

PROPOSAL RESPONSE REQUIREMENTS

This document provides the information necessary to aid interested firms in formulating a thorough response. Responses must be submitted electronically, through Bidnet/Rocky Mountain E-Purchasing at (www.bidnetdirect.com) for consideration, by the response date and time of 2:00p.m., December 8, 2026.

Completely address and provide all of the RFP evaluation criteria identified and describe in items (1-9). Attached additional information as it relates to each item, as necessary. Responses shall include at minimum the following items (1-9) in sequential order:

1. Company Information
    a. Provide Company overview
    b. Company headquarters and Colorado service and support locations
    c. Number of years in business under the current company name.
2. Company Qualifications and Experience
    a. Provide information regarding the following minimum qualifications, but not limited to:
        - Licensed and insured in the State of Colorado and project related jurisdictions.
        - Number of years of experience performing Boiler installation & commissioning.
        - Experience with code authorities of the State of Colorado and/or related jurisdictions.
        - Previous experience providing Boiler replacements projects.
3. Staff/Team Qualifications and Experience.
    a. Provide the employee names, qualifications and experience who will be assigned to each of these 2 projects.
4. Proposed Solution, **by Site**
    a. Provide a detailed solution that best meets at minimum the information contained in the RFP Scope of Work, Specifications and Requirements, for each of the 2 projects.
5. Project Schedule & Timeline.
    a. Provide proposed schedules and timelines to complete each of the 2 projects.
6. References:
    a. Provide (3) three references for whom you have provided similar work for in a K12 environment and the work that was performed.
        - Include: Company Name, Contact Name, Contact Phone, Contact Email Address and Work/Services relevant to this solicitation.
        - Previous experience working with St. Vrain Valley School District RE-1J
7. Warranty/Guarantee:
    a. Provide product warranty and installation work warranty/guarantee information, for the equipment and services being proposed, for each of the 2 projects.
8. Proposed Pricing **by Site**, to include but not limited to all permits, labor, material, equipment, tools, rentals, lifts, ladders, site protection, job cleanup, trash and hazardous material removal and proper disposal for a full and complete replacement and startup.
    a. **Black Rock Elementary Project:**
        - Provide a complete detailed/itemized pricing breakout of all costs including discounts and/or savings to perform both locations.
    b. **Eagle Crest Elementary Project:**
        - Provide a complete detailed/itemized pricing breakout of all costs, including discounts and/or savings to perform both locations.
9. Other information: Submittal may include other information you may wish to include.

Issuance of this Request for Proposal in no way constitutes a commitment by the St. Vrain Valley School District to award a contract. St. Vrain Valley School District reserves the right to waive any informality, or to reject any or all proposals received in response to this RFP, or to cancel the RFP, if it is determined to be in the best interest of the District.

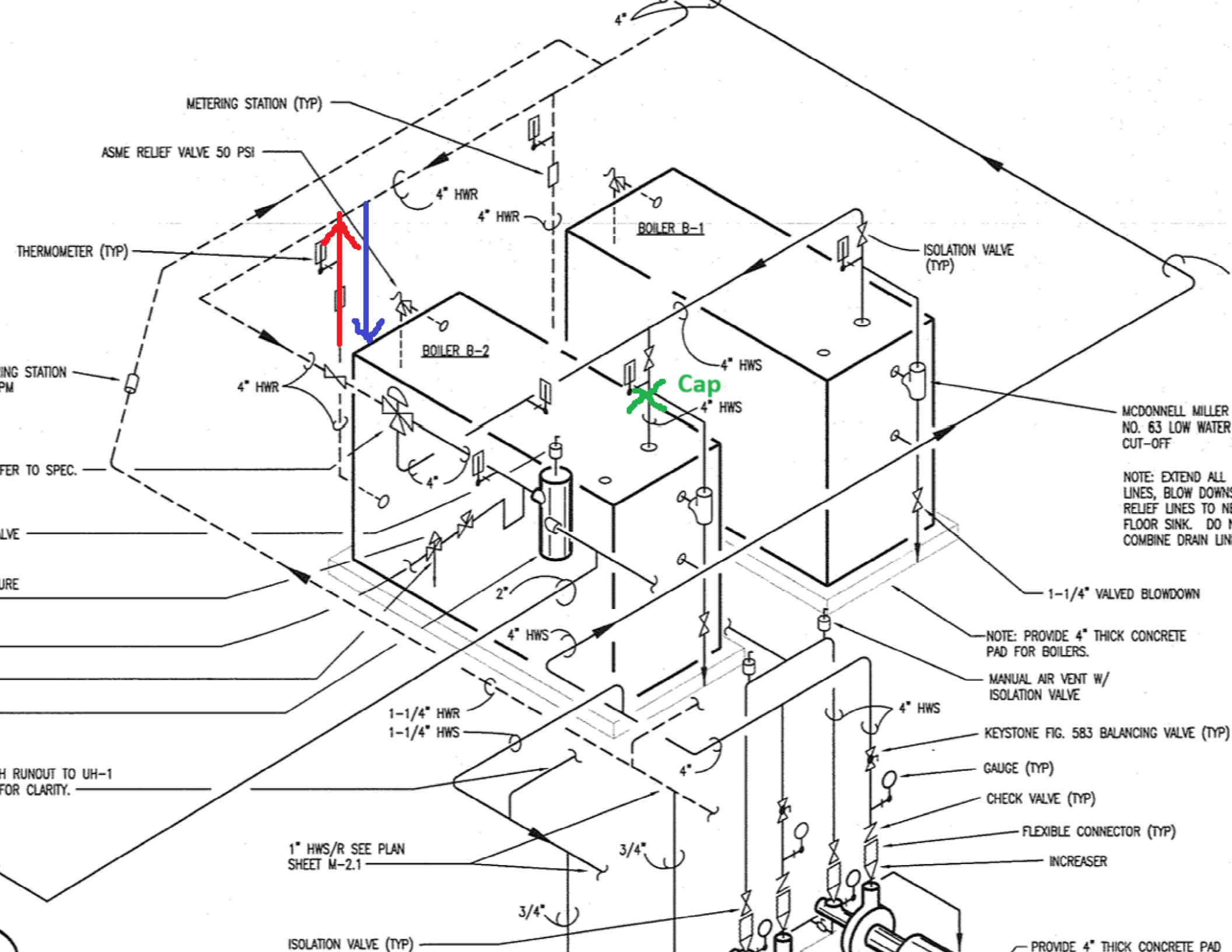

METERING STATION (TYP)
ASME RELIEF VALVE 50 PSI
THERMOMETER (TYP)
4" HWR
4" HWR
BOILER B-1
BOILER B-2
ISOLATION VALVE (TYP)
4" HWS
Cap
4" HWS
4" HWR
4"
2"
4" HWS
1-1/4" HWR
1-1/4" HWS
4"
1" HWS/R SEE PLAN SHEET M-2.1
3/4"
3/4"
ISOLATION VALVE (TYP)
MCDONNELL MILLER NO. 63 LOW WATER CUT-OFF
NOTE: EXTEND ALL LINES, BLOW DOWN RELIEF LINES TO FLOOR SINK. DO COMBINE DRAIN LIN
1-1/4" VALVED BLOWDOWN
NOTE: PROVIDE 4" THICK CONCRETE PAD FOR BOILERS.
MANUAL AIR VENT W/ ISOLATION VALVE
4" HWS
KEYSTONE FIG. 583 BALANCING VALVE (TYP)
GAUGE (TYP)
CHECK VALVE (TYP)
FLEXIBLE CONNECTOR (TYP)
INCREASER
PROVIDE 4" THICK CONCRETE PAD
ING STATION
PM
FER TO SPEC.
LVE
URE
H RUNOUT TO UH-1
FOR CLARITY.

EXHIBIT #1

RFP RESPONSE FORM

ALL PRICING SHALL REMAIN VALID FOR NINETY (90) DAYS

RFP No. 2027-011
Boiler Replacement Projects – 2 Locations

REQUIRED ATTACHMENTS: (submitted in the following order as one single electronic document).

1. RFP Response Form with Signature, Exhibit 1
2. Proposal Response Requirements (Items 1-9, as defined)
3. Statement of Insurance, Exhibit 2
4. District Contract Acknowledgement Form, Exhibit 3
5. Addendum(s) if any
6. Any other pages or documents where signatures are required.

Original Signature by Authorized Officer/Agent	Vendor's Tax Number (FEIN)
Type or printed name of person signing	Company Name
Title	Phone Number
Vendor Mailing Address	Email address
City, State, Zip	Price Valid Until
Payment Terms (Less than Net 30 will not be accepted)	Lead time ARO
DUNS #	Principal/Owner

Variances to Terms and/or Conditions:
The Respondent shall identify all variations and exceptions taken to the General Terms and Conditions, the Special Conditions and any Specifications in the space provided below; provided, however, that such variations are not expressly prohibited in the solicitation documents. For each variation listed, reference the applicable section of the solicitation document. If no variations are listed here, it is understood that the Respondents Proposal fully complies with all terms and conditions. It is further understood that such variations may be cause for determining that the Respondents Proposal is non-responsive and ineligible for award:

Sect_____Variance_____
Sect_____Variance_____
Sect_____Variance_____

Signature constitutes acceptance of all terms and conditions and documents attached.

EXHIBIT #2
TO BE RETURNED WITH YOUR RFP RESPONSE

STATEMENT OF INSURANCE

The Respondent agrees to maintain, on a primary basis and at its sole expense, at all times during the life of this project the following insurance coverages, limits, including endorsements described herein. The requirements contained herein, as well as the District's review or acceptance of insurance maintained by the Respondent, is not intended to and shall not in any manner limit or qualify the liabilities or obligations assumed by the Respondent under this project.

Commercial General Liability: The Respondent agrees to maintain Commercial General Liability at a limit of liability not less than $2,000,000 Each Occurrence. Coverage shall not contain any endorsement(s) excluding nor limiting Product/Completed Operations, Contractual Liability, or Cross Liability.

Business Automobile Liability: The Respondent agrees to maintain Business Automobile Liability at a limit of liability not less than $1,000,000 Each Occurrence. Coverage shall include liability for Owned, Non-Owned & Hired automobiles. In the event the Contractor does not own automobiles, the Contractor agrees to maintain coverage for Hired & Non-Owned Auto Liability, which may be satisfied by way of endorsement to the Commercial General Liability policy or separate Business Auto Liability policy.

Worker's Compensation Insurance & Employers Liability: The Contractor agrees to maintain Worker's Compensation Insurance & Employers Liability insurance in accordance with Colorado Revised Statutes.

Additional Insured: Respondent agrees to include the District as an Additional Insured to the Commercial General Liability. The Additional Insured shall read "St. Vrain Valley School District RE-1J."

Waiver of Subrogation: The Respondent agrees to a Waiver of Subrogation for each required policy herein. When required by the insurer, or should a policy condition not permit the Respondent to enter into a pre-loss agreement to waive subrogation without an endorsement, then the Respondent agrees to notify the insurer and request the policy be endorsed with a Waiver of Transfer of Rights of Recovery Against Others, or its equivalent. This Waiver of Subrogation requirement shall not apply to any policy, which includes a condition specifically prohibiting such an endorsement, or voids coverage should the Respondent enter into such an agreement on a pre-loss basis.

Certificate(s) of Insurance: The Respondent agrees to provide the District a Certificate(s) of Insurance evidencing that all coverages, limits, and endorsements required herein are maintained and in full force and effect. Said Certificate(s) of Insurance shall include a minimum thirty (30) day endeavor to notify due to cancellation or non-renewal of coverage. The Certificate Holder address shall read:

St. Vrain Valley School District RE-1J
Attn: Procurement Manager
Procurement Department
825 Delaware Ave., Suite 400A.
Longmont, CO 80501

Umbrella or Excess Liability: The Respondent may satisfy the minimum liability limits required above for Commercial General Liability or Business Auto Liability under an Umbrella or Excess Liability policy.

There is no minimum Per Occurrence limit of liability under the Umbrella or Excess Liability; however, the Annual Aggregate limit shall not be less than the highest “Each Occurrence” limit for either Commercial General Liability or Business Auto Liability. The Respondent agrees to include the District as an “Additional Insured” on the Umbrella or Excess Liability unless the Certificate of Insurance states the Umbrella or Excess Liability provides coverage on a “Follow-Form” basis.

Right to Revise or Reject: The District reserves the right, but not the obligation, to revise any insurance requirement, not limited to limits, coverages, and endorsements based on insurance market conditions affecting the availability or affordability of coverage; or changes in the scope of work/specifications affecting the applicability of coverage. Additionally, the District reserves the right, but not the obligation, to review and reject any certificate of insurance failing to meet the criteria stated herein or any insurer providing coverage due to its poor financial condition or failure to operate legally.

I HEREBY CERTIFY THAT I HAVE READ AND UNDERSTAND THIS DOCUMENT AND THAT THE AFOREMENTIONED INSURANCE IS IN EFFECT. THE CERTIFICATE IS ENCLOSED.

_____
Company Name

_____
Signature

_____
Print Name

_____
Title

_____
Date

EXHIBIT #3
TO BE RETURNED WITH YOUR RFP RESPONSE

DISTRICT CONTRACT

The attached word document is a sample of either the (a) Vendor Contract, (b) AIA Contract Document Supplemental Terms, or (c) SV A-001 Construction Service Agreement (collectively, the "District Contract") template. Once the Bid or RFP has been awarded, a final District Contract similar to this will be sent for your signature.

IF YOUR COMPANY NORMALLY SEEKS LEGAL REVIEW OF CONTRACT TERMS, THIS PROCESS SHOULD OCCUR PRIOR TO SUBMISSION OF YOUR RESPONSE AND YOUR OFFER MUST IDENTIFY ANY TERM OR CONDITION YOU ARE QUESTIONING AND HOW YOU WOULD SUGGEST IT BE RESOLVED.

By submitting a response to the Bid or RFP you acknowledge that you accept the terms and conditions of the District Contract as the primary agreement. Other agreements that may be provided by the contractor will follow in priority of interpretation.

Please check one and return with your response:

- ☐ I accept the District Contract and have no changes.
- ☐ I have attached a redlined version in MS Word with questions, comments, or suggested changes to the District Contract.

IMPORTANT NOTE: Exceptions requested, or alternative language suggested, will be treated as a "counter-offer." The willingness and/or ability of the vendor to accept or negotiate such suggested changes will be considered on a case-by-case basis. Vendor, please note, IF you do offer suggested modifications, the decision to reject, accept, or further negotiate requested changes will be at the District's discretion.

Please check if you have an additional agreement for District to consider

- ☐ I have specific contracts and agreements that will follow the priority in interpretation that we would like you to review and consider as part of the overall agreement if awarded. IMPORTANT NOTE: Please add to the end of this submission (after any redlines to the District Contract) in MS Word format.

ST. VRAIN VALLEY SCHOOLS
academic excellence by design

Technical Guidelines

Edition **2026.1**
Issued January 2026

2026 Updates are indicated by gray highlight

St. Vrain Valley Schools
Construction & Maintenance Department
303-682-7269
395 S. Pratt Pkwy
Longmont, CO

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APPENDICES

Appendices supplement the information provided in the Technical Guidelines, and can be found on the District Bluebeam Project, under the Guidelines folder.

Appendix A Sustainability Targets
~~Appendix B~~ ~~HVAC & Building Controls, Detailed Guidelines~~ *Division 23 has been incorporated into this document.*
Appendix C Communications, Technology, and Security Guidelines
Appendix D SVVSD Door & Hardware Installation Guide, Wood and Hollow Metal
Appendix E SVVSD Door & Hardware Installation Guide, Aluminum

DESIGN BULLETINS

Design bulletins are space or function specific requirements, and can be found on the District Bluebeam Project under the Guidelines folder, as well as at the end of this document.

Current Design Bulletins include the following, other may be added as developed:

- 2018-05-17 Design Bulletin, Calm Rooms
- 2019-09-17 Design Bulletin, Kilns & Kiln Rooms
- 2024-11-07 Design Bulletin, Automated Emergency Systems
- 2024-11-07 Design Bulletin, Monument Signs
- 2025-03-27 Design Bulletin, Building Services
- 2025-10-08 Design Bulletin, Classroom AV & AGILE

INSTRUCTIONS TO THE DESIGN TEAM

Intent and Use of this Document

This document is intended to guide the development of project specifications. District standards and preferences for performance criteria, products, materials, systems, etc. are contained within. This document is intended to function as a companion to the District's Educational Specifications. It is assumed that the design professionals preparing the contract documents will develop designs and specifications to conform to all locally adopted building codes and regulations, as well as industry standards and best practices, therefore the intent of this document is not to reiterate those requirements, but to describe requirements unique to this District.

The absence of any material, manufacturer, or product from this document does not preclude its incorporation into a project. The components listed are intended to describe preferences, however the District encourages design professionals to introduce and recommend alternate products where applicable and beneficial.

Definitions

Words or phrases defined in this section are identified throughout this document by the use of *italicized* text.

Acceptable Components or Products
Acceptable products are those materials, products, or other components of construction which are compatible with District infrastructure, systems, or procedures. Components identified as *acceptable* shall only be used if budget or other constraints prevent the use of the District's *preferred components*.

Discouraged Components or Products
District preference is to avoid these components or products and use must be approved by the District.

District Standard Components or Products
District Standard Components or Products coordinate directly with District-Wide systems, accounts, service agreements, or other items which necessitate compatibility with new components.

Preferred Components or Products
Preferred components are the District's first choice and shall always be the initial consideration. Where budget or other constraints preclude the incorporation of the preferred components into a project, *acceptable components* may be specified.

Unacceptable Components or Products
Unacceptable components shall not be specified. Unacceptable components are designated as such due to an incompatibility with existing District infrastructure, systems, or procedures.

Architectural & Engineering Services

The architect and their consultants shall provide standard A/E services as defined in the agreement between the District and the Architect. To expand upon the definitions of services contained within the agreements the architect shall coordinate the following items with the District's project manager as they relate specifically to school design and District procedures.

Design Advisory Groups (DAGs)

The architect shall coordinate with the District's project manager to form, schedule, and facilitate DAG teams and meetings.

Record Drawings
At project close-out the architect shall submit Record Drawings as required by the Owner-Architect Agreement.

DIVISION 00 – PROCUREMENT AND CONTRACT PROCEEDURES

00 00 00 Primary District Requirements

Where required for Design-Bid-Build project delivery, the contents of Division 00 Procurement and Contracting Requirements will be prepared and distributed by SVVSD. The architect shall coordinate as required with the District's Project Manager.

DIVISION 01 – GENERAL REQUIREMENTS

01 26 00 Contract Modification Procedures

A. Contract modification shall be administered by the District's Project Manage, using the typical AIA documents.

01 50 00 Temporary Facilities and Controls

A. PAYMENT FOR TEMPORARY UTILITIES:
   1. The contractor shall provide and maintain all necessary temporary utilities required for the work and for contractor project trailers.
   2. Payment for temporary utilities shall be coordinated with the District. Generally:
      a. The contractor shall pay for all temporary utilities for new construction, addition, and major renovation projects.
      b. The District will pay for utilities within existing buildings where construction work will proceed while the building is occupied.

B. OPERATION OF PERMANENT HVAC EQUIPMENT
   1. The building's permanent HVAC equipment shall not be operated prior to Substantial Completion unless all of the following criteria are met:
      a. A plan for operation has been prepared by the project's Engineer of Record. Plans shall address at minimum:
         1. Procedures to protect the equipment from damage during construction.
         2. Procedures to prevent construction dust and debris from entering the system, from which damage may not occur until after occupancy.
         3. Procedures to repair or replace damaged or consumable components prior to substantial completion.
      b. Written approval from the District has been received.
      c. The warranty period for all equipment shall begin upon Substantial Completion.

01 74 19 Construction Waste Management

A. See **Appendix A - Sustainability Targets** for Construction Waste Recycling requirements

01 78 39 As-Built Drawings

A. The contractor shall be required to provide the District with A-Built drawings for deferred design systems including, but not limited to:
   1. Fire Sprinkler
   2. Fire Alarm
B. See Owner-Architect Agreement for As-Built requirements by the architect.

01 80 00 Attic Stock

A. District preference is to limit attic stock to reasonable and usable amounts. Coordinate with District Maintenance staff.

DIVISION 02 – EXISTING CONDITIONS

02 00 00 Primary District Requirements

A. SALVAGED MATERIALS
    1. Coordinate any desired salvaged materials with the District's Project Manager.

B. CONTROLLED MATERIALS, ASBESTOS, LEAD PAINT, ETC.
    1. This District has on file Asbestos Hazard Emergency Response Act (AHERA) documentation available for most existing buildings. AHERA reports may be found on the Districts Bluebeam "Project" or by contacting the District's Environmental Compliance Manager.

DIVISION 03 – CONCRETE

03 30 00 Cast-In-Place Concrete

A. REINFORCEMENT
   1. Fibrous reinforcement preferred for on-grade interior and exterior flat work.
   2. Specify epoxy-coated rebar for areas expected to receive de-icing materials (i.e. stairs, ramps, loading docks, etc.).

B. CAST-IN-PLACE CONCRETE
   1. No specific requirements.

C. CONCRETE PAVING
   1. Defer to project engineer or City of Longmont "Public Improvement Design Standards and Construction Specifications".

D. CONCRETE FINISHES
   1. Colored concrete shall not be employed except with specific approval by the District.
   2. Polished Concrete
      a. Final polish sheen: Satin

E. UNDERSLAB VAPOR BARRIER
   1. Install vapor barrier in all slab on grade interior applications. Vapor barrier must be contact with underside of slab.
   2. Clean substrates of substances harmful to insulation or vapor retarders, including removing projections capable of puncturing vapor retarders or of interfering with insulation attachment.
   3. Installation shall be in accordance with manufacturer's instructions and ASTM E 1643-98.
   4. Unroll Vapor Barrier with the longest dimension parallel with the direction of the pour.
   5. Overlap joints 6 inches and seal with manufacturer's tape.
   6. Seal all penetrations (including pipes) per manufacturer's instructions.
   7. Available Products: Subject to compliance with requirements, products that may be incorporated into the Work include, but are not limited to, the following:
      a. Stego Industries, LLC; Stego Wrap (15 mil) Vapor Barrier (Basis-of-Design).
      b. Stego Tape.
      c. Stego Mastic.
      d. Or District approved equals
      e. Reef Industries, Inc; VaporGuard 15 mil
      f. Epro Services, Inc; Ecoshield E-15
      g. Fortfiber Building Systems Group; Moistop Ultra 15
      h. Insulation Solutions, Inc.; Viper VaporCheck II – 15 mil
      i. Meadows, W.R., Inc; Perminator 15 mil

03 50 00 Polished Concrete

A. General District Preferences
   1. Grind
      a. See table below.
   2. Stain:
      a. District experience has been that darker stains show more acid etching and other imperfections than light and medium stains.
   3. Polish
      a. See table below.
   4. Densifier
      a. Ensure that densifier is fully apply, to full rejection.
      b. Recommend densifiers with greater than 20% solids content.
   5. Impregnator
      a. Provide impregnator product to minimize the absorption of the concrete.

Primary District Prefference
Secondar District Preference
Variations from these preferences are acceptable with District approval.

Aggregate Exposuer Class

Class	*Name*	*AKA*	*Aggregate Size*	*Appearance Description*
A	Cement Fines	Cream	Very little	Little aggregate exposure.
B	Fine Aggregate	Salt & Pepper	1/16"	Fine aggregate exposure with little or no medium aggregate at random locations.
C	Coarse Aggregate	Medium	1/8"	Medium aggregate exposure with little or no large aggregate exposure at random locations.
D	Large Aggregate		1/4"	Large aggregate exposer with little or now fine aggregate exposure.

Gloss or Polish Level

Level	*Grit*	*Sheen*	*Gloss Reading*	*Appearance Description*
1	Below 100	None to very low	n/a	Flat. Floor has little if any reflectivity.
2	100 to 400	Low to medium	40-50	Satin or matte appearance with or without slight diffused reflection.
3	800 & higher	Medium to High	50-60	Semi-polished. Objects being reflected are not quite sharp and crisp, but can be easily identified.
4	800 & higher (additional passes)	High	60-80	Highly polished. Objects being reflected are sharp and crisp, with mirror like clarity.

DIVISION 04 – MASONRY

04 20 00 Unit Masonry

A. BRICK
   1. Brick is a *preferred* exterior finish material.
   2. Quality: FBX
   3. Surface Texture
      a. Smooth texture is preferred. Rough or raked textures are discouraged.
   4. Graffiti prevention
      a. Water based sealers.

B. CONCRETE MASONRY UNITS (CMU)
   1. CMU is a *preferred* exterior finish material.
   2. Provide Integrated Water Repellant (IWR)
   3. Grout door frame set in CMU walls.
   4. Smooth face textures are preferred to aid in graffiti removal.
   5. Graffiti prevention
      a. Water based sealers.

C. STONE
   1. Stone is a preferred exterior finish material.

D. ADHERED BRICK AND STONE
   1. Adhered brick and stone are *acceptable* exterior finishes in limited quantities, full be masonry is preferred.
   2. All joints in adhered brick and stone installation shall receive mortar, “dry stack” installation is discouraged.

E. UNACCEPTABLE MASONRY MATERIALS
   1. Adobe masonry

DIVISION 05 – METALS

05 52 00 Primary District Requirements

A. HANDRAILS AND GUARDRAILS
    1. Materials and Finishes
        a. Handrails and other contact surfaces
            1. *Preferred* finishes
                a. Uncoated Steel (i.e. stainless or blackened)
                b. Wood
                c. Galvanized (at exterior)
            2. *Acceptable* finishes
                a. Powder coated
            3. *Discouraged* finishes
                a. Painted
        b. Balusters and other non-contact surfaces.
            1. *Preferred* finishes
                a. Natural (i.e. stainless steel,)
                b. *Powder coated*
            2. *Acceptable finishes*
                a. *Painted*
B. Therapy Swing Anchors
    1. The District has experienced problems with therapy swing anchors when lateral braces are attached with a screw to floor or roof decking. The screws work loose and release. Engineer to consider this and provide a more adequate solution.

DIVISION 06 – WOOD, PLASTIC, AND COMPOSITES

06 41 00 Casework

A. TYPICAL CABINET / CASEWORK CONSTRUCTION
   1. Provide American Woodworkers Institute (AWI) "Custom Grade" cabinets and millwork.
      a. Requirement AWI certification may be omitted with approval of the Project Manager.
   2. Plastic Laminate (PLAM) surfacing, except where Solid Surfacing is preferred, see below.
   3. Metallic laminates are discouraged.
   4. Shelves: ¾" thickness min.
   5. Countertops
      a. Plastic laminate countertops are only to be used in dry locations (no counters with sinks or window sills).
      b. At PLAM counters laminate edge banding is not permitted. Hardwood edging is *preferred*; vinyl or other heavy duty edge banding is *acceptable*.
   6. Hinges
      a. Stainless steel concealed hinges.
      b. Min. 140-degree swing.
   7. Pulls
      a. Metal only, no plastic pulls allowed.
   8. Locks
      a. Disc tumbler or cam locks prohibited. Use CompX / National M2-3718-351 / 26d finish pin tumbler deadbolt type locks with ¾" throw. Add "Advantage Plus" feature.
      b. Provide locks on all drawers and doors.
      c. Keying
         1. Key each room separately.
         2. Provide (2) keys per room or lock group.
         3. Key all cabinets for master
         4. Provide (3) master keys
            a. Intent is (1) for SVVSD lockshop and (2) for building administration.
      d. Keying schedule
         1. School Name / Date of mfg.
         2. Shop Drawing Plan / Elevation Reference / Room Number
         3. Group Key Alike Number / Key number / Door / Drawer number.

B. SOLID SURFACING
   1. All wet counters (those with sinks or other plumbing) shall be solid surfacing.
   2. Window sills shall be solid surface or similar material.
   3. Min. thickness of material: ½"

C. SCIENCE LAB COUNTERS
   1. Provide chemical resistive epoxy-resin material according to NEMA LD 3, Test Procedure 3.4.5. Color: Black
   2. Sink shall be min. depth of 8-10", size of sink shall be no less than 22 x 18. Architect to coordinate sink configuration and size with overall layout intentions of the lab design.
   3. Fabricate with factory cutouts for sinks, holes for service fittings and accessories, and with butt joints assembled with epoxy adhesive and concealed metal splines.
   4. Provide with polypropylene strainers and tailpieces.
   5. Provide integral sinks in epoxy countertops, bonded to countertops with invisible joint line.
   6. Epoxy: Factory-molded, modified epoxy-resin formulation with smooth, nonspecular finish.

7. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:
    a. Durcon Company (The).
    b. Epoxyn Products.
    c. Laboratory Tops, Inc.

DIVISION 07 – THERMAL & MOISTURE PROTECTION

07 30 00 Low Slope Roofing

A. Quality Assurance
   1. Manufacturer: Minimum of 10-years of experience and ISO certified.
   2. Installer: Minimum of 5-years of experience with specified systems.
   3. Warranty inclusions:
      a. Manufacturer to be on site a minimum of 4 days per week.
      b. 24 hour response time for leak calls post-construction.
      c. Compulsory, no cost, yearly inspection and PM training.

B. Warranty
   1. All roofing shall be provided with a 30 or VE 20 year non-prorated, no-dollar-limit (NDL), warranty. Term shall begin at Substantial Completion.
   2. Provide Manufacturer's standard or customized form, without monetary limitation, in which manufacturer agrees to repair or replace components of built-up roofing that fail in materials or workmanship within specified warranty period. Special warranty includes built-up roofing membrane, base flashings, roof insulation, fasteners, cover boards, roofing accessories, roof pavers, and other components of built-up roofing.

C. Preferred Systems
   1. Modified Built-Up Roofing Systems, at low sloped roofs.
      a. General Built-Up Roofing System Description
         1. Aluminized or flood and gravel coating
         2. 80 mil SBS Modified Cap Sheet
         3. (2) Layers of felt in asphalt
         4. Substrate and insulation per project requirements.

   2. EPDM Membrane Roofing
      a. 90 mil field thickness, min. discuss mil thickness variation at detail locations.
      b. Consider fleece backed membrane products
      c. Fully Adhered
      d. 30 year warranty

07 40 00 Metal Roofing and Wall Panels

A. Warranties:
    1. 45 year finish, minimum on painted or coated systems.
    2. At galvanized finish provide manufacture's standard warranties where applicable.
    3. 40 or VE 30 year NDL waterproof warranty.
B. General
    1. Concealed fastener, standing seam.
    2. 2" min. standing seam height.
    3. Provide high temperature ice and water shield under all metal roofing.
C. Colors:
    1. Natural finishes and muted colors are preferred. The most common metal panel maintenance issue for the District is fading of finishes prior to the useful life of the panel. Finishes that will not fade or where fading will be less noticeable over a 50-year period are preferred.

07 50 00 Siding

A. FIBER CEMENT SIDING
    1. Fiber Cement Siding may be considered as an accent material, but masonry or metal panel product are preferred for the majority of the building enclosure.
    2. Products shall be limited to those designed and marketed for commercial applications.
    3. Min. material thickness: 5/16”

B. WOOD SIDING
    1. Wood siding may be considered as an accent material at high profile locations.
    2. Products such as Western Red Cedar which poses inherent durability and rot resistance are preferred.

DIVISION 08 – OPENINGS

08 11 13 Doors and Frames

A. SVVSD INSTALLATION GUIDE
   1. See **Appendices D, SVVSD Door & Hardware Installation Guide, Wood and Hollow Metal**
   2. See **Appendices E, SVVSD Door & Hardware Installation Guide, Aluminum**

B. HOLLOW METAL FRAMES
   1. Construction
      a. Welded frames, knock-down frames are discouraged.
      b. 16ga @ Interior, 14ga @ Exterior
      c. Provide continually welded high frequency hinge reinforcement at top hinge, min.
      d. Galvanneal frames @ exterior openings and at all masonry openings.

C. HOLLOW METAL DOORS
   1. 14 ga. Min.
   2. All exterior door to be galvannealed.
   3. All door lites to be held 8" min from top and sides of door edge and 10" min from door bottom.
   4. Provide inverted channel at top and bottom of door.

D. WOOD DOORS
   1. Door vertical stiles to be 8" inches minimum
   2. On full light doors, top rail to be 8" inches, Center rail to be 8" inches, Bottom rail to be 10" inches.
   3. On full light doors, center rail should align horizontally with the center line of the exit device or lock set from finished floor.
   4. Mineral Core Fire doors are discouraged.
   5. Solid-Core Doors: Structural Composite Lumber Core (Particleboard Core is strictly prohibited)
   6. Do not veneer door edges.

E. LARGE INTERIOR OPENINGS
   1. At medium/large interior opening, such as from classrooms to adjacent common spaces, consider the following:
   2. Preferred: Multi-slide doors (Kawneer 1010 Sliding Mall front or similar)
      a. This style of door is easily operable, quiet, minimizes finger pinching concerns, and can be quickly closed and locked if needed.
   3. Acceptable
      a. Folding glazed doors, overhead sectional door, and other options may be acceptable, pending review and budget alignment.

08 40 00 Aluminum Storefront and Curtain Wall

A. PRE-INSTALLATION MEETING
   1. Prior to installation of the storefront or curtainwall assemblies, conduct on-site meeting to review the configuration of materials, products, and assemblies in coordination with other work along with any special conditions specific to the specific project. See latest version of the SVVSD Aluminum Door & Hardware Installation Guide. Owner (SVVSD Lockshop) representative must attend and address past installation or product issues.

B. ALUMINUM FRAMING SYSTEMS
   1. Immediate Storefront Door Frame or Curtainwall Sub-frame Members must be "Heavywall" (.1875 inch (3/16") – 4.7625 mm) minimum thickness matching the doors.
   2. Thermally broken systems are preferred.
   3. Door closer shoes and removable mullion installations are to be mounted below the blade stop (as applicable) and must have a solid aluminum block spacer for full support. Plastic spacers are NOT approved for mounting to aluminum door frames. Wood or self-tapping screws are strictly prohibited!
   4. Flashing: Manufacturer's standard corrosion-resistant, nonstaining, nonbleeding flashing compatible with adjacent materials. Form exposed flashing from sheet aluminum (.0625" minimum thickness) finished to match framing and of sufficient thickness to maintain a flat appearance without visible deflection..
   5. Framing System Gaskets and Sealants: Manufacturer's standard recommended by manufacturer for joint type.
   6. Finish: Selected from manuf. standard paint colors or clear anodic finish.
   7. Provide frame reinforcement/backing at:
      a. Hinges
      b. Closer arm attachment at head

C. ALUMINUM DOORS
   1. Heavy-Duty Doors: Manufacturer's heavy duty glazed doors, for manual swing operation.
   2. Door Construction:
      a. ALL doors to be modeled as left and right hand single doors with beveled edges, including double doors.
      b. Meeting stile weather seals on door edges are prohibited! Mullion seals will provide weather seal function.
      c. 2- to 2-1/4-inch (50.8- to 57.2-mm) overall thickness, with minimum (.1875 inch (3/16") – 4.7625 mm) thick, extruded-aluminum tubular rail and stile members.
      d. Door vertical stiles to be 5" inches minimum.
      e. Top rail to be 8" inches, Center rail to be 8" inches, Bottom rail to be 10" inches. Any other dimensions must be approved in writing by Owner (SVVSD Lockshop).
      f. Mechanically fasten corners with clips, SIGMA deep penetration plug welds and 1-1/8" long fillet welds inside and out-side of all four corners. Anodized finish must be removed prior to welding.
      g. Glazing stops shall be hook-in type with EPDM glazing gaskets reinforced with non-stretchable cord.
      h. Accurately fit and secure joints and corners. Make joints hairline in appearance.
      i. Prepare component with internal reinforcement for door hardware.
      j. All butt hinges to be mounted on (minimum 5/32" thick) zinc-plated steel back-up plates secured with stainless steel screws (stainless steel mandatory on exterior doors only). (Stainless steel back up plate mounting screws should be secured with a Loctite thread-locking – blue or similar product)
   3. Hinges: Hinges on Aluminum doors to be equal or better than the specifications for those specified in hardware schedule:
      a. Exterior doors should have 5" x 4.5" heavy weight, four ball-bearing, stainless steel hinges, w/ non-removable pins (630).
      b. Interior doors should have 5" x 4.5" heavy weight, four ball-bearing, brushed steel hinges (626).
      c. Doors over 36" wide must be approved in writing by Owner (SVVSD Lockshop).
      d. Doors over 7"-6" high must be approved in writing by Owner (SVVSD Lockshop).
      e. Doors over 7'-0" high must have 4 hinges each.
   4. All surface mounted hardware on the door must be installed with the primary use of sex bolts.
      a. NOTE: Template should be used to ensure square installation of sex bolts.

b. Holes should be properly sized for snug fit. Loose sex bolts will be rejected.
c. Any secondary attachments may be drilled and tapped.

5. Thresholds and Door sweeps:
   a. DO NOT use manufacturer's thresholds.
   b. Use thresholds specified in Hardware Division 08 70 00 Section J.1.Q.
   c. Attach thresholds with stainless steel (2" minimum) screws. DO NOT use screws provided with threshold.
   d. DO NOT use manufacturer's door sweeps.
   e. Use sweeps specified in Hardware Division 08 70 00 Section J.1.P.
   f. Pre-drill holes and attach with stainless steel screws only.
   g. NOTE: If floor surface does not slope sufficiently for sweep to clear, raise threshold and cut door as needed.
6. Drip Caps:
   a. Use on ALL exposed exterior doorways with less than 4' of overhang or with an overhang higher than 4' from the door.
   b. Use drip caps specified in Hardware Division 08 70 00 Section J.1.N.
   c. Mount with stainless steel screws only.
7. Arrange fasteners and attachments to conceal from view.
8. Finish: Clear anodized or light colored finish. Thermal expansion of frames and doors an issue.

D. GLAZED ALUMINUM CURTAIN WALL ASSEMBLIES
1. Framing Members: Manufacturer's standard extruded- or formed-aluminum framing members of thickness required and reinforced as required to support imposed loads. Thermally broken.
2. Concealed Flashing: Manufacturer's standard corrosion-resistant, nonstaining, nonbleeding flashing compatible with adjacent materials, ASTM A 240/A 240M of type recommended by manufacturer.
3. Provide caulked cosmetic seals in addition to critical seals at perimeter where curtain wall assembly meets masonry or other wall surfaces.
4. Finish: Clear anodized or light colored finish. Thermal expansion of frames and doors an issue.
5. Basic Wind Speed: 110 mph, Exposure C.
6. WARRANTY
   a. Special Assembly Warranty: Standard form in which manufacturer agrees to repair or replace components of glazed aluminum curtain walls that do not comply with requirements or that fail in materials or workmanship within specified warranty period. Warranty Period: Five years from date of Substantial Completion.
   b. Special Finish Warranty: Standard form in which manufacturer agrees to repair finishes or replace aluminum that shows evidence of deterioration of factory-applied finishes within specified warranty period. Warranty Period: 10 years from date of Substantial Completion.

E. MANUFACTURERS OF ALUMINUM STOREFRONT AND CURTAINWALL ASSEMBLIES
1. Available Manufacturers: Subject to compliance with requirements, use one of the following:
   a. EFCO Corporation
   b. Vistawall
   c. Kawneer Company
   d. Tublite
   e. Other as approved by Owner

08 50 00 Windows

A. SUMMARY
   1. Steel Windows: Not permitted.
   2. Aluminum Windows: *Preferred*.
   3. Stainless Steel Windows: Not permitted.
   4. Wood Windows: Preferred (aluminum clad).
   5. Plastic or Vinyl Windows: Not permitted.
   6. Fiberglass Windows: Considered with District approval.

B. GENERAL
   1. Operable sash with insect screen preferred for most exterior applications in assignable spaces. Doors and Windows.
   2. Natural light preferred throughout the exiting system, especially stairs.
   3. For purposes of energy conservation, glazing should be limited to 20% or less of the total exterior wall area.
   4. Locate openings to minimize solar heat gain.
   5. For glazing over 10'-0" above first floor elevation: Inside glazing preferred.
   6. Verify method(s) for cleaning and replacement.
   7. Utilize tinted glazing to minimize solar heat gain and eye strain to occupants.
   8. Largest single opening for glazing shall be 8’ x 8’.

C. WARRANTY
   1. Extended warranty: Underwritten by window manufacturer, 10 years minimum term from the date of project acceptance covering defects in materials and workmanship.
   2. Ten (10) years minimum term from the date of project acceptance covering specified performance standards.

D. ALUMINUM WINDOWS
   1. Thermal-break design required, PVC materials in concealed locations only. Internal and external drainage to exterior.
   2. Reinforce frame at hardware and door locations.
   3. All points of contact between the operable and non-operable components of the door shall be weatherstripped.
   4. Operable Units: Windows to have cam handle locks and 4-bar hinges with 15° limit stops. Operable Units should not have screens.
   5. Provide not less than 2 operable window for each classroom/teaching space.

E. WOOD AND PLASTIC WINDOWS
   1. All window styles and configurations to be reviewed and approved by Owner during design process.
   2. Wherever possible, attempts should be made to match existing windows adjacent to new construction, unless otherwise stated by Owner.
   3. Wood framed window assembly (including glass, sash and frame) U-factor shall be rated, certified and labeled in accordance with NFRC 100.
   4. Wood: Ponderosa pine or similar fine-grained lumber.
   5. Finger joints permitted only for clad assemblies; prohibited for surfaces to be painted.
   6. Use FSC-certified sustainably harvested wood.
   7. Cladding: Extruded aluminum with custom-color baked acrylic enamel, Kynar or equivalent, required for all exterior surfaces.
   8. Provide not less than 2 operable window for each classroom/teaching space.

08 60 00 Roof Windows and Skylights

A. SUMMARY
   1. Section Includes
   2. Tubular daylighting device, consisting of roof dome, reflective tube, and diffuser assembly
   3. Plastic (Lexan) Unit Skylights.
   4. Metal Framed Skylights.

B. APPLICATIONS/RESTRICTIONS
   1. For reasons of acoustics, security, and maintenance; vertical glazing, monitors, or clerestories are preferred over skylights.
   2. When provided, limit skylights to non-assignable, public, circulation areas equipped with security sensors.
   3. Review Maintenance Issues with District personnel regarding appropriate size limits, accessibility for repair/replacement and overall weather resistance and material life of application.

C. METAL FRAMED SKYLIGHTS
   1. Engineered by manufacturer to prevent wind blow-off and live/dead loads per IBC with
   2. minimum air filtration per ASTM E283 and no uncontrolled water leakage per ASTM E331
   3. testing.
   4. Water test skylights upon completion. Thoroughly wet the entire skylight area in accordance
   5. with referenced test and check for uncontrolled water leakage.
   6. Manufacturers: Subject to compliance with requirements, but are not limited to the following:
   7. Vistawall - HP-1175 sloped glazing with integral condensation gutters and 1" double sealed insulating glass with tempered outside glass and laminated glass inside face. Finish of aluminum as approved by the School District.
   8. Kawneer

D. TUBULAR DAYLIGHTING DEVICES
   1. Tubular Daylighting Devices General : Transparent roof-mounted skylight dome and self-flashing curb, reflective tube, and ceiling level diffuser assembly, transferring sunlight to interior spaces; complying with ICBO/ICC AC-16.
   2. SolaMaster Series: Solatube Model 750 DS-C & O, 330 DS C & O 21 inch (530 mm) Daylighting System, Model 750DS Dome
   3. LightTracker Reflector, made of aluminum sheet, thickness 0.015 inch (0.4 mm) with Spectralight Infinity. Positioned in the dome to capture low angle sunlight.
   4. Roof Flashing Base: One piece, seamless, leak-proof flashing functioning as base support for dome and top of tube.
   5. Interior Finish: high reflectance specular finish on exposed reflective surface. Specular reflectance for visible spectrum (400 nm to 760 nm) greater than 99.7 percent. Total solar spectrum reflectance (400 nm to 2500 nm) less than 93 percent.
   6. Reflective extension tube, Type EXX, Notched for Open Ceiling diffuser attachment, 24 inches (610 mm) long
   7. Lens: Type L2, Prismatic lens designed to maximize light output and diffusion. Visible Light Transmission shall be greater than 90 percent at 0.100 inch (2.5 mm) thick. Classified as CC2.
   8. Local Dimmer Control: Provided with dimmer switch and cable.
   9. Daylight Dimmer: Type D Electro-mechanically actuated daylight valve; for universal input voltages ranging between 90 and 277 V at 50 or 60 Hz; maximum current draw of 50 ma per unit; controlled by low voltage, series Type T02: circuited, 4 conductor, size 22 cable; providing daylight output between 2 and 100 percent. Provided with dimmer switch and cable.
   10. Switch: Type SW, Manufacturer-specific low voltage DC DP/DT switch (white) required to operate Daylight Dimmer. Note: only one switch is required per set of synchronously controlled dimmers.
   11. Warranty: Manufacturer's standard warranty for 10 years.
   12. Manufacturers: Subject to compliance with requirements, but are not limited to, the following (Other products / companies to be reviewed and pre-approved prior to bid):

a. Solatube International, Inc.

E. PLASTIC SKYLIGHTS

1. Skylights shall be double-dome plastic skylights and shall be engineered by the manufacturer to revent wind blow off and to provide a minimum carrying capacity of 40 lb. live load per square foot of skylight.
2. Glazing shall be one piece continuous cast acrylic resin plastic, with tinted outside dome and clear inside dome. Mounted on thermal barrier extruded aluminum frame with integral condensate gutter.
3. Water test skylights upon completion. Conduct test using a hose without nozzle and thoroughly wet the entire skylight area.
4. Manufacturers: Subject to compliance with requirements, but are not limited to the following:
5. Skyline Products, Inc.
6. Naturalite/EPI
7. Plasticrafts
8. A.I.A. Industries

08 71 00 Door Hardware

A. GENERAL
   1. Hardware submittals shall be reviewed by the District.
   2. Hardware schedules shall be prepared by or under the supervision of a DHI Certified Architectural Hardware Consultant (AHC) or a Certified Door Consultant (CDC).
   3. Power transfers @ electrified hardware:
      a. Von Duprin EPT-10, door frame to door edge.
      b. Electrified hinges are discouraged, but can be used where the use of a power transfer is not possible.
      c. Electric door loops are prohibited.

B. EXECUTION AND INSTALLATION
   1. See **Appendices D and E, SVVSD Door & Hardware Installation Guides.**
   2. Include SVVSD Technical representatives in pre-construction meetings.
   3. Furnish the District with a complete set of specialized tools and maintenance instructions.

C. KEYING SYSTEMS
   1. *District Standard* Products:
      a. Small Format Interchangeable Core (SFIC)
         1. Schlage Everest B SFIC or
         2. Best Access Control Coremax X-Series SFIC
      b. Cylinder core housing
         1. Schlage SFIC Rim and Mortise 7-pin
         2. Best Access SFIC Rim and Mortise 7-pin
   2. Special Instructions
      a. Contact Owner (SVVSD Lockshop) to finalize keying requirements and obtain keying instructions in writing.
      b. Key System will be coordinated by SVVSD Lockshop with Supplier's AHC representative and school facility directly.
      c. Provide the latest software upgrade for Sitemaster for tracking the new key system to the SVVSD Lockshop.
      d. Provide one complete digital copy of the bitting list (specific to this current school project only) of key cuts capable of being imported into Sitemaster Software. This should be emailed directly to the SVVSD Locksmith.
      e. Furnish temporary interchangeable core construction keying system for each new lock and exit device during construction period. This will allow installer to verify the function of each lockset and exit device PRIOR to the installation of the permanent cores by Owner.
      f. Owner shall remove all existing cores (if applicable; for remodel and rekeying projects) and install all new permanent keyed cores and return temporary interchangeable construction cores to General Contractor for return to the hardware supplier.
      g. Master keying pin combinations will be generated by manufacturer's (Schlage Lock / Best Access) keying system, and coordinated with Owner (SVVSD Lockshop).
      h. Permanent Cores: Keying shall be performed by Lock manufacturer. Furnish blank cylinder face, no logo; finish face to match lockset; interchangeable cores –
      i. Cores shall be factory keyed and combinated as directed by Owner (SVVSD Lockshop).
      j. Keys and cores shall be stamped with the applicable key mark for identification.
      k. Key identification stamping to be approved by Owner (SVVSD Lockshop).
      l. Key Material: Furnish keys of nickel silver.
      m. Do not package permanent keys with locks. Package keys separately from locksets and cores.
      n. Failure to comply with these requirements may be cause to require replacement of all or any part of the cores and keys involved as deemed necessary at no additional cost to the Owner.

o. Deliver all permanent keys, key blanks and other security items direct to Owner (SVVSD Lockshop) from lock manufacturer by secure courier, return receipt requested.
p. Key Quantity: Furnish keys in the following quantities:
    1. 15 each Temporary construction keys to General Contractor.
    2. 10 each Temporary construction keys to Owner (SVVSD Lockshop).
    3. 5 each Construction control keys to General Contractor.
    4. 5 each Construction control keys to Owner (SVVSD Lockshop).
    5. 20 each Master Keys for each master used, to Owner (SVVSD Lockshop).
    6. 2 each Change keys per core to Owner (SVVSD Lockshop).
q. Additional Key blanks to Owner (SVVSD Lockshop):
    1. 250 Key blanks for High Schools.
    2. 125 Key blanks for Middle Schools
    3. 100 Key blanks for Elementary Schools
r. Key Storage System: Provide wall mounted metal key storage cabinet.
s. Elementary or Middle Schools:
    1. Use HPC key cabinet - KEKAB-KK95X (capacity 95 expandable to 265 with KEKAB-DA expansion leaf)
t. High Schools or K8/P8 Schools:
    1. Use HPC key cabinet – KEKAB-KK265

D. HINGES

1. *Preferred* types and materials
    a. Butt type (5 knuckle) are preferred.
    b. Base Material:
        1. @ Exterior and high humidity locations: stainless steel 630 finish
        2. @ Interior: Steel, brushed chrome w/ non-rising steel pin (626 / 26d) finish.
        3. Screws to match hinge material.
2. *Preferred* Products

Bommer:	BB5000	BB5004	BB5006
Hager:	BB1279	BB1168	BB1199
Ives:	5BB1	5BB1HW	5BB1HW
Stanley:	FBB179	FBB169	FBB199
McKinney:	TA2714	T4A3786	T4A3386

3. *Discouraged* types and materials
    a. Continuous hinges
4. Hinge Pins: Except as otherwise indicated, provide hinge pins as follows:
    a. Exterior Out-Swing Doors with Locks: Non-removable pins.
    b. Interior Doors: Non-rising pins.
    c. Tips: Flat button and matching plug, with a 3/32" hole, finished to match leaves.
5. Number of Hinges:
    a. Provide number of hinges indicated but not less than 3 hinges for door leaf for doors 84" or less in height
    b. Provide 4 hinges per door leaf greater than 84" in height.
    c. Provide 4 hinges per door leaf greater than 36" in width.
6. Hinges Weight and function:
    a. All hinge open widths shall be minimal, but of sufficient size to permit door to swing 180 degrees.
    b. Unless otherwise specified, hinge size for doors through 3'-0" shall be standard weight, 0.134 gage, 4.5" x 4.5".
    c. For doors over 3'-0" wide shall be 5" x 4.5" four ball- bearing, heavy weight, 0.190 gage.
    d. ALL Exterior doors will use heavyweight (0.190 gage) ball-bearing (NRP) hinges

e. Areas with “Heavy Use” doors will also use heavyweight (0.190 gage) hinges. These areas include but may not be limited to:
    1. Elementary Schools
        a. Main interior access doors to the cafeteria
        b. Main interior access doors to the gymnasium
        c. Restroom doors
        d. Cross corridor security doors
    2. Middle Schools
        a. Main interior access doors to the cafeteria
        b. Main interior access doors to the gymnasium
        c. Restroom doors
        d. Main interior access doors to the auditorium
        e. Cross corridor security doors
    3. High Schools
        a. Main interior access doors to the cafeteria
        b. Main interior access doors to the gymnasium
        c. Restroom doors
        d. Main interior access doors to the auditorium
        e. Locker rooms
        f. Weight / Wrestling rooms
        g. Cross corridor security doors

E. LOCKSETS, LATCHSETS AND DEADBOLTS
1. *District Standard* Products:
    a. Schlage Lock Company, ND Series
        1. Sparta style lever.
        2. SFIC
    b. Schlage Lock Company, L Series Mortise
    c. Schlage Lock Company, M Series Deadbolts
    d. Acceptable alternate: Best Access Systems, products equivalent to District Standard.
    e. At locksets with electronic access control, electric strikes are preferred over electrified locksets.
2. Locations and Functions
    a. EXTERIOR
        1. Storage, mechanical, custodial.
            a. Storeroom function – Lever is fixed, entry by key only.
        2. Card access, single door.
            a. Storeroom function – Lever is fixed, entry by key only, or electronic release.
        3. Exit only door
            a. Exit Function – Outside lever always fixed, inside lever always free for immediate egress.
    b. INTERIOR
        1. Classrooms / Staff Offices / Generally any room that can be occupied in Lockdown scenario:
            a. Entrance function – Outside key retracts latch when inside button is pushed and turned. Inside button locks outside lever by pushing, or by pushing and turning. Inside lever is always free for immediate egress)
        2. Multi-User (common restrooms) at Middle and High Schools (Elementary may use Push/Pull hardware without locking or latching)
            a. Classroom Function – Outside lever is locked or unlocked by key only.
        3. Communicating doors between classrooms and offices.
            a. Communicating function – Both levers unlocked only by key.
        4. Storage or mechanical rooms / janitor’s closets.
            a. Storeroom function – Exterior lever is fixed, entry by key only.
        5. Faculty Restrooms

a. Mortise lock w/ Occupancy Indicator.
    i. Schlage L9496 or Best 45H-7-T-14-S-VIN

F. EXIT DEVICES
1. *District Standard* Products:
    a. Exterior: Von Duprin XP98 Series (XP = Extra Protection)
    b. Interior: Von Duprin 98 Series
2. Dogging
    a. Less Dogging @ all exterior doors.
    b. Cylinder dogging only where specifically requested, spaces may include:
        1. Interior: Libraries, Auditoriums, Cafeterias, Gymnasiums, Corridors.
    c. Hex dogging is prohibited.
3. Provide Quiet Electric Latch (QEL) at electrified doors.
4. Provide Pushpad Monitor Switch
5. Locations and Functions
    a. EXTERIOR
        1. Main or Bus loop exterior entrance with card access (ALL doors must be electrified)
            a. Active Leaf: [RX-LX-LD-QEL-XP98NL x 990NL] (only one leaf at each entry needs key override cylinder)
                i. Function/Trim: Night Latch
            b. Inactive Leafs: [RX-LX-LD-QEL-XP98DT x 990DT]
                i. Function/Trim: Dummy
        2. Accessory exterior entrances with card access (only one leaf must be electrified)
            a. Active Leaf: [RX-LX-LD-QEL-XP98NL x 990NL] (only one leaf at each entry needs key override cylinder)
                i. Function/Trim: Night Latch
            b. Inactive Leaf [LD-XP98EO x 990EO No pull]
                i. Function/Trim: Exit Only
        3. Exit only doors (Auditorium, Gym, locker room exits, or classrooms that are exit only)
            a. [LD-XP98EO x 990EO No Pull]
                i. Function/Trim: Exit Only, no pull
        4. Exterior Alarmed exit door (possibly SPED areas)
            a. [ALK-XP98NL x 990NL] Alarmed exit device w/ power supply
                i. Function/Trim: Night Latch
    b. INTERIOR
        1. ALL non-rated occupied rooms requiring exit devices (except large gathering rooms listed next)
            a. [LD-98NL (device) w/ 990NL] fixed pull (Entry by key only)
                i. Function/Trim: Night Latch, Fixed Pull
        2. ALL non-rated (large gathering) occupied rooms requiring exit devices (Auditoriums, Gymnasiums, Library, and non-rated corridor doors)
            a. [CD-98NL (device) w/ 990NL] fixed pull (Entry by key only or push/pull when dogged)
                i. Function/Trim: Night Latch
                ii. Inside thumbturn rim cylinder will be Schlage 09-900 NH 626 w/ B502-191 cam
                iii. Blocking ring will be Schlage 36-079-037 626
                iv. or Best Access Systems equivalent
        3. ALL Fire-rated occupied rooms requiring self-latching exit devices
            a. [98L-2-F x 996L x 17] lever trim w/ Double cylinder
                i. Inside cylinder will be Schlage XB11-979 thumbturn rim cylinder
                ii. Provide Von Duprin Lock/Unlock Label #971-220-00
                iii. NOTE: Cover will need to be scored for label to adhere properly
        4. Mechanical rooms requiring exit devices
            a. [LD-98L-NL (device) w/ 990NL] fixed pull (Entry by key only)

i. Function/Trim: Night Latch

5. Library Alarmed exit
    a. [ALK-98NL x 990NL] Alarmed exit device w/ power supply
        i. Function/Trim: Night Latch

G. CLOSERS / HOLDERS

1. District Standard Products:
    a. Door Closers:
        1. LCN 4041 XP Series EDA ST3596 (HEDA or SHCUSH for most door hold open needs)
            a. No spring cush arm
    b. Magnetic Door Hold-Opens:
        1. Recessed mounting: Rixson Model 994M, total projection 3-3/4" from wall
        2. Surface mounting: Rixson Model 991M
    c. Mechanical Door Hold-Opens:
        1. Wall mounted mechanical door holders are prohibited.
        2. Overhead door mount: Glynn Johnson 90H
        3. Wherever possible, use hold open closer arm at door closer.
2. Special Instructions
    a. ALL closers to be mounted on the push side, parallel arm mounting.
    b. When replacing closers on existing doors DO NOT use the existing holes unless they are correct for the conditions.
    c. Furnish extra duty arms, EDA, to protect against excessive force.
    d. Furnish TBWMS fasteners; drill and tap for machine screws in all fully reinforced hollow metal doors. Use through bolts in all wood doors, wood fire doors, and non-reinforced hollow metal applications.
    e. Refer to door and frame details, furnish accessories such as drop plates, special templates, spacers and supports as required to correctly install door closers on a solid, stable surface. (Example: Doors with projecting light trim requiring special blocking)
    f. Install closers to allow maximum degree of opening, position backcheck to activate well in advance of the stop position to cushion the opening swing and prevent door and frame damage.
    g. DO NOT use the standard EDA door closer arm to stop door travel. Acceptable methods of door stop function in order of preference are as follows:
        1. <u>Exterior doors:</u>
            a. Door swings to wall bumpers installed in adjacent wall set in epoxy (Ives FS18S)
            b. Door swings full 180º to frame assembly with hard rubber wall bumper or floor stop set in epoxy (Ives FS18S)
            c. Door swings to Bollards with high density rubber stops. (D-Style dock bumpers) (See Division 05 50 00 / G.1-4)
            d. Door swings to overhead stop – This is ONLY for ADA operated doors with no bollard stop possible.
            e. SCUSH EDA arm on door closer.
        2. <u>Interior doors:</u>
            a. Wall bumpers installed on adjacent wall. **(Verify Blocking)**
            b. SCUSH EDA arm on door closer.
            c. Overhead door stop (Glynn-Johnson 90 series) Use only on doors with no closer.
            d. High density rubber floor stop. (Ives FS18S, FS18L) This is ONLY for doors with no wall bumper stop possible.
    h. Hold Open Function applications – Closer arm hold functions are as follows:
        1. <u>Exterior doors:</u>
            a. No hold open functions on any exterior door that allows access to interior common areas or classrooms. (Exception: Rear Kitchen Entry)
        2. <u>Interior doors:</u>
            a. No hold open functions on fire rated openings or doors specified to have magnetic hold opens.

b. Hold open (HEDA) arms should be used on ALL interior doors where holding the door does not conflict with applicable codes.
c. Spring Cush Hold Open (SHCUSH) arms may be used in the above listed applications when wall bumpers are not possible.

i. Doors swinging into ANY corridors should fully swing to the wall 180º as needed.
j. Adjust all door closers to compensate for final operation of heating and ventilating equipment.
k. Minimum spring tension settings on the closer should be used to comply with the referenced ADA accessibility requirements while ensuring the door will self-latch.

H. ADA AUTO OPERATORS AND ACTUATORS

1. District Standard Products
   a. LCN: 4642-CS Series with ST-3420 (mount on the push side only)
   b. ST-3420 special template on double doorways with mullions
2. Special Instructions
   a. Use 'Flush Ceiling' mount in conditions where wall bumper or bollard stop is not possible. This allows space for overhead stop installation.
   b. Provide key operated power disconnect within 6' of operator.
      1. Use Best Access Systems [1W7C2 SPST] key switch mounted in single or double gang electrical box with metal finish cover.
   c. General or Electrical Contractor to install all wiring to operator on a separate circuit breaker routed into header as specified below.
   d. All wiring and/or connections made to the operator shall be routed through the hollow metal frame or aluminum in to the back of the unit. No surface conduit or flex conduit is to be used. If the frames are not accessible, wiremold or other metal raceway that matches the door frame can be used.
   e. Actuators
      1. Use two per doorway mounted within 4' of ADA door.
      2. Jamb mounted actuators are not permitted without approval from Owner (SVVSD Lockshop).
      3. Actuators should be connected with low voltage wiring from actuators to operator wherever possible. Wireless digital transmitters/receivers can be used if hard wire option is not possible.
      4. Use 4" round button with depressed marking and "Push to operate door" with wheelchair logo.
      5. Locate buttons near the card reader if card reader is within 4' of ADA door as noted above.
      6. Mount "Caution Automatic Door" decals, centered on the glass at approximately 52"a.f.f.

I. MULLIONS

1. District Standard Products:
   a. Von Duprin KR Series
2. Special Instructions
   a. Keyed security removable mullions shall accept security cylinders of specified cylinder manufacturer and be removed by use of the cylinder. Mullions to be furnished with a self-locking mechanism for reinstallation without the use of the cylinder.
   b. Provide Cylinder with Schlage blocking ring 36-079-012-626 1/8" blocking ring. NO Compression Rings!
   c. Provide mullions without strike prep.
   d. Provide a self-adhesive mullion seal (minimum 3/16" thick) on all doorways requiring a mullion.
   e. Equip each mullion with mullion stabilizers to maintain integrity between door and mullion to prevent vandalism.
   f. All removable mullions should be supplied with a storage bracket or device to hold mullion when removed from doorway. These brackets must be attached to masonry walls or metal frames. No drywall mounted locations are permitted without solid blocking.

J. PROTECTIVE PLATES

1. Preferred Products:
   a. Ives 8400, (0.050) B3E

b. Rockwood K1062, (0.050) B3E
c. Or Similar

2. Special Instructions
   a. Fabricate protection plates, armor, kick or mop, not more than 2 inches less than door width on stop side and not more than 1 inch less than door width on pull side. Mount ¼" from bottom of door.
   b. Size plates to provide clearance for grills, louvers and door lites.
   c. Metal Plates: Stainless steel plates (0.050).
   d. Provide Armor plates on the doors at the following locations:
      1. Band classrooms and corridor doors.
      2. Kitchen area entry cafeteria access doors. Mount with stainless steel screws only.
      3. Custodial closets and work areas Mount with stainless steel screws only.
      4. Provide mop plates on pull side of the following locations: Mount with stainless steel screws only.
      5. Custodial closets with mop sinks that swing into the room.
      6. Restroom doors that swing into the room.
      7. Any wood door that swings over a hard surface.

K. THRESHOLDS, SWEEP, AND DOOR BOTTOMS
   a. Exterior doors: All doors must have thresholds designed to comply with ADA standards.
      1. Preferred Products
         a. Thresholds:
            i. Pemko 2005AT Door stop/seal threshold Clear Aluminum or
            ii. Equivalent by NGP
         b. Door Bottom Sweeps
            i. Preferred type: Nylon brush
            ii. Discouraged type: Neoprene
            iii. Manufacturer: Pemko or sim.
      2. Special Instructions
         a. All doors must have sweeps that make contact with the threshold AND ample clearance over concrete.
         b. ALL doors must have threshold at sufficient height allowing sweeps ample clearance over the concrete.
            i. NOTE: This may require additional modular ramp components to compensate for existing conditions.
         c. ALL thresholds must be installed with stainless steel screws (2" minimum length).
         d. ALL sweeps must be installed with stainless steel screws where exposed to weather.
         e. ALL thresholds must be sealed in full bed of butyl-rubber or polyisobutylene mastic sealant complying with requirements specified in Division 7 Section "Joint Sealers."
   b. Interior Doors: For sound mitigation
      1. *Preferred* Products:
         a. Thresholds:
            i. Pemko: 2001AT Stop threshold Clear Aluminum
            ii. Equivalent by NGP
         b. Door Bottoms:
            i. Pemko: 222AV or 220AV door bottoms Clear Aluminum
            ii. Equivalent by NGP
      2. Locations: Use ADA compliant interior coverplates and thresholds on the following:
         a. Music, Vocal, and Band rooms, any classrooms or labs in close proximity to loud rooms like those listed above.
      3. Automatic door bottoms are prohibited.

L. DRIP CAPS, PERIMETER, AND MULLION SEALS
   1. *Preferred* Products:

a. Drip Caps: Use over ALL exposed exterior doorways. Mount with stainless steel screws only.
    1. Pemko: 346C – mounted and sealed to exterior door frame header in Clear Aluminum
    2. Equivalent by NGP
b. Perimeter Seals:
    1. Exterior doors: All doors must have weather seals installed with stainless steel screws.
        a. Pemko: 303CS – Silicone (bulb-type) Clear Aluminum
        b. Equivalent by NGP
    2. Interior doors: For sound mitigation
        a. Use Pemko: 305CR EPDM sponge type Clear Aluminum
        b. Equivalent by NGP
c. Smoke Seals:
    1. Pemko: S44 or S88 series self-adhesive seals
    2. Equivalent
d. Meeting Stiles:
    1. Pemko: 18041ANB – Nylon brush Clear Aluminum
    2. Equivalent by NGP
e. Mullion Seals:
    1. Use Pemko: 5110BL
    2. NGP 3/16” thick equivalent

08 80 01 Glazing Security Film

A. Preferred products
    1. Clear Armor
        a. SL-15
        b. SL-31
    2. Substitutions will be considered, but require approval from the SVVSD office of Safety and Security.
B. General Glazing Security Film Locations, coordinate per project requirements.
    1. SL-15, or similar:
        a. Exterior windows up to 6’-0” AFF.
        b. Interior glazing at sensitive locations, coordinate with SVVSD Safety and Security.
    2. SL-31, or similar
        a. Glazing at main entrances.
        b. Transaction windows at secure waiting rooms.

DIVISION 09 – FINISHES

09 00 00 Primary District Requirements

A. PAINTING
1. Furnish Owner with clean, new, one gallon cans of paint used on the project. Containers shall be well marked with the color name and construction document designation (i.e. PNT-1)
2. Standard Colors:
    a. White
        1. Sherwin Williams: Natural Choice White
    b. Other colors shall be unique to each project; however, the District preference is to limit the number of colors in a building to 5.
3. Paint types:
    a. All paint selections shall conform to manufacturer's recommendations or Master Painters Institute (MPI) systems for appropriate surface preparation and substrate suitability.
    b. District prefers Sherwin Williams paint.
4. Sheen
    a. District preference for all interior surfaces is **semi-gloss**.
    b. Provide highest sheen possible for dry-fall paint.
    c. Other sheens may be specified where specific requirements exist.
5. Texture
    a. District preference is for a light orange peel texture.

B. WALLS
1. Wall protection

Preferred	***Acceptable***	***Discouraged***
Ceramic Tile	Mechanically Fastened solid panel, ½" min. thickness, i.e. plastic laminate faced MDF, veneered plywood, or similar.	Adhered panels Vinyl wall coverings or wall paper Panels less than ½" thickness

2. Writable Walls
    a. Provide magnetic surfaces where possible.
    b. Consider projection quality.
    c. Preferred Systems
        1. Porcelain enamel boards or panelized writable walls, by Claridge or a similar company.
        2. Glass writable walls, by Claridge or a similar company.
    d. Acceptable Systems
        1. Writable Plastic Laminate
        2. Writable Wall Covering (i.e. "Wall Talker")
    e. Discouraged Systems
        1. Writable paint
3. Ceramic Tile
    a. Terminations and edge protection
        1. Provide details and products that avoid sharp edges and corners.
        2. Preferred product: Schluter Rondec or similar
            a. Provides smooth outside 3-way corner (e.g. for tops of wainscot outside corners)

C. CEILINGS

1. Acoustical Ceiling Tiles
    a. Minimum Noise Reduction Coefficient (NRC): 0.70
    b. Minimum Ceiling Attenuation Class (CAC): 35
    c. Minimum Reflectance: 0.75

09 60 00 Flooring

A. FLOOR FINISH PREFERENCE MATRIX
   1. This matrix is intended to reflect general District preferences and should be viewed as a starting point.
   2. ***The District encourages designers to present alternative options.***

Space	***Finish Preferences***		
	Preferred	***Acceptable***	***Discouraged***
Admin./Office	-Carpet -Resilient Sheet	-Woven Textile Composite -LVT	
Cafeterias	-Polished Concrete -Flexible Terrazzo Tile	-LVT -Resilient Sheet	-VCT/BBT
Classrooms	-Carpet -Resilient Sheet	-LVT -Woven Textile Composite	-VCT/BBT
Corridors (elementary)	-Carpet	-Polished Concrete -Woven Textile Composite -LVT	-VCT/BBT
Corridors (middle & high)	-Polished Concrete -Carpet	-Woven Textile Composite -LVT	-VCT/BBT
Gyms	-Wood Athletic Flooring (HS/MS) -Multi-Sport Sheet (ES)	-Fluid-Applied Athletic Flrg	-Carpet
Kitchens	-Resinous Flooring -Quarry Tile		
Restrooms, Locker Rooms	-Polished Concrete -Resinous -Ceramic Tile -Resilient Sheet	-Rubber Sheet	-Sealed concrete -VCT/BBT
Science / Laboratory / Art	-Polished Concrete	-Resilient Sheet -Resinous	-VCT
Vestibules / Entries	-Walk-Off Carpeting (broadloom) -Polished Concrete with: -Recessed Matting		

B. CARPET
   1. General
      a. Avoid solid colors
      b. Select designs with patterns to obscure stains and wear.
      c. Consult SVVSD Project Manager and Carpentry Department for carpet tile vs broadloom preference for each project.
   2. Texture Appearance Retention Rating (TARR)
      a. 3.0 TARR minimum.
   3. Fiber
      a. Preferred: Nylon 6,6 or
      b. Type 6 nylon must be approved by the District for low impact locations or temporary installations.
   4. Gauge
      a. 1/12 inch min.
   5. Face Weight
      a. 14oz or greater

6. Density
    a. 5,000 min.
7. Backing
    a. Moisture barrier backing
8. Adhesive
    a. High moisture adhesives only.
    b. Modular adhesive tabs are prohibited.
9. Warranty
    a. 15 year minimum, non-prorated
10. Products
    a. Preferred Products:
        1. Mannington
        2. Shaw Contract
        3. Tarkett
    b. Other manufacturers able to submit products complying with the requirements of this section.

C. CERAMIC & PORCELIN TILE
1. Tile
    a. Domestic products are strongly preferred.
    b. Preferred recycled content, greater percentages and consideration for Post-Consumer content are encouraged.
        1. Pre-Consumer: 25%
2. Mortar & Grout
    a. Select dark colored grouts to obscure staining.
    b. Epoxy grout at all flooring applications.
3. Accessories
    a. Stone Thresholds: Install stone thresholds at locations where mortar bed (thickset) would otherwise be exposed above adjacent floor finishes, set thresholds in latex-portland cement mortar (thin set).
    b. Metal Edge Strips: Install at locations indicated or where exposed edge of tile flooring meets carpet, wood, or other flooring that finishes flush with top of tile.
    c. Waterproofing and Crack Isolation Membrane: Nonplasticized, chlorinated polyethylene faced on both sides with nonwoven polyester fabric.

D. FLEXIBLE TERRAZZO TILE
1. Preferred Manufacturers
    a. Fritztile, or similar

E. FLUID APPLIED ATHLETIC FLOORING
1. Polyurethane Flooring Over Rubberized Base Mat System: Total Thickness: Minimum 1/4 in; with minimum.07 in polyurethane.
2. Base Mat: Prefabricated rubber mat of recycled rubber granules in polyurethane binder.
3. Sealer: Manufacturer's standard two-component polyurethane compound designed to seal base mat before application of resin topcoat.
4. Resin: Two-component, solid, pigmented, self-leveling polyurethane without fillers, zero mercury formulation, with properties as follows:
5. Tensile strength: Minimum 1000 psi, per ASTM D 412.
6. Hardness: Minimum 70, when tested in accordance with ASTM D 2240 using Type A durometer.
7. Ultimate Elongation: Minimum 100 percent, per ASTM D 412.
8. Manufacturers: All products by the same manufacturer.
    a. Connor Sports Flooring; "Elastiplus": www.connorfloor.com.
    b. Robbins Sports Surfaces; "Pulastic 9+2": www.robbinsfloor.com.
    c. Action Floor Systems; "Hurculan MF": www.actionfloors.com.

9. Verify that concrete sub-floor surfaces are dry enough and ready for resilient flooring installation by testing for moisture emission rate and alkalinity in accordance with ASTM F 710.
10. Add language to specify the the slab prep for high moisture content as a base bid product or look at Concure like product in Division 3.

F. LUXURY VINLY TILE (LVT)
1. Minimum Wear layer thickness:
    a. Low Traffic Areas: 20 mil (0.5mm)(0.02")
    b. High Traffic Areas: 40 mil (1.0mm)(0.04")
2. Provide scratch and stain resistant protective coating.

G. MULTI-SPORT SHEET
1. Designed for multi-use, including: sports, assembly, tables & chairs
2. Sheet material with heat welded seams.
3. Preferred
    a. Gerfloor Taraflex Multi-Use

H. POLISHED CONCRETE
1. See Division 03

I. RECESSED WALK-OFF MATTING
1. Recessed entrance system with aluminum frame and heavy duty carpet inserts.
2. Fold-Back or Roll-Up designs preferred for cleaning.
3. Products: Ronick PediTread LP Entrance System, or similar.

J. RESILIENT SHEET
1. Homogeneous Vinyl Sheet & Tile
2. Preferred Products
    a. Johnsonite iQ Optima
    b. Gerfloor Taralay
3. Unacceptable
    a. Marmolium

K. RESILIENT STAIR ACCESSORIES
1. Resilient Stair Treads Standard: ASTM F 2169, Type TS (rubber, vulcanized thermoset). Class 1, Raised-rib design.
2. Nosing Style: Square, adjustable to cover angles between 60 and 90 degrees.
3. Nosing Height: 2 inches (51 mm).
4. Size: Lengths and depths to fit each stair tread in one piece.
5. Manufacturers: Subject to compliance with requirements, but are not limited to, the following:
    a. Johnsonite
    b. Burke Mercer Flooring Products; Division of Burke Industries, Inc.
    c. Mondo Rubber International, Inc.
    d. Musson, R. C. Rubber Co.
    e. Roppe Corporation, USA.

L. RESINOUS FLOORING
1. Slip resistant but cleanable wearing surface.
2. Provide integral base in rooms with floor drains
3. Do not provide integral base in rooms without floor drains
4. 100% Solids
5. Total thickness: 1/8" to ¼"

6. Manufacturers
    a. Dur-A-Flex
    b. Silikal
    c. Stonehard

M. RUBBER IMPACT FLOORING
1. Preferred Components
    a. Mondo, Advance + Sport Impact
    b. Acer Flooring
    c. Connor Sports Flooring
    d. Robbins Sports Surfaces
    e. Regupol

N. RUBBER SHEET FLOORING
1. Review finish requirements with manufacturer and Project Manager.
    a. Preferred Product: Johnsonite, Arcade

O. STAGE FLOORING SYSTEM
1. Per project requirements

P. VINYL COMPOSITION TILE (VCT) AND BIO-BASED COMPOSITION FLOOR TILE (BBT)
1. Tile Standard: ASTM F 1066, Class 1, solid-color tile or Class 2, through-pattern tile.
2. Wearing Surface: Smooth.
3. Moisture Testing: Per manufacturer's requirements
4. Available Manufacturers: Subject to compliance with requirements, manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following
    a. Armstrong World Industries, Inc.; BIOBASED TILE Migrations
    b. AB ColorPlus, American Biltrite (Canada) Ltd.
    c. Mannington Mills, Inc.
    d. Tarkett, Inc.
    e. Vinylasa Tile, Distributed by American Tile Inc.
5. **Installation Note:**
    a. Contractor shall be required to prepare new tile for initial waxing (removal of factory coating if required) and provide initial waxing. Upon project completion the new flooring shall be finished such that additional stripping and waxing shall not be required until the next annual cycle.

Q. WALK-OFF CARPETING
1. Type: Broadloom or roll-good Walk-off carpeting
2. Warranty: 15 year min.
3. Products: Provide a product marketed specifically for walk-off or entryway flooring.

R. WOOD ATHLETIC FLOORING
1. Preferred Manufacturer: Aacer, substitutions acceptable with District approval.
2. ~~Plywood Underlayment: APA rated, C-D Plugged, exterior glue, tongue and groove, 15/32 inch thick.~~
3. ~~Wood Sleepers: Standard grade; 48 inches long; kiln dried Eastern hemlock, fir, pine, or spruce.~~
4. ~~Preservative Treatment: Clear, penetrating, water-repellent wood preservative that protects against mold, mildew, staining, and decay fungi; complying with MFMA's written recommendations and applied by immersion.~~
5. ~~Resilient Pads: With air voids for resiliency and installed at manufacturer's standard spacing for product designation indicated above.~~
6. ~~Type: Manufacturer's PVC standard, Thickness: 7/16 inch.~~
7. ~~Resilient Underlayment: Flexible, multicellular, closed-cell, expanded polyethylene-foam sheet; nominal 2-lb/cu. ft. density.~~

~~8. Vapor Retarder: ASTM D 4397, polyethylene sheet not less than 6 mils (0.15 mm) thick.~~

S. WOVEN TEXTILE COMPOSITE FLOORING

1.	Wear layer	Woven Solution Dyed Polyester
2.	Backing	Polyester Felt Cushion
3.	Weight	4.5 oz. min.
4.	Recyclability	100%
5.	Recycled Content	>55% (post consumer/industrial)
6.	NRC	0.30 min.
7.	Products	Kinetex or similar

DIVISION 10 – SPECIALTIES

10 10 00 Interior Specialties

A. INFORMATION DISPLAYS
   1. Markerboards
      a. Material: Porcelain-Enamel, three-ply
      b. Accessories:
         1. Continuous map rail
         2. Continuous tray
         3. Music staff lines in music room.

B. TOILET PARTITIONS
   1. Material: Phenolic or Solid High Density Polyethylene (HDPE)
   2. Style: Overhead braced (at toilets and urinals)
   3. Hinges: Full height
   4. Wall Brackets: Full Height

C. CORNER GUARDS
   1. Material: Stainless Steel
   2. Length: 48" min.
   3. Profile: Type with light bend near the end of each leg, may be referred to as wings, huggers, or beveled.

D. TOILET ACCESSORIES
   1. Toilet Tissue Dispenser: Owner furnished, Contractor installed.
   2. Paper Towel Dispenser: Owner furnished, Contractor installed. (warm air dryers preferred)
   3. Liquid-Soap Dispenser: Owner furnished, Contractor installed.
   4. Grab Bar: 1-1/4 inch Stainless steel with Flange mount with concealed fasteners. Basis-of-Design Product: Bobrick Washroom Equipment, Inc.; Concealed Mounting with Snap Flange, Item #B-6806.
   5. Sanitary-Napkin Disposal Unit: Stainless steel, Surface mounted. Self-closing, disposal-opening cover,removable receptacle. Product: Bobrick Washroom Equipment, Inc.; ConturaSeries® Surfact-Mounted Sanitary Napkin Disposal, Item #B-270.
   6. Sanitary-Napkin & Tampon Vendor
      a. Free vend (no charge)
      b. Stainless steel
      c. Surface mount or recessed, pending project requirements.
      d. Bradley 4017-114000, or similar
   7. Mirror Unit: Stainless-steel angle, 0.05 inch (1.3 mm) thick or stainless-steel channel. Product: Bobrick Washroom Equipment, Inc.; Series Angle-Framed Mirrors, Item #B-290.
   8. Shower Curtain Rod & Curtain: 1-inch (25.4-mm) OD; fabricated from nominal 0.0375-inch- (0.95-mm-) thick stainless steel. Bobrick Washroom Equipment, Inc.; ClassicSeries™ Heavy-Duty Shower Curtain Rod, Item #B6107. Vinyl, minimum 0.008 inch (0.2 mm) thick, opaque, matte with integral antibacterial agent. Color: White. Minimum size 42 inch width by 72 inches (1828 mm) high. Bobrick Washroom Equipment, Inc.; Vinyl Shower Curtain, Item B204-2.
   9. Folding Shower Seat: L-shaped seat, designed for wheelchair access. Seat shall be phenolic composite of slat-type, Mounting Mechanism shall be Stainless steel, No. 4 finish (satin). Product: Bobrick Washroom Equipment, Inc.; Reversible Folding Shower Seat, Item #B-5181.
   10. Robe Hook: Double-prong unit. Stainless steel, Bright polished. Product: Bobrick Washroom Equipment, Inc.; ClassicSeries™ Double Robe Hook, Item #B-672.

11. Mop and Broom Holder: Unit with shelf, hooks, holders, and rod. Length: 36 inches (914 mm). Product: Bobrick Washroom Equipment, Inc.; Shelf with Mop and Broom Holders and Rags, Item #B-224.

E. Custodial Specialties
    1. Chemical Dispensers
        a. Hillyard Arsenal Quick Change, or sim
        b. Verify 2 or 4 bottle option with SVVSD Building Services Department
    2. Hand Sprayer (at mop sinks)
        a. See 22 40 00 Plumbing Fixtures
    3. Mop Sink
        a. See 22 00 00 Primary District Requirements
    4. Utility Shelf
        a. Bradley Model 9934 or sim.
            1. Stainless Steel
            2. 8"+/- shelf
            3. (4) rubber cam mop/broom holders
            4. (5) hooks

F. WARM AIR HAND DRYERS
    1. Preferred products:
        a. American Specialties: Model 0165
            1. 115VAC-50/60Hz
            2. 14.4 Amps
            3. 1653 Watts
    2. Acceptable Products:
        a. Dyson Airblade V

G. FLAG POLE
    1. Material: Aluminum
    2. Height
        a. Elementary: 30 feet
        b. High and Middle School: 35 feet
    3. Internal halyard, winch system: Manually operated winch system.

10 14 00 Signage

A. INTERIOR ROOM SIGNS
   1. Materials
      a. Panel:
         1. 1/8" non-glare acrylic with either 2nd surface or integrated color per manufacturer's standard range.
      b. Raised Lettering:
         1. 1/32" matte or non-glare acrylic with integrated color per manufacturer's standard range, surface applied with time-bond adhesive.
      c. Braille:
         1. Grade 2, per ADA guidelines.
      d. Clear Cover:
         1. 1/16" non-glare clear acrylic.
   2. Typical room signage
      a. SVVSD Typical Detal # 10-1
   3. ADA Symbol
      a. Where available, provide the "active" or "italicized" ADA symbol:
         1. 

B. BUILDING PLAQUE
   1. Material: Cast Bonze
   2. Size: 18"w x 24"h
   3. Information:
      a. Date
      b. Project Name
      c. School Board Members
      d. Architect
      e. Contractor

C. ROOM EVACUATION SIGNS
   1. Provide 8.5"h x 11"w acrylic sign holder in all student occupied rooms.
   2. Coordinate paper insert graphics with District.

D. EXTERIOR SIGNAGE
   1. Coordinate fonts, SVVSD logo types and other features with the SVVSD Branding Guide, which can be requested from the Project Manager and SVVSD Communications Department.

10 18 00 Changing Stations

1. Pre-School
    a. Verify the need for pre-school changing station with the SVVSD Early Childhood Education (ECE) Coordinator and Physical Therapy staff.
    b. 250 lbs. min. capacity.
    c. Products, coordinate selection with ECE Coordinator
        1. Brocar by Foundations 10-SSE Adult Changing Station (60" minimum width), or similar.
    d. Consider the need for a Special Needs style changing table to aid in transfer, see spec below.

2. Special Needs "Changing" Table
    a. "Changing" tables are appropriate for most needs in the District, however in some cases a deeper "mat" table is necessary, coordinate with District Special Education staff.
    b. Motorized height adjustment.
    c. 300 lbs. min. capacity
    d. 72" minimum width
    e. Coordinate free-standing vs. wall mounted options based on room size and typical use.
    f. Coordinate clearance at the base of the table. The table must allow a lift (Hoyer or similar) to roll beneath for transfer, this may require special kits, specifications, or upgrades to base models.
    g. Changing tables shall be Contractor Furnished and Contractor Installed
    h. Preferred Manufacturers/Products
        1. Armedica Hi-Lo Changing Table
        2. Bailey Economy Hi-Low Electric Mat Table
        3. Smirthwaite Hi/Low Riser Changing Bench
        4. Or similar
3. Special Needs "Mat" Table
    a. Deeper "mat" tables are beneficial for students with significant mobility limitations or with requirements for "mat time" while at school. Coordinate requirements for mat tables vs. changing tables with District Special Education staff.
    b. Foot switch operated motorized height adjustment.
    c. 300 lbs. min. capacity
    d. Minimum Dimensions: 84"L x 48"D
    e. Coordinate clearance at the base of the table. The table must allow a lift (Hoyer or similar) to roll beneath for transfer.
    f. Preferred Manufacturers/Products
        1. Armedica Hi-Lo Mat Treatment Table, AM-640
        2. Or Similar

10 40 00 FD & PD Lock Boxes

A. FIRE DEPARTMENT LOCK BOX
    1. Products Requirements:
        a. Preferred manufacturer is Knox, substitutions require District approval.
        b. Provide dual key access, allowing access by both the Fire Department and District maintenance.
    2. Special Instructions
        a. Recessed mount shall be furnished with recessed mounting kit (RMK) for new concrete or masonry construction.

B. POLICE DEPARTMENT LOCK BOX
    1. General Requirements
        a. Coordinate with SVVSD Low Volt Department, including but not limited to the following:
            1. Data drop(s)
            2. Card reader access
    2. Typical Product
        a. Knox Corporation, 551-ACS, Medvault

10 50 00 Lockers

A. Academic Lockers
    1. Construction:
        a. All-welded formed metal lockers (knock-down construction is not allowed)
        b. One-piece side frame, locker front, and back
        c. 18 ga. min. for all components
        d. Back stiffener panel required at all doors.
        e. Hinges: 3 inch, five knuckle or piano (continuous)
        f. Sloped top (where exposed)
    2. Venting
        a. Louvers
    3. Latching:
        a. Single point recessed (with back stiffener)
            1. Intent is to minimize moving and mechanical parts.
        b. Magnetic catch to hold door without lock preferred.
    4. Locking:
        a. Discuss the two options below with SVVSD Project Manager:
            1. Integral “Master Lock” locking mechanism with lifetime warranty.
            2. “Keyless” Keyless360 lock.
    5. Size: 15”w x 15”d x 72”h+/-, or as determined by project requirements.
    6. Arrangement: Two tier, typical
    7. Finish: Powder coat
B. Athletic Lockers
    1. Construction:
        a. All-welded angle iron frame OR all-welded formed metal lockers
        b. 18 ga. min. for all components
        c. Formed metal doors
        d. Hinges: 3 inch, five knuckle or piano (continuous)
        e. Sloped top
    2. Venting
        a. Door: Diamond punched
        b. Sides: Mesh
    3. Latching:
        a. Fully welded turn handle latch, with top and bottom vertical rods.
    4. Typical Locking (coordinate with individual project requirement):
        a. Athletic
            1. Discuss the two options below with SVVSD Project Manager:
                a. Integral “Master Lock” locking mechanism with lifetime warranty.
                b. “Keyless” Keyless360 lock.
        b. Physical Education or visiting team
            1. Student provided pad lock or single use combination.
    5. Size: As determined by design.
    6. Arrangement: Single or two tiered, as determined by design.
    7. Finish: Powder coat
C. Accessories
    1. Number Plates
    2. Coat hooks, (2) per
D. Preferred Manufacturers:
    1. All-Welded
    2. DeBourgh Mfg.
    3. Lyon Worspace Products

4. Penco Products

DIVISION 11 – EQUIPMENT

11 00 00 Primary District Requirements

A. WALL SAFE MANUFACTURERS
   1. Provide wall safe in Administration area
   2. Manufacturers: Subject to compliance with requirements, but are not limited to the following:
      a. Gardell; Economical Wall Safes, Model WS-1317-T-K
      b. Safesetc.com
      c. Sentry Group

B. PEDESTRIAN CONTROL DEVICES
   1. Removable posts, floor sockets and barrier tape heads at food service area of kitchen.
   2. Review design at food service lines with school district prior to final placement of floor sockets.
   3. Removable Posts: Lawrence 320-1S, satin chrome plated steel posts.
   4. Floor Sockets: Lawrence 421-1S-S sockets with cap, satin chrome finish.
   5. Barrier Tape Heads: Lawrence 695-1S/43 “TensaBarrier” head with tape, satin chrome head with color tape. Provide 695-RCVR-1S satin chrome receiver heads at ends of barriers.
   6. Manufacturers: Subject to compliance with requirements, but are not limited to the following:
      a. Lawrence Metal Products, Inc.

11 30 00 Residential Appliances

A. Manufacturers: Subject to compliance with requirements, but are not limited to, the following:
    1. General Electric Company (GE)
    2. KitchenAid; a division of Whirlpool Corporation.
    3. Maytag; a division of Whirlpool Corporation.
    4. Sears Brands LLC (Kenmore).
    5. Bosche
    6. Whirlpool
    7. Others with District approval

B. Refrigerators
    1. Water Connection / ice maker
        a. Typically not provided, but is acceptable to SVVSD Maintenance if the project benefits from inclusion.
    2. Door Configuration
        a. Side-by-side or top-bottom are acceptable, but models with more than 2 doors are discouraged.

11 40 00 Food Service Equipment: Revised 2026

A. **Can Rack**
1. Provide rack design that allows cans to dispense first-in-first-out (FIFO)
2. Heavy duty and easily cleanable materials required.
3. **Preferred Product**: Metro, MetroMax Q Industrial Plastic Shelving Roll Under C-Rack

B. **Combi Oven**
1. Provide gas-fired, stacked configuration where required by project.
2. Self-cleaning
3. 2-year minimum warranty
4. **Preferred Product**: Alto-Shaam, CTP7-20G Combitherm®, or current equivalent.
    a. Sample specification, designer to coordinate with unique project requirements.
        1. Gas, boiler-free, (8) 18" x 26" full size sheet or (16) 12" x 20" full size hotel pan (1/1 GN) capacity, PROtouch control with steam/convection/combi and retherm cooking modes, programmable cool-down, SafeVentTM steam venting, removable single-point temperature probe, (3) power levels, (4) cooking modes, CombiClean PLUSTM with (5) cleaning levels, (2) side racks with (8) non-tilt support rails, CoolTouch3TM glass window, door hinged right, high efficiency LED lighting, stainless steel construction, adjustable stainless steel legs, 98,000 BTU, EcoSmart®, cULus, UL EPH Classified, CE, IPX5, Gastec, ENERGY STAR®, EAC Remote Pre Assembled Refrig. System Model RFO250L4SDA 208-230/60/1 or approved equivalent Low Temperature, base, weather hood, winter controls, Scroll, Air-cooled, R448A, Std. Defrost Kit (32 MCA, 45 MOPD, 14.7 Compressor RLA) with RL6E077DDASC 208-230/60/1 coil (1.0 fan amps, 9.8 heater amps) with Dual Speed EC motor. Accessories: 1 ea Low ambient kit with heated and insulated receiver, low pressure bypass relay. 5031203 Reverse Osmosis System. Chef Connect capability.Provide fully programmable controls.

C. **Dish Machine**
1. 2-year minimum warranty
2. **Preferred Product:** Hobart AM16T-BAS-2
    a. Sample Specification, designer to coordinate with unique project requirements.
        1. Dishwasher, door type, tall chamber (27”), high temp sanitizing, 208-240/60/3 (field convertible to single phase), 60 racks/hour, straight-thru or corner, user-friendly smart touchscreen controls, Wi-Fi connectivity with SmartConnect app,Sense-A-Temp™ booster, electric tank heat, pumped rinse, pumped drain, auto-fill, stainlesssteel tank, frame, doors & feet, sheet pan rack, cULus, NSF, ENERGY STAR® WWST-STARTUP0NO Without startup DWT-AM16 Drain water tempering (single valve) kit with Pumped Drain Air Gap for BAS and Tall models

D. **Disposal**
1. Provide units sized appropriately for dish areas.
2. **Preferred Product**: InSinkerator

E. **Dunnage Racking**
1. Provide heavy duty floor racks in all storage areas including dry, cooler, and freezer areas.
2. Provide racks easily disassembled for cleaning.
3. **Preferred Product**: Fermod

F. Frost Top
1. Provide size capable of holding a sheet tray.
2. **Preferred Product:** USECO

G. **Heated Cabinet**

1. Pass through type
2. Humidity Control (ability to hold wet or dry heat)
3. Insulated stainless steel construction
4. **Preferred Product:** CresCor, Aquatemp Hot Cabinet, Model H-138-PWS-1834D, or sim.

H. Heated Merchandizer

1. Custom with serving line build

I. **Lockers**

1. ½ size lockers with safety view door

J. Milk/Beverage Merchandizer/Cooler

1. Prefer merchandisers over traditional coolers
2. **Preferred Product:** Delfield

K. **Mixer, Floor Type**

1. Provide commercial floor mixer sized for school food service use.
2. Provide mixing attachments as required by project.
3. Emergency stop feature
4. Integrated timer
5. Gear driven motor, with up to 3 variable speeds
6. Stainless steel bowl with mobile carrier
7. Include: dough hook, whip, and flat beater attachments
8. 2-year minimum warranty
9. **Preferred Product**: Hobart, Legacy+ Floor Mixer, 30qt
10. Alternate Products: Globe SP30, Doyon SM300

L. **Racking and Shelving**

1. Provide tool-free adjustable shelves
2. Provide easily disassembled systems for cleaning.
3. **Preferred Product**: Fermod

M. **Refrigeration, Reach-In**

1. Provide stainless steel construction.
2. **Preferred Product**: True Mfg., General Foodservice Model No T-49-HC
    a. Sample Specification, designer to coordinate with unique project requirements.
        1. reach-in, two-section, (2) stainless steel doors, (6) PVC coated adjustable wire shelves, interior lighting, stainless steel front, aluminum sides, aluminum interior with stainless steel floor, 4"castors, R290 Hydrocarbon refrigerant, 1/2 HP, 115v/60/1-ph, 5.4 amps, NEMA 5-15P, cULus, UL EPH Classified, Made in USA, ENERGY STAR®

N. Serving Line / Point of Sale

1. Adjustable sneeze guards preferred
2. Built-in (no casters)
3. Coordinate locations of tray dispensers, salad bars, rails and heights and other components with SVVSD Nutrition Services for specific age groups.
4. Prefer frost tops and built-in heated merchandisers over drop-in wells.
5. Provide approved artwork wrapping on exposed serving lines.
6. **Preferred Products:**
    a. Multiterra
    b. Counter Craft

c. BSI

O. Tilting Skillet

1. Self draining
2. **Preferred Product:** Cleveland, Cleveland Range Model No SGL40TR Dura Pan, or sim.
   a. Sample specification: Designer to coordinate with unique project requirements:
      1. Gas, 40-gallon capacity, modular open base, standard with hydraulic hand tilt with quick lowering feature, stainless steel construction, includes spring-assisted cover, gallon markings and electronic spark ignition, stainless steel level adjustable feet, 130,000 BTU, CE, NSF, IPX6. Gas type to be specified. 120v/60/1-ph, 1.8 amps NEMA 5-15P, standard. PT1 Power Tilt, with hand tilt override. TD2SK 2" tangent draw-off valve, front mounted left side (Note: May require additional lead time, contact factory). TDDC Drain Cup Assembly, includes: 8' length of 2" ID hose , stainless steel cup assembly & hose clamp. SLD Sliding Drain Drawer. Single Pantry Faucet (SPS14), includes Faucet Mounting Bracket; Hot & Cold Water Pre-Rinse Spray Head with Hose (PRS-S). Food Strainers for pouring spout (FS). Vegetable Steamers (VS).

P. Vector Oven

1. **Preferred Product:** AltoSham, VMC-F4G Vector® F Series Multi-Cook Oven:
   a. Sample specification: Designer to coordinate with unique project requirements.
      1. Gas, (4) individually controlled cooking chambers, holds (8) half-size sheet pans (4) full sized sheet pans, programmable touch screen controls with Structured Air Technology®, double pane glass door, interior LED cavity lighting, includes: (1) wire rack & (2) jet plates per cooking chamber, USB port, stainless steel construction, cULus, UL EPH Classified. Chef Connect capability.

Q. **Walk-in Cooler**

1. **Preferred Product**: Thermo-Kool

R. **Walk-in Freezer**

1. **Preferred Product**: Thermo-Kool

S. **Wells (Hot and Cold/Frozen)**

1. Provide (1) well per serving counter, between heated merchandiser and frost top, for salad bars, (4) cold/frozen wells required.
2. **Preferred Product:** Vollrath

T. **Work Tables**

1. Stainless steel
2. 30” width, typ.
3. 16 gauge
4. Galvanized legs
5. 5” locking casters
6. (2) draws per side
7. 800 pound top capacity
8. 450 pound undershelf capacity
9. Customizable height preferred

11 50 00 Education, Art, and Scientific Equipment

A. PROJECTION SCREEN
   1. Acceptable manufacturer and type (for High School stage and/or commons): Da-Lite Senior Electrol automatic projection screen or equal product of other acceptable manufacturer. Provide key operated recessed three position control switch in box with cover plate. Screens shall be flame retardant, mildew resistant, glass beaded with black border. Size: 18' x 18' with 3'-6" black extension.
   2. Acceptable Manufacturer and type for elementary schools: Da-Lite Cosmopolitan or equal of other acceptable manufacturer. Provide key operated recessed three position control switch. Screen shall be flame retardant, mildew resistant, glass beaded with black border. Picture surface shall be directed towards gym. Size: 9' x 9'
   3. Manufacturers: Subject to compliance with requirements, but are not limited to the following:
      a. Da-Lite Screen Company, Inc.
      b. Draper Shade and Screen Co., Inc.
      c. Knox
      d. Stewart Filmscreen Corporation
      e. Bretford Manufacturing, Inc.

B. GOGGLE SANITIZER CASES
   1. Goggle sanitizer cases with UV lamp: As manufactured by Ker-Kau Manufacturing Company.

C. KILNS
   1. See Design Bulletin titled “Kilns”.

11 60 00 Theater Equipment

A. SUMMARY
   1. This Section includes
      a. Stage Drapery
      b. Stage Dead-hung Rigging System.
      c. Motorized Hoist Systems.
      d. Performance and House Lighting and Dimming System.
      e. Orchestra Pit Cover to have rope lights, keyed switch.

B. STAGE DRAPERY
   1. Inherently Flame Retardant (IFR) drapery products are required at all SVVSD schools unless the exception below applies.
      a. Flame Retardant (FR) drapery products require costly annual maintenance to reapply flame retardant products, but may be considered at elementary school platforms and location not defined as a “Stage” by the IBC or local AHJ.
   2. Warranty for Rigging Equipment: Manufacturer's standard form in which manufacturer agrees to repair or replace components of rigging equipment that fail in materials or workmanship within specified warranty period. Warranty Period: One year from date of Substantial Completion.

C. MOTORIZED BATTEN HOISTING SYSTEM
   1. Provide 4 PowerLift “T” series hoists each with SureTarget controller.
   2. Provide battens, fittings, and related equipment as required for a functional system.
   3. Hoists shall be the PowerLift as manufactured by J. R. Clancy or approved equal.
   4. Travel: 65 feet
   5. Gear motor and Primary Brake - The motor, primary brake, and gearbox shall be an integrated unit from a single manufacturer. For enhanced reliability, a continuous shaft shall link the brake, motor armature, and the first stage pinion gear without the use of couplings. Brakes with a lower retarding torque shall not be permitted.
   6. Drum: The drum shall be helically grooved to accept a single layer of cable
   7. Sheaves: All load bearing sheaves, both internal and external, shall have a minimum 26:1
   8. Loft Blocks: shall have 12-gauge steel (minimum) side plates that shall fully enclose the sheave. Base angles shall be a minimum 1 ½" x 1 ½" x 3/16" angle punched with a universal hole pattern for easy installation.
   9. Limit Switches: Hoists shall have positively actuated limit switches for normal end of travel indication. These switches shall open the control circuit in the drive or starter to stop any further movement in the direction of travel.
   10. Battens - Battens shall have the ability to accept standard lighting fixture clamps, scenery chains, and pipe clamps, shall be 1 1/2" nominal diameter, schedule 40 steel pipe
   11. Products: Subject to compliance with requirements, provide one of the following:
       a. Automatic Devices Company;
       b. Autodraper Model No. 1454
       c. H & H Specialties Inc.
       d. Tru-Roll, Inc.

D. THEATRICAL AND HOUSE LIGHTING DIMMER SYSTEMS
   1. Architectural Lighting Control System:
      a. The system shall be capable of controlling individual dimmers via patchable zones.
      b. The system shall support LCD, slider and pushbutton stations.
      c. The system shall provide “snapshot” recording of lighting control console output data top reset on stations.
      d. The system shall have capability for uploading and downloading system software configurations.
   2. Dimming and Distribution Equipment shall include the following major components:
      a. Operator's Console.
      b. House Light Master Stations.
      c. House Light Entry Stations.

d. Console and Network Node Receptacle Stations and Accessories.
e. Performance Lighting Dimmer Racks and Dimmer Modules.
f. Distribution Equipment.

3. Equipment Manufacturers:
   a. Electronic Theatre Controls (ETC)
   b. Strand Lighting is considered an acceptable alternate to the ETC equipment listed above.
   c. Rigging Innovators, ETC and Strand Lighting are considered acceptable alternates to the SSRC equipment listed above.

E. ORCHESTRA PIT COVER

1. Removable orchestra pit cover at High Schools.
2. Manufacture and installation of the orchestra pit cover and storage truck shall be such quality as guided by professional staging practices. The pit cover shall be capable of supporting, without failure, a uniform load of 125 pounds per square foot. The pit cover shall consist of lift-out sections.
3. Storage trucks for pit cover sections shall be provided. Overall width of storage trucks shall not exceed 32 inches to allow easy passage through a 3'-0" doorway. Each truck shall have four (4) ball bearing hub casters with minimum 5" diameter, hard rubber wheels
4. with load rating of 500 lbs. Each. Two (2) casters shall be fixed and two (2) casters shall be of the swivel type.
5. Products: Subject to compliance with requirements, provide one of the following:
   a. SECOA
   b. Stage Engineering International, Ltd.
   c. Wenger Corporation
   d. Norcostco

11 65 00 Athletic Equipment, Indoor

A. SUMMARY
   1. This Section includes the following gymnasium equipment:
      a. Basketball equipment.
      b. Volleyball equipment.
      c. Badminton equipment.
      d. Exercise equipment.
      e. Safety pads.
      f. Batting cages.
      g. Scoreboards
   2. Field Measurements: Verify position and elevation of floor inserts and layout for gymnasium equipment.
   3. Coordinate installation of floor inserts with structural floors and finish flooring installation and with court layout and game lines and markers on finish flooring.
   4. Coordinate layout and installation of overhead-supported gymnasium equipment and suspension system components with other construction including light fixtures, HVAC equipment, fire-suppression-system components, and partition assemblies.

B. WARRANTY
   1. Special Warranty: Manufacturer's standard form in which manufacturer agrees to repair or replace components of gymnasium equipment that fail in materials or workmanship within specified warranty period. Warranty Period: Five years from date of Substantial Completion.

C. BASKETBALL BACKSTOPS
   1. Acceptable backstops (high schools and middle schools):
      a. Forward Fold - Rear Braced Main Court Backstops in main gym and auxiliary gym: Porter No. 90950-000 “Center Strut”, ceiling suspended, or equal of other acceptable manufacturer. Provide Porter Model No. 00224-300 rectangular glass bank with Porter Model No. 00224- 900 Pro-Strut adapter; No. 00227-000 bottom padding; and No. 00245-500 Power-Flex goal with net.
      b. Forward Fold - Rear Braced Practice Court Backstops: Porter No. 90950-W00 “Center Strut”, ceiling suspended, or equal of other acceptable manufacturer. Provide Porter Model No. 00228-300 rectangular Duralam bank; No. 00229-000 bottom padding; and No. 00245-500 Power-Flex goal with net.
   2. Acceptable backstops (elementary schools):
      a. Folding Backstop: Porter Model No. 90617-000 motorized Forward Fold or equal of other acceptable manufacturer; adjustable height goal, Model No. 00228-300 rectangular wood bank, No. 229-000 bottom padding and No. 00225-000 goal with net.
      b. Fixed Backstop: Porter Model No. 90918-000 or equal of other acceptable manufacturer; stationary ceiling suspended rear braced with goal height adjustable from 8' to 10'; Model No. 00228-300 rectangular wood bank, No. 00229-000 bottom padding, No. 00225-000 goal with net.
      c. Practice Backstops: Porter Model No. 312 Standard Stationary Wall Mount or equal of other acceptable manufacturer. Rectangular wood bank No. 00228-300 and No. 00225-000 goal with net. Flush mount to wall.
   3. Accessories - all schools: The following shall be provided for each backstop. Provide and install remote electric winches incorporating emergency provisions for manual operation in case of power failure. Winch design shall incorporate integral up and down limit switches. Safety disconnects shall be provided by backstop manufacturer. Winch motors shall be sized as required, 115V, single phase.
   4. Provide Porter Model No. 10798-000 Safety Strap and Hoist Cable Retractor to prevent accidental dropping of backstop in case of winch failure and to retract the hoist cable from area of play.
   5. Provide recessed type, three-position key selection
   6. Available Manufacturers:
      a. Porter Athletic Equipment Company
      b. Douglas Industries, Inc.

c. Draper Inc.
d. Jaypro Sports, LLC.
e. Performance Sports Systems.

D. GYMNASIUM DIVIDER CURTAIN
1. Roll-fold divider curtain in main gymnasium (high schools and middle schools):
2. Support System: Gym divider supplier shall provide all required and necessary support system for divider to be secured to roof structure.
3. Manufacturer and Type: Porter No. 90675-100 Roll-Fold Divider Curtain or equal of other acceptable manufacturer. Entire curtain fabric to be anti-bacterial treated, fungi-resistant, and flame retardant. Curtain shall have wire polyester open mesh with PVC coating upper section and solid 8'-0" high vinyl polyester reinforced fabric lower section. Provide continuous bottom batten.
4. Provide and install remote, single phase electric winch and integral limit switches to automatically control upper and lower limits of travel. Safety disconnect shall be provided by gym divider installer. Winch motor shall be sized as required. Provide recessed three position key switch and cover plate. Key switch shall have “spring return” to center.
5. Control: Porter Athletic Equipment Company; 3/4 H.P. Electric Operator for Divider Curtain
6. Acceptable manufacturers - gym divider curtain:
    a. Porter Equipment Company
    b. National Gym Equipment
    c. Aalco
    d. Architectural Design Products

E. MAT MOVERS
1. Motorized wrestling mat mover and storage system at gymnasium
2. Provide necessary framing and support system to attach to roof and wall structure.
3. Porter 91101-200 (91101-100 at middle school) overhead supported mat mover or equal. Mat mover system shall consist of fully welded steel carriage structure with rails and fully assembled drive and lifting mechanism, standard of manufacturer. System shall include suspension trolleys, travel beam, load bar with sling, control panel, electric motors for hoist and tractor, drive tractor,
4. key switches with covers, cables and pulleys, and other standard components of manufacturer required for fully operational system.
5. Controls: Two (2) lock box at control station, located as directed by the Architect.
6. Load Bar and Sling: Designed to support two (2) 42’ x 42’ mat for high schools, and (1) 42’ x 42’ mat for middle schools.
7. Manufacturer shall provide two (2) year written warranty for material and installation.
8. Acceptable manufacturers - mat mover system:
    a. National Gym Equipment (NGE)
    b. Porter Athletic Equipment Company
    c. Mat Press, Inc.

F. ELECTRONIC SCOREBOARDS
1. **Note:** The District has installed video boards in lieu of standard scoreboard in some instances. Review this option with the Design Team.
2. Coordinate shot clocks and end of period lights with scoreboard system.
3. Type: Interior, multi-purpose basketball/volleyball/wrestling electronic scoreboard with two integral horns, changeable captions, LED displays for time, scores, period, number of player fouling with personal fouls, team fouls, bonus and double bonus indicators, and next possession arrows.
4. Size: 8’-0” (L) by 6’0” (H) by 8 inches (D). Provide stats boards at high schools.
5. Location: 2 at Main Gymnasium and 1 at Auxiliary Gymnasium.
6. Wireless only.

7. Control Console: Provide one (1) control console in Main Gymnasium capable of controlling both scoreboard panels in Main Gym from console on Scorer's table. Provide one duplicate control console with cord and carrying case to operate second scoreboard in Main Gymnasium (verify second scoreboard in middle schools) or both scoreboards in Auxiliary Gymnasium. Provide sufficient control cable to reach from scorer's tables in Main and Auxiliary Gymnasium bleachers to junction boxes on walls. Verify if controls can be wireless.
8. Available Manufacturers: Subject to compliance with requirements, but are not limited to the following:
    a. Nevco; Model No. 2700
    b. Fair-Play Scoreboards
    c. Daktronics
    d. Digital Scoreboards

G. FLOOR INSERTS
1. Floor Insert: Chrome-finished steel floor plate; and steel pipe sleeve, concealed by floor plate, with capped bottom end, sized with ID to fit post standards, not less than 9 inches (230 mm) long to securely anchor pipe sleeve below finished floor in concrete footing; with anchors designed for securing floor insert to floor substrate indicated; one per post standard.
2. Available Manufacturers: Porter Athletic Equipment Company; Model No. 00872-100 or 0087-200 for 3-1/2 inch diameter posts with 8 inch diameter cover

H. SAFETY PADS
1. Wall Safety Pads: Padded wall wainscot panels designed to be attached in a continuous row; each panel section consisting of fill laminated to backer board with visible surfaces fully covered by seamless fabric covering, free of sag and wrinkles and firmly attached to back of backer board.
2. Size: Each panel section, 24 inches (600 mm) wide by not less than 72 inches (1800 mm) long.
3. Corner Wall Safety Pads: Wall corner pad consisting of not less than 1-1/4-inch- (32-mm-) thick, multiple-impact-resistant, closed-cell polyethylene-foam filler, covered on both sides and all edges by fabric covering with backer board and manufacturer's standard anchorage to wall
4. Available Manufacturers:
    a. Draper Inc.; EcoVision Gym Wall Pads
    b. American Athletic, Inc.
    c. Jaypro Sports, LLC.
    d. Performance Sports Systems.
    e. Porter Athletic Equipment Company.

I. BATTING CAGES (Interior)
1. Size: 10'-0" high by 50'-0" long.
2. Supporting Frame: 1-7/8-inch O.D. heavy-wall electro-plated tubing with cross spreaders located at 14'-0" centers for hoist cable attachments.
3. Hoist Cables: Six 1/8-inch diameter, vandal-proof galvanized cables, 2,100-pound breaking strength each cable.
4. Bottom End of Each Cable: Terminate in sling arrangement for proper support of frame assembly.
5. Line Shaft: Driven by heavy-duty, 3/4-hp, single-output shaft, gear motor furnished with automatic overload protection.
6. Recessed switch box: Up and down, with stainless steel cover, to prevent improper operation of system
7. Available Manufacturers: Subject to compliance with requirements, but are not limited to the following: Porter Athletic Equipment Company; Ceiling-Suspended Batting Cage, Item #90920-000

11 65 10 Athletic Equipment, Outdoor

A. ATHLETIC SURFACING

1. See Section 32 10 20

B. BASEBALL & SOFTBALL

1. Nail Drag
    a. Nail only, no bolts
    b. Preferred products:
        1. Beacon Athletics, Adjustable Weight Nail Drag
        2. Or similar.
2. Bases, baseball
    a. Coordinate with SVVSD standard imbed depth and sleeve size
    b. Preferred products:
        1. Hollywood Bases, Jack Corbett Bases, Original MLB Base Set
        2. Or similar
3. Bases, softball
    a. Coordinate with SVVSD standard imbed depth and sleeve size
    b. Coordinate with governing body for color (orange/white) requirements
    c. Preferred products:
        1. Hollywood Bases, Schutt Hollywood Bases
        2. Or similar.
4. Home Base
    a. Double sided
    b. Preferred products:
        1. Bulldog, Double-Sided Home Plate
        2. No substitutions, District Standard
5. Pitching Rubber
    a. 3” core
    b. 40 lbs. approx. weight
    c. Preferred products
        1. Bulldog 3” Pitching Rubber
        2. No substitutions, District Standard
6. Portable Pitching Mound
    a. Where required at combination softball/baseball fields
    b. Preferred products:
        1. Beacon Athletics, Proper Pitch Portable Game Mound
        2. Or similar
7. Foul Poles
    a. Metal wing/mesh. Netting or other less substantial materials are not allowed.
    b. Preferred products:
        1. Beacon Athletics: Standard Foul Poles
8. Batting Cages
    a. All wheels must rotate
    b. Capable of being moved through a 12 foot gate.
    c. Preferred products:
        1. Beacon Athletics, Big League Bomber, Portable All-Star Batting Cage
9. Batter’s Eye
    a. Include in projects:
    b. Preferred product:
        1. Coordinate with District.
10. Backstops
    a. Backstop netting terminating at a low wall (48”+/-) is preferred over chain link solutions.

C. SOCCER
    1. Goals
        a. Specify portable goals, to allow movement for practice to reduce wear patterns.
        b. Specify goals designed for synthetic turf
        c. Preferred products:
            1. Kwik Goal, Fusion 120 Soccer Goal
            2. Or similar
D. FOOTBALL
    1. Goals
        a. Preferred products, similar to those installed at Niwot HS and Skyline HS
E. TRACK & FIELD
    1. Discus Cage
        a. With curved or standoff poles
        b. Preferred pfoducts:
            1. Gill Athletics, High School Aluminum Discus Cage

11 70 00 Athletic Therapy Equipment

A. SUMMARY
   1. This Section includes the following therapy equipment:
   2. Stationary sports whirlpool.
   3. Icemaker/storage bin.

B. STATIONARY SPORTS WHIRLPOOL
   1. Provide stainless steel whirlpool with the capacity of 110 gallons.
   2. Complete with optional thermostatic mixing valve/vacuum breaker and tailpiece. Provide piping from tailpiece to floor drain provided by Mechanical.
   3. Available Manufacturers: Subject to compliance with requirements, but are not limited to the following:
      a. Source 1 Medical; S-110-2 Whirlpool
      b. Morley Athletic Supply Company Inc.
      c. MSEC Medical
      d. Supplies & Equipment Company.

C. ICE MACHINE/STORAGE BIN
   1. Provide Ice Machine with a production capacity of 540 lbs. per 24-hour period and storage capacity of 430 lbs. Ice Machine located in Training Room.
   2. Provide water filter at each ice machine, preferred product:
      a. OptiPure FXI-11
   3. Available Manufacturers: Subject to compliance with requirements, but are not limited to the following:
      a. Preferred
         1. Manitowoc; S-Series Ice Cube Machine on B-570 Bin, Model SY-0504A-B570
      b. Acceptable Alternates
         1. Ice-O-Matic.
         2. Kold-Draft.
         3. Scotsman.

DIVISION 12 – FURNISHINGS

12 20 00 Window Blinds

A. DISTRICT PREFERENCE
   1. Roller Shades are typically preferred over horizontal louver blinds; however, project requirements may supersede this preference.

B. HORIZONTAL LOUVER BLINDS, ALUMINUM SLATS
   1. Mounting: Inside mounting, permitting easy removal and replacement without damaging blind or adjacent surfaces and finishes; with spacers and shims required for blind placement and alignment indicated.
   2. Slats: Aluminum; alloy and temper recommended by producer for type of use and finish indicated; with crowned profile and radiused corners. Width: 1”

C. ROLLER SHADES
   1. Opacity
      a. Provide translucent or light filtering shades at typical exterior windows.
      b. Provide opaque or blackout shades where a fully darkened room is required for video or other functions.
      c. Provide opaque or blackout shades at interior windows and sidelites where the function of the shade is to limit views during lockdown scenarios.
   2. Size
      a. Avoid large roller shade units, this can lead to breakage and maintenance issues.
         1. Generally, shades over 48” in width are discouraged. Tall shades, 10’-0” or more in height, may benefit from reduced widths.

12 20 00 Multiple Seating

A. FIXED AUDIENCE SEATING
   1. Fixed Audience Seating: Molded-plastic chairs with upholstered panel inserts.
   2. Self-Rising Seat Mechanism: Spring-actuated.
   3. Armrests: Plastic with rounded edges, concealed mounting.
   4. Preferred Manufacturers:
      a. Irwin Seating Company
      b. Hussey Seating Company
      c. Interkal
      d. Alternates of similar quality to those listed above will be considered by the District.

B. TELESCOPING BLEACHERS
   1. General: Operable systems of multiple-tiered seating on interconnected folding platforms that close, without being dismantled, into a nested stack for storing. Stand units permit opening and closing of adjacent rows, allow individual and collective rows to be locked open for use, and close with vertical faces of upper skirts on the same vertical plane.
   2. Any retractable seating used for non-athletic use should be provided with skirting on both sides, where exposed to view.
   3. Access behind telescoping bleachers is required for maintenance, without obstruction, when bleachers are in retracted position.
   4. Operation: Automatic, friction-type, integral power unit.
   5. Provide access panel in bleacher deck to access traction motor.
   6. Safety Rails: Structural steel, finished with manufacturer's standard powder coat system. Removable and folding mid-aisle handrails located at centerline of each vertical aisle with seating on both sides.
   7. End rails (guards) that are telescoping and self-storing.
   8. Preferred Manufacturers
      a. Irwin Telescopic Seating Company
      b. Hussey Seating Company
      c. Interkal
      d. Alternates of similar quality to those listed above will be considered by the District.

DIVISION 13 – SPECIAL CONSTRUCTION

13 10 00 Exterior Bleachers

A. Type / Material
    1. Aluminum, complying with all applicable Codes.
B. Standard capacities
    1. Football / Track: 800 (home) 480 (away) seats
    2. Varsity Baseball & Softball: 150 seats
    3. Jr. Varsity Baseball & Softball: 84 seats
    4. Tennis Courts: 100 seats
    5. Middle School Fields: 100
    6. Elementary: no requirements
C. Provide approx.. 8'x18'press box integrated in bleacher section on home side of football fields.

13 20 00 Portable Stairs, Ramps, and Railings

A. ALUMINUM ACCESS COMPONENTS
    1. Preferred Manufacturers
        a. REDD Team
    2. Walking Surfaces
        a. Provide walking surfaces that promote drainage and provide maximum traction in wet and icy conditions.
        b. Preferred Components
            1. REDD Team Dura-Grip or similar.
        c. Unacceptable Components
            1. Diamond plate surfaces.
    3. Handrails
        a. Preferred Components
            1. Provide welded or monolithic single-piece construction where possible.
        b. Unacceptable Components
            1. Handrail components comprised of multiple pieces that screw or bolt together on site are discouraged.

DIVISION 14 – CONVEYING EQUIPMENT

14 24 00 Elevators

A. Acceptable Manufacturers
    1. Kone Elevator Corp.
    2. Otis
B. Discouraged Manufacturers
    1. ThyssenKrupp Elevator, due to service challenges.
C. Unacceptable Manufacturers
    1. Schindler, manufacturer not supported by District service contract.
D. Flooring
    1. Hard surface to coordinate with adjacent circulation space.

DIVISION 21 – FIRE SUPPRESSION

21 10 00 Water Based Fire Suppression System

A. SUMMARY
   1. This Section includes
   2. Sprinkler Piping and Specialties
   3. Piping
   4. Valving
   5. Specialties

B. GENERAL SUMMARY
   1. The entire building shall be sprinklered. Sprinkler system shall have a separate room, with emergency backup generator and consist of the following types: Wet and dry pre-action. A zone which serves the exterior of the building shall be dry preaction type. All other zones such as kitchen area, office area, crawl space and classrooms shall be wet type.

C. SYSTEM DESCRIPTION:
   1. Fire protection system is a "wet-pipe" system employing automatic sprinklers attached to a piping system containing water and connected to a water supply so that water discharges immediately from sprinklers opened by fire.
   2. Fire protection system for Stage areas are "Class III, Standpipe and Hose" system which is an arrangement of piping, valves, hose connections and allied equipment.
   3. Elevator Shafts and Machine Rooms: Sprinklers shall be installed in elevator machinery rooms, at the top of elevator shafts, and at the bottom of elevator shafts.
   4. Sprinkler coverage shall be designed for Ordinary Hazard Group One. Sprinkler heads shall be high temperature classification (286 degrees F.).
   5. Deviations from approved "working plans" for sprinkler piping, require written approval of the authority having jurisdiction. Written approval shall be on file with the Engineer prior to deviating from the approved "working plans."
   6. There shall be no fire sprinkler piping in electrical rooms (other than piping serving sprinklers directly in that room) or installed over any electrical panels.
   7. Install pressure gauges on city and system sides of fire entry valve assembly.
   8. Install hangers straight and true and piping parallel to building lines.

D. HYDRAULIC DESIGN:
   1. The design for the Fire Protection system shall be designed by an assigned specialty consultant by the District to the design team. Fees for the fire protections consultant shall be paid for under the Architect/Owner agreement.

E. MATERIALS AND PRODUCTS:
   1. General: Provide piping materials and factory-fabricated piping products of sizes, types, pressure ratings, temperature ratings, and capacities as indicated. Where not indicated, provide proper selection as determined by Installer to comply with installation requirements. All equipment used on this project shall be new and UL listed unless noted or specified otherwise.
   2. Manufacturer: Subject to compliance with requirements, provide fire protection system products from one of the following:
      a. Gate Valves:
         1. Nibco
         2. Kennedy Valve, Div. of ITT Grinnell Valve Co., Inc.

3. Mueller
4. Stockham
5. Grinnell
6. Milwaukee

b. Swing Check Valves:
1. Central
2. Mueller
3. Kennedy Valve, Div. of ITT Grinnell Valve Co., Inc.
4. Star Sprinkler Corp.
5. Viking
6. Victaulic
7. Globe
8. Potter Roemer

c. Butterfly and Ball Valves:
1. Grinnell
2. Mueller
3. Victaulic
4. Milwaukee
5. Kennedy
6. NIBCO
7. Victaulic Co. of America

d. Grooved Mechanical Couplings:
1. Gruvlok
2. Victaulic Company of America
3. Central Sprink, Inc.

e. Double Check Valve Assembly:
1. Watts
2. Fire Protection Specialty Valves
3. Reliable Automatic Sprinkler Co., Inc.
4. Star Sprinkler Corporation
5. Viking Corporation
6. Central
7. Globe
8. Grinnell/Gem

f. Fire Department Connection:
1. Croker
2. Potter-Roemer
3. Elkhart
4. Grinnell/Gem

g. Sprinkler Heads:
1. Central Sprinkler Corp.
2. Reliable Automatic Sprinkler Co., Inc.
3. Star Sprinkler Corp.
4. Viking Corp.
5. Globe
6. AFAC Inc.
7. Firematic Sprinkler Devices, Inc.
8. Tyco Fire Suprression & Building Products

h. Flexible Sprinkler Head Assemblies:
1. FlexHead Industries
2. Reliable Automatic Sprinkler Co., Inc.
3. Star Sprinkler Corp.

4. Viking Corp.
5. Globe

F. PIPE AND TUBING MATERIALS (INSIDE BUILDING):
   1. American Tube Company "Dyna-Thread-40" and "Dyna-Flow" and Allied Tube and Conduit
      a. Gruvlock
      b. Shurjoint Piping Produts
      c. Tyco Fire & Building Products LP
      d. Victaulic Company
   2. Corporation "Super Flo" are acceptable to Schedule 40 pipe. The Corrosion Resistance Ratio of the pipe shall be 1.00 or greater
   3. Schedule 5 pipe shall not be allowed.
   4. Schedule 10 pipe shall only be allowed for pipe sizes 2-1/2inches and larger.
   5. Provide galvanized, schedule 40, piping system for preaction system and drain risers.

G. FLEXIBLE SPRINKLER HEAD ASSEMBLIES:
   1. FlexHead braded sprinkler heads allowable.
   2. Flexible connectors shall have materials suitable for system fluid. Include 200-psig maximum working-pressure rating.
   3. Stainless-Steel-Hose/Steel Pipe, Flexible Connectors: Corrugated, 304 stainless-steel, inner tubing covered with stainless-steel wire braid. Include steel nipples welded to hose; 1 inch NPT for attachment to the sprinkler system branch piping. Minimum bend radius: 12 inches.
   4. Flexible Hose Attachment: Removable hub type with set screw.

H. FITTINGS (INSIDE BUILDING):
   1. Cast-Iron Threaded fittings: ANSI B16.4, Class 125 standard pattern, for threaded joints. Threads shall conform to ANSI B1.20.1.
   2. Malleable-Iron Threaded Fittings: ANSI B16.3, Class 300, standard pattern, for threaded joints. Threads shall conform to ANSI B1.20.1. Install steel pipe with threaded joints and fittings for 2inches and smaller.
   3. Steel Fittings: ASTM A234, seamless or welded, for welded joints.
   4. Fittings: All mechanical fitting and couplings shall be grooved for the entire fire protection system shall be of the same manufacturer as submitted in shop drawing equipment review.
   5. Cast-Iron Threaded Flanges: ANSI B16.1, Class 250; raised ground face, bolt spot faced.
   6. Cast Bronze Flanges: ANSI B16.24, Class 300; raised ground face, bolt holes spot faced.
   7. Plain end, hooker type, or push-on fittings or couplings shall not be allowed.
   8. Bushings and reducing couplings shall not be allowed.

I. GENERAL DUTY VALVES:
   1. Gate Valves - 2 Inch and Smaller: Body and bonnet of cast bronze, 175 pound cold water working pressure - non-shock, threaded ends, solid wedge, outside screw and yoke, rising stem, screw-in bonnet, and malleable iron handwheel.
   2. Gate Valves - 2-1/2 Inch and Larger: Iron body; bronze mounted, 175 pound cold water working pressure - non-shock. Valves shall have solid taper wedge; outside screw and yoke, rising stem; flanged bonnet, with body and bonnet conforming to ASTM A 126 Class B.
   3. Ball Valves: 1-1/2inches and smaller shall be threaded, forged brass construction, with teflon seats and blow out proof stem. Ball shall be full port with chrome plated ball.
   4. Ball Valves: 2inches to 3inches shall be listed to 300 p.s.i. with optional internal tamper switch. Body shall be ductile iron with corrosion resistant coating. Ball shall be 316 stainless steel, standard port design.
   5. Swing Check Valves: MSS SP-71; Class 175, cast iron body and bolted cap conforming to ASTM A 126, Class B; horizontal swing, with a bronze disc or cast iron disc with bronze disc ring, and flanged ends. Valve shall be capable of being refitted while the valve remains in the line.

J. ALARM DEVICE AND FIRE PROTECTION SPECIALTIES:
    1. Water Flow Indicators: Vane type water flow detector, rated to 250 psig; designed for horizontal or vertical installation.
    2. Supervisory Switches: SPST, normally closed contacts, designed to signal valve in other than full open position.
    3. Pressure Switch: Indicating low pressure trouble in sprinkler system.
    4. Pressure switch: Indicating flow in sprinkler system.

K. AUTOMATIC SPRINKLERS:
    1. Sprinkler Heads: Fusible link or frangible bulb type, quick-response; style as indicated or required by the application: Upright, Pendent and Sidewall Styles: Factory brass, rough bronze finish for heads in unfinished spaces. Heads shall be stainless steel where installed exposed to acids, chemicals, or other corrosive fumes.
    2. Concealed Style (Corridors only): Rough brass, adjustable, with painted white cover plate in finished spaces. (GEM Models FR946 and F946 Clean Line Sprinklers are not acceptable.)
    3. Recessed Style: recessed sprinklers are not acceptable.

L. FIRE VALVE CABINETS:
    1. Construction: Manufacturer's standard enameled steel box, with trim, frame, door and hardware to suit cabinet type, trim style, and door style indicated. Weld all joints and grind smooth. Miter and weld perimeter door frames.
    2. Provide standard equipment "Croker" Series 2750, or "Potter Roemer" Series 1880 valve and extinguisher cabinets.

M. FIRE DEPARTMENT CONNECTIONS:
    1. Polished chrome cast brass, wall type, with wall escutcheon and having 5” storz inlet connection and NPT outlet, as specified in NFPA 1963. Inlet shall have a clapper valve, and cap and chain. Unit shall have wall escutcheon of cast brass, finish to match connections, with words "Standpipe - Fire Dept. Connection" or "Auto Spkr. - Fire Dept. Connection" or "Auto Spkr. and Standpipe - Fire Department Connection" in raised letters.

DIVISION 22 – PLUMBING

22 00 00 Primary District Requirements

A. DESIGN PREFERENCES

1. Provide dual backflow preventers with Wye strainer at domestic water entry.
2. Isolate hot and cold water lines from main line for each room with multiple plumbing fixtures, (i.e. toilet rooms, science classrooms, or kitchen).
3. Provide two-way floor cleanouts for bathroom groups.
4. PVC piping is acceptable below grade, where allowable by Code.
5. Domestic water piping 3" and larger shall be brazed
6. Hubless cast iron pipe and fittings are unacceptable for underground sanitary and storm sewer.
7. Heavy Duty No-Hub couplings shall be used on above grade piping 5" and larger.
8. Lavatories in countertops are preferred over wall mounted, where feasible.

B. PREFERRED FIXTURES

1. Mop Sinks
   a. Provide floor level mop sinks with reduced font edge to accommodate dumping.
   b. Corner orientation preferred, where applicable.
   c. Marble & cement preferred.
   d. 32"+/- sides
   e.  Example product.
2. Water Closets
   a. Floor mount with a flange.

C. FLUSH AND FAUCET OPERATION

1. *Consider operation of flush valves and faucets in the event of power outages.*
2. Water Closets & Urinals: Manual flush valve
3. Lavatories: 3-hole sink and faucet with proximity (I.R./sensor) with self charging battery.

D. WATER CONSERVATION

1. Maximum allowable flow rates:
   a. Lavatory 0.5 gpm
   b. Sink 2.5 gpm
   c. Mop Sink 1.6 gpm
   d. Water Closets 1.28 gpm (siphon jet)
   e. Urinals 0.125 gpm (siphon jet)
   f. Shower Heads 1.25 gpm

E. TRACER WIRE

1. To assist the District with future utility locates, tracer wire is required on all exterior underground piping for water, sewer, and storm drain per section 33 50 00.

F. PROJECT COMPLETION

1. Camera main sanitary sewer lines with water flow after backfill is complete, and provide video to school district.

22 10 00 Water Distribution

A. PRODUCTS

1. Shock Arrestors: provide on water piping for hood system.
   a. Provide isolation valve and access panel at each shock arrestor.
   a. Shock Arrestors - Bellows Type: Stainless steel casing and bellows, rated for 250 psi, 200 degrees F, tested and certified in accordance with PDI Standard WH-201.
   b. Shock Arrester - Piston Type: 60 psi precharge, sealed from system water with free sliding piston and EPDM Orings. Suitable for up to 150 psi and 180 degrees F thread connections.
2. Hose Bibbs:
   a. Finished Rooms with Floor Drains: Polished chrome plated bronze body, with renewable composition disc, tee handle, 3/4 inch NPT inlet, 3/4 inch hose outlet. Basis of design: Chicago Faucet No. 293
   b. Unfinished and Equipment Rooms: Rough bronze body, renewable composition disc, wheel handle, vacuum breaker, 3/4 inch NPT inlet, 3/4 inch hose outlet with vacuum breaker. Basis of design: Chicago Faucet No. 952-293-6
   c. Finished Rooms: Polished chrome with flush mounting lockable wall box. ¾" NPT inlet, ¾" hose outlet. Basis of design: Woodford No. B65.
3. Wall Hydrants:
   a. Flush with wall, non-freeze, box type, wall hydrant; all brass with nickel bronze box cover and frame, "T" handle loose key, key lock cover, integral vacuum breaker, self draining body and shank, ¾ inch male hose thread outlet, ¾ inch male or female I.P.S. thread inlet, renewable seat; shank length to extend thru primary exterior wall surface sufficient distance to prevent freezing. Basis of design: Woodford Model B65
4. Backflow Preventers:
   a. Design Note: Provide Dover/Hydro "1951 Pressure Bleed Device" in mop sink room with chemical dispensers.
   b. Design Note: Split back flow preventers for domestic water.
   c. Design Note: Provide pressure gauges before and after backflow preventer.
   d. Vacuum Breaker, Atmospheric: Atmospheric type, all brass, angle pattern with disc float that closes the atmospheric vent for temperatures up to 210 degrees F.; full size orifice for maximum flow; female thread inlet and outlet; rough brass exterior finish; unit to be approved by National Sanitary Foundation. Basis of design: Watts No. 288
   e. Vacuum Breaker Pressure Type: Spill resistant design, suitable for indoor or outdoor installation, with vent seal diaphragm designed to seal air vent prior to opening the check valve. Suitable for use with downstream valves. Provided with ball valve shut-offs and test cocks. Basis of design: Watts Series 008QT
   f. Double Check Assembly: Sizes 3/4inch through 2inches include two spring-loaded, Y-pattern bronze check valves, two bronze ball valves and four testcocks for field testing. All valves are threaded type and the unit is shipped completely assembled. Sizes 2-1/2inches through 10inches include two spring loaded Y-pattern check valves with epoxy coated ductile iron bodies and bronze trim, two cast iron non-rising stem gate valves and four testcocks for field testing. Compact style, inlet flow vertical up, outlet flow vertical down or up as shown on

drawings. All valves are flanged type and the unit is shipped completely assembled; unit to be approved by National Sanitary Foundation, U.S.C. Foundation for Cross Connection Control, A.S.S.E. State and or Local authorities. Basis of design: Watts Series 757

g. Reduced pressure type: All bronze (3/4inch-2inch)/stainless steel (2-1/2inch – 10inches) body with two independently operating, spring loaded check valves and one differential relief valve with automatic intermediate atmospheric vent. Pressure in intermediate zone to activate relief valve when there is a 2 psig. differential between the zone and the upstream side of the first check valve. The relief valve shall remain open until a positive pressure differential is re-established. Assembly to be furnished with fullport, positive shut off non-rising stem, isolation valves, in-line strainer, union connections, funnel, and all test cocks. Compact style, inlet flow vertical up, outlet flow vertical down or up. Assembly to have approval of National Sanitary Foundation, U.S.C. Foundation for Cross Connection Control, A.S.S.E. State and or Local Authorities. Basis of design: Watts Series 795 and 909, as required by size.

5. Pressure Reducing Valves (Direct Acting):
   a. Low and Medium Capacity (15 to 120 GPM): All bronze pressure reducing valves, sensitive spring and diaphragm for accurate pressure control; manual adjustment for outlet pressure integral strainer, female thread connections. See detail on mechanical plans for size, capacity and piping arrangement. Basis of design: Watts No. 223
   b. High Capacity (50 to 1800 GPM): Epoxy coated cast iron body pressure reducing/sustaining valve, stainless steel seats, sensitive spring and large Buna-N diaphragm area for accurate pressure control; pilot valve with manual adjustment for outlet water pressure; female thread connections for sizes up to 2-1/2inches and flanged connections for sizes 3inches and larger. Provide in-line inlet water strainer; inlet and outlet valves and threaded or flanged connections for easy valve removal. See detail on mechanical plans for size, capacity and piping arrangement. Basis of design: Watts 115-2A
6. Water Tempering Valves: Pressure Equalizing with Thermometer: Concealed type, 3-port, all bronze, pressure balancing mixing valve with integral angle check stops, maximum temperature limit stop, stainless steel balancing piston, chrome plated finished wall plate with hot and cold water marking; ¾ inch inlet and outlet pipe sizes and volume control. Unit to hold temperature steady and constant with pressure fluctuations up to 85 percent. Unit shall have integral or separate flush with wall, 3-color dial thermometer with scale from 70 degrees F to 130 degrees F and chrome plated trim ring for separate thermometer. Provide stop and check valves.
   a. Preferred Products:
      1 Lawler Series 66
      2 Watts LFN170
7. Thermostatic Mixing Valves: Exposed type, all bronze thermostatically controlled mixing valve with stainless steel piston, fail safe automatic shut-down if either hot or cold water pressure fails; union connection, polished chrome finish. Basis of design: Powers LFMM430, Lawler 805.
8. Pressure/Temperature Relief Valves (PTRV): Fully automatic, all bronze pressure/temperature relief valve with test lever and extension thermostatic element; temperature relief setting at 210 degrees F and pressure setting at 150 psig; valve to meet ASME Standards and comply with the latest AGA ratings. Relief opening to be piped to an indirect connection at nearest floor drain.
9. Water Meter: Provide water meters with registration in gallons.
   a. Water Meter - 2 Inches and Smaller: Disc type conforming to AWWA C700.
   b. Water Meter - 3 Inches and Larger: Compound type, conforming to AWWA C702.
   c. Remote Registration System: Utility company standards.
10. Domestic hot water expansion tanks: Fabricated steel designed and constructed in accordance with ASME Section VIII, Division I. Heavy duty removable Butyl bladder. Working pressure of 125 psig/Maximum operating temperature of 240 degrees F. Tank liner of FDA approved materials.

B. GENERAL INSTALLATION

1. Run piping as direct as possible to required connections, and slope to drain valves at low points for complete system draindown. Locate drain valves at accessible points within the system. Coordinate routing with Construction and O&M personnel of the District.
2. Provide ball valves for branch lines off main, sub-main takeoffs and main take-offs.
3. Provide building shut-off and separate hose end drain valves with vacuum breaker at main service entry.

C. PRESSURE REDUCING INSTALLATION:

1. Install backflow preventers on plumbing lines where contamination of domestic water may occur and on boiler make-up lines, hose bibbs and flush valves.
2. Install pressure reducing valves to limit maximum static pressure at plumbing fixtures to 65 psig.
3. Install water hammer arresters in domestic water piping system on each set of flush valves and where hydrostatic shock pressures could occur.

D. SERVICE ENTRANCE INSTALLATION:

1. Extend water distribution piping to connect to water service piping, of size and in location indicated for service entrance to building.
2. Install sleeve and mechanical sleeve seal at penetrations through foundation wall for watertight installation as detailed on mechanical drawings.
3. Install shutoff valve at service entrance inside building; complete with strainer, pressure gauge, and test tee with valve.
4. Install water meter in accordance with utility company's installation instructions and requirements.
5. Size meter and arrange piping and specialties to comply with utility company's requirements.

E. INSTALLATION OF VALVES:

1. General Duty Valve Applications: The drawings should indicate valve types to be used, where specific valve types are not indicated the following requirements apply:
2. Ball Valve manufacturer: Nibco – District Preference
3. Shut-off duty: Use ball and butterfly valves.
4. Throttling duty: Use globe, ball, and butterfly valves.
5. Sectional Valves: Install sectional valves on each branch and riser, close to main, where branch or riser serves 2 or more plumbing fixtures or equipment connections, and elsewhere as indicated. For sectional valves 2inch and smaller, use ball valves; for sectional valves 2- 1/2inch and larger, use butterfly valves.
6. Shutoff Valves: Install shutoff valves on inlet of each plumbing equipment item and on inlet of each plumbing fixture, and elsewhere as indicated. For shutoff valves 2inch and smaller, use ball valves; for shutoff valves 2- 1/2inch and larger, use butterfly valves.
7. Drain Valves: Install drain valves on each plumbing equipment item, located to completely drain equipment for service or repair. Install drain valves at the base of each riser, at low points of horizontal runs, and elsewhere as required to completely drain distribution piping system. For drain valves 2inch and smaller, use ball valves; for drain valves 2-1/2inch and larger, use butterfly valves.
8. Check Valves: Install non-slam spring loaded check valves on discharge side of each pump, and elsewhere as indicated.
9. Balance Cocks: Install in each hot water recirculating loop, discharge side of each pump, and elsewhere as indicated.
10. Hose Bibbs & Wall Hydrants: Install where indicated with vacuum breaker.

22 15 00 Consumable Water Filtration

A. INTENT

1. Provide filtration of consumable drinking water for the following contaminants:
    a. Required
        1. E. coli
        2. Lead
        3. Copper
    b. Preferred, if feasible
        1. Per- and polyfluoroalkyl substances (PFAS)
2. Filter locations
    a. If feasible and aligned with budget considerations a single point of filtration at the domestic water entry is preferred.
    b. If a single point of filtration is not feasible, individual filters at each fixture providing consumable drinking water are desired.

22 20 00 Sanitary and Storm Sewer

A. CAST-IRON SOIL PIPES AND PIPE FITTINGS:

1. Hubless Cast-Iron Soil Pipe: FS WW-P-401 and CISPI Standards 301 and 310. Pipe and fittings shall be marked with the collective trademark of the cast iron soil pipe institute or receive prior approval of the engineer.
2. Cast-Iron Hub-and-Spigot Soil Pipe: ASTM A 74. Pipe and fittings shall be marked with the collective trademark of the cast iron soil pipe institute or receive prior approval of the engineer.
3. Hubless Cast-Iron Soil Pipe Couplings: Neoprene gasket complying with ASTM C564, CISPI Standard 310 and stainless steel clamp holding band.
4. Heavy Duty Hubless Cast Iron Soil Pipe Couplings: Shall be used on pipe 5" and larger. Neoprene gasket coupling with ASTM C564. 304 stainless steel shield, minimum 0.15 inches thick, minimum 3 inches wide with 4 sealing bands up to 4 inch pipe, minimum 9 inches wide with 6 sealing bands up to 10 inch pipe.
5. Cast-Iron Hub-and-Spigot Soil Pipe Fittings: Match soil pipe units; complying with ASTM A 74.
6. Neoprene Compression Gaskets: ASTM C 564.

B. SANITARY AND STORM SEWER SYSTEM INSTALLATION

1. Do not install vents within 2' of roof edge, parapet or wall line of an "on-the-roof" structure.
2. Do not install vents within 40' of outside air intakes.
3. Camera main lines.

C. SUB-SURFACE DRAINAGE TILE AND PIPE, FITTINGS, AND ACCESSORIES:

1. Clay Drain Tile: ASTM C 4, Standard Class, unless otherwise indicated; lengths as indicated.
2. Perforated Polyvinyl Chloride Pipe (PVC): ASTM D 2729; perforated except where standard sections of pipe are indicated.
3. Perforated Clay Pipe: ASTM C 700; Standard Strength, unless otherwise indicated.
4. Fittings and Accessories for Drainage Tile and Pipe: Unless otherwise indicated, provide fittings matching and of same material as pipe units; comply with same standards, where applicable, except fittings need not be perforated where pipe is required to be perforated.
5. Tile Joint Accessories: Where open-joint drainage tile is indicated, provide joint accessory units as follows:
6. Heavy-mesh burlap joint covers, with copper wire ties.
7. Coal-tar saturated felt covers, with copper wire ties.
8. Copper screen covers, 18-14 mesh, with copper wire ties.
9. Corrosion-resistant metal spacer-band covers.

D. PLUMBING PUMPS

1. General: Provide factory-tested pumps, thoroughly cleaned, and painted with one coat of machinery enamel prior to shipment. Type, size, and capacity of each pump is listed in pump schedule. Provide pumps of same type by same manufacturer.
2. Power supply wiring from power source to power connection on pumps. Include starters, disconnects, and required electrical devices, except where specified as furnished, or factory-installed, by manufacturer.
3. Interlock wiring between pumps; and between pumps and field-installed control devices.
4. Interlock wiring specified as factory-installed is work of this section.
5. Provide the following electrical work as work of this section, complying with requirements of Division-26 sections:
6. Control wiring between field-installed controls, indicating devices, and pump control panels.

7. In-line Circulation Pumps: Provide in-line recirculation pumps where indicated, and of capacities as scheduled.
   a. Type: Single stage, wet rotor type, designed for 145 psi working pressure, 140 degrees F (60 degrees C) continuous water temperature, and specifically designed for quiet operation.
   b. Pump volute: bronze.
   c. Impeller, impeller seal ring, rotor can, bearing plate, motor shaft and rotor cladding shall be constructed of stainless steel.
   d. Impeller shall be secured directly to the motor shaft be means of a stainless steel tapered split cone and locking nut.
   e. Manufacturer: Subject to compliance with requirements, provide products by one of the following: 1) Grundfos Pumps Corp.
8. Submersible Sump Pumps: Provide submersible sump pumps as indicated, of size and capacity as scheduled.
   a. Pump: Cast-iron shell, bronze impeller, stainless steel shaft, factory-sealed grease lubricated ball bearings, ceramic mechanical seal, and perforated steel strainer.
   b. Motor: Hermetically sealed, capacitor-start, with built-in overload protection, electrical characteristics as scheduled. Provide 10 feet of 3-conductor PVC cord and molded grounding plug.
   c. Manufacturer: Subject to compliance with requirements, provide products by one of the following: 1) Goulds Pumps, Inc.
      2) Weil Pump Co. 3)
      Zoeller Co.
9. Elevator Sump Pump: Provide Stancor Oil Minder System for elevator sump as manufactured by Stancor, Inc.
   a. Stancor Oil Minder control system with built in audible and visual alarms. System will lock out pump on detection of oil or high amperage condition. Provide silencing button for audible alarm. Panel shall have alarm contact for remote monitoring. Provide required cables and accessories for complete installation.
   b. Manufacturer: Subject to compliance with requirements, provide products by one of the following: 1) Stancor, Inc.

E. FACITLITY DRAINAGE AND VENT SYSTEMS
1. General: Section specifies building sanitary drainage, storm drainage and vent piping systems, including drains and drainage specialties.
2. Backwater Valves: Valve assembly shall be bronze fitted cast-iron, with bolted cover. Flapper shall provide a maximum 1/4inch clearance between flapper and seat for air circulation. Valve ends shall suit piping. Basis of design: Jay R. Smith Fig. 7012
3. Trap Primers (For equipment room floor drains, restrooms and as otherwise noted): Bronze body valve with automatic vacuum breaker, with 1/2inch connections matching piping system. Connections: Inlet 1/2inch male NPT; outlet 1/2inch female NPT. Install trap primers with piping pitched towards drain trap, minimum of 1/8 inch per foot (1 percent). Adjust trap primer for proper flow.
4. Trap Guards (Only used with Owner approval. Coordinate acceptable locations with Owner): ASSE 1072, neoprene diaphragm and sealing gasket with ABS plastic housing body for insertion into gravity drain outlet connections of sizes 2, 3, and 4-inch diameter piping with compression fit at 80 durometer. Basis of design: SureSeal Manufacturing Co.
5. Valves shall be easily adjusted to high or low pressure and shall be automatically activated whenever any faucet is opened in the building, causing a pressure drop. Basis of design: "Prime-Rite" Trap Primer Valve as manufactured by Precision Plumbing Products, Inc
6. Floor Cleanout: Round, cast iron body with recessed bronze closure plug; scoriated polished bronze frame and cover plate.
7. Wall Cleanout: Cleanout tee with raised head brass plug tapped for 1/4-20 thread; flat style chrome plated wall cover plate with holes for 1/4inch bolt; 1/4-20 threaded bolt with chrome plated flat head.
8. Surface Cleanout: Cast iron body ferrule with raised head brass plug. Medium duty cast iron manhole cover and ring 12inch diameter to be set in concrete pad, Neenah No. R-1791-A.
9. Line Cleanout: Cast iron tapped cleanout ferrule with raised head brass plug.

10. Floor Drains:
    a. Toilet Rooms, Showers and Finished Areas - Round cast iron body with flashing collar and cast iron ring, 6 inch round nickel bronze adjustable strainer head with secured square hole grate, bottom waste outlet. Provide a deep seal trap for each floor drain installed in Locker Rooms and Shower Rooms. Basis of design: Jay R. Smith Fig. 2005-A
    b. Boiler and Mechanical Rooms - Square cast iron, light duty, shallow body drain with flashing collar and cast iron ring, 8inches round tractor type non-tilt slotted grate and sediment bucket, bottom waste outlet. Basis of design: Jay R. Smith Fig. 2220
11. Floor Sinks:
    a. Indirect Waste Drain - Square, cast iron, porcelain enameled interior, sump body drain 6inch deep x 8inch square with flashing collar and cast iron ring, 8inch square nickel bronze removable half top grate, dome button strainer, bottom waste outlet. Basis of design: Jay R. Smith 3100-12
    b. Floor Sink - Square cast iron, porcelain enameled interior, sump body drain, 16inch square x 12inch deep with 16inch square nickel bronze removable half top grate, bottom waste outlet. Basis of design: Jay R. Smith 320012
12. Roof and Overflow Drains:
    a. Type 1 Drain - Cast iron bodies with sump, removable cast iron vandal-proof dome strainer, combined flashing clamp and gravel stop for roof drain and combined flashing clamp and gravel stop with 3-1/2" high internal water dam standpipe for overflow drain. Deck plate with securing holes provided to receive both drains. Basis of design: Jay R. Smith Fig. 1800–U-CID with Husky bands.
    b. Type 2 Drain - Scupper drain with cast iron bodies with flashing clamp and secured cast iron grate with vandalproof screws. Basis of design: Jay R. Smith Fig. 1510
13. Vent Caps: Provide cast-iron, vandal-proof, hooded vent caps for all vent terminations. Vent cap shall be held in place with recessed allen head set screws. Install on all vents.
14. Non-Metallic Trench Drains: Polyester resin and quartz aggregate or glass-filled polyester fiberglass, precast, interlocking design, with bottom radius and 0.6 percent slope. Provide catch basin with sediment bucket constructed of same material as trench drain and connected to trench drain; ductile iron solid cover (catch basin applicable to 3D Art only).
15. Manufacturer: Subject to compliance with requirements, provide drainage and vent systems from one of the following:
    Josam Mfg. Co.
    Smith (Jay R) Mfg. Co.
    Wade
    Precision Plumbing Products
    Quazite Corp.

F. INTERCEPTORS

1. Grease Interceptors: Provide grease interceptors, of size and capacity as indicated. Construction shall be precast concrete, two compartment, designed for $H2_O$ traffic loading. Water proofed by manufacturer for continuous interior submergence.
2. Variations: Provide the following construction feature variations:
    a. Cast iron manhole rings.
    b. Gas tight manhole cover and frame.
3. Solids Interceptors: Provide solids interceptors of size and capacity as indicated. Construction shall be cast-iron body, aluminum cover, aluminum sediment basket, and removable bronze screws.
4. Manufacturers: Subject to compliance with requirements, provide products by one of the following: a.
    Interceptors:
    Josam Mfg Co.
    Rockford Sanitary Systems Corp.
    J.R. Smith Mfg Co.

Wade Div; Tyler Pipe.

b. Grease Interceptors:
Copeland Enterprises, Inc.
Colorado Precast Concrete, Inc.

G. CHEMICAL WASTE DRAINAGE SYSTEMS

1. General: Provide piping materials and factory-fabricated piping products of sizes, types, pressure ratings, temperature ratings, and capacities as indicated. Where not indicated, provide proper selection as determined by Installer to comply with installation requirements. Provide sizes and types matching piping and equipment connections; provide fittings of materials which match pipe materials used in laboratory drainage piping systems. Extent of chemical-waste drainage systems work shall be indicated on drawings.
2. Codes and Standards:
   a. PDI Compliance: Fabricate and install chemical-waste drainage systems in accordance with PDI "Code Guide 302".
   b. UPC Compliance: Fabricate and install chemical-waste drainage systems in accordance with IAPMO "Uniform Plumbing Code".
3. Acid Waste Neutralizing Sumps: Provide acid waste neutralizing sumps, of size and capacity as indicated, constructed of acid resisting material, complete with inlet, outlet, and vent connections of indicated size. Locate outside of building.
   a. Provide submerged type inlet, complete with down pipe or baffle, and 2" minimum cleanout at top of vertical drop.
   b. Permanently mark fill line on inside of sump, 1" below outlet invert.
   c. Manufacturers: Subject to compliance with requirements GSR Fuseal
      Enfield Industrial Corporation
4. Acid Waste Piping and Fittings:
   a. Polypropylene Pipe and Pipe Fittings: Flame retardant polypropylene DWV piping and fittings, which meet ASTM D635 flame test and ASTM D-1785 dimensional requirements, with electric fusion/resistance joints. Piping shall be Schedule 40.
   b. Manufacturer shall provide vertical expansion joints with Viton O-Rings for vertical risers, where required to compensate for expansion and riser movement.
   c. Traps and Tailpieces: Provide traps and tailpieces for each fixture inlet connection to chemical-waste drainage system, constructed of same material and weight as piping system. Trap assemblies shall be mechanical joint. Mechanical joints for fused polypropylene piping systems may not be used for any other purpose.
   d. Manufacturers: Subject to compliance with requirements
      a. GSR Fuseal
      b. Enfield Industrial Corporation
      c. Charlotte Chem Drain
5. High Silicon Cast Iron Pipe & Pipe Fittings Systems:
   a. For pipe sizes 4 inches and smaller: Service weight, 14-1/2percent high silicon content cast-iron, hubless fittings, stainless steel 2-bolt couplings with one-piece sintered non-porous teflon inner sleeve liner.
   b. For pipe sizes 6inches and larger: Service weight, 14-1/2percent high silicon content cast-iron, ball and spigot fittings, lead and special acid-resistant rope packing, caulked joints.
6. Install acid waste and vent piping systems as indicated:

All sizes, below grade (burieid) inside building	Duriron or Poly-propylene
All sizes, below grade (buried) outside building	Duriron
For sizes 4” and smaller, above grade, inside bldg.	Duriron or Poly-propylene For
sizes 6” and larger, above grade, inside bldg.	Duriron

22 30 00 Plumbing Equipment

A. SUMMARY
   1. This Section includes
      a. Water Heaters

B. WARRANTY
   1. Warranty on Coil, Heat Exchanger, and Burner: Provide written warranty, signed by manufacturer, agreeing to replace/repair, within warranty period, coils, heat exchangers, and burners with inadequate or defective materials and workmanship, including leakage, breakage, improper assembly, or failure to perform as required; provided manufacturer's instructions for handling, installing, protecting, and maintaining units have been adhered to during warranty period. Replacement is limited to component replacement only, and does not include labor for removal and reinstallation. Warranty Period: 2 years from Date of Substantial Completion.

C. Water Heaters
   1. Commercial Gas Fired Water Heater: Provide commercial gas-fired water heaters of sizes and capacities as indicated on schedule. Provide certification of design by AGA under Volume III tests for commercial water heaters for delivery of 180 degrees F water. Provide water heater components which are UL-listed and labeled.
      a. Heater: Construct for working pressure of 150 PSI; boiler type hand hole cleanout; magnesium anode rod; 3/4inch tapping for relief valve; glass lining on internal surfaces exposed to water.
      b. Jacket: Insulate tank with vermin-proof glass fiber insulation. Provide outer steel jacket with baked enamel finish over bonderized undercoating.
      c. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
         Commercial Gas-Fired:
            Bardford - White tank type only.
            Lochinvar, Shield or Armor Series.
         Electric:
            Bradford – White – District Preference Rheem
            Water Heater Div; City Investing Co.
            Ruud Water Heater Div; City Investing Co.
            State Industries, Inc.
   2. Electric Water Heaters: Construct for working pressure of 150 PSI; magnesium anode rod; glass lining on internal surfaces exposed to water.
      a. Heating Elements: Low watt density with zinc plated copper sheath; double element, non-simultaneous operation.
      b. Jacket: Equip with full size control compartments with front panel opening. Insulate tank with vermin-proof glass fiber insulation. Provide outer steel jacket with baked enamel finish.

22 40 00 Plumbing Fixtures

A. FIXTURES:

1. Water Closets
    a. Floor Mount
    b. Material: vitreous china, and stainless steel satin #4 finish, gauge and rim type as specified.
    c. China fixture color: white.
    c. Action: siphon jet 1.28 g.p.f. maximum, gravity, vacuum or pressure assist as appropriate.
    d. Rim: elongated unless specified otherwise.
    e. Operating handle for manual flush valve to be located and operable on the wide side of the ADA toilet compartment. Standard shall have flush valve supplied from the right side, handle or button on the left side.
    f. Fully glazed trapways, partial glazing is not acceptable.
    g. Closet flange for mounting
    h. Flush valves: Manual - Sloan or Toto
2. Urinals:
    a. Wall mounted, white vitreous china.
    b. Gallons per flush: to match specified urinal flush rating and appropriate manual flush valve.
    c. Carrier mounted with cast iron nipple and galvanized steel pipe using floor mounted hanger and bearing plate type carrier.
    d. Flush valves: Manual - Sloan or Toto
3. Lavatories:
    a. Sink
        i Three hole faucet layout is preferred.
        ii White vitreous china, or type 302/304 stainless steel.
        iii Mounting: wall hung from floor-mounted top and bottom bearing plate or, concealed arm carrier; (wall mounted wood or metal blocking or hangers, will not be permitted).
        iv Counter mounting, self-rimming, or composite counter top with integral bowl will be permitted when specified. C. All lavatory faucets shall be provided with ceramic disc seats.
    b. Faucet
        i Configuration: Three hole faucet layout preferred.
        ii Operation: Proximity sensor (I.R., motion, etc.) preferred.
            1 Self-charging battery power preferred.
        iii Preferred product: Chicago EQ with mixing valve.
4. Sinks:
    a. Stainless Steel, Type 302/304-18-8, 18 gauge, #4 satin finish.
    b. *Preferred* Manufacturers:
        i Elkay
        ii Just
        iii Bradley
    c. *Acceptable* Manufacturers:
        i Advance Tabco, with District approval
    d. Sinks not located in casework shall be wall mounted and carrier mounted.
    e. All sink strainers shall be “Spin-N-Grin” type models unless specifically specified otherwise.
    f. All sink faucets shall have ceramic disc seats, base plate, with manual or automatic controls; types include hand, foot, knee and infra-red or heat sensing type operations.
    g. Stainless Steel sinks to be sound deadened with “Sound Guard”® undercoating.

h. Provide backflow for chemical dispensers at mop sinks by General Contractor. Chem. dispensing unit shall be provided by Owner.

5. Water Fountains, Bottle Fillers:
   a. No filter light.
   b. Preferred products:
      i Single: Elkay EZSG8WSLK
      ii Double: Elkay EZSTL8WSLK
      iii At athletic fields: Self-Draining.
   c. Backflow Preventers and Vacuum Breakers: Hose End Nidel #37HD1 Only.
6. Hand Sprayer (at mop sinks in custodial support spaces).
   a. Chicago Pre-Rinse Fitting, 512, or sim.
7. Fixture Carriers:
   a. Josam
   b. J.R.Smith
   c. Wade
   d. Zurn.
8. Emergency Showers (floor mounted) and Eye Washes (integrated with shower or sink mounted):
   a. Bradley or Guardian with Approved Thermostatic Mixing Valve complying with ANSI Z-358.1, latest edition.
   b. Provide floor sink with ¼ grate (Bradley District Preference)
9. Lead Free Faucets: (Manual & Specialty),
   a. Standard and Single Lever with ceramic discs
   b. American Standard, Chicago Faucet
   c. District Preference: Chicago EQ-A1ZB-23ABCP
10. Flexible Supplies 302-18-8 flexible stainless steel tube with braided S.S jackets, McGuire “convertible” only.
11. Garbage Disposers:
    a. Vibration isolators and sound attenuation
    b. Insinkerator Badger
12. Laboratory Faucets and Cocks with ceramic discs:
    a. Cambridge – Water Saver – T&S Brass – Royal Brass, Chicago Faucet, with serrated ends and vacuum breaker spouts and integral or added check valves for both faucet and gas cocks inlets.
13. Plumbing Fixture & Trim: T & S Brass, Chicago Faucet.
14. Plumbing Fixtures: Vitreous China: American Standard – Toto – Kohler.
15. Plumbing Fixtures: Molded Stone - Florestone – Crane/Fiat – Stern- Williams.
16. Shower: Individual or accessory Pressure Balanced Water Mixing Valves and equalizing valves less integral thermometers, Eljer, Kohler, Symmons, Powers, Lawler, Sloan. Symmons only where integral thermometers are specified.
17. Shower: Individual Standard and A.D.A. Individual Thermostatic and Pressure Balanced Water Mixing Valves, with or without integral thermometers – Lawler,- Leonard.
18. Solids Interceptor (at art rooms)
    a. Zurn Z1189, or similar.
    b. Designer to confirm appropriateness for project.
    c. Install such that lids or other maintenance opening are accessible and workable.
    d. Install with unions to allow removal for maintenance.
19. Supply Stops: Quarter Turn, Ball type with “convertible” Wheel Handle-Loose Key combination, ¼ turn. Brass Craft or Watts.

20. Thermostatic Water Mixing Valves for Volume Flow: Symmons products as specified – (Limited to a maximum operating temperature differential to 10□F or less, and minimum rate of controlled flow to .5 (1/2) G.P.M. @ 10 PSI pressure drop.
21. Trap Wrap: A.D.A Seamless: McGuire only.
22. Washer Wall Boxes: Guy Gray, IPS Corp., Symmons
23. Circulating Pumps: Grundfos, Taco

DIVISION 23 – HEATING VENTILATION & AIR CONDITIONING (HVAC)

23 00 00 Primary District Requirements

A. General Comments
   1. **Hydronic copper fittings, valves, etc. above 1" shall be brazed, not soldered.**
      a. Consider placing this note in a visible location on the hydronic piping drawing sheets.
   2. **Hydronic Water Treatment**
      a. See section 23 21 00 District for requirements.
   3. Final system selected needs to be low maintenance
   4. More and more summer school is likely because 80% of the schools are occupied year-round, all day in the summer. Need to assume summer school will occur within the design objectives.
   5. Direct evaporative cooling is not desired at all due to maintenance issues.
   6. All spaces are to be actively cooled except for the kitchen.
   7. RTU and chiller and cooling tower fans need to be quiet.
   8. Duct socks are not desired.
   9. See other sections for District preferred pumps, VFDs, and Rooftop Units.
   10. Manual isolations with drain valves for branches or risers off main piping runs are required in many locations of hydronic piping to allow for ease of hydronic system repairs.
   11. K-12 buildings are generally heating dominated buildings, so heating energy efficiency is imperative. Also, with increased summer school activities, cooling efficiency is still important.
   12. HVAC design solutions should allow for flexibility around future classroom size modifications, as well as to allow small portions of the buildings to be occupied.

B. Air Handling Units and Rooftop Unit Guidelines
   1. See HVAC Air Distribution section for District preferences for Air Handling and Rooftop Units.
   2. Administrative area should be on its own HVAC unit. The unit should have a hot water heating coil and DX cooling (with variable capacity inverter scroll compressors) so that a central chiller plant is not required.
   3. Heat recovery air handlers / RTUs with heat wheels are desired and work well and save energy. These are only to be used in spaces with high requirements for outside air.
   4. If a geo-exchange system is to be installed, SVVSD would like to have boiler plant backup and chiller backup.
   5. Ideally the projects would have major mechanical equipment in a "crawl space" that is 12' tall.
   6. Direct evaporative cooling is not desired at all due to maintenance concerns.
   7. Insure the CO2 demand controlled ventilation does not open the outside air dampers more than ASHRAE 62 requires for design occupancy.

C. Zone Level Equipment Guidelines
   1. Displacement air systems are not desired; therefore, the supply air should be overhead air distribution only.
   2. Return air intakes located at floor level are desired to improve air circulation for large spaces such as gym and cafeteria.
   3. Hydrotemp heat pumps are not desired. Climate master heat pumps are acceptable, but generally speaking SVVSD would rather avoid heat pumps because of high failure rate.
   4. Heat pumps, or FCUs are not allowed to be above ceilings because SVVSD facilities staff need to have full access to all sides of the HVAC equipment. The only HVAC equipment that is allowed in a classroom ceiling plenum are VAV boxes. Radiant floor tubing and baseboard radiators are allowed within classrooms, but SVVSD does not want any PEX water lines in concrete.
   5. VAV boxes take minimal maintenance and are easy to fix and are desired by SVVSD.

6. Radiant heating for pre-school and kindergarten is desired. Baseboard radiators are preferred over Runtal radiators (thin wall panel) to prevent occupant burns. BBR should be designed for max 160F HWS.
7. Many kitchens do not get enough use relative to what is required to be designed for exhaust and make-up air, so modulating exhaust hood fans and a modulating supply fan on the MAU can make sense. The kitchen MAU shall have return air and a modulating economizer, with a gas-fired or hot water heat section. Hot water is preferred if possible. There is to be no evaporative cooling on the MAU.

D. Heating Plant Guidelines
1. Boilers are good and a heating plant is preferred over gas fired RTUs.
2. For heating plant, boiler control isolation valves are not desired– instead design for a small boiler circulation pump for each boiler or primary pumps and then a secondary pumping system.
3. Central heating plant ideally consists of two boilers with associated variable speed circulation pumps, and each boiler is sized for 2/3 capacity, with 2 building pumps on VFDs. This allows for redundancy throughout the heating plant.
4. Air/Dirt Separator shall be installed on all heating hydronic systems. Sprirotherm is the District's preferred manufacturer.

E. Mechanical Cooling Guidelines
1. Boilers and chillers are good and air cooled and centrifugal chillers are both acceptable. Chillers should have a very large turn-down so that a small portion of the school can be occupied at a time.
2. SVVSD would consider a water cooled chiller with a water-side economizer (FPHX).
3. AHUs/ RUTs each with their own DX cooling equipment is nice over a chiller plant because it provides redundancy and you don't lose the whole school if the chiller is down. A flat plate heat exchanger for free cooling could overcome this concern, but should only be considered on middle and high schools.

F. Building Automation System Guidelines
1. SVVSD division 23 technical guidelines have updated sequences for nearly all HVAC equipment for use by the design engineers and contains technical guidelines on all building automation. Full adherence to SVVD Building Automation System Guidelines is expected.
2. All controls to be via the Building Automation System. RTUs, boiler management systems, RTU's, chillers, etc. are all to be directly controlled by the BAS. SVVSD prefers all equipment to be field controlled. All factory safeties may remain.
3. Floating point 24VAC power is acceptable for valves on VAV box reheat coils.
4. Demand controlled ventilation is desired in the gym and cafeteria, but not in the classrooms.
5. Small bathrooms should use the light status to run the EF. Use gravity backdraft dampers instead of motorized dampers on most bathroom and locker room and science room fans.
6. Unit heaters are to be on standalone control – not DDC, and no aquastats.
7. MDF (IT) rooms – prefer a FCU with economizer and split DX or DX RTU with economizer, not a mini split. No heating required and no minimum OA required. Sometimes just a transfer fan will work if the loads are low.
8. Electric rooms with transformers only need exhaust fans to roof – these spaces do not require cooling, or AHUs/RTUs.
9. DDC temperature sensors are desired cooler and freezer monitoring only.
10. Domestic hot water pump to be on the DDC system with a start/ stop schedule.

G. Labeling
1. Label ceiling grid or other surface where HVAC equipment is located above ceilings (e.g. VAVs, pumps, valves, etc.)
2. Label thermostats with the controlled equipment (e.g. VAV#, RTU#, etc.)

H. Kiln Room Design Guidelines
1. See Design Bulletin titled "Kilns".

23 05 13 Motors

A. Torque characteristics shall be sufficient to satisfactorily accelerate the driven loads with a time limit acceptable to the motor manufacturer. Motors shall be capable of starting the driven equipment while operating at 90 percent rated terminal voltage.
B. Motor sizes shall be large enough so that the driven load will not require the motor to operate in the service factor range.
C. Temperature Rating: Rated for 40 degrees C environment with maximum 80 degrees C temperature rise for continuous duty at full load (Class B Insulation). Provide Class F insulation for variable frequency drive motors.
D. Service Factor: 1.15 for poly-phase motors, and 1.35 for single phase motors, and 1.0 for inverter duty motors.
E. Motor construction: NEMA Premium Standard MG 1, general purpose, continuous duty, Design "B", except "C" where required for high starting torque. Design "E" shall not be used.
F. No shaded pole.
G. All motors specified for VFD applications have shaft grounding devices to protect the bearings.
H. Ball bearings with inner and outer shaft seals.
I. Ball bearing construction for 2 HP and larger.
J. No adjustable pitch pulleys for 3 HP and larger.
K. Omitted
L. Re-greasable, except permanently sealed where motor is normally inaccessible for regular maintenance.
M. Bearings shall be rated for minimum L-10 life of 40,000 hours.
N. Designed to resist thrust loading where belt drives or other drives produce lateral or axial thrust in motor.
O. Enclosure Type: Open drip-proof motors for indoor use where satisfactorily housed or remotely located during operation; Guarded drip-proof motors where exposed to contact by employees or building occupants.
P. Overload protection: Built-in thermal overload protection for all single phase motors and, where indicated, internal sensing device suitable for signaling and stopping motor at starter.
Q. Noise rating: "Quiet".
R. Nameplate: indicate the full identification of manufacturer, ratings, characteristics, construction, special features and similar information.
S. Phases and Current Characteristics: Unless indicated otherwise, provide squirrel-cage induction polyphase motors for 3/4 hp and larger, and provide capacitor-start single-phase motors for 1/2 hp and smaller, except 1/6 hp and smaller may, at equipment manufacturer's option, be split-phase type. Tri-voltage motors are not acceptable. Do not purchase motors until power characteristics available at building site have been confirmed by contractor.
T. All equipment specified to operate with variable frequency drives shall be provided with inverter-duty motors specifically designed for variable speed operation with high efficiency at part load conditions and constructed with Class F inverter grade insulation. Inverter duty motors shall meet requirements of NEMA MG-1 part 31.4 O.4.2.
U. All motors which will be operated by a variable frequency drive shall be warranted against any damage or defects as a result of being used with a variable frequency drive.
V. Magnetic Starters: Unless otherwise indicated, provide NEMA style, sized and rated magnetic starters including contacts and coils for motors 3/4 hp and larger and for smaller motors where interlock or automatic operation is indicated or required:
W. All 3-phase motors 2 hp and larger shall be protected against loss of phase (single phasing protection) wired into the starter. Reset shall be manual.
X. Motor connections: Flexible conduit, except where plug-in electrical cords are specifically indicated.
Y. Exterior, non-metal flexible connections should be UV rated.
Z. V-Belt Drives: Capacity of V-Belt Drives at rated RPM shall be not less than 150 percent of motor nameplate horsepower rating.
AA. Motors and Fan Wheel Pulleys: Adjustable pitch for use with motors through 15 HP; fixed pitch for use with motors larger than 15 HP. All fixed pitch sheaves, including single groove fan sheaves, shall be of the bushed type. Fixed bore sheaves will not be acceptable for adjustable pitch sheaves.
BB. Manufactures: Subject to compliance with requirements, provide products by one of the following manufacturers for each type of product:
CC. O. Smith
DD. Baldor
EE. U.S. Motors
FF. Leland Faraday

23 05 14 Variable Frequency Drives

A. Enclosure: Mount VFD and all components within a NEMA 1 metal enclosure. Provide floor stand where building walls are not suitable for mounting drive. VFDs are not to be located outdoors, and when located in a RTU, are to be in a filtered air section.
B. Provide filtered, fan powered ventilation for drive cooling. Fan shall be sized for "dirty filter" condition, at project altitude.
C. Drive and by-pass enclosure doors shall have provisions for locking with a padlock or integral lock, keyed to the building standards.
D. Switches and pilot lights shall be labeled with engraved plastic laminate tags riveted or similarly permanently fastened.
E. Disconnects: Provide a door fused disconnect with Type "J" fuses, with a through-the-door handle, pad lockable in the OFF position. The door interlock shall be defeatable with a tool to allow service access without de-energizing the system. The disconnect shall remove power from all components within both compartments.
F. Provide a three pole motor overload relay connected to shut down the motor.
G. 120V control power transformer with fused secondary and primary. Primary power shall be derived ahead of the VFD input contactor.
H. Input Power: The drive shall be capable of accepting facility power as specified on the drawings. Variations of up to plus or minus 10 percent of line voltage and plus or minus 2HZ of line frequency shall be permitted without the drive shutting down on a fault.
I. Power line interruptions of up to 0.5 seconds shall be permitted without the drive shutting down on a fault. 3-phase monitors shall be provided on the main line of the feed into the device for RTUs, VFDs, 3-phase exhaust fans.
J. Manufacturer shall guarantee that harmonic voltage and current distortion, on the line side (input terminals) of the VFD does not exceed 5 percent total voltage Harmonic distortion, and 15 percent total current Harmonic distortion.
K. Manufacturer shall review electrical drawings to determine optimum characteristics of the reactor/filter system.
L. Output Power: The variable frequency AC drive shall convert 3 phase, 60 HZ input power to an adjustable AC frequency and voltage for controlling the speed of any standard NEMA B Design, AC squirrel cage motors driving variable torque loads. The drive shall be rated for continuous duty at the NEC standard full load current of it's associated motor.
M. The VFD shall be arranged to provide automatic restart after a trip condition resulting from over-voltage or under-voltage.
N. For safety, the drive shall shut down and require manual reset and restart if the automatic reset/restart function is not successful within a maximum of five attempts.
O. Belt driven fans: All VFD fans shall be specified with synchronous belts and pulleys.
    1. Gates
    2. Browning
P. VFD shall include a "Bus Charged" warning indicator, and shall be provided with automatic circuitry to discharge the bus within 120 seconds after main power is disconnected.
Q. Warranty: All variable frequency drives shall be warranted for 24 months after the building has been accepted by the Owner. This warranty shall include all parts, labor, materials, shipping cost, travel, lodging and meals with no cost to the Owner.
R. Manufactures: Subject to compliance with requirements, provide products by one of the following manufacturers for each type of product:
    1. Variable Frequency Drives:
    2. ABB
    3. Mitsubishi
    4. Yaskawa Electric America
    5. Eaton

23 05 15 Piping Specialties

A. Hydronic System Safety Relief Valves: Diaphragm operated, cast-iron or brass body valve, with low inlet pressure check valve, inlet strainer removable without system shut-down, and noncorrosive valve seat and stem. Select valve size, capacity, and operating pressure to suit system. Valve shall be factory-set at operating pressure to suit system and have the capability for field adjustment. Safety relief valve shall be designed, manufactured, tested and labeled in accordance with the requirements of Section IV of the ASME Boiler and Pressure Vessel Code. Valve body shall be cast-iron, with all wetted internal working parts made of brass and rubber; 125 psig working pressure and 250 degrees F maximum operating temperature. Select valve to suit actual system pressure and BTU capacity. Set valve to relieve at 10 psi above operating pressure. Acceptable manuf. of Combination Pressure and Temperature Relief Valves:
    1. Bell & Gossett ITT; Fluid Handling Div.
    2. Taco, Inc
    3. Watts Regulator Co.

B. Pressure Reducing Valves: Diaphragm operated, bronze or brass body valve, with low inlet pressure check valve, stainless steel inlet strainer removable without system shut-down, and stainless steel valve seat and stem. Select valve size, capacity, and operating pressure to suit system. Valve shall be factory-set at operating pressure and have the capability for field adjustment. Acceptable manuf. of Pressure Reducing Valves:
    1. Amtrol, Inc.
    2. Taco, Inc.
    3. Watts Regulator Co

C. Coin Operated Manual Air Vent: Bronze body and nonferrous internal parts; 150 psig working pressure, 212 degrees F operating temperature; manually coin operated and having discharge outlet connection and 1/8 inch NPT male connection.

D. Automatic Air Vent: 100 psi working pressure, 240 degrees working temperature, stamped brass body and non-metallic float, with threaded outlet connector for "safe waste" discharge pipe. Acceptable manuf. of Air Vents:
    1. Hoffman Specialty ITT; Fluid Handling Div.
    2. Amtrol 703
    3. Spirotherm, Inc.

E. Air Eliminator and Dirt Separator: Furnish and install as shown on the drawings combination coalescing type air eliminator and dirt separators. Pipe size is not a factor and all units shall be selected per the manufacturer's recommendations. All combination units shall be fabricated steel, rated for 150 psig working pressure with entering velocities not to exceed 4 feet per second at specified GPM. Units specifically designed for high velocity systems may have an entering velocity of up to 10 feet per second. Units shall include an internal bundle filling the entire vessel to suppress turbulence and provide high efficiency. The bundle shall consist of a copper core tube with continuous wound copper medium permanently affixed to the internal element. Each eliminator shall have a separate venting chamber to prevent system contaminants from harming the float and venting valve operation. At the top of the venting chamber shall be an integral full port float actuated brass venting mechanism. Units shall include a valved side tap to flush floating dirt or liquids and for quick bleeding of large amounts of air during system fill or refill. Air separators shall be capable of removing 100% of the free air, 100% of the entrained air and up to 99.6% of the dissolved air in the system fluid. Dirt separation shall be at least 80% of all particles 30 micron and larger within 100 passes.
    1. Acceptable manufacturers:
        a. Spirotherm, Inc.
        b. Caleffi, Inc.

F. Bladder-Type Expansion Tanks: Construct of welded carbon steel for 125 psig working pressure, 240 degrees F maximum operating temperature. Separate air charge from system water to maintain design expansion capacity, by means of a flexible heavy duty bladder securely sealed into tank. Expansion tank shall have replaceable Bladder. Bladder shall be suitable for glycol service and system water treatment chemicals. Provide taps for pressure gauge and air charging fitting, and drain fitting. Support vertical tanks with steel legs or base; support horizontal tanks with steel saddles. Tank, with taps and supports, shall be constructed, tested, and labeled in accordance with ASME Pressure Vessel Code, Section VIII, Division 1. Acceptable manuf. of Bladder-Type Expansion Tanks:
    1. Amtrol, Inc.
    2. Armstrong Pumps, Inc.
    3. Bell and Gossett ITT; Fluid Handling Div.

4. Taco, Inc.

G. Pump Suction Diffusers: Cast-iron body, with threaded connections for 2 inch and smaller, flanged connections for 2-1/2 inch and larger; 175 psig working pressure, 300 degrees F maximum operating temperature. Inlet vanes with length 1-1/2 times pump suction diameter or greater. Cylinder strainer with 3/16 inch diameter openings with total free area equal to or greater than 5 times cross-sectional area of pump suction, designed to withstand pressure differential equal to pump shutoff head. Acceptable manuf. of Pump Suction Diffusers:
   1. Armstrong Pumps, Inc.
   2. Bell & Gossett, ITT
   3. Taco
   4. Aurora Pumps

H. Pressure Independent Automatic Flow Control Valves to be used on all terminal coils, AHUs, RTUs, FCUs. Valves shall be: class 150, cast iron housing, stainless steel operating parts; threaded connections for 2 inch and smaller, flanged connections for 2-1/2 inch and larger. Factory set to automatically control flow rates within plus or minus 5 percent design, while compensating for system operating pressure differential. Provide quick disconnect valves for flow measuring equipment. Provide a metal identification tag with chain for each valve, factory marked with the zone identification, valve model number, and rate flow in GPM. Acceptable manuf. of Automatic Flow Control Valves: Danfoss

I. Low Pressure Strainers to be installed on all coils: Provide strainers full line size of connecting piping, with ends matching piping system materials. Select strainers for 125 psi working pressure, with Type 304 stainless steel screen. Two inches and smaller steam and liquid strainers shall have 20 mesh screens. Provide 3/64 inch perforations for 2-1/2 inch and 3 inch steam and liquid strainers. Provide 1/8 inch mesh perforations for 4 inches and larger liquid strainers. Provide 1/16 inch mesh perforations for 4 inches and larger steam strainers. Acceptable manuf. of Low Pressure Strainers:
   1. Bell & Gossett, ITT
   2. Hoffman Specialty ITT; Fluid Handling Div.
   3. Watts Regulator Co.

J. Dielectric Waterway: Zinc electroplated nipple with non-metallic lining for use in service indicated, which effectively isolate ferrous from non-ferrous piping (electrical conductance), prevent galvanic action, and stop corrosion. Union style not acceptable. Acceptable manuf. of Dielectric Waterways
   1. Perfection Corp.
   2. Flow Design Inc.
   3. Perfection Corp.
   4. Rockford-Eclipse Div.

K. Mechanical Sleeve Seals: Modular mechanical type, consisting of interlocking synthetic rubber links shaped to continuously fill annular space between pipe and sleeve, connected with bolts and pressure plates which cause rubber sealing elements to expand when tightened, providing watertight seal and electrical insulation. Acceptable manuf. of Mechanical Sleeve Seal:
   1. Thunderline Corp.
   2. Metra Flex.

23 05 19 Meters & Gauges

A. PHOTO VOLTAIC DIGITAL THERMOMETERS: Utilize in central plants for temperature gages. High image ABS, with photovoltaic power cell and digital readout. Selectable between -40-300 degrees F/ -40-150 degrees F, displayed to 0.1 degrees. Display update: 10 seconds. to compliance with requirements, provide products by one of the following:
    1. Versa Gauge
    2. Weiss Instruments
B. DIAL TYPE INSERTION THERMOMETERS: Provide dial type insertion thermometers of materials, capacities, and ranges indicated, designed and constructed for use in service indicated. Bi-metal, stainless steel case and stem, 1 inch diameter dial, dust and leak proof, 1/8 inch diameter stem with nominal length of 5 inch. Range shall be 0 degrees - 220 degrees F. Subject to compliance with requirements, provide products by one of the following:
    1. Trerice (H.O.) Co.
    2. Weiss Instruments, Inc.
C. TEMPERATURE GAUGE CONNECTOR PLUGS: Provide temperature gauge connector plugs pressure rated for 500 psi and 200 degrees F. Construct of brass and finish in nickel-plate, equip with 1/2inch NPS fitting, with self-sealing valve core type neoprene gasketed orifice suitable for inserting 1/8 inch O.D. probe assembly from dial type insertion thermometer. Equip orifice with gasketed screw cap and chain. Provide extension, length equal to insulation thickness, for insulated piping. Subject to compliance with requirements, provide products by one of the following:
    1. Peterson Equipment Co.
    2. Universal Lancaster
    3. Sisco
D. PRESSURE GAUGES: Provide general use, 1 percent accuracy, ANSI B40.1 grade A, phospher bronze bourdon type, bottom connection. Include snubbers prior to sensor pickup. Across pumps, use one pressure gauge with two isolation valves for pump pressure readings. Case shall be drawn steel or brass, glass lens, 4-1/2inch diameter. Range: Water: 0 - 100 psi. Subject to compliance with requirements, provide products by one of the following:
    1. Ametek/U.S. Gauge.
    2. Trerice (H.O.) Co.
    3. Weiss Instruments, Inc.
E. PRESSURE GAUGE COCKS: Provide pressure gauge cocks between pressure gauges and gauge tees on piping systems. Snubber: ¼ inch brass bushing with corrosion resistant porous metal disc, through which pressure fluid is filtered. Select disc material for fluid served and pressure rating. Subject to compliance with requirements, provide products by one of the following:
    1. Peterson Equipment Co.
    2. Universal Lancaster
    3. Sisco
F. VENTURI TUBE FLOW MEASURING ELEMENTS: Within central plants, primary flow measuring elements shall consist of venturi tubes. Arrange piping in accordance with manufacturer's published literature. In horizontal pipes, place connections slightly above horizontal centerline of pipe. Provide venturi tubes of solid brass or bronze. Tubes larger than 2 inches shall have flanges or butt weld connections and may be cast iron or steel. Steel tubes may be fabricated or cast with cadmium or zinc-plating. Line throats of cast iron tubes with bronze and plate cast iron portion with cadmium. Subject to compliance with requirements, provide products by one of the following:
    1. Gerand
    2. Barco/Aeroquip Corp.
    3. Presso
    4. Flow Design Inc.
G. BALANCE VALVES: In central plants, provide calibrated balance valves equipped with readout valves to facilitate connecting of differential pressure meter to balance valves. Equip each readout port with a quick connect valve designed to minimize system fluid loss during monitoring process. Subject to compliance with requirements, provide products by one of the following:
    1. Armstrong+
    2. Flowset - Flow Design, Inc.
    3. Presso.

4. Tour and Anderson, Inc.
5. Gerand "Balvalve Indicator"

23 05 20 Valves

A. General Valve Features: Unless otherwise indicated, provide valves of same size as upstream pipe size. (Control valves shall be sized for required flow.)
B. Lever Handle on quarter turn valves 6 inch and smaller, except plug valves. Provide a wrench for every plug valve.
C. Chainwheel operators for valves 2-1/2 inch and larger that are installed 96 inches or higher above finished floor elevation. Provide chains to an elevation of 6'-0" above finished floor elevation.
D. Extended Stems: Where insulation is indicated or specified, provide extended stems to allow full operation of the valve without interference by the insulation.
E. BALL VALVES:
    1. Ball Valves – 1inch and Smaller: 150 WSP, 600 WOG, rated for 150 PSI at 350 degrees F, two piece end entry body style, bronze body conforming to ASTM B584, full port chrome plated brass ball, 15 percent glass reinforced PTFE seats, PTFE packing, adjustable packing nut blow-out proof stem, vinyl covered steel handle. Provide solder ends or threaded ends to match piping system. Basis of design: Apollo 77-100/200
    2. Ball Valves 1-1/4 inch through 2 inches and for all silver soldered or brazed lines: ANSI B16.34, 150 WSP, 600 WOG, rated for 150 PSI at 350 degrees F. Three piece body style, bronze body conforming to ASTM B584, full port, chrome plated brass ball and stem of ASTM A276 type 316, 15 percent glass reinforced RTFE seats, RTFE packing and blow out proof stem, vinyl coated steel handle. Provide solder ends or threaded ends to match piping material system.
    3. Ball Valves Manufacturers:
        a. Conbraco (Apollo 82-100/200)
        b. Jomar
        c. Nibco
        d. Red and White
F. BUTTERFLY VALVES:
    1. Hydronic Service Butterfly Valves - 2-1/2 inches to 12 inches: MSS SP-67, cast iron body conforming to ASTM A126 class B, nylon coated ductile iron disc, EPDM seat non-sulfur high purity peroxide cured EPDM, ie: Bray, upper and lower bronze bearing, non-metallic bushing and stem seal, ANSI class 125 flange, rated for 200 psi pressure differential, 200 psi drop-tight shut off dead end service, with downstream flange removed. Provide extended neck for 2 inches thick insulation. All valves shall be factory tested to 110 percent of pressure rating. All butterfly valves shall be full lugged body, drilled and tapped. Refer to “MOTORIZED VALVE ACTUATORS” below for butterfly actuator specifications.
        a. Butterfly Valve Manufacturers:
            1. Bray (basis of design)
            2. Centerline
            3. Nibco
G. CHECK VALVES:
    1. Swing Check Valves - 2-1/2 Inch and Smaller: MSS SP-80; Class 125/150 WSP 200/300, cast bronze body and cap conforming to ASTM B 62, ASTM B61 for 200/300 bronze, horizontal swing design, Y-pattern, with a bronze/teflon disc, stainless steel pin and having threaded or solder ends. Class 150 valves meeting the above specifications may be used where pressure requires or Class 125 are not available.
    2. Swing Check Valves - 2-1/2 to 3 Inch: MSS SP-71; Class 125 (Class 175 FM approved for fire protection piping systems), cast iron body and bolted cap conforming to ASTM A 126, Class B; horizontal swing, with a bronze disc or cast iron disc with bronze disc ring, and flanged ends. For sewage ejector and sump pump discharge swing check valves 2-1/2 inches and larger, provide outside lever with weight or spring to assist disc to close rapidly.
    3. Non-Slam Check Valves - 2 Inch and smaller: Bronze body, 200 psi @ 250 degrees F., threaded ends, resilient seats, center guided spring loaded disk.

4. Non-Slam Check Valves - 2-1/2 Inch and Larger: Class 125/ 250 cast iron or stainless steel body, replaceable lapped bronze seat and balanced twin bronze flappers or bronze center guided disc and stainless steel trim. Valve shall be designed to open and close at approximately one foot differential pressure. Twin flappers or center guided disc shall be loaded with a stainless steel spring to assure even non-slam checking action. Seals shall be EPDM.
    a. Swing Check Valve Manufacturers:
        1. Conbraco
        2. Grinnell Corp.
        3. Nibco
        4. Watts (screwed)
    b. Non-Slam Check Valves:
        1. Centerline
        2. Keystone
        3. Metraflex
        4. Nibco
        5. Milwaukee

H. SOLENOID VALVES:
1. Solenoid valves shall consist of a solenoid (electro-magnet) with its core and a valve body containing one or more orifices. The solenoid shall be mounted directly on the valve body. Response time from fully open or closed to fully closed or open shall be slow acting. Solenoid valves shall be 2 way type valves. Solenoid valves shall be normally open or normally closed operation as required for proper operation of the system for protection against freeze, fire and safety.
    a. Solenoid Valve Manufacturers:
        1. Asco
        2. Magnatrol Valve Corp.

I. CONTROL VALVES
1. Valves under 4" used for control shall be modulating ball valves with characterized discs, and shall be sized for 2-5psi pressure drop. Control isolation valves shall be line-sized. All valves to have NPT threads. Power to be 24VAC, generally no valve position feedback is required. Control signal to be 0-10VDC or 2-10VDC, except VAV reheat coil valves can be 24V floating point. If 3-way valves are required, engineer to clearly note where they are required. Fail positions to be based on service.
2. BBR control valves to be located in BBR enclosure.
    a. Valve Manufacturers:
        1. Griswold
        2. Belimo

J. MOTORIZED VALVE ACTUATORS:
1. 4" valves and larger: Valve actuator shall consist of a 120 volt, single phase, permanent split capacitor, reversible type electric motor which drives a compound epicyclic gear. A manual override handwheel shall be integral to the unit. The electric actuator shall have visual mechanical position indication, readable from a distance of 25 feet, showing output shaft and valve position. Unit shall be capable of mounting directly to butterfly valve without brackets and adapters.
2. The actuator shall have an integral terminal strip, for wiring to power supplies. Cable entry shall be by means of two (2) 1 inch NPT threaded connections. Cable entries shall have UL recommended gland stops within the NPT hole to prevent glands from being screwed in too far and damaging cable. The environmental temperature range of the actuator shall be -20 degrees F. to +140 degrees F.
3. The electric actuator shall have a pressure die-cast, hard anodized aluminum base and cover. The compound gear shall be made of die-cast, hard anodized aluminum or steel. An alloy steel worm gear shall be provided for manual override and torque limiting. Bearings for gears shall be of the ball and needle type; bronze bearings shall be used on the shafting parts.
4. Fail positions to be based on service. Fail mechanism shall be from a spring, not a capacitor.
    a. Valve Manufacturers:
        1. Bray

2. Belimo
3. Griswold

K. VALVE INSTALLATIONS:

L. Locate valves for easy access and provide separate support where necessary.

M. Install valves and unions for each fixture and item of equipment in a manner to allow equipment removal without system shut-down. Unions are not required on flanged devices.

N. Install 3-valve bypass around each pressure reducing valve using throttling type valves.

O. Globe valves shall be installed with the stem in the upright position. In overhead horizontal piping, ball valves shall be installed with the handle in the side or bottom of the piping. Butterfly valves shall be installed with the stem within 45 degrees of the horizontal position. The handle of quarter turn valves shall open in the direction of flow. Quarter turn valves with hand wheels or chain wheels shall be located so that the position indicator is visible from the floor without the use of a ladder or climbing on equipment or piping.

P. Installation of Check Valves: Install for proper direction of flow as follows:

SERVICE	VALVE TYPE
Plumbing Water Services; 3" or smaller	Ball Valve
Plumbing Water Services; 2-1/2" and larger	Butterfly Valves
Plumbing and HVAC Pressure Reducing Bypass; all sizes	Globe Valve
Plumbing Balancing Service; 3" and smaller	Calibrated Balancing Ball Valve See Section 15135
Plumbing Balancing Service; 3" and larger	Butterfly Valve
HVAC Hydronic Piping; 3" and smaller	Ball Valve
HVAC Hydronic Piping; 2-1/2" and larger	Butterfly Valve
HVAC Hydronic Balancing valve; 3" and smaller	Calibrated Balancing Ball Valve
HVAC Hydronic Balancing Valve; 2" and larger	Butterfly Valve.
HVAC & Plumbing Check Valves; 2" and smaller	Swing Check
HVAC & Plumbing Pump Discharge Check Valve;	Non-Slam Spring Loaded
HVAC Control Valves 1-1/4” and smaller	Ball Valve with Characterizing Disc
HVAC Control Valves 1” and larger and All 3 Way Valves	Globe Valve
Natural Gas Line 2-1/2” and smaller	Ball Valve
Natural Gas Line 2-1/2” and larger	Eccentric Plug Valve
Sump Pump	Swing Check, (Provide outside weighted lever or spring for 2-1/2" and larger)

Q. Swing Check Valves: Install in horizontal position with hinge pin level.

R. Wafer Check Valves: Install between 2 flanges in horizontal or vertical upward flow position.

23 05 30 Electric Heating Cables

A. Pipe Freeze Protection Heating Cables: Provide self-regulating heat-tracing electric heating cable and components which are suitable for freeze protection of metal or plastic piping. Select heater consisting of 2-16 AWG tinned-copper bus wires parallel embedded in a radiation cross-linked conductive polymer core that varies its power output to respond to temperature along its length, allowing the heater to be crossed over itself without overheating, to be cut to length in the field, and to be used directly on plastic pipe. Cover heater with a radiation cross-linked modified polyolefin dielectric jacket. Provide heater with a self-regulating factor of greater than 90 percent to conserve energy and to prevent overheating.

B. Roof and Gutter De-icing Heating Systems: Provide self-regulating electric heating cable and components which are suitable for use on roofs and in gutters for snow and ice melting. Select heater consisting of 2-16 AWG tinned-copper bus wires embedded in parallel in a radiation cross-linked conductive polymer core that varies its heat output to respond to temperature along its length, allowing the heater to be crossed over itself without overheating, to be used safely in contact with wood, PVC, EDPM, neoprene and asphalt building materials, and to be cut to length in the field. Cover heater with a radiation cross-linked modified polyolefin dielectric jacket and protect with a tinned-copper braid and a polyolefin outer jacket. Provide self-regulating heater with a power output of 10 watts per foot where it contacts ice and snow and 5 watts per foot in air.

C. Electric Heating System Controls: Provide electric radiant heating cable thermostats, with temperature range from 35 to 95 deg. F (2 to 32 deg. C), operating voltage of 125, 60 Hz, and with manually operated OFF-position switch, as recommended by cable manufacturer.

D. Manufacturers: Subject to compliance with requirements, provide electric heating cables of one of the following (for each type of cable):
    1. Easy-Heat WireKraft Div; Bristol Corporation.
    2. Thermon
    3. Raychem Corporation.
    4. Smith-Gates Corporation.
    5. Chromalox Div; Emerson Electric Company

23 05 40 Vibration Control

A. All mechanical equipment, piping and ductwork as noted or in the specification shall be mounted on vibration isolators to prevent the transmission of vibration and mechanically transmitted sound to the building structure. Vibration isolators shall be selected in accordance with the weight distribution so as to produce reasonably uniform deflections.

B. Manufacturer: Subject to compliance with requirements, provide products by:
    1. Amber/Booth Company, Inc.
    2. Kinetics Noise Control
    3. Mason Industries, Inc.

23 05 50 HVAC Identification

A. DUCTWORK IDENTIFICATION:
   1. Identify air supply, return, exhaust ductwork and duct access doors with duct markers; or provide stenciled signs and arrows, showing ductwork service and direction of flow, in black or white (whichever provides most contrast with ductwork color).
   2. Access Doors: Provide duct markers or stenciled signs on each access door in ductwork and housings, indicating purpose of access (to what equipment), other maintenance and operating instructions, and appropriate safety and procedural information.
   3. Plastic Duct Markers: Provide manufacturer's standard laminated plastic, duct markers. For hazardous exhausts, use colors and designs recommended by ANSI A13.1.
   4. PIPING SYSTEM IDENTIFICATION:
   5. Install pipe markers of the following type on each system indicated to receive identification, and include arrows to show normal direction of flow. Locate pipe markers and color bands as follows wherever piping is exposed to view in occupied spaces, machine rooms, accessible maintenance spaces (shafts, plenums, etc.) and exterior non-concealed locations.
   6. Near each valve and control device.
   7. Near each branch, excluding short take-offs for fixtures and terminal units; mark each pipe at branch, where there could be question of flow pattern.
   8. Near locations where pipes pass through walls or floors/ceilings, or enter non-accessible enclosures.
   9. At access doors and similar access points which permit view of concealed piping.
   10. Near major equipment items and other points of origination and termination.
   11. Spaced intermediately at maximum spacing of 25 feet along each piping run, except reduce spacing to 15' in congested areas of piping and equipment.
   12. Plastic Pipe Markers: Provide manufacturer's standard pre-printed, semi-rigid snap-on, color-coded pipe markers, complying with ANSI A13.1. Furnish 1 inch thick molded fiberglass insulation with jacket for each plastic pipe marker to be installed on uninsulated pipes subjected to fluid temperatures of 125 degrees F. (52 degrees C.) or greater. Cut length to extend 2inches beyond each end of plastic pipe marker. Print each pipe marker with arrows indicating direction of flow, either integrally with piping system service lettering (to accommodate both directions), or as separate unit of plastic.

B. UNDERGROUND PIPING:
   1. Not allowed without specific District approval.

C. EXPOSED ABOVE GROUND PIPING:
   1. General: Not preferred and limit scope.
   2. Where required, protect pipe from unauthorized access (limit ability for public to access piping).

D. VALVE IDENTIFICATION:
   1. General: Provide valve tag on valves in each piping system. List each tagged valve in valve schedule for each piping system.
   2. Building services main shut-off valves.
   3. Each individual system main shut-off valves.
   4. Each individual system riser shut-off valves.
   5. Each individual system floor shut-off valves.
   6. Each individual system major branch shut-off valves.
   7. Valve Tags: Provide 19-gage polished brass valve tags with stamp-engraved piping system abbreviation in 1/4inch high letters and sequenced valve numbers 1/2inch high, and with 5/32inch hole for fastener. Provide 1-1/2inch diameter tags, except as otherwise indicated.

E. MANUFACTURERS:
   1. Manufacturer: Subject to compliance with requirements, provide products by one of the following:
      a. Allen Systems, Inc.
      b. Brady (W.H.) Co.; Signmark Div.
      c. Brimar Industries, Inc.

d. Industrial Safety Supply Co., Inc.
e. Seton Name Plate Corp.
f. PVC Specialties

23 05 90 Testing and Balancing

A. Pre-Construction test shall be performed on air handlers, duct work, pumps and piping when equipment is to remain and be re-used:
B. Test, adjust and balance the following mechanical systems and the mechanical equipment associated with these systems:
    1. Air Side Systems and Equipment:
        a. Supply/Return Air Systems
        b. General Exhaust/Supply Fans
        c. Air Handling Units
        d. General Exhaust Systems
        e. Cabinet/Unit Heaters
        f. Kitchen Systems
        g. Science Classroom Systems
    2. Hydronic Systems and Equipment:
        a. Heating water systems
        b. Hydronic coils
        c. Baseboard
        d. Hydronic Pumps
        e. Heating Water Boilers
        f. Terminal Units
        g. Air Handling Units
    3. Plumbing:
        a. General Piping
        b. Sump Pumps
        c. Booster
        d. Water heaters
    4. Gas Fired Systems and Equipment:
    5. Electrical Components:
        a. Manual and magnetic starters
        b. Variable frequency drives
        c. Control Systems and Equipment
    6. Life Safety Systems and Equipment:
        a. Fire/smoke dampers and detection systems
        b. CO^2 and Freon Detectors
C. BELTS, SHEEVES, IMPELLERS:
    1. Impellers shall be trimmed or replaced by the mechanical contractor and shall be correctly re-sized and coordinated by the Test and Balancing Contractor per the hydronic systems and equipment portion of this section.
    2. The Test & Balance Contractor shall determine the fan belt and sheeve replacement necessary for final balance condition for specified air quantity when the VFD is operating in the by-pass mode for final field conditions, without placing the motor over its nameplate amp rating.
D. PRELIMINARY PROCEDURES:
    1. Testing and balancing shall not begin until the system has been completed and is in full working order and the following project conditions have been determined suitable for start of work.
    2. Preliminary Testing & Balancing Contractor requirements shall be ascertained prior to the commencement of work through a review of the project plans and specifications. Conduct TAB conference with owner, CxA and contractors prior to commencement of work for review and approval of proposed TAB procedures and strategies.

3. Before any air balance work is done, the system shall be checked for duct leakage (obtain pressure test results), assure filters are installed, verify filters are changed if they are dirty, check for correct fan rotation, equipment vibration, and check automatic dampers for proper operation. All volume control dampers and outlets shall be wide open at this time.

E. GENERAL SYSTEM AND EQUIPMENT PROCEDURES:
1. Balance all air and water flows at terminals within +10% to -5% of design flow quantities.
2. Pressure relationships indicated on drawings shall take priority over air quantities.
3. Measure, adjust and report equipment running motor amps and power factor, KW, rated motor amperage, listed motor power factor, voltage, and all nameplate data. Perform these measurements for all equipment operational modes.
4. Check and adjust equipment belt tensioning.
5. Verify that all equipment safety and operating controls are in place, tested, adjusted and set prior to balancing.
6. Verify that manufacturer start-up has occurred per specification prior to balancing.

F. AIR SIDE SYSTEMS AND EQUIPMENT PROCEDURES:
1. Filters shall be restricted to increase pressure drop to 50% of span between initial pressure drop and final recommended pressure drop for setting final airflows for fans. Check fan motor amps with clean filters and simulated loaded filters, and report for each piece of equipment. Equipment shall be supplied with clean filters upon completion of balance. Balance and report air quantities.
2. Supply/Return Air Systems: Balance and report supply and return diffuser/grille quantities. Air diffusion patterns shall be set as noted on drawings and to minimize objectionable drafts and noise. Report design air device inlet or outlet size, actual inlet or outlet size, design and actual velocity through the orifice, for each terminal in the system. Balance and report the above measurements in all system operational/modes, and to include diversity when applicable.
3. General Exhaust/Supply Fans: Adjust CFM to system requirements. For belt drive include sheave and belt exchange to deliver airflow within limits of installed motor horsepower and mechanical stress limits of the fan. Determine the limiting fan tip speed before increasing RPM. Final fan speed setting shall allow for filter loading (as applicable) and shall establish proper duct pressures for operation of zone CFM regulators. For direct drive with speed taps: Set fan speed on tap which most closely approaches design CFM by adjusting the speed control After adjustment, check fans ability to re-start after powering down. Increase setting if required for proper starting.
4. In addition to the direct measuring of airflow quantities, measure and record outside air, return air and mixed air temperatures. Adjusting and setting the outside air quantity as a percentage of damper position will not be acceptable.
5. General Exhaust Systems: Balance and report exhaust grille quantities. Report objectionable noise. Provide full pitot traverses at each individual exhaust riser and at each exhaust fan. Balance and report. Report design air device inlet or outlet size, actual inlet or outlet size, design and actual velocity through the orifice, for each terminal in the system.
6. Cabinet/Unit Heaters: Balance and report entering and leaving air temperature. Report airflow. Balance all coils per hydronic equipment portions of this section.
7. Science Room Hood Exhaust System Performance Evaluation: The Balancing Contractor shall demonstrate that the exhaust system and pressure relationships are performing properly under the conditions listed below. Verify that the exhaust fans on the roof are operating properly. The discharge dampers should be performing as specified and the fans should be operating smoothly without surging or vibrating excessively.

G. HYDRONIC SYSTEMS AND EQUIPMENT:
1. Hydronic Systems With Meters: The system shall be balanced proportionally using the flow meters. On completion of the balance, the following information shall be recorded in the report: Flow meter size and brand, required flow rate and pressure drop, valve settings on meters with a readable scale, flow rate in both full coil flow and full bypass modes. Contractor shall verify the meters are installed per the manufacturer's recommendations and shall notify the Mechanical Contractor of any deficiencies before utilizing meter.
2. Sample heating water and report on freeze and burst temperatures of the glycol water solution. Report glycol percentage.
3. Hydronic Coils: Balance, measure and report inlet and outlet air temperatures for heating design air quantities. Balance, measure and report coil water flow, inlet and outlet water pressure and temperature.

4. Fintube Baseboard /Terminal Units: Balance, measure and report GPM, inlet temperature and outlet temperature/pressure at full heating.
5. Hydronic Pumps: Adjust and balance pumps to provide design system flow rate, and design flow to most remote system location. Trim or replace impellers as necessary to achieve this. Do not induce false head to achieve balance results, without the prior approval of SVVSD. Report impeller size, flow rate, inlet and outlet water pressure and pump shut-off head. Provide pump curve and operating point in final report. Include compensations for temperature and percentage glycol.
6. Heating Water Boilers: Verify that the boiler has been properly cleaned, flushed and the burner is operating properly prior to balancing. Check and report on proper operation of boiler feed pumps. Balance, measure and report boiler water flow rate, inlet and outlet temperature and pressure.

H. PLUMBING SYSTEMS AND EQUIPMENT:
1. General: Check, adjust and set temperature control devices to 110□F domestic hot water temperature and as indicated on drawings. Adjust pressure-reducing stations, report downstream system static pressure. Verify proper location and operation of ASME pressure and temperature relief valves. Measure and report residual pressure at full flow at most remote plumbing fixture; requiring highest operating pressure (usually flush valve water closets).
2. Sump Pumps: Verify proper discharge of sump pump systems. Verify that units are meeting system demands. Report.
3. Water Heaters: Measure, set and report inlet and outlet temperatures. Balance and report per gas portions of this Section.

I. GAS FIRED SYSTEMS AND EQUIPMENT:
1. General: Record gas pressures downstream of each pressure-reducing valve. Verify that pressures match those on construction documents and report. Measure and report flue temperature at all stages of heating.

J. ELECTRICAL COMPONENTS ASSOCIATED WITH MECHANICAL SYSTEMS:
1. Manual and Magnetic Starters: Check all new and existing thermal overloads. Identify improperly protected equipment in report. Furnish and exchange thermals as required for proper motor protection.
2. Motor Control Center Magnetic Starters: Check for correct sizing. Notify Electrical Contractor of discrepancies.
3. Variable frequency drives: Coordinate balance process with equipment manufacturer start-up representative. Insure that VFD settings match equipment limitations and specifications, as well as design parameters.

K. CONTROL SYSTEMS AND EQUIPMENT:
1. General: Operate all temperature control systems with the temperature control contractor's representative for proper sequence of operation. Be responsible for calibration of flow measurement devices used as input to the temperature control system. All air system flow measurement stations including VAV terminals shall be calibrated against a Pitot tube traverse or air diffuser capture hood. Balancing Contractor shall assure accuracy of all flow measurement devices or shall report their failure to be accurate.
2. Work with the Controls Contractor to set minimum outside air damper positions at various fan speeds as required.
3. Work with the Controls Contractor to optimize VAV duct static pressure, VFD pump hydronic system pressure differential and building pressure.

L. LIFE SAFETY SYSTEMS AND EQUIPMENT:
1. Fire/Smoke Dampers and Detection Systems: Verify that each fire/smoke damper closes when the associated duct or space detector is tripped. Verify that air handlers shut down and outside air dampers close as dictated by the control sequence.
2. Verify that air supply units shut down when smoke is detected by the associated duct detector. Verify that outside air dampers and system fire/smoke dampers close as dictated by the control sequence.
3. Report any detectors or dampers that are malfunctioning. Report any discrepancies from the control sequence.

M. SOUND AND VIBRATION:
1. Sound Inspection and Testing: Prior to sound testing, all equipment that can potentially impact sound testing shall be put into operation. Examples include fan coil units, air handling units, and equipment in adjacent mechanical spaces. VFD systems shall be placed at 80% of full speed.
2. Prior to sound testing the mechanical test and balance of all systems shall be completed.
3. Perform sound testing on the following rooms within the project area: Auditorium
4. Vibration Inspection and Testing: Prior to vibration testing, all equipment shall be put into operation. On variable speed equipment, testing shall occur at low, medium and high speeds.

N. GUARANTEE OF WORK:

1. The Testing & Balancing Contractor shall guarantee the accuracy of the tests and balance for a period of 90 days from date of final acceptance of the test and balance report. During this period, the Testing & Balancing Contractor shall make personnel available at no cost to the Owner to correct deficiencies that may become apparent in the system balance. TAB contractor to work with Cx or Owner's representative to verify work as a spot check of a sample of air and water flows.

23 07 00 Mechanical Insulation

A. PLUMBING PIPING SYSTEM INSULATION:
   1. Cold Piping: Insulate the following cold plumbing piping systems:
   2. Potable cold water piping.
   3. Interior above-ground horizontal storm water piping including elbow up & down.
   4. Roof drain bowls and roof drain leader to horizontal piping.
   5. Overflow roof drain bowls and first 10 feet of overflow piping.
   6. Insulate each piping system specified above with the following types and thicknesses of insulation:
   7. Fiberglass; 1/2 inch thickness for pipe sizes up to and including 1 inch, 3/4 inch thick for pipe sizes from 1-1/4 inch to and including 1-1/2 inch; 1 inch thick for pipe sizes over 2 inches.
   8. Fiberglass; 1 inch thickness Above Ground Inside Building.
   9. Hot Piping: Insulate the following hot plumbing piping systems:
   10. Potable hot water and tempered recirculating piping.
   11. Insulate each piping system specified above with the following types and thicknesses of insulation:
   12. Fiberglass; 1 inch thick.
   13. Flexible Elastomeric: 1/2 inch thickness; below ground runouts to Individual fixtures - for pipe sizes up to 2 inches.

B. HVAC PIPING SYSTEM INSULATION:
   1. Insulation Omitted: Omit insulation on hot piping unions, flexible connections, and expansion joints. Insulation may be omitted inside of cabinet unit heaters, convectors and fan coils for hot piping. Hot and cold piping routed inside air handler units shall be insulated. Omit insulation on strainers in heating water strainers operating below 200 degrees F.
   2. Heating System Piping (to 200 degrees F):
   3. Hot water supply and return piping - Fiberglass: 1 inch thick for pipe sizes up to and including 1-1/2 inches in size, 2 inch thick for pipe from 2 inches to over 8 inches.

C. DUCTWORK SYSTEM INSULATION:
   1. Lined ductwork not allowed on project except as noted on drawings. Provide exterior ductwrap as specified below.
   2. Application Requirements: Insulate the following ductwork:
   3. HVAC supply ductwork between fan discharge, or HVAC unit discharge, and room terminal outlet.
   4. Rigid oval or round supply air ductwork.

D. PIPING INSULATION MATERIALS:
   1. Fiberglass Piping Insulation: ASTM C 547, Class 1 unless otherwise indicated. "K" factor shall be maximum 0.24 at 75 degrees F. mean temperature, jacket with tensile strength of 35 lbs/in, mullen burst 70 psi, beach puncture 50 oz. in/in, permeability .02 perm factory applied vapor barrier jacket and adhesive self-sealing lap joint.
   2. Flexible Closed Cell Piping Insulation: ASTM C 534, Type I. "K" factor shall be maximum 0.27 at 75 degrees F. mean temperature, with a water vapor permeability of 0.10 perm inches or less. Insulation shall be pre-installed on piping, or un-slit to be slipped over piping as a single piece.
   3. Jackets for Piping Insulation: ASTM C 921, Type I for piping with temperatures below ambient, Type II for piping with temperatures above ambient. Type I may be used for all piping at Installers option.
   4. Fitting Covers: UV resistant PVC, pre-molded fitting covers, flame spread 25, smoke developed 50. PVC tape for cold systems, serrated tacks or PVC tape for hot systems.

E. DUCTWORK INSULATION MATERIALS:
   1. Rigid Fiberglass Ductwork Insulation: ASTM C 612, Class 1, 450 degrees F temperature limit, density of 3 pcf. "K" value shall be maximum 0.23 at 75 degrees F. mean temperature, facing of 7 mil, foil reinforced with glass mesh and laminated to 40 lb kraft.
   2. Round Surface Semi-Rigid Fiberglass Blanket Insulation: ATSM C 612, Class 1, 450 degrees F temperature limit, 2.5 PCF density “K” value of .25 max at 75 degrees F mean temp, foil-skrim-kraft facing. Orientation of fibers shall be perpendicular to facing to facilitate application on round surfaces.

3. Flexible Fiberglass Ductwork Wrap Insulation: ASTM C 553, Type I, 3/4 lbs per cu. ft. density. "K" value shall be maximum 0.25 at 75 degrees F. mean temperature, 250 degreed F temperature limit, vapor transmission rating shall not exceed 0.02 perms, facing of .7 mil foil reinforced with glass mesh and laminated to 40 lb kraft.

F. EQUIPMENT INSULATION MATERIALS:
   1. Rigid Fiberglass Equipment Insulation: ASTM C 612, Class 2. "K" factor shall be maximum 0.28 at 200 degrees F. mean temperature, 3.0 lb. density, 850 degrees F temperature limit.
   2. Flexible Fiberglass Equipment Insulation: ASTM C 553, Type I, "K" factor shall be maximum 0.45 at 250 degrees F. mean temperature. 850 degrees F temperature limit.
   3. Flexible closed cell elastomeric insulation: ASTM C534, Type I, "K" valve shall be a maximum of 0.27 at 75 degrees F mean temp, 220 degrees F temperature limit, water vapor permeability of 0.10 perm inches or less.

G. DUCT LINER:
   1. Limited to 20 feet of supply and return ductwork from air handling unit. Limited to first 5 feet downstream of VAV boxes. Leading edges of duct liner to be sealed and glued securely.
   2. ASTM G21
   3. Rated velocity on coated side for air erosion 5,000 fpm.
   4. Minimum Noise Reduction Coefficient:
      a. inch thickness: 0.45
      b. ½ inch thickness: 0.60
   5. **Sweeps versus 90 degree turns in the ducts.

H. Manufacturer: Subject to compliance with requirements, provide product by one of the following:
   1. Mechanical Insulation:
      a. Johns Manville Corp.
      b. Owens-Corning Fiberglas Corp.
      c. Knauf Fiber Glass
      d. Manson
      e. Armstrong World Industries, Inc.
   2. Jacketing & Covering Products:
      a. Childers
      b. Ceel-Co
      c. Zeston
      d. Alpha Associates, Inc.
   3. Sound Lagging/Insulation:
      a. Soundseal
      b. Vibro-Acoustics
      c. Schuller
      d. Owens-Corning
      e. Certainteed
   4. Duct Liner:
      a. Knauf Insulation
      b. Johns Manville Corp.
      c. Owens Corning Corp.
      d. Certainteed

23 08 00 Mechanical Commissioning Requirements

A. Commissioning agent to be independent of the contractor or design team.
B. Responsibilities: The responsibilities of various parties in the commissioning process are provided in this section. The responsibilities of the mechanical contractor, TAB, and controls contractor are in Division 23 and those of the electrical contractor in Division 26. It is noted that the services for the project Manager, Design Team, the TAB contractor, and Commissioning Agent are not provided for in this contract. That is, the Contractor is not responsible for providing their services. The commissioning responsibilities applicable to each of the mechanical, controls and TAB contractors of Division 23 are as follows (all references apply to commissioned equipment only):
    1. Construction and Acceptance Phases
        a. Attend a commissioning scoping meeting and other meetings necessary to facilitate the Cx process.
        b. Contractors shall provide the CA with normal cut sheets and shop drawing submittals of commissioned equipment.
        c. Provide additional requested documentation, prior to normal O&M manual submittals, to the CA for development of start-up and functional testing procedures.
        d. Contractors shall assist (along with the design engineers) in clarifying the operation and control of commissioned equipment in areas where the specifications, control drawings or equipment documentation is not sufficient for writing detailed testing procedures.
        e. Develop a full start-up and initial checkout plan using manufacturer's start-up procedures and the prefunctional checklists from the CA for all commissioned equipment. Submit to CA for review and approval prior to startup.
        f. During the startup and initial checkout process, execute the mechanical-related portions of the prefunctional checklists for all commissioned equipment. Perform and clearly document all completed startup and system operational checkout procedures, providing a copy to the CA.
        g. Address current A/E punch list items before functional testing. Air and water TAB shall be completed with
        h. Perform functional performance testing under the direction of the CA for specified equipment.
        i. Correct deficiencies (differences between specified and observed performance) as interpreted by the CA, PM and A/E and retest the equipment.
        j. Prepare O&M manuals according to the Contract Documents, including clarifying and updating the original sequences of operation to as-built conditions.
        k. Provide training of the Owner's operating personnel as specified.
        l. Coordinate with equipment manufacturers to determine specific requirements to maintain the validity of the warranty.
    2. Warranty Period
        a. Execute seasonal or deferred functional performance testing, witnessed by the CA, according to the commissioning plan.
        b. Correct deficiencies and make necessary adjustments to O&M manuals and as-built drawings for applicable issues identified in any seasonal testing.
C. Systems to be Commissioned: The following checked systems will be commissioned in this project.

	Equipment and System		Equipment and System
X	Chillers / Condensing Units	X	Testing, Adjusting and Balancing work
X	Pumps		Chemical treatment systems
X	Cooling tower	X	HVAC control system
X	Boilers		Fire and smoke dampers
X	Piping systems		Generator
X	Ductwork		Equipment sound control
X	Variable frequency drives		Equipment vibration control
X	Air handlers		Egress pressurization
X	Packaged units (AC and HP)	X	Domestic Hot water heaters and pump
X	Terminal units (air)	X	Automated Lighting

	Equipment and System		Equipment and System
X	Unit heaters		Irrigation systems
X	Heat exchangers		Kitchen Equipment
	Lab room pressures	X	Fume hoods
	Specialty fans	X	Heat Pumps

D. Meetings: Within 60 to 90 days of commencement of construction, the CA will schedule, plan and conduct a commissioning scoping meeting with the entire commissioning team in attendance. Meeting minutes will be distributed to all parties by the CA. Information gathered from this meeting will allow the CA to revise the Commissioning Plan.

E. Reporting: The CA will provide regular reports to the PM with increasing frequency as construction and commissioning progresses. The CA will regularly communicate with all members of the commissioning team, keeping them apprised of commissioning progress and scheduling changes through memos, progress reports, etc. A final Commissioning Report is compiled which summarizes the procedures, findings, conclusions, and recommendations of the commissioning process.

F. Submittals: The CA will provide appropriate contractors with a specific request for the type of submittal documentation the CA requires to facilitate the commissioning work. The Commissioning Agent will review submittals related to the commissioned equipment for conformance to the Contract Documents as it relates to the commissioning process, to the functional performance of the equipment and adequacy for developing test procedures. This review is intended primarily to aid in the development of functional testing procedures. The CA may request additional design and operations narrative from the A/E and Controls Contractor.

G. Start-up Checklist and Initial Checkout:
    1. General - Pre-functional Checklists are developed and completed by the installing contractor for all major equipment and systems being commissioned. The checklist captures equipment nameplate and characteristics data, confirming the as-built status of the equipment or system. These checklists also ensure that the systems are complete and operational, so that the functional performance testing can be scheduled.
    2. Start-up and Initial Checkout Plan - The CA shall assist the commissioning team members responsible for startup of any equipment in developing detailed start-up plans for all equipment. The primary role of the CA in this process is to ensure that there is written documentation that each of the manufacturer-recommended procedures have been completed.
    3. Sensor Calibration - Calibration of all sensors shall be included as part of the pre-functional checklists performed by the Contractors.
    4. Execution of Pre-functional Checklists and Startup -
    5. Subs and vendors schedule startup and checkout with the PM, GC and CA.
    6. The CA shall observe, at minimum, the procedures for each piece of primary equipment, unless there are multiple units, (in which case a sampling strategy may be used as approved by the PM).
    7. For lower-level components of equipment, (e.g., VAV boxes, reheat coils), the CA shall observe a sampling of the pre-functional and start-up procedures. The sampling procedures are identified in the commissioning plan.
    8. The Subs and vendors shall execute startup and provide the CA with a signed and dated copy of the completed start-up and pre-functional checklists.
    9. Only individuals that have direct knowledge and witnessed that a line item task on the pre-functional checklist was actually performed shall initial or check that item off.
    10. Deficiencies, Non-Conformance and Approval in Checklists and Startup.
    11. The Subs shall clearly list any outstanding items of the initial start-up and pre-functional procedures that were not completed successfully, at the bottom of the procedures form or on an attached sheet. The procedures form and any outstanding deficiencies are provided to the CA within two days of test completion.
    12. The CA reviews the report and recommends approval to the PM. The CA shall work with the Subs and vendors to correct and retest deficiencies or uncompleted items. The CA will involve the PM and others as necessary.

H. Functional Performance Testing:
    1. Objectives and Scope - The objective of functional performance testing is to demonstrate that each system is operating according to the documented design intent and Contract Documents. In general, each system should be operated through all modes of operation (seasonal, occupied, unoccupied, warm-up, cool-down, part- and full-

load) where there is a specified system response. Verifying each sequence in the sequences of operation is required. Proper responses to such modes and conditions as power failure, freeze condition, low oil pressure, no flow, equipment failure, etc. shall also be tested.

2. Development of Test Procedures - Before test procedures are written, the CA shall obtain all requested documentation and a current list of change orders affecting equipment or systems, including an updated points list, program code, control sequences and parameters. The CA shall develop specific test procedures and forms to verify and document proper operation of each piece of equipment and system. Prior to execution, the CA shall provide a copy of the test procedures to the Sub(s) who shall review the tests for feasibility, safety, equipment and warranty protection. The CA may submit the tests to the A/E team for review. The CA shall review owner-contracted or factory testing which the CA is not responsible to oversee and shall determine what further testing may be required to comply with the Specifications. Redundancy of testing shall be minimized. The test procedure forms developed by the CA shall include the following information:
3. System and equipment or component name(s).
4. Equipment location and ID number.
5. Date.
6. Project name.
7. Participating parties.
8. Reference to the specification section describing the test requirements.
9. A copy of the specific sequence of operations.
10. Instructions for setting up the test.
11. Special cautions, alarm limits, etc.
12. Specific step-by-step procedures to execute the test.
13. Acceptance criteria of proper performance with a Yes / No check box.
14. A section for comments.
15. Signatures and date block for the CA.
16. Test Methods - Functional performance testing and verification may be achieved by manual testing (persons manipulate the equipment and observe performance) or by monitoring the performance and analyzing the results using the control system's trend log capabilities or by stand-alone dataloggers. The CA will determine which method is most appropriate.
17. Setup. Each function and test shall be performed under conditions that simulate actual conditions as close as is practically possible. The Sub executing the test shall provide all necessary materials, system modifications, etc. to produce the necessary flows, pressures, temperatures, etc. necessary to execute the test according to the specified conditions. At completion of the test, the Sub shall return all affected building equipment and systems, due to these temporary modifications, to their pre-test condition.
18. Sampling - Multiple identical pieces of non-life-safety or non-critical equipment may be functionally tested using a sampling strategy. The sampling strategy will be developed by the CA and approved by the PM. If, after three attempts at testing the specified sample percentage, failures are still present, then all remaining units are tested at the contractors' expense.
19. Coordination and Scheduling - The Subs shall provide sufficient notice to the CA regarding their completion schedule for the pre-functional checklists and startup of all equipment and systems. The CA will schedule functional tests through the PM, GC and affected Subs. The CA shall direct, witness and document the functional testing of all equipment and systems. The Subs shall execute the tests.
20. Problem Solving - The CA will recommend solutions to problems found, however the burden of responsibility to solve, correct and retest problems is with the GC, Subs and A/E team.

I. DOCUMENTATION, NON-CONFORMANCE AND APPROVAL OF TESTS

1. Documentation - The CA shall witness and document the results of all functional performance tests using forms developed for that purpose. Prior to testing, these forms are provided to the PM for review and approval.
2. Non-Conformance - The CA will record the results of the functional test on the procedure or test form. All deficiencies or non-conformance issues shall be noted and reported to the PM on a standard form.
3. Corrections of minor deficiencies identified may be made during the tests at the discretion of the CA. In such cases the deficiency and resolution will be documented on the procedure form.

4. Every effort will be made to expedite the testing process and minimize unnecessary delays, while not compromising the integrity of the procedures. However, the CA will not be pressured into overlooking deficient work or loosening acceptance criteria to satisfy scheduling or cost issues, unless there is an overriding reason to do so at the request of the PM.
5. As tests progress and a deficiency is identified:
    a. When there is no dispute on the deficiency and the responsibility to correct it:
        1. The CA documents the deficiency and the Sub's response and intentions the testing continues. The Sub corrects the deficiency and notifies the CA that the equipment is ready to be retested.
        2. The CA reschedules the test and the test is repeated.
    b. If there is a dispute about a deficiency or who is responsible:
        1. The deficiency shall be documented on the non-compliance form and a copy given to the PM and GC.
        2. Resolutions are made at the lowest management level possible. Other parties are brought into the discussions as needed. Final interpretive authority is with the A/E team. Final acceptance authority is with the Project Manager.
        3. The CA documents the resolution process.
    c. Once the interpretation and resolution have been decided, the appropriate party corrects the deficiency and notifies the CA that the equipment is ready to be retested. The CA reschedules the test and the test is repeated until satisfactory performance is achieved.
6. Cost of Retesting - The cost for the Sub to retest a pre-functional or functional test, if they are responsible for the deficiency, shall be theirs. If they are not responsible, any cost recovery for retesting costs shall be negotiated with the GC.
7. Approval - The CA notes each satisfactorily demonstrated function on the test form. Formal approval of the functional test is made later after review by the CA. The CA recommends acceptance of each test to the PM. The PM gives final approval on each test, providing a signed copy to the CA and the Contractor. The CA shall be consulted as to whether or not the contractors of the commissioned systems have substantially completed their work, prior to awarding the project with a substantial completion status. This is also true for final completion.

J. OPERATION AND MAINTENANCE MANUALS
1. Commissioning Record in O&M Manuals - The CA is responsible to compile, organize and index all commissioning data by equipment into labeled, indexed and tabbed, three-ring binders and deliver it to the PM. Three copies of the manuals will be provided. The manuals shall include the Commissioning Plan, Final Commissioning Report, System Type, Design Intent narrative, Startup and Pre-functional checklists, Functional performance tests, trending and analysis, approvals and corrections, training plan, records, and approvals.

K. TRAINING OF OWNER PERSONNEL
1. The GC shall be responsible for training coordination and scheduling and for ensuring that training is completed.
2. The CA shall be responsible for overseeing and approving the content and adequacy of the training and to be reviewed and approved by the Architect/Engineer and the District, of Owner personnel for commissioned equipment.
3. Each Sub and vendor responsible for training will submit a written training plan to the CA for review and approval prior to training. The plan will cover the following elements:
    a. Equipment (included in training)
    b. Intended audience
    c. Location of training
    d. Objectives
    e. Subjects covered (description, duration of discussion, special methods, etc.)
    f. Duration of training on each subject
    g. Instructor for each subject
    h. Methods (classroom lecture, video, site walk-through, actual operational demonstrations, written handouts, etc.)
    i. Instructor and qualifications

L. DEFERRED TESTING
1. Unforeseen Deferred Tests. If any check or test cannot be completed due to the building structure, required occupancy condition or other deficiency, execution of checklists and functional testing may be delayed upon

approval of the PM. These tests will be conducted in the same manner as the seasonal tests as soon as possible. Services of necessary parties will be negotiated.

2. Seasonal Testing. During the warranty period, seasonal testing (tests delayed until weather conditions are closer to the system's design) shall be completed as part of this contract. The CA shall coordinate this activity. Tests will be executed, documented and deficiencies corrected by the appropriate Subs, with facilities staff and the CA witnessing. Any final adjustments to the O&M manuals and as-builds due to the testing will be made.

23 09 00 Building Automation System (BAS) / Direct Digital Control Systems (DDS)

A. GENERAL REQUIREMENTS

1. The Building Automation System (BAS) shall be capable of integrating multiple building function including equipment supervision and control, alarm management, energy management, and historical data collection and archiving.
2. The BAS shall typically consist of the following:
    a. Standalone DDC panels
    b. Fully Configurable Controllers
    c. High Speed Communication Network (LAN) – BAS contractor shall use the SVVSD Ethernet network.
    d. Factory stand-alone controls are not acceptable. All devices must be DDC controlled.
3. The Control Contractor will be responsible for all installation, programming, commissioning, testing and performance verification. The Controls Contractor will be responsible for providing a fully integrated Building Automation System (BAS), incorporating direct digital control (DDC) for energy management, equipment monitoring and control, and subsystems with open communications capabilities as herein specified.
    a. Provide open communications system, in accordance with District preferred protocol with the capabilities to support a multi-vendor environment. To accomplish this effectively, system shall be capable of utilizing standard protocols as follows as well as be able to integrate third-party systems via existing vendor protocols.
    b. No wireless DDC equipment allowed.
    c. All systems and components must integrate with the District Front End system and allow the user to have a single seat from which to perform daily operation.
    d. All graphics, PX images, overrides, and text boxes must be approved by the HVAC department before implemented.
    e. All Home screen graphics must use and or modify existing district images, to include all schedules.
    f. The controls contractor will be responsible for setting up the following schedules based on a provide SVVSD template.
        1. Master schedule: Shuts down all RTU's and AHU's, boiler and chiller plant will not be effected.
        2. Parent teacher schedule: Starts all RTU's and AHU's with the exception of gyms and cafeteria.
        3. Each AHU and RTU will have its own schedule.
        4. All schedules will be in Tridium and will use the district standard template.
        5. Optimum start stop, morning warmup and night set-back will use the district standard object.
4. Proprietary Information
    a. All custom software, programs, code, databases, graphic files and drawings (whether hard copy or CAD based files) prepared for this system shall be the exclusive property of the Owner and shall not be reproduced or distributed without prior written permission from the Owner.

B. QUALITY ASSURANCE

1. The BAS system shall be designed and installed, commissioned and serviced by factory trained personnel. BAS contractor shall:
    a. Have a support facility within 100 miles of the site with technical staff, spare parts inventory and necessary test and diagnostic equipment.

b. Provide full-time, on site, experienced project manager for this work, responsible for direct supervision of the design, installation, start up and commissioning of the BAS.
c. Be regularly engaged in the installation and maintenance of District's preferred systems & requirements and shall have a minimum combined staff experience of thirty (30) years of demonstrated technical expertise and experience in the installation and maintenance of products to be installed on this project.
d. Have no less than 6 personnel factory trained in the District's front-end system.
e. Maintain a service organization consisting of factory trained service personnel and provide a list of 5 projects, similar in size and scope to the project completed within the last five years.

2. Warranty

a. The Warranty period shall begin on the date of substantial completion as authorized by the Architect/Engineer and Owner in writing. Substantial Completion shall not occur before the Contractor has performed the tests required.

3. Training
   a. The Contractor shall provide 40 hours of training for the building operators, split into two sessions; 1) at the time of substantial completion, and 2) at SVVSD HVAC department discretion.
4. Contractor shall provide to the district after job completion, one of each variety of Component Level Controllers.

C. SVVSD BAS SYSTEM, COMPONENTS, & REQUIREMENTS
1. SVVSD Front End System
   a. Tridium Niagra 4
      1. All system components must integrate into an N4 Supervisor.
      2. Verify current release or version with SVVSD.
2. Devices / Controllers
   a. Preferred Controls Communication Protocol
      1. BACnet Protocol
      2. Other protocols which may be used with District permission only:
         a. LonWorks
         ~~b. Modbus~~
   b. Building Level Controllers
      1. Tridium JACE
   c. Component Level Controllers
      1. Distech
3. Max sub-LAN controllers on a N4 JACE shall not exceed 85 devices, and on a JACE 8200 shall not exceed 200 devices. On a N4 JACE with LonWorks, no more than 800 binds will be acceptable. This will ensure the district can add more building level devices and programming logic to either JACE in the future. Every DDC controller will have at least two spare points. One configurable input, and one output.

D. LICENSING
1. The MSIP "Multi-system Integration Platform" Interface Device provided, shall be an N4 JACE that is open licensed to the SVVSD N4 WEB Supervisor, and will also include the Embedded Workbench feature that allows SVVSD to access the programming and interface for the N4 JACE directly through a browser. The District standard JACE / UNC shall be a JACE 8000 with 1gig of ram, and the 5 year maintenance agreement. Controls contractor will provide the proper modules to allow communication to all district controls. Controls contractor is required to purchase additional licenses needed for system integration to building level controls. In addition, all DDC field controllers must be fully programmable, by either Niagara software or controller's native software. Controls contractor will provide 2 copies of native controller software to SVVSD.
   a. The system shall be modular in nature, and shall permit expansion of both capacity and functionality through the addition of sensors, actuators, standalone DDC panels, and operator devices. Standalone DDC panels shall be able to access any data from any other device on the network and share information amongst them, and must be able to integrate into Niagara.

b. BMS contractor must have the ability to program both Tridium, and accepted controls with in-house personnel. Subcontracting of any controls programming,graphics is unacceptable.
c. Device alarming shall be handled by Tridium. Each DDC panel shall operate independently by performing its own specified control. The failure of any single component or network connection shall not interrupt the execution of control strategies at other operational devices.
d. Tridium will have priority control of ALL units or devices. To include set points, overrides, resets, Optimum start stop and Night set back. Night setback and morning warm up will use the districts standard Tridium OSS object. If a point can be monitored or controlled via the building level controller's, it will be brought out in Tridium, and placed on graphics depending on point function, at SVVSD discretion. BAS specifications shall include reference to SVVSD graphics standards and naming conventions.
e. All logic for resets, global set points, shall be programmed in Tridium and passed to the building level controls via the network. Contact HVAC department for programming clarification.
f. BAS shall allow third party software to operate on personal computer workstation without any degradation to the controls operating normally.
g. BAS graphics will mirror current district standards including mapping to desired pages.

23 09 01 Control Device Requirements

A. The following is direction from SVVSD on control device details.
   1. Low limit sensors to be auto reset.
   2. Differential pressure sensors to include snubbers on the pipes to reduce pressure fluctuations.
   3. Mixed air temperature sensors to be averaging sensors.
   4. Dirty filter alarms points are not desired or needed.
   5. VAV reheat coil valves will be floating point control.
   6. BBR valves to be located in BBR enclosure.
   7. Space sensors – No LCD screen, no occupant override buttons, no occupant temperature control.
   8. Guards are to be used to protect DDC sensors locations in the Gyms.
   9. Flat stainless plate temperature sensors are to be used in hallways.
   10. Each hydronic system is to have a low pressure safety switch monitored by the BAS.
   11. All TC contractor's DDC controls will reside in a panel below grid, no DDC controls will be mounted inside ore reside in any unit above the ceiling grid.
   12. No stand-alone factory controls are acceptable. All devices must be deviced by control contractors building level controllers. Factory equipment must provide terminal strips for control contacts.

23 09 02 Sequence of Operation

A. SUMMARY This section includes general requirements that apply to Sequences of Operation for controlled equipment. The following sequences pursue the design intent of the owner which is to produce systems that are as energy efficient as possible without sacrificing indoor environmental quality or thermal comfort. The Mechanical Engineer and installing contractor can modify the Sequence of Operation to further satisfy the design intent. All setpoints are adjustable. All resets will be adjustable in Tridium. All changes to the Sequences shall be recorded in as-built documentation. All initial occupancy schedules shall be 7am to 3 pm, Monday through Friday.

B. GENERAL HOT WATER SYSTEM (all setpoints and time periods are adjustable)

1. General note: All heating plant programming and control to be done through the BAS. No boiler management systems allowed.
2. System Enable: The BAS control system shall enable the hot water heating under the following conditions:
    a. The outside air temperature (OAT) drops below 50F. Disable with the OAT rises above 60F.
    b. The HWS Schedule is enabled.
3. Hot Water Return Reset (non-condensing boilers): The hot water valve shall modulate to maintain the Hot Water Return Temperature (HWRT) at the hot water return setpoint (sp-hwrt).
    a. SP-hwrt is defined by the following reset schedule when any air handling unit is enables: If the air handling units are disabled, sp-hwrt shall be 140 degrees F.

OAT	sp-hwrt
10 degrees F	165 degrees F
58 degrees F	140 degrees F

(If three-way valve control strategy is not used, hot water supply temperature may be reset by sending a hot water supply temperature signal to the boiler control panel. IN this case, the HWS setpoint shall be equal to the values above plus 15 degrees F).

4. Hot Water Return Reset (condensing boilers): The hot water valve shall modulate to maintain the Hot Water Return Temperature (HWRT) at the hot water return setpoints (sp-hwrt). Sp-hwrt is defined by the following reset schedule (assuming a loop design HWS temp of 140 degrees F)

OAT	sp-hwrt
10 degrees F	125 degrees F
58 degrees F	90 degrees F

(If three-way valve control strategy is not used, hot water supply temperature may be reset by sending a hot water supply temperature signal to the boiler control panel. IN this case, the HWS setpoint shall be equal to the values above plus 15 degrees F).

5. Boiler Start/Stop (single boiler): Boiler shall be enabled when the system enable is on, the HWS schedule is enabled, pump status is on (primary loop, secondary loop and circulation pumps as available) and the HWRT is 10 degrees F below the sp-hwrt. If the HWRT is 5 degrees F above the sp-hwrt for more than 10 minutes the boiler shall be disabled.
6. Boiler Lead/Lag (multiple equally-sized boilers): The two hot water boilers shall be staged through the DDC. The lead boiler shall be enabled when the system enable is on, pump status is on (primary loop, secondary loop and circulation pumps as available) and the HWRT is 10 degrees F below the sp-hwrt. The lag boiler shall be enabled when the HSRT falls more than 10 degrees F below the sp-hwrt for more than one hour “Adjustable”. The lead/lag selector shall rotate the boilers every other Wednesday at 12:00 pm. Upon failure of the lead boiler, the lag boiler shall be enabled and alarm shall be generated at the DDC.
7. Boiler Lead/Lag (multiple different sized boilers): The two hot water boilers shall be staged through the DDC. The lead boiler shall be enabled upon system enable and pump status. Boiler B-1 is sized for X% heating capacity, Boiler B-2 is sized at 100-X% capacity (B-1 is the larger of the two). When the outside air temperature is between

58 degrees F and 40 degrees F Boiler B-2 will operate and Boiler B-1 will be disabled. When the temperature falls below 40 degrees F, Boiler B-1 will be enabled, and Boiler B-2 will be enabled if HWRT falls more than 10 degrees F below the sp-hwrt for more than one hour.

8. Boiler Staging (multiple equally-sized boilers; most efficient at low firing rate, high turn-down ratio): The two heating water boilers shall be staged through the DDC. The lead boiler shall be enabled and its associated isolation valve shall open (with end switch indication) when the system enable is on and the distribution pump status in on. The Lead/Lag selector shall rotate the boilers every other Tuesday at 8:00 am. Upon failure of the lead boiler, the lag boiler shall be enabled and alarm shall be generated. Boilers shall be staged on in response to building heating demand as follows:

Stage	Boiler Operation
1	Both boilers off
2	Lead boiler 0 to 25% fire / Lag boiler off
3	Lead boiler 25% fire / Lag boiler 0 to 25% fire
4	Both boilers modulate 25% to 100% fire

9. Pump Lead/Lag (if multiple pumps): The lead pump shall be enabled upon system enable. The lead/lag pump selector shall rotate the pumps every other Wednesday at 11:00 am. Upon failure of the lead pump, the stand-by pump shall start automatically and an alarm shall be generated to the DDC. The pumps shall remain enabled for a period of 10 minutes after the heating system is disabled.
10. Variable Speed Pumping (If VFD's are available): The hot water pump VFD speed (HWP VFD SPEED) shall modulate the hot water differential pressure (via a differential pressure transducer located by the design engineer) to maintain the hot water differential pressure setpoint (sp-hw loop p.s.i.) set by the balance contractor. Combustion air dampers (if available): Combustion air damper shall open upon an enable command for the associated appliance, and must prove open via a DDC Status, upon sensing the combustion damper is open, DDC will then enable the appliance to fire. Hard wired damper interlock to boiler enable will be unacceptable.
11. Alarms:
    a. EPO shutdown (if available): If any EPO switches are activated the enable control signal to the heating boilers and the domestic water boiler shall be disabled and an alarm generated. The heating hot water pumps may remain in service.
    b. Additional alarms (as available): Any of the following alarm/failure points shall generate an alarm in the DDC.
        1. Glycol pumping station low level alarm (1 minute delay, adj)
        2. Hot water pump alarm (Command ON and status OFF, Command OFF and Status ON (15 minute delay, adj).
        3. General boiler alarm: The general boiler alarm shall be a factory provided dry contact alarm that shall provide a summary alarm of any boiler failure condition (low water cut off, high limit temperature, flame failure, high gas pressure alarm or low gas pressure alarm).
        4. Hot water supply/return temperature (10 degrees F less than or greater than setpoint (15 minute delay, inhibit when heat is disabled). Adjustable logic done in Tridium.

C. GENERAL CHILLED WATER SYSTEM (All setpoints and time periods are adjustable)

1. System enable: The DDC control system shall enable the chilled water system whenever the outside air temperature is above 65 degrees F, and the CHWS Schedule is enabled.. Chilled water system shall be disable when the outside air temperature is below 63 degrees F or when building CWS schedule is off.
2. Chiller Lead/Lag (if available, equally-sized chillers): The chillers shall be staged through the DDC control system. The lead chiller shall be enabled with system enable and proof of secondary pump status. The lag chiller shall be enabled if the lead chiller has been enabled for at least one hour and the chilled water return temperature rises above 58 degrees F (lag chwrt max) for a period of more than 5 minutes (lag chiller delay) or in the event of a lead chiller failure. If both chillers are operating, the lag chiller shall disable when the load of both chillers drops below 50%. The lead-standby chiller selector shall rotate the chillers every Tuesday at 8:00 am.

3. Pump Lead/Lag (if available): The pumping system shall be enabled with system enable. The lead/lag pump selector shall select a lead pump and a lag pump, rotating the pumps every Wednesday at 12:00 pm. Upon failure of a lead pump, the stand-by pump shall start automatically and an alarm shall be generated.
4. Chiller Operation: The chiller will be enabled upon system enable and proof of all pump status, via current sensing relay. Chiller shall operate under its own factory controls to maintain 45 degrees F leaving chilled water temperature set point. If a reset point is available, the chiller will use the following reset.

OAT	Sp
85 degrees F	45 degrees F
65 degrees F	55 degrees F

5. Condenser water control (if available): The three-way diverting tower bypass valve (if available) shall be closed to the tower when the system is disabled. Upon system enable the valve shall modulate to maintain a sump condenser water setpoint equal to the outdoor wetbulb temperature (temp to come from internet/local weather station) +10degF with a lower limit of 60F (adj). The cooling tower VFD (if available) shall be enabled upon condenser pump status and allowed to maintain sump condenser water setpoint.
6. Variable Pressure Pumping (if available): The chilled water pump VFD speed (CHWP VFD SPEED) shall modulate the chilled water differential pressure (via a differential pressure transducer located by the design engineer) to maintain the differential pressure setpoint (sp-chw loop p.s.i
7. Alarms (as available):
    a. Chilled water pump alarm (Command ON and status OFF, Command OFF and status ON (15 minute delay)).
    b. Chilled water supply temperature (10 degrees F less than or greater than setpoint (15 minute delay, inhibit when cooling is disabled)).
    c. Condenser water supply temperature (10 degrees F less than or greater than setpoint (15 minute delay, inhibit when cooling is disabled))
    d. Chiller general alarm (15 minute delay)
    e. Sump low level alarm (15 minute delay)

D. GENERAL SWITCH-OVER SEQUENCE FOR TWO-PIPE HEATING/COOLING SYSTEMS
    1. Note: No two pipe system will be acceptable.

E. GENERAL CONSTANT VOLUME UNIT (single-zone, no VFD, all setpoints and time periods are adjustable)
    1. Optimum Start: An optimum start sequence shall be used to start the unit early enough (2 hour maximum) to achieve an average zone temperature of 70 degrees F (optimum warm up) or 74 degrees F (optimum cool down) at the beginning of the occupied scheduled start-time. In optimum warm up the outside air dampers shall remain closed. In optimum cool down the outside air dampers shall be allowed to modulate and all mechanical cooling shall not be allowed to run. Optimum cool shall only be used if the outside air temperature is greater than 50 degrees F and less than the average zone temperature – 4 degrees F.
    2. Occupied Cycle (7am to 5 pm):
    3. During occupied hours the unit runs continuously.
    4. Provide extra occupied schedule points for the following zones to allow for District scheduling.
    5. Administration, Music, Gym, Cafeteria, Classrooms (per wing).
    6. Space Temperature Control: The space temperature setpoint shall be as follows.

Clg Sp	HTG SP
74 “ADJ”	70 “ADJ”

    7. Discharge Air Temperature Setpoint (Econ /HW / CHW only): The discharge air temperature control shall modulate between 50-90 degrees F based space temperature reset at the space temperature setpoint. The hot water valve shall modulate to maintain the DATsp – 2 degrees F. The CHW valve shall modulate to maintain the DATsp + 2 degrees F. If the economizer is enabled, the mixed air damper shall control to maintain a mixed air temperature equal to the effective discharge air temperature setpoint.

RMT	DAT
68	90
76	50

8. DX Control: DX shall be allowed to operate when the outside air temperature rises above 60 degrees F "ADJ" (economizer enabled and OA damper is 100% open, or when economizer is disabled), and locked out when the outside air temperature drops below 58 degrees F. The first DX stage shall enable when the space temperature is equal to the space temperature setpoint + 2 degrees F and disable when the space temperature reaches setpoint. The second DX stage (if applicable) shall enable when the space temperature is equal to the space temperature setpoint +3 degrees F, or if the first stage of DX is on and the space temperature does not reach setpoint after 15 minutes. The second DX stage shall disable when the first stage is disabled.
9. Economizer Control: The economizer mode shall be enabled if the building is occupied and the outside aire temperature is 2 degrees F less than the return air temperature. The economizer will stay enabled until the outside air temperature is equal to the return air temperature. If the economizer is enabled, the mixed air damper shall control to maintain a mixed air temperature equal to the effective discharge air temperature setpoint.
10. If CHW is available, the mixed air damper shall modulate to maintain the sp-dat – 2 degrees F.
11. Mixed Air Damper Control: The mixed air damper shall be off (full return air) unless the supply fan is running, optimum warm is off, night setback is off and the mixed air temperature is not in alarm. Mixed air damper enable shall be subject to a 5 minute delay on make relay. When the unit is enabled and not in economizes or CO2 control mode, the mixed air damper position shall be reset with fan speed to maintain the scheduled quantity of OA air at all fan speeds.
12. CO2 Control (Gym and Cafeteria Units): A return air, duct mounted, CO2 sensor shall monitor CO2 content of the return air. The outside air damper shall modulate to maintain less than 1200 ppm. A MAT low limit will have priority over C02 control.
13. Unoccupied Cycle: During the unoccupied cycle, when night setback is not active, the supply fan is off, the mixed air dampers area set to full return air and the hot and chilled water valves are closed. If the hot water coil is in the unit, when the OA temperature is below 40 degrees F, the hot water valve (if present) shall be allowed to modulate to maintain a mixed air temperature setpoint of 50 degrees F.
14. Night Setback Mode: If the average temperature served by the unit drops below 60 degrees F "ADJ", the unit shall energize and operate as described in the Occupied cycle EXCEPT the dampers shall remain open to full return. Night setback mode shall run until all space temperature are above 63 degrees F "ADJ". Heating in this mode shall be maintained per the occupied sequence.
15. Safeties: The unit is disabled upon activation of the smoke detector or low temperature detection thermostat (hardwire interlocks). Upon a fan failure alarm, low limit alarm or mixed air alarm, the dampers shall go to full return. Provide high duct static safeties where fire/fire-smoke dampers are directly downstream of AHUs.
16. Alarms:
    a. Any of the following conditions shall generate an alarm within the DDC and at the district central monitoring facility. Any safety alarm must trigger a supply fan alarm to the DDC.
        1. Supply Fan Failure (Command ON and status OFF, Command OFF and status ON (15 minute delay))
        2. Low temperature detection thermostat trip.
        3. Duct high limit.
        4. Any safety alarm must trigger a supply fan alarm to the DDC.

F. GENERAL CONSTANT VOLUME UNIT (multiple zones, all setpoints and time periods are adjustable)
1. Optimum Start "Done in Tridium": An optimum start sequence shall be used to start the unit early enough (2 hours maximum) to achieve an average zone temperature of 70 degrees F (optimum warm up) or 75 degrees F (optimum cool down) at the beginning of the occupied scheduled start-time. In optimum warm up the outside air dampers shall remain closed. In optimum cool down the outside air dampers shall be allowed to modulate and all mechanical cooling shall not be allowed to run. Optimum cool shall only be used if the outside air temperature is greater than 50 degrees G and less than the average zone temperature – 4 degrees F.
2. Occupied Cycle (7 am to 5 pm "ADJ"):
3. During Occupied Hours, the unit runs continuously.
4. Building will be occupied based on one master occupancy schedule

5. Provide extra occupied schedule points for the following zones to allow for district scheduling:
6. Administration, Classrooms (per wing).
7. Discharge Air Temperature Setpoint (Econ/HW/CHW only): The discharge air temperature control shall modulate between 50-90 degrees F based space temperature reset at the average space temperature. The hot water valve shall modulate to maintain the DATsp – 2 degrees F. The CHW valve shall modulate to maintain the DATsp + 2 degrees F. If the economizer is enabled, the mixed air damper shall control to maintain a mixed air temperature equal to the effective discharge air temperature setpoint.

RMT	DAT
68	90
74	50

8. DX control (if applicable): DX shall be allowed to operate when the outside air temperature rises above 60 degrees F (economizer enabled and OA damper is 100% open, or when economizer is disabled), and lock out when the outside air temperature drops below 58 degrees F. The first DX stage shall enable when the average space temperature is equal to the space temperature is equal to the space temperature setpoint +2 degrees F, and disable when the space temperature reaches the setpoint. The second DX stage (if applicable) shall enable when the space temperature is equal to the space temperature setpoint + 3 degrees F, or if the first stage of DX is on and the space temperature does not reach the setpoint after 10 minutes. The second DX stage shall disable when the first stage is disabled.
9. Modulating DX Control (if applicable): A digital scroll is modulating compressor capable of 10-100% capacity. When using digital compressor to maintain DAT the following criteria is applicable:
    a. Stage 1 – Modulate digital compressor #1
    b. Stage 2 – Turn ON digital ON /OFF compressor #2
    c. Stage 3 – Turn ON digital ON/OFF compressor #3
    d. Stage 4 – Turn ON digital ON/OFF compressor #4
    (The main idea behind this sequence, is compressor with the most wear starting and stopping. Once a modulating compressor is turn on it stays on and modulates until DX is disabled. The other ON/OFF compressors can start and stop for larger steps in capacity).
10. Economizer Control: The economizer mode shall be enabled if the building is occupied and the outside air temperature is 2 degrees F less than the return air temperature. The economizer will stay enabled until the outside air temperature is equal to the return air temperature. When the economizer is enabled the MAD will modulate to maintain the discharge air temperature setpoint.
11. Mixed Air Damper Control: The mixed air damper shall be off (full return air) unless the supply fan is running, optimum warm is off, night setback is off and the mixed air temperature is not in alarm. Mixed air damper enable shall be subject to a 5 minute delay on make relay. When the unit is enabled and not in economizer or CO2 control mode, the mixed air damper position shall be reset with fan speed to maintain the scheduled quantity of OA air at all fan speeds.
12. CO2 Control: A return air, duct mounted, CO2 sensor shall monitor the CO2 content of the return air. The outside air damper shall modulate to maintain the less than 1200 ppm. A MAT low limit will have priority over C02 control.
13. Unoccupied Cycle: During the unoccupied cycle, when night setback is not active, the supply fan is off, the mixed air dampers are set to full return air and the hot and chilled water valves are closed. If the hot water coil is in the unit, when the OA temperature is below 40 degrees F, the hot water valve (if present) shall be allowed to modulate to maintain a mixed air temperature setpoint of 50 degrees F.
14. Night Setback Mode: If the average temperature served by the unit drops below 60 degrees F, the unit shall energize and operate as described in the Occupied cycle EXCEPT the dampers shall remain open to full return. Night setback mode shall run until all space temperatures are above 63 degrees F. Heating in this mode shall be maintained per the occupied sequence.
15. Safeties: The unit is disabled upon activation of the smoke detector or low temperature detection thermostat (hardware interlocks or high static safety switch). Upon a fan failure alarm, low limit alarm or mixed air alarm, the dampers shall go to full return. Upon a low limit alarm, the hot water valve shall open.

16. Alarms: Any of the following conditions shall generate an alarm within the DDC and at the district central monitoring facility. Any safety alarm must trigger a supply fan alarm to the DDC.
    a. Supply Fan Failure (Command ON and status OFF, Command OFF and status ON (15 minute delay))
    b. Low temperature detection thermostat trip.
    c. Duct high limit.
    d. Any safety alarm must trigger a supply fan alarm to the DDC.

G. GENERAL VARIABLE AIR VOLUME UNIT (FPVAV & VAV, all setpoints and time periods are adjustable)

1. Optimum Start: AN optimum start sequence shall be used to start the unit early enough (2 hour maximum) to achieve an average zone temperature of 70 degrees F (optimum warm up) or 75 degrees F (optimum cool down) at the beginning of the occupied scheduled start-time. In optimum warm up the outside air dampers shall remain closed. In optimum cool down the outside air dampers shall be allowed to modulate and all mechanical cooling shall not be allowed to run. Optimum cool shall only be used if the outside air temperature is greater than 50 degrees F and less than the average zone temperature – 4 degrees F.
2. Occupied Cycle (7 am to 5 pm): During occupied hours, the unit runs continuously, DDC adjustable. At all times when the fan is running, the supply fan VFD shall modulate to maintain the duct static setpoint set by balancer.
3. Discharge Air Temperature Control: The discharge air temperature control will enable the gas fired heat (or modulate the HW valve if available) (and modulate the CHW valve if available) to maintain a discharge air temperature between 55 degrees F – 72 degrees F based on a reset working to maintain the average zone temp. The hot water valve shall modulate to maintain DAT – 2 degrees F. The CHW valve shall modulate to maintain DAT + 2 degrees F.

RMT	DAT
70	70
74	55

4. DX Control (if applicable): DX shall be allowed to operate when the outside air temperature rises above 60 degrees F, and locked out when the outside temperature drops below 58 degrees F. The first DX stage shall enable when the highest space temperature is equal to the space temperature setpoint + 2 degrees F and disable when the highest space temperature reaches setpoint. The second DX stage (if applicable) shall enable when the highest space temperature is equal to the space temperature setpoint +3 degrees F, or if the first stage of DX is on and the space temperature does not reach setpoint after 10 minutes. The second DX stage shall disable when the first stage is disabled.
5. Space Temperature Control: Setpoint for space temperature will be as follows.

Clg Sp	HTG SP
74 “ADJ”	70 “ADJ”

6. Economizer Control: The economizer mode shall be enabled if the building is occupied and the outside air temperature is 2 degrees F less than the return air temperature. The economizer will stay enabled until the outside air temperature is equal to the return air temperature. Mixed Air Damper Control: The mixed air damper shall be off (full return air) unless the supply fan is running, optimum warm is off, night setback is off and the mixed air temperature is not in alarm. Mixed air damper enable shall be subject to a 5 minute delay on make relay. When the unit is enabled and not in economizer or CO2 control mode, the mixed air damper position shall be reset with fan speed to maintain the scheduled quantity of OA air at all fan speeds.
7. Building Static Control: Any exhaust dampers and/or relief dampers shall modulate to maintain the building static pressure at the building static pressure setp0int of 0.05” wc (adj).
8. Mixed air plenum pressurization control: If the return fan is modulating (inlet vane or VFD) building static pressure is monitored and return plenum pressure is monitored, a mixed air plenum pressurization control scheme will be implemented. The return fan shall be controlled from a pressure transducer locate in the return plenum and a

return plenum pressure setpoint of 0.15" wc. The exhaust air dampers will modulate to maintain building static pressure setpoint.

9. Unoccupied Cycle: During the unoccupied cycle, when night setback is not active, the supply fan is off, the mixed air dampers are set to full return air and the hot and chilled water valves are closed. If the hot water coil is in the unit, when the OA temperature is below 40 degrees F, the hot water valve (if present) shall be allowed to modulate to maintain a mixed air temperature setpoint of 50 degrees F.
10. Night Setback Mode: If the average VAV space temperature served by the unit drops below 60 degrees F, the unit shall energize and operate as described in the Occupied cycle EXCEPT the dampers shall remain open to full return. Night setback mode shall run until all space temperatures are above 63 degrees F. Heating in this mode shall be maintained per the occupied sequence. If the unit serves Fan Powered Boxes the Night Setback Mode will be accomplished through the FPVAV sequence.
11. VAV boxes will not have an Unoccupied set point in programming or graphics.
12. Safeties: The unit is disabled upon activation of the smoke detector or low temperature detection thermostat (hardware interlocks). Upon a fan failure alarm, low limit alarm or mixed air alarm, the dampers shall go to full return.
13. Alarms: Any of the following conditions shall generate an alarm within the DDC and at the district central monitoring facility. Any safety alarm must trigger a supply fan alarm to the DDC.
    a. Supply Fan Failure (Command ON and status OFF, Command OFF and status ON (15 minute delay))
    b. Low temperature detection thermostat trip.
    c. Any safety alarm must trigger a supply fan alarm to the DDC.

H. HEAT RECOVERY ADD-ON FOR AIR HANDLING UNITS (all setpoints and time periods are adjustable)

1. Same as other sequences except add the following:
2. Heat Recovery: The heat recovery wheel assembly shall be enabled during occupied hours only, whenever the OAT is less than 45 degrees F (adj) or above 78 degrees F (adj) (3 deg dead-band on each). When the heat wheel is enabled, the outside air and exhaust air heat recovery by-pass dampers shall close, and when the heat wheel is disabled, the outside air and exhaust air heat recovery by-pass dampers shall be open. When the heat wheel is enabled, the outside air and return air dampers shall modulate together to maintain the mixed air temperature set point (mixed air temperature set point = DAT set point minus 2 degrees F). During the transition to occupied mode and startup of the heat recovery wheel, the supply fan will be started and the OA/RA dampers set to full RA and the bypass dampers shall be closed, then the heat wheel shall be started, next the exhaust fan shall be enabled, and after a 3 minute delay, the OA damper shall open and be allowed to control.
3. Unoccupied operation: Heat recovery deactivated, return air damper is fully open, outside air and exhaust air dampers are fully closed. Exhaust and supply fans are off.
4. Morning Warm-up: Heat recovery is deactivated, return air damper is fully open, outside air damper is fully closed, bypass dampers are fully closed. Supply fan is on operating to duct static pressure set point, exhaust fan is off, and exhaust damper is closed.
5. Morning Cool-down: The exhaust fan is activated and shall operate at full speed, the heat recovery wheel is deactivated, the outside air damper is fully open, the return air damper is closed and the outside air and exhaust air by-pass dampers are open to provide maximum ventilation air pre-cooling.
6. Frost-control operation: Frost control operation shall be initiated upon sensing an exhaust air temperature (return air leaving the heat recovery wheel) of 34 degrees F or less. The exhaust fan is activated, the heat recovery wheel is activated, the outside air by-pass damper modulates open to maintain an exhaust air temperature (leaving the heat recovery wheel) of 35 degrees F, and the exhaust air by-pass damper is fully closed.

I. FAN POWERED BOX (FPVAV, all setpoints and time periods are adjustable)

1. Occupied Cycle (Fan powered box shall be energized prior to the RTU or AHU to prevent fan from spinning backwards When in the cooling mode, the primary air damper shall modulate between its maximum and minimum setting to maintain the occupied cooling setpoint. When in the heating mode the primary air damper shall close to the cooling minimum position and below the heating occupied setpoint, the induced air fan shall be energized. Upon a further drop in space temperature, the reheat coil alve shall modulate to maintain the heating occupied setpoint. Upon a rise in space temperature, the reverse shall occur.
2. Setpoint for space temperature will be as follows.

Clg Sp	HTG SP
74 “ADJ”	70 “ADJ”

3. Space temperature will be maintained by corresponding reheat coil
4. Unoccupied Mode (based on fan status of associated air handler): If the air handler is not running (no fan status) the damper and HW reheat valve shall both close, fan shall be commanded off.
5. Night Setback: Air handler stays off. If the space temperature served by the box falls below 60 degrees F the fan shall start, the damper shall close and the valve shall modulate to maintain the night setback setpoint. Night setback mode shall run until the space temperature is above 63 degrees F.

J. VARIABLE AIR VOLUME BOX (VAV, all setpoints and time periods are adjustable)
1. Occupied Cycle: The VAV box terminal damper shall modulate in sequence with the terminal box heating valve to maintain space temperature setpoint of 70 degrees F and 74 degrees F cooling. As the space temperature rises above cooling setpoint, the VAV damper shall modulate open and vary the supply air to the space from cooling minimum to maximum to maintain setpoint. As the space temperature fall below the heating setpoint, the VAV box damper shall modulate back to it minimum air position and the heating valve shall open to maintain heating temperature setpoint. During MWU and NSB, and sensing a DAT above 80 degrees F, “through the DDC” the VAV damper will modulate to maximum cooling setpoint until sp is reached.
2. Setpoint for space temperature will be reset based on outside air temperature

Clg Sp	HTG SP
74 “ADJ”	70 “ADJ”

3. Unoccupied Mode: Unoccupied mode for the box is determined from the HVAC mode block of Tridium of the associated air handler. If the air handler is not running the damper and HW reheat valve shall both be closed.
4. All VAV’s or Reheat Coils will have a delivered air sensor installed, to include fan powered boxes.

K. ENABLE AND MONITOR (Package Controlled Unit, all setpoints and time periods are adjustable)
1. Optimum Start: An optimum start sequence shall be used to start the unit early enough to achieve an average zone temperature of 70 degrees F (heating) or 74 degrees F (cooling) at the beginning of the occupied scheduled start-time. Optimum start schedules shall be a slave schedule from the master building occupied schedule. If the building occupied mode is off the optimum start must also be off.
2. Occupied Cycle (7am to 5 pm):
3. During Occupied hours or activation of the 4 hour override push button, the unit runs continuously, DDC adjustable. Factory unity control shall operate heating and cooling to maintain room setpoint.
4. DDC shall monitor temperatures, fan and safety status.
5. DDC shall enable associated exhaust fans when unit is enabled.
6. Unoccupied Cycle: During the unoccupied cycle, when night setback is not active, the supply fan is off and the mixed air dampers are set to full return air.
7. Night Setback Mode: If any of the zone temperatures served by the unit drops below 60 degrees F, the unit shall energize and operate as described in the Occupied cycle EXCEPT the outside air dampers shall remain closed. Night setback mode shall run until all space temperatures are above 63 degrees F.
8. If the economizer is enabled, the mixed air damper shall control to maintain a mixed air temperature equal to the effective discharge air temperature setpoint.

L. GENERAL MULTI-ZONE AIR HANDLING UNIT CONTROL (with DX cooling, all setpoints and time periods are adjustable)
1. Optimum Start: An optimum start sequence shall be used to start the unit early enough (2 hour maximum) to achieve an average zone temperature to 68 degrees F (optimum warm up) or 75 degrees F (optimum cool down) at the beginning of the occupied scheduled start-time. In optimum warm up the outside air dampers shall remain

closed. In optimum cool down the outside air dampers shall be allowed to modulate and all mechanical cooling shall not be allowed to run. Optimum cool shall only be used if the outside air temperature is greater than 50 degrees F and less than the average zone temperature – 4 degrees F.

2. Occupied Cycle (7 am to 5 pm, Adj)
    a. During occupied hours,
    b. The DDC system shall reset the HD temperature setpoint from the average zone damper position (0% = full heating) as follows:

Avg. Zone Damper Position	Hot Deck Setpoint
30%	100 degrees F
85%	75 degrees F

    c. The DDC system shall reset the CD temperature setpoint based on the average zone damper position (100% = full cooling) as follows:

Avg. Zone Damper Position	Cold Deck Setpoint
80%	55 degrees F
10%	65 degrees F

    d. The economizer shall be enabled when the RAT is 2 degrees F less than the OAT, and disabled when the RAT equals the OAT. When enabled, the MA damper shall modulate to maintain the CD temperature setpoint. When the economizer is disabled the OA damper shall return to minimum position.
    e. The DX cooling shall be enabled if the OAT is greater than 60 degrees F. When enabled the DX shall be staged as follows:

DX STAGE	Enable stage when Avg. Zone Damper is greater than:	Disable stage when Avg. Zone Damper is less than:
DX1	55%	40%
DX2	80%	55%

    f. Below 58 degrees F OAT the DDC system shall lock out the mechanical cooling system.
    g. Unoccupied Cycle: During the unoccupied cycle, when night setback is not active, the supply fan is off, the mixed air dampers are set to full return air and the hot and chilled water valves are closed. If the hot water coil is in the unit, when the OA temperature is below 40 degree F, the hot water valve (if present) shall be allowed to modulate to maintain a mixed air temperature setpoint of 50 degrees F.
    h. Night Setback Mode: If any of the zone temperatures served by the unit drops below 60 degrees F, the unit shall energize and operate as described in the Occupied cycle EXCEPT the outside air dampers shall remain closed. Night setback mode shall run until all space temperatures are above 63 degrees F. Heating in this mode shall be maintained per the occupied sequence.
    i. Safeties: The unit is disabled upon activation of the smoke detector or low temperature detection thermostat (hardware interlocks). Upon a fan failure alarm, low limit alarm or mixed air alarm, the dampers shall go to full return.
    j. Alarms: Any of the following conditions shall generate an alarm within the DDC and at the district central monitoring facility. Any safety alarm must trigger a supply fan alarm to the DDC.
        1. Supply Fan Failure (Command ON and status OFF, Command OFF and status ON (15 minute delay, Adj)
        2. Low temperature detection thermostat trip.
        3. Duct high limit.
        4. Any safety alarm must trigger a supply fan alarm to the DDC.

M. CABINET UNIT HEATERS: The cabinet unit heater valve shall open and the fan shall cycle on to maintain space temperature setpoint of 65 degrees F. Unit will be controlled by a line voltage stat. Aquastats are not to be used.
N. KITCHEN EXHAUST FAN INTERLOCK CONTROL: The exhaust fan will be not be interlocked with MAU unit. Exhaust fan shall be energized from a local switch with pilot light, and when exhaust fan schedule is enabled. Upon sensing exhaust fan status, and the MAU unit schedule is enabled, the unit will be forced into the occupied sequence. If the Ansul monitoring system alarm contact closes, the MAU shall be disabled and the exhaust hood shall be enabled via a 120V interlock wired by the Electrical Contractor.
O. KILN SEQUENCE See Kiln Design Bulletin
P. BASE BOARD RADIANT HEATERS: The base board valve shall open to maintain space temperature setpoint of 65 degrees F (adjustable via the DDC). BAS controlled.
Q. CRAWL SPACE HUMIDITY CONTROL: The crawl space shall be ventilated by an exhaust fan interlocked with a 120V humidistat and an intake damper. This interlock shall be wired by the electrical contractor.
R. MDF ROOM CONTROL: Packaged Room Air Conditioning Equipment controls shall be enabled through DDC /BAS unit controls. To include the following points. Use of district standard MDF programming templet required.
    1. DAT 5. Supply fan enable 9. DAT Low limit to be done in programming.
    2. RAT 6. Supply fan status
    3. MAT 7. Cooling Enable
    4. RMT 8. Economizer, 2-10VDC
S. EXHAUST FANS: Unit and building exhaust fans will be controlled via a schedule from the DDC. Restroom exhaust fans will be enabled or disabled via occupancy sensors serving that space with a delay off. Exhaust fans will not run during morning warm up or night sight back. An override point along with fan status will be available on the DDC graphics.
T. DISHWASHER EXHAUST FAN: Fan shall be interlocked with dishwasher equipment to operate during dishwasher operation. Fan shall remain on for a period of 8 minutes after dishwasher shuts off.
U. FREEZER/ COOLERS: Walk-in freezers and coolers are both to have DDC temperature sensors for temperature monitoring, and alarming. This system is separate from the requirement to provide alarming to the Burglar Alarm System.
V. BAS shall have control of exterior building and parking lights. Each will have its own scheduled, BAS to have the following points:
    1. Enable
    2. Status
    3. Alarm
    4. Override

W. GEOTHERMAL LOOP SEQUENCE OF OPERATION: The geothermal loop provides energy from the ground to the heat pumps within the building for heating and cooling. The loop includes a variable flow ground source loop (primary) and a variable flow heat pump water distribution loop (secondary). The geothermal loop has a pump set that is comprised of an operating pump and a standby pump, operated in a lead/lag sequence. The heat pump distribution loop has a pump set that is comprised of three pumps, one low use pump for reduced flow conditions during summer operations when the administration offices and summer activities heat pumps are operating, or when low use is experienced, and two full flow pumps, an operating pump and a standby pump, operated in a lead/lag sequence.
    1. The geothermal loop is not controlled by the unoccupied modes of the DDC operation.
    2. Pump start/stop: The operation of the pumps are determined by the position of their variable frequency H-O-A switch.
    3. On general power loss and restart, provide a 60 second delay (adjustable) before restart.
    4. Ground Source Loop Pump Control
        a. The DDC shall start the ground source lead pump whenever the building heat pump system operates.
        b. Once started, the pump shall run for a minimum of 30 minutes (adjustable)
        c. The DDC shall modulate the variable frequency pump speed control to maintain an absolute temperature difference of 10 degrees F (adjustable) between the heat pump supply water temperature sensor and the heat pump return water temperature sensor. The pump speed shall increase when the absolute temperature difference rises above setpoint and decreases when the absolute temperature difference falls below the setpoint.

d. The pump shall not modulate below 50% (adjustable) of full speed.

23 10 00 Facility Fuel Systems

A. DESIGN NOTES
   1. Design or construction team is responsible to calculate and submit all required information to the gas utility for gas service permits and service.

B. NATURAL GAS SYSTEMS
   1. Install gas cock upstream and within 6 feet of gas appliance. Install a union or flanged connection downstream from the gas cock to permit removal of controls.
   2. Sediment Traps: Install a tee fitting with the bottom outlet plugged or capped as close to the inlet of the gas appliance as practical. Drip leg shall be a minimum of 3 pipe diameters in length.
   3. Install valves in accessible locations, protected from physical damage. Tag valves with a metal tag attached with a metal chain indicating the piping systems supplied.
   4. Install pressure relief or pressure limiting devices so they can be readily operated to determine if the valve is free; so they can be tested to determine the pressure at which they will operate; and examined for leakage when in the closed position. Pipe atmospheric vent to outdoors.
   5. Label natural gas piping per current fuel gas code requirements. Yellow with black letters.

C. NATURAL GAS PIPING SPECIALTIES:
   1. Gas Meters: Diaphragm-type, positive displacement gas meters with aluminum cases, temperature compensated, with internal corrosion-resistant components; threaded ends for 2 inch and smaller, flanged ends for 2-1/2 inch and larger; for gas working pressures, specific gravity, and volume flow indicated. Provide pulse monitor for interface to DDC system.
   2. Flexible Connectors: Corrugated type 304 stainless steel flexible pipe with stainless steel braid and heavy flexible armor shield.
   3. Quick Couplers: One way quick coupler with gas rating in cubic feet per hour equal to equivalent gas appliance rating.
   4. Valve cabinets: For gas safety valves, No. 4 stainless steel finish, fully recessed, Series S construction, style K stainless steel door with key lock, 4 keys per cabinet, all cabinets keyed alike.
   5. Use of megapress components is discouraged.

D. VALVES:
   1. Solenoid Valves: Aluminum body, 120 volts AC, 60 Hz, Class B continuous duty molded coil; NEMA 4 coil enclosure; electrically opened/electrically closed; dual coils; normally closed; UL and FM approved and labeled.
   2. Gas Line Pressure Regulators: Single stage, steel jacketed, corrosion-resistant gas pressure regulators; with atmospheric vent, elevation compensator; with threaded ends for 2 inch and smaller, flanged ends for 2-1/2 inch and larger; for inlet and outlet gas pressures, specific gravity, and volume flow indicated.
   3. Gas Safety Valves: Gas safety valve latched open when energized, free handle design, manual reset, and a visual position indicator.
   4. Shutoff Valve: Bronze or aluminum body solenoid valve with Buna “N” seat, 120 VAC, 60 Hz, Class F continuous duty molded coil. The valve shall be normally closed, UL listed for 0 to 5 psi natural gas safety shutoff, automatic reset. Threaded end for NPS 2 (DN 50) and smaller, and flanged end for NPS 2-1/2 (DN 65) and larger. Design Basis: ASCO Model 8215, size as indicated on the drawings.
   5. Valve Control Station: Provide an 8” x 8” x 4” (200 mm x 200 mm x 100 mm) deep NEMA 1, UL listed gas valve control panel with brushed stainless steel cover for flush mounting. The control station shall provide a red pushbutton switch for emergency valve shutoff, a key operated spring-return switch for manual reset, a latching type 120 VAC control relay and a red “on” pilot light.
   6. Valve Ceiling Enclosure: Provide a 2’ x 2’ x1’ (600 mm x 600 mm x 300 mm) deep solenoid valve enclosure box with a perforated face and a hinged cover suitable for installation in a T-bar grid ceiling (see plans for detail and locations).
   7. Manufacturer: Subject to compliance with requirements, provide products by one of the following:
      a. Jenkins Bros.
      b. Lunkenheimer Co.
      c. Nibco, Inc.
      d. Powell Co.

e. Stockham.

23 21 00 Hydronic Piping and Pumps

A. BASIC PIPING MATERIALS AND METHODS

1. Piping Materials: Provide pipe and tube of type, pressure and temperature ratings, capacities, joint type, grade, size and weight (wall thickness or Class) indicated for each service. Where type, grade or class is not indicated, provide proper selection as determined by Installer for installation requirements, and comply with governing regulations and industry standards.
2. Pipe/Tube Fittings: Provide factory-fabricated fittings of type, materials, grade, class and pressure rating indicated for each service and pipe size. Provide sizes and types matching pipe, tube, valve or equipment connection in each case. Where not otherwise indicated, comply with governing regulations and industry standards for selections, and with pipe manufacturer's recommendations where applicable.
3. Provide isolation valves for all equipment, coils, valves, gauges, etc. so component can be repaired or replaced without relieving pressure or draining system.
4. Mark as existing if plan is to re-use piping.
5. Isolate system utilizing drain and air bleed valves so entire building doesn't have to be drained to repair leak.
6. No flanged unions.
7. **Hydronic copper fittings, valves, etc. above 1" shall be brazed not soldered.**
8. When connected to boilers, chillers, pumps, RTU's, AHU's, etc. support of piping needs to be from the floor in addition to the ceiling.
9. Dielectric Unions should be nipple type and provided with ball valve isolation.
10. Domestic water bibs to be located next to all glycol feeders.
11. System safety relief valve should be installed before any PRV.
12. Define acid flush procedure.
13. Vibration isolation: Braided stainless steel only.
14. Auto air venting piped to return loop or glycol feeder.

B. STEEL PIPES AND PIPE FITTINGS:

1. Black Steel Pipe: ASTM A 53, Grade B, type E
2. Steel Water Pipe: AWWA C200 for pipe 6 inches and larger.
3. Coal Tar Protective Coatings and Linings for Steel Water Pipe: AWWA C203 for enamel and tape, hot applied.
4. Cast-Iron Flanged Fittings: ANSI/ASME B16.1, including bolting (Class 125 and 250).
5. Cast-Iron Threaded Fittings: ANSI/ASME B16.4 (Class 125 and 250).
6. Malleable-Iron Threaded Fittings: ANSI/ASME B16.3; plain or galvanized as indicated (Class 125 and 300).
7. Malleable-Iron Threaded Unions: ANSI B16.39, Class 150, 250 or 300; selected by Installer for proper piping fabrication and service requirements, including style, end connections, and metal- to-metal seats (iron, bronze or brass); plain or galvanized as indicated (Class 150, 250 and 300).
8. Threaded Pipe Plugs: ANSI/ASME B16.14.
9. Steel Flanges/Fittings: ANSI/ASME B16.5, ASTM A234 (Fire Protection) including bolting and gasketing of the following material group, end connection and facing, except as otherwise indicated.
10. End Connections: Buttwelding.
11. Facings: Raised-face.
12. Steel Pipe Flanges for Waterworks Service: AWWA C207 (water service piping only).
13. Forged-Steel Socket-Welding and Threaded Fittings: ANSI B16.11, except MSS SP-79 for threaded reducer inserts; rated to match schedule of connected pipe (up to 4 inch pipe size).
14. Wrought-Steel Buttwelding Fittings: ANSI B16.9, except ANSI B16.28 for short-radius elbows and returns; rated to match connected pipe.
15. Cast-Iron Threaded Drainage Fittings: ANSI B16.12.
16. Forged Branch-Connection Fittings: Except as otherwise indicated, provide type as determined by Installer to comply with installation requirements.
17. Pipe Nipples: Fabricated from same pipe as used for connected pipe; except do not use less than Schedule 80 pipe where length remaining unthreaded is less than 1-1/2 inches, and where pipe size is less than 1-1/2 inches, and do not thread nipples full length (no close-nipples).

C. COPPER TUBE AND FITTINGS:
    1. Copper Tube: ASTM B 88; Type K or L as indicated for each service; hard-drawn temper, except as otherwise indicated.
    2. DWV Copper Tube: ASTM B 306.
    3. Cast-Copper Solder-Joint Fittings: ANSI B16.18.
    4. Wrought-Copper Solder-Joint Fittings: ANSI B16.22.
    5. Cast-Copper Solder-Joint Drainage Fittings: ANSI B16.23 (drainage and vent with DWV or tube).
    6. Wrought-Copper Solder-Joint Drainage Fittings: ANSI B16.29.
    7. Cast-Copper Flared Tube Fittings: ANSI B16.26.
    8. Bronze Pipe Flanges/Fittings: ANSI B16.24 (Class 150 and 300).
    9. Copper-Tube Unions: Provide standard products recommended by manufacturer for use in service indicated.
    10. Pro-Press fitting use and parameters.
D. PIPING INSTALLATION:
    1. Install piping at right angles or parallel to building walls. Diagonal runs are not permitted, unless expressly indicated.
    2. Conceal all pipe installations in walls, pipe chases, utility spaces, above ceilings, below grade or floors, unless indicated to be exposed to view.
    3. Install piping tight to slabs, beams, joists, columns, walls, and other permanent elements of the building. Provide space to permit insulation applications, with 1 inch clearance outside the insulation. Allow sufficient space above removable ceiling panels to allow for panel removal.
    4. Locate groups of pipes parallel to each other, spaced to permit applying insulation and servicing of valves.
    5. Install drains in pressure pipe systems at all low points in mains, risers, and branch lines consisting of a tee fitting, ¾ inch ball valve, and short ¾ inch threaded end nipple and cap with chain.
    6. Anchor piping to ensure proper direction of expansion and contraction.
E. PIPING APPLICATION:
    1. Accessible Gas Piping:
        a. Exposed Location: 2 Inches and Smaller: Schedule 40, black steel pipe, beveled ends, with 150 lb. malleable iron fittings and threaded joints. Over 2 Inches: Schedule 40 black steel with butt weld fittings and welded joints.
        b. Inaccessible Location: Schedule 40 black steel pipe, beveled ends, with socket weld fittings same thickness as pipe; welded joints.
        c. Below Grade: Schedule 40 seamless black steel with 150 lb. forged steel fittings and welded joints. Provide machine applied, coated and wrapped pipe in accordance with local code and utility company requirements. Provide cathodic protection as called for on the drawings.
    2. Hydronic Piping:
        a. All piping joining, coupling, fittings, etc. shall be designed of appropriate materials intended for the fluid type and temperature ranges of the system.
        b. 2 Inches and Smaller:
            1. Schedule 40, black steel with 125 lb. cast iron or 150 lb. malleable iron threaded fittings (OR) Type L copper, hard drawn copper wrought copper or bronze fittings.
        c. 2-1/2 Inches and Larger:
            1. Schedule 40, seamless or ERW (std. weight 12 inches and over) black steel.
            2. Type L copper, hand drawn, wrought copper or bronze fittings.
        d. Allowable Joining Methods
            1. Welded
                a. Fittings: Standards weight / extra strong, seamless steel, butt weld type.
            2. Grooved
                a. Manufacturer: Victaulic, no substitutions
                b. Grooved mechanical couplings and grooved-end fittings permitted.
            3. Flanged: 150 lb. 300 lb. forged steel slip-on or welding neck type.
                a. Bolting: Regular square head machine bolts with heavy hexagonal nuts.

b. Gaskets: Thickness, material and type suitable for fluid to be handled, and design temperature and pressures.

e. Equipment Drains and Overflows: Type "M" or "DWV" copper.

F. HVAC PUMPS:

1. Provide factory-tested pumps, thoroughly cleaned, and painted with one coat of machinery enamel prior to shipment. Type, size, and capacity of each pump shall be listed in pump schedule. Provide pumps of same type by same manufacturer.
2. Pump motor shall be sized so as not to be overloaded at any point along impeller curve for specified performance. All pump couplers shall be suitable for both constant speed and variable speed operation.
3. In-line Circulator Pumps: Single stage, wet rotor type, designed for 145 psi working pressure, 140 degrees F (60 degrees C) continuous water temperature, and specifically designed for quiet operation.
    a. Pump volute: cast iron.
    b. Impeller, impeller seal ring, rotor can, bearing plate, motor shaft and rotor cladding shall be constructed of stainless steel. Impeller shall be secured directly to the motor shaft be means of a stainless steel tapered split cone and locking nut.
    c. The motor shaft shall have tungsten carbide bearing journals and shall be supported by two aluminum oxide ceramic bearings. The motor shaft shall have a stainless steel mounted carbon thrust ring.
    d. Manufacturer: Subject to compliance with requirements, provide products by one of the following:
        1. Grundfos
        2. Wilo
4. Frame-mounted End Suction Pumps: Horizontal mount, single stage, vertical split case, flexible coupling, base mounted, designed for 175 psi working pressure. Cast iron, 125 psi ANSI flanges, tappings for gauge and drain connections. Pumps shall be laser aligned by installing contractor or manufacturer's representative.
    a. Shaft: Steel with replaceable shaft sleeve.
    b. Bearings: Regreaseable sleeve bearings.
    c. Seal: Mechanical, with carbon seal ring and ceramic seat.
    d. Impeller: Bronze enclosed type, hydraulically and dynamically balanced, keyed to shaft and secured with locking screw. Assembly components shall be 304 stainless steel.
    e. Coupling:
        1. Timpken Quick Flex

G. WATER TREATMENT

1. Contractor shall be required provide a submittal for water treatment designs for hydronic systems as well as testing data confirming the installed material. Submittals shall be reviewed by the Engineer and/or Commissioning Agent to ensure alignment with equipment manufacturer requirement and industry best practices.
2. Engineering shall align water chemistry with boiler manufacturer recommendations.
3. Total dissolved solids shall be 10 ppm or less.
4. 30% Glycol, when required.
5. Use of nitrites as corrosion inhibitor is prohibited.

23 30 00 HVAC Air Distribution

A. METAL DUCTWORK

1. Designer Note: Mark existing duct work as such if plan is to re-use.
2. Designer Note: Install building relief hood when static control is used.
3. Designer Note: Locate and provide fire damper schedule.
4. Extent of metal ductwork shall be indicated on drawings and in schedules, and by requirements of this section.

DUCT SERVICE	TYPE/CONSTRUCTION
Supply air between fan and terminal boxes (medium and high).	Galvanized steel, spiral, round or rectangular.
Rectangular supply air from discharge of terminal box/fan to air devices (low pressure).	Galvanized sheet metal /spiral round and oval or rectangular (wrapped or lined as noted on drawings.)
Return air ductwork.	Galvanized steel (lined where noted on drawings); factory or shop fabricated.).
General building exhaust.	Galvanized sheet metal (lined as noted on drawings); factory or shop fabricated.).
Transfer ducts.	Internally lined galvanized sheet metal as described above for low pressure supply; factory or shop fabricated.
Sound elbows for R.A. grilles	Galvanized sheet metal (internally lined).
Fume hood exhaust risers, mains and branch ducts including fume hoods.	PVC coated ductwork; factory or shop fabricated.
Dishwasher exhaust.	316 stainless steel all welded construction
Kitchen grease exhaust.	Concealed From View: Min 16 GA, carbon steel, all welded construction. Exposed To View: Type 304 stainless steel, min. 18 GA all welded construction, with welds ground smooth for a #4 finish.
Shower, locker room exhaust.	Aluminum with silicone sealant.

5. Protection: Protect shop-fabricated and factory-fabricated ductwork, accessories and purchased products from damage during shipping, storage and handling. Prevent end damage and prevent dirt and moisture from entering ducts and fittings. By providing end caps on all open sections, bagging small fittings, surface wrapping and shrink wrapping.
6. Storage: Store ductwork inside elevated from floor on pallets. At no time shall the inside surfaces be exposed, or stored with open ends and protect from weather.
7. Routing: Locate ductwork runs, except as otherwise indicated, vertically and horizontally and avoid diagonal runs wherever possible. Locate runs as indicated by diagrams, details and notations or, if not otherwise indicated, run ductwork in shortest route which does not obstruct useable space or block access for servicing building and its equipment. Hold ducts close to walls, overhead construction, columns, and other structural and permanent enclosure elements of building. Limit clearance to 1/2" where furring is shown for enclosure or concealment of ducts, but allow for insulation thickness, if any. Where possible, locate insulated ductwork for 1" clearance outside of insulation. Wherever possible in finished and occupied spaces, conceal ductwork from view, by locating in mechanical shafts, hollow wall construction or above suspended ceilings. Do not encase horizontal runs in solid partitions, except as specifically shown. Coordinate layout with suspended ceiling and lighting layouts and similar finished work.

8. Electrical Equipment Spaces: Do not route ductwork through transformer vaults and their electrical equipment spaces and enclosures.
9. Slope shower, locker room, and high moisture ductwork down to air device.
10. Penetrations: Where ducts pass through fire rated walls and do not contain fire or smoke dampers, protect with fire stop material installed in accordance with its listing. Where ducts pass through interior partitions or exterior walls, and are exposed to view, conceal space between construction opening and duct or duct insulation with sheet metal flanges of same gauge as duct. Overlap opening on all four sides by at least 1-1/2". Fasten to duct only. Where ducts penetrate non-fire rated, mechanical, electrical or acoustically sensitive walls, provide 1" annular space between duct and wall, pack annular space with mineral wood insulation, and caulk both sides with non-hardening acoustical sealant.
11. Ductwork Materials:
    a. Sheet Metal: Except as otherwise indicated, fabricate ductwork from galvanized sheet steel complying with ASTM A 527, lockforming quality; with G 90 zinc coating in accordance with ASTM A 525; and mill phosphatized for exposed locations. Provide flat seam construction where standing seams are a hazard to the Owner's operation personnel.
    b. Duct Liner: Fibrous glass, complying with Thermal Insulation Manufacturers Association (TIMA); Unless otherwise noted, provide 1" thick, 1-1/2 lb density, fiberglass duct liner meeting ASTM C1071 Type I, NFPA 90A and 90B and TIMA (AHC-101) with minimum NRC (noise reduction coefficient) of 0.70 as tested per STM C 423 using an "A" mounting with minimum "K" factor of 0.25.
    c. Flexible Ducts: Flexible air ducts shall be listed under UL-181 standards as Class I Air Duct Material and shall comply with NFPA Standards 90A and 90B. Flexible air ducts shall have a flame spread rating 25 or less and a smoke-developed rating not higher than 50. Minimum operating pressure rating shall be 6" W.C. through a temperature range of -20o to 150oF; minimum working velocity rating shall be 4000 f.p.m. Contractor shall assume responsibility for supplying material approved by the authority having jurisdiction.
        1. All insulated flexible ducts shall be constructed of a metalized ripstop reinforced laminate inner core, 1" thick, 3/4 lb. density fiberglass insulation with "C" factor of 0.23 or less and an outer jacket made exclusively of fire retardant reinforced aluminized material.
        2. Non-insulated flexible ducts shall be constructed from dead soft aluminum sheet, spiral corrugated, or aluminum construction over a steel spring helix.
    d. Duct Take Off Fittings to Individual Air Inlets & Outlets: Provide conical spin-in fittings at flexible or round sheet metal duct takeoffs. Where specifically shown on drawings, where the duct dimension does not allow for a conical spin-in, or at Contractor's option, provide 45° inlet rectangular to round duct take off fittings, with factory applied gasket. Fittings shall include butterfly type manual volume damper with regulator, and dual locking device. Dual locking device shall consist of two shaft mounted wing nuts, one on each side of the damper. Wing nuts shall tighten on shafts to lock butterfly in place. Shafts shall be solid metal; rolled metal shafts are not acceptable.
    e. Duct Pressure Testing: All high and medium pressure duct to be pressure tested prior to installation of VAV boxes or other terminal devices. Allowable leakage shall follow current SMACNA standards as presented in the SMACNA HVAC Air Duct Leakage Test Manual.
    f. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
        1. Duct Liner:
            a. CertainTeed Corp.
            b. Manville Products Corp. (Schuller)
            c. Owens-Corning Fiberglas Corp.
            d. Pittsburgh Corning Corp.
        2. Flexible Ducts:
            a. Flexmaster
            b. Thermaflex
        3. Duct Take Off Fittings
            a. Hercules Industries
            b. Flexmaster
            c. Thermaflex

d. Ominair

4. Round Ductwork (low, medium, and high pressure):
    a. Semco Mfg., Inc.
    b. United Sheet Metal Div., United McGill Corp.
    c. Sheet Metal Products Co.
    d. Hercules Industries

B. MANUAL VOLUME DAMPERS:
1. Low Pressure Rectangular Dampers (less than 2000 FPM and under 2" W.C. S.P. Differential):
    a. For 12" in height or larger, use multiple opposed blade type and close fitted to ducts. The frame and blades shall be constructed of 16 ga. galvanized steel with plated steel shaft mounted with synthetic bearings. Linkage shall be in-jamb fixed type located outside the airstream made of plated steel tie bar and crank plates, with stainless steel pivots. Damper panels shall not exceed 48" wide. Provide jack shafting when duct size required is greater than 48" wide. Provide notched shaft end indicating damper position, locking quadrant to fix damper position and handle. Provide stand off bracket for insulated ducts. For flat oval and round ductwork, provide type C housing.
    b. For ducts less than 12" in height, frame shall be 18 ga. blade galvanized steel, steel axle with synthetic bearings locking quadrant handle and notched shaft end indicating damper position. Provide stand off bracket for insulated ducts.
2. Low Pressure Round Dampers (less than 1800 FPM and under 1" W.C. S.P. differential):
    a. For low pressure spin-in fitting dampers serving individual returns/diffusers.
    b. Dampers 4" diameter through 18" diameter shall be 20 ga. galvanized steel frame and blade, utilize multi-blade square dampers with transitions for ducts over 18" diameter.
    c. Axle shaft shall be plated steel with retainers mounted on synthetic bearings with notched end shaft indicating damper position, locking quadrant and handle. Provide stand off brackets for insulated ducts.
3. Medium/High Pressure Rectangular Dampers (less than 4000 FPM and under 6" W.C. (48" wide or less) S.P. or 8" W.C. S.P. (36" wide or less)):
    a. Dampers shall be opposed blade for volume control and parallel blade for isolation/shut-off service.
    b. Frame shall be 16 ga. galvanized steel with welded corners or 1/8" thick 6063-T5 alloy aluminum frame. Blades shall be double skin galvanized steel with single-lock seam, or .081" thick 6060-T5 extruded aluminum, airfoil shape. Blade edge seals shall be vinyl, silicone, or other approved synthetic and metallic compression seals at the jambs. Axles shall be hexagonal or square plated steel mounted on bronze oilite or synthetic (ACETAL) bearings. Linkage shall be in-jamb type located outside the airstream. Maximum damper size shall be 48" wide and 60" high. For isolation or shut-off duty, damper leakage shall not exceed 9.5 CFM/Ft2 at 4" W.C. S.P. differential. Provide extended shaft with notched end indicating damper position, locking quadrant and handle. Provide stand off brackets for insulated ducts.
4. Medium/High Pressure Round Dampers (less than 3000 FPM and under 4" W.C. S.P. differential):
    a. Damper frame construction shall be as follows:
        1. ROUND Under 6" dia. 12 Gauge
           6" to 18" dia 14 Gauge
5. Manufacturer: Subject to compliance with requirements, provide products by one of the following:
    a. Greenheck
    b. Arrow Louver and Damper; Div. of Arrow United Industries, Inc.
    c. Louvers & Dampers, Inc.
    d. Nailor
    e. Penn Ventilator Co.
    f. Pottoroff
    g. Ruskin

C. FIRE DAMPERS:
1. Fire Dampers: Provide dynamic rated type B or C fire dampers except as noted on drawings. Construct sleeve of galvanized steel with bonded red acrylic enamel finish, gauge as required by the listing. All fire dampers shall be

UL labeled. Provide fusible link rated at 160 to 165 deg. F unless otherwise indicated. See architectural drawings for the separations and listings. Provide 2 fire damper access locations at each fire damper, one on the bottom and one on the side of the duct. Provide horizontal mounted fire damper with positive lock in closed position, and with the following additional features:

a. Damper Blade Assembly: Curtain Type.
b. Blade Material: Galvanized steel.
c. Provide integral sleeve type G fire dampers for sidewall air devices terminating at fire rated walls. Ruskin DIBD20-G or equivalent.

2. Manufacturer: Subject to compliance with requirements, provide products by one of the following:
   a. Greenheck
   b. Nailor
   c. Phillips Industries, Inc. Conaire Division
   d. Ruskin
   e. Pottoroff

D. SMOKE DAMPERS:

1. Rectangular Motor-Driven Smoke Dampers: Frame constructed of 16-ga. steel, type 304 stainless steel side seals, silicone edge seals, bronze oilite or stainless steel sleeve bearings, airfoil shaped galvanized steel formed interlocking blades, with factory mounted actuator motor, motor mounting bracket. Out of air stream plated steel linkage.
2. Round Motor-Driven Smoke Dampers 18" and Under: Frame constructed of 20 gauge galvanized steel, 2 layers of galvanized steel butterfly blade equivalent to 14 gauge, silicone rubber seal sandwiched between blade layers. Stainless steel sleeve bearings pressed into frame.
   a. Electric Damper Actuators shall have microprocessor based motor controller providing:
      1. Electronic cut off at full open so that no noise can be generated while holding open. Holding noise level shall be inaudible.
      2. Shall be incapable of burning out if stalled before full rotation is reached.
      3. Housing shall be steel and gears shall be permanently lubricated.
      4. The actuators shall be direct coupled and employ a steel toothed clamp for connecting to damper shafts. Aluminum clamps or set-screw attachment are not acceptable.
      5. Actuator shall carry a manufacturer's 5-year warranty and be manufactured under ISO 9001 quality control
      6. Damper actuator shall fail open/close upon loss of power.
3. Manufacturer: Subject to compliance with requirements, provide products by one of the following:
   a. Greenheck
   b. Arrow Louver and Damper; Div. of Arrow United Industries, Inc.
   c. Louvers & Dampers, Inc.
   d. Nailor
   e. Penn Ventilator Co.
   f. Pottoroff
   g. Ruskin

E. COMBINATION FIRE/SMOKE DAMPERS:

1. Rectangular Fire/Smoke Dampers: 16 gauge galvanized steel frame, type 304 stainless steel side seals, combination silicone/galvanized steel edge seals, bronze oilite or stainless steel sleeve bearings, airfoil shaped galvanized steel parallel acting blades, square or horizontal plated steel axles, out of airstream in-jamb linkage with stainless steel pivots, factory sleeve, caulked and attached to damper in accordance with UL fire damper requirements.
2. Round Fire/Smoke Dampers 18" Diameter and Smaller: 20 gauge galvanized steel frame/integral sleeve, 2 layer galvanized steel butterfly blade equivalent to 14 gauge, silicone rubber seal sandwiched between blade layers, stainless steel sleeve bearings pressed into frame, retaining plates in accordance with the UL listing.

Execution: Contractor to test all fire, smoke and fire-smoke dampers prior to air balancing.

F. DUCT ACCESS DOORS:

1. Access Doors for Low Pressure Rectangular Duct: Construct of same or greater gauge as ductwork served, provide double wall insulated doors for insulated ductwork. Duct access doors to be located upstream of every in duct coil, airflow station, fire damper, smoke damper and at heating coils. For fire dampers, 2 access doors are required; one on the bottom of the duct and one on the side in the same general location. Exposed insulation adhered to door is not acceptable. Provide flush frames for uninsulated ductwork, extended frames for externally insulated duct. All access doors shall have gasket and will be air tight. Provide one side hinged, other side with one handle-type latch for doors 12" high and smaller, 2 handle-type latches for larger doors. Where a hinged door cannot be fully opened a removable door may be used.
2. Access Doors for Medium and High Pressure Rectangular Duct: Insulated double wall round door and frame arranged for "Spin-In" installation, with continuous gasket in frame for door. Leakage of less than .5 cfm at 6" W.G.
3. Design based on load calculations
4. Design Parameters:
    a. 160° water temperature
    b. -10° ambient temperature
5. Manufacturer: Subject to compliance with requirements, provide products by one of the following:
    a. Kess
    b. Greenheck
    c. Flexmaster
    d. Cesco-Advanced Air
    e. Duro Dyne Corp.
    f. Ventfabrics, Inc.

G. FLEXIBLE CONNECTIONS:
1. Provide flexible duct connections of neoprene-coated flameproof fabric crimped into duct flanges for attachment to duct and equipment. Make airtight joint. Provide adequate joint flexibility to allow for thermal, axial, transverse, and torsional movement, and also capable of absorbing vibrations of connected equipment. Shelf life shall be verified to not exceed six (6) months. Any sign of cracking on interior or exterior shall be cause for replacement immediately.
2. Provide flexible connections immediately adjacent to equipment in ducts associated with fans and equipment subject to forced vibration. Provide matching flanged backing frame with flexible connector where flanged fan connections are provided.
3. Use the following product types for each application accordingly:
    a. Indoor Equipment Non-Corrosive Air Systems: Heavy glass fabric, double-coated with DuPont's NEOPRENE, non-combustible fabric, fire retardant coating with good resistance to abrasion and flexing. Fabric shall be 30 oz per square yard, capable of operating at -10oF to 200oF, waterproof, air tight, 6 inches wide, complies with NFPA 90 and UL Standard #214. Design Basis: "Ventglas" Model as manufactured by VentFabric, Inc.
    b. Indoor Corrosive Air System: Heavy glass fabric coated with DuPont's teflon fluorocarbon resins, capable of operating between temperatures of - 20oF and 500oF. Fabric shall be 14 oz per square yard, watertight, air tight, chemically resistant to most chemicals including but not limited to sulfuric acid, acetic acid, chlorine, dimethyl ether, xylene, hexane, ozone, nitric acid, butyl acetate, ammonia gas and liquid, acetone, mercury, cyclohexane, methanol, 6 inches wide. Design Basis: "Ventel" model as manufactured by VentFabrics, Inc.
4. Manufacturer: Subject to compliance with requirements, provide products by one of the following:
    a. Duro Dyne Corp.
    b. Ventfabrics, Inc.
    c. General Rubber Corp. (Process & Exhaust Only)

H. AIR HANDLING FANS
1. Extent of air handling equipment work required shall be indicated on drawings and schedule.
2. Provide fans that are factory fabricated and assembled, factory tested, and factory finished, statically and dynamically balanced and designed for continuous operation at the maximum rated fan speed and motor horsepower.

3. Belt Drives: Factory mounted, with final alignment and belt adjustment made after installation.
4. Motors and Fan Wheel Pulleys: Adjustable pitch for use with motors through 15 HP; fixed pitch for use with motors larger than 15 HP. Select pulley so that pitch adjustment is at the middle of the adjustment range at fan design conditions. Provide energy efficient motor.
5. Shaft Bearings: Provide type indicated, having a median life "Rating Life" AFBMA L10 of 100,000 medium duty calculated in accordance with AFBMA Standard 9 for ball bearings and AFBMA Standard 11 for roller bearings.
6. Warranty: Provide 2 year warranty on all fans.
7. Demonstration Services: Arrange and pay for a factory-authorized service representative to train Owner's maintenance personnel on procedures and schedules related to start-up and shutdown, troubleshooting, servicing, preventative maintenance, and how to obtain replacement parts.

I. UTILITY SET FANS:
1. Belt-driven, centrifugal fans consisting of housing, wheel, fan shaft, bearings, motor and disconnect switch, drive assembly, and accessories.
2. Housings: Fabricated from heavy-gauge steel with side sheets fastened to scroll sheets by means of welding or deep lock seam.
3. Housings Discharge: Adjustable to 8 standard positions. Slip-joint duct connection.
4. Fan Wheels: Single-width, single-inlet, welded to cast-iron or cast-steel hub and spun steel inlet cone, with hub keyed to the shaft.
5. Blade Type: Backward-curved, die-formed.
6. Backdraft Dampers: Gravity-actuated with counterweight and interlocking aluminum blades and felt edges in steel frame installed on fan discharge.
7. Access Doors: Gasketed doors with latch-type handles.
8. Inlet Screens: Removable, heavy wire mesh.
9. Drain Connections: 3/4-inch, threaded coupling drain connection installed at lowest point of housing.
10. Weather Hoods: Weather-resistant with stamped vents over motor and drive compartment.
11. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
    a. Acme Engineering And Mfg. Corp.
    b. Buffalo
    c. Loren Cook Co.
    d. New York Blower Co.
    e. Twin City Fan and Blower Co. (up to 24”)
    f. Greenheck

J. CENTRIFUGAL ROOF VENTILATIORS:
1. Belt-driven or direct-drive as indicated, centrifugal consisting of housing, wheel, fan shaft, bearings, motor and disconnect switch, drive assembly, curb base, and accessories.
2. Housing: Heavy-gauge, removable, spun-aluminum, dome top and outlet baffle; square, one-piece, hinged, aluminum base with venturi inlet cone.
3. Upblast Units: Provide spun-aluminum discharge baffle to direct discharge air upward, with rain and snow drains.
4. Fan Wheel: Aluminum hub and wheel with backward-inclined blades.
5. Belt-Driven Drive Assembly: Resiliently mounted to the housing, with the following features:
6. Shaft Bearings: Permanently lubricated, permanently sealed, self-aligning ball bearings.
7. Disconnect Switch: Nonfusible type, with thermal overload protection mounted inside fan housing, factory-wired through an internal aluminum conduit.
8. Bird Screens: Removable 1/2-inch mesh, 16-gauge, aluminum or brass wire.
9. Dampers: Gravity backdraft dampers for bathroom and general exhaust, larger dampers to be motor-operated, parallel-blade, volume control dampers mounted in curb base.
10. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
    a. Acme Engineering and Manufacturing Corp.
    b. Carnes Company, Inc.
    c. Loren Cook Co.
    d. Greenheck
    e. Twin City

K. HEAT AND SMOKE VENTS:
    1. Hatch-Type Heat and Smoke Vents: Manufacturer's standard single-leaf, hatch-type heat and smoke vents with integral insulated curbs and frame, with welded or sealed mechanical corner joints, integral condensation gutter, and cap flashing. Fabricate with insulated double-wall lid, continuous weathertight perimeter lid gaskets, and equip with automatic self-lifting mechanisms, UL-listed fusible links rated at 165 deg F and corrosion-resistant or hot-dip galvanized hardware including hinges, hold-open devices, and independent manual-release devices for inside and outside operation of lids.
    2. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
        a. Babcock-Davis; a Cierra Products Inc. Company.
        b. Bilco Company (The).
        c. Nystrom, Inc.
        d. ThyCurb; Div of Thybar Corporation.
        e. Wasco Products, Inc.
L. PREFABRICATED ROOF CURBS:
    1. Furnish and install roof curbs as scheduled for duct openings through the roof and for exhaust fan support. The curbs shall be galvanized steel self-flashing type. If the curbs are to have sound attenuation qualities, they shall be not less than those catalogued for the equipment specified.
    2. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
        a. Custon Curb, Inc.
        b. Pate Co.
        c. S & L Manufacturing Co.
        d. ThyCurb Div.; Thybar Corp.
M. AIR TERMINAL UNITS
    1. Types of air terminals include Central Air Terminals - Shutoff Single Duct/Reheat
    2. Provide factory-fabricated and tested air terminals as indicated, selected with performance characteristics which match or exceed those indicated on schedule.
    3. Air terminal units shall be low pressure drop, single duct throttling type pressure independent and suitable for use in medium pressure variable volume air distribution systems.
    4. Internal damper blade shall be extruded aluminum or 18 gauge steel with keyed fit shaft and nylon bushing. Damper shall seal against gasketed stops maximum 2% leakage at 3.0" S.P. All mechanical parts shall be galvanized or non-ferrous. Alternate damper design as produced by the Trane Company as acceptable.
    5. Hot water heating coils shall be designed for 200 psig maximum working pressure and 200oF maximum operating temperature. Coil shall be serpentine-type, constructed of 1/2" O.D. copper tubes mechanically bonded to aluminum fins; galvanized steel casing.
    6. Provide label on each air terminal unit, indicating plan designation, unit size, cfm range and settings and calibration curve.
    7. Provide a pressure independent pneumatic cross or ring-shaped flow sensor with velocity pressure pickup points for measuring inlet airflow. The sensor shall maintain control accuracy with the same size inlet duct in any configuration. Single point hot wire anemometer or straight line pneumatic sensors are not acceptable. Provide gauge ports in flow sensor tubing.
    8. Air terminals shall be selected so required dB sound levels are not exceeded at 1.5” w.g. inlet pressure. Both unit casing radiated sound levels and discharge sound levels shall be considered in terminal unit selection.
    9. Control devices shall be factory set and calibrated for operation with direct acting/reverse acting room thermostat.
    10. Temperature Control contractor to furnish DDC controls including all actuators and controllers. Terminal unit controls shall be factory mounted and wired.
    11. Manufacturer: Subject to compliance with requirements, provide products by one of the following:
        a. Krueger
        b. Envirotech
        c. Metal-Aire
        d. Nailor
        e. Price
        f. Titus Products Div.; Philips Industries, Inc.

g. Trane (The) Co.

N. AIR OUTLETS AND INLETS
   1. Types of air outlets and inlets required for project include the following:
      a. Ceiling air diffusers.
      b. Wall registers and grilles.
      c. Louvers.
   2. CEILING AIR DIFFUSERS: Performance: Provide ceiling air diffusers that have, as minimum, temperature and velocity traverses, throw and drop, and noise criteria ratings for each size device as listed in manufacturer's current data.
   3. REGISTERS AND GRILLES: Provide registers and grilles that have, as minimum, temperature and velocity traverses, throw and drop, and noise criteria ratings for each size device as listed in manufacturer's current data.
      a. Manufacturer: Subject to compliance with requirements, provide products by one of the following:
         1. Price (displacement diffusers)
         2. Anemostat Products Div.; Dynamics Corp. of America.
         3. Carnes Co.; Div. of Wehr Corp.
         4. Krueger; Div. of Philips Industries, Inc.
         5. Titus Products Div.; Philips Industries, Inc.
         6. Metal-Aire
         7. Nailor
   4. LOUVERS: Performance: Provide louvers that have a minimum of 60% free area, and a maximum pressure drop through the free area of not more than 0.075" for each type as listed in manufacturer's current data. On inside face of exterior louvers. Provide 1/2" square mesh anodized aluminum wire bird screens mounted in removable extruded aluminum frames.
      a. Manufacturer: Subject to compliance with requirements, provide products by one of the following:
         1. Air Balance
         2. Arrow United Industries, Inc.
         3. Pottoroff
         4. Louvers & Dampers, Inc.
         5. Penn Ventilator Co., Inc.
         6. Ruskin

O. NON-METAL DUCTWORK
   1. Types of non-metal ductwork required for this project include Fabric Air Dispersion products.
   2. Warranty: Manufacturer must provide a 5 Year Product Warranty for products supplied for the fabric portion of this system as well as a Design and Performance Warranty.

23 50 00 Central Heating Equipment

A. DESIGNER NOTES
   1. Trane equipment is discouraged
   2. Bell and Gossett products are discouraged
   3. Existing roof ladders not meeting current OSHA standards to be replaced.
   4. No motion control lighting in mechanical rooms with roof ladder access.
   5. Foot pads at the top of all ladders.
   6. Swamp coolers are discouraged.
   7. Power Panel location labeling on all roof top equipment.
   8. Accessible water source on roof for cleaning equipment.
   9. Anything located above hard ceiling shall have access immediately below.

B. BREECHINGS, CHIMNEYS, AND STACKS
   1. Special Warranty: Manufacturer's standard form in which manufacturer agrees to repair or replace components of venting system that fail in materials or workmanship within specified warranty period. Failures include, but are not limited to, structural failures caused by expansion and contraction. Warranty Period: 15 years from date of Substantial Completion.
   2. SPECIAL GAS VENTS: Double-wall metal vents tested according to UL 1738 and rated for 480 deg F continuously, with positive or negative flue pressure complying with NFPA 211.
      a. Construction: Inner shell and outer jacket separated by at least 1/2-inch airspace.
      b. Inner Shell: ASTM A 959, Type 29-4C stainless steel.
      c. Outer Jacket: Aluminized steel.
      d. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
      e. Heat-Fab, Inc.
      f. Metal-Fab, Inc.
      g. Selkirk Inc.; Selkirk Metalbestos and Air Mate.
      h. cap designed to exclude minimum 90 percent of rainfall.
   3. GREASE DUCTS: Double-wall metal vents tested according to UL 1978 and rated for 500 deg F continuously, or 2000 deg F for 30 minutes; with positive or negative duct pressure and complying with NFPA 211.
      a. Construction: Inner shell and outer jacket separated by at least a 2-inch annular space filled with high-temperature, ceramic-fiber insulation.
      b. Inner Shell: ASTM A 666, Type 316 stainless steel.
      c. Outer Jacket: Aluminized steel where concealed. Stainless steel where exposed.
      d. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
         1. Selkirk Inc.; Selkirk Metalbestos and Air Mate.
         2. Simpson Dura-Vent Co., Inc.; Subsidiary of Simpson Manufacturing Co.
         3. Van-Packer Company, Inc.

C. BOILERS
   1. Each boiler shall be; UL Listed, CSD-1 approved, ASME coded and stamped, and incorporate a gas train designed in accordance with FM & IRI. Each boiler shall have an input of 3,000mbh with a gross output of 2,610 to 2,880mbh (dependent upon return water temperature) when fired with natural gas. The boiler manufacturer must publish known part load value efficiencies; the thermal efficiency must increase as the firing rate decreases. Electrical service to each unit shall be as indicated on electrical plans. The boiler control panel shall be proprietary in design and incorporate the functions of temperature control, combustion safeguard control, message annunciation, and fault diagnostic display, on individual field replaceable circuit boards mounted within a single housing. Each boiler shall have a footprint of no more than 28" W, 64" L, and 79" H with a UL Listing for zero side wall clearance. The boiler installed weight shall not exceed 2170 lbs. dry. The boiler shall have an ASME approved relief valve setting.
   2. Construction: Boiler modules shall be natural gas fired, condensing fire tube design with a modulating forced draft power burner and positive pressure vent discharge.

3. Modulating Air/Fuel Valve and Burner: The boiler burner shall be capable of a 15 to 1 turndown ratio of the firing rate without loss of combustion efficiency or staging of gas valves. The burner shall produce <30ppm of NOx corrected to 3% excess oxygen. The burner shall be metal fiber mesh covering a stainless steel body, with spark ignition and flame rectification. All burner material exposed to the combustion zone shall be of stainless steel construction. There shall be no moving parts within the burner itself. A modulating air/fuel valve shall meter the air and natural gas input. The modulating motor must be linked to both the gas valve body and air valve body with a single linkage. The linkage shall not require any field adjustment. A variable frequency drive (VFD) controlled cast aluminum pre-mix blower with a nylon impeller shall be utilized to ensure the optimum mixing of air & fuel between the air/fuel valve and the burner.
4. Pressure Vessel/Heat Exchanger: The boiler shall be capable of handling return water temperatures down to 40 F without any failure due to thermal shock or fireside condensation. The heat exchangers shall be ASME stamped for a working pressure not less than 160 psig. The pressure vessels shall have a maximum water volume of 49 gallons. The boiler water pressure drop shall not exceed 5.5 psig at 258 gpm. The boiler water connections shall be 4" flanged 150 lb. ANSI rated. The pressure vessels are constructed of SA53 carbon steel, with a 0.25" thick wall and 0.50 in. thick upper head. Inspection openings in the pressure vessels shall be in accordance with ASME Section IV pressure vessel code. The boiler shall be designed so that the thermal efficiency increases as the boiler firing rate decreases. The heat exchangers shall be constructed of 316L stainless steel fire tubes and tube sheets with a one-pass combustion gas flow design. The fire tubes shall be 5/8" OD with no less than 0.065" wall thickness.
5. Exhaust Manifold: The exhaust manifold shall be of corrosion resistant stainless steel with an 8"diameter flue connection. The exhaust manifold shall have a gravity drain for the elimination of the condensation with collecting reservoir.
6. Controls Interoperability: Boilers will be controlled by field devices, no boiler management system may be used, all heating plant programing and control to be done through the BAS. Provide demand load capabilities compatible with District BAS for the following points: Start/Stop, and firing rate, general alarm.
7. Warranty: The pressure vessel/heat exchanger of the boiler shall carry a prorated 10-year warranty against failure due to condensate corrosion, thermal stress, mechanical defects or workmanship. The six individual circuit boards of the "C-More" control panel assembly shall carry a 2 year warranty against failure due to defective materials or workmanship. A Warranty Certificate must be issued to the owner from the manufacturer and a copy of warranty must be submitted for engineer's approval.
8. Services: After testing and inspection is complete, provide the services of an authorized factory service representative to perform start-up and operation demonstration service.
9. Start-up: Perform services in accordance with manufacturer's written start-up instructions. Test controls and demonstrate compliance with requirements. Replace damaged or malfunctioning controls and equipment.
10. Provide Combustion Test Record for each Boiler.
11. Preferred Manufacturer for Condensing Boilers:
    a. Advanced Thermal Hydronics (ATH), KN Series
12. Acceptable Manufacturers for Condensing Boilers
    a. Fulton,
    b. Evo Hamiltion,
    c. Lochinvar
    d. Raypak
    e. Riello
13. Preferred Manufacturer for Non Condensing Boilers:
    a. Buderus,
    b. Lochivar.

23 70 10 Roof Top Units (RTUs)

A. General
   1. The District's strong preference is for hydronic heat at RTUs, gas fired heat is typically prohibited without District approval.

B. Products
   1. ***Note to Design Teams:*** To maintain consistency of maintenance and repair across the various District facilities and to continue to realize the performance and service offered by the manufacturer and other supporting companies, the District's strong preference is for the RTU manufacturer indicated below. However, in the interest of providing the best value, other acceptable products shall be evaluated and compared based on a balance of performance and cost.

      a. Preferred Manufacturer
         1. AAON
      b. Acceptable Manufacturers, Tier 1 (Must provide minimum performance requirements listed below)
         1. Daikin
         2. Annexair
      c. Acceptable Manufacturers, Tier 2 (Pending District approval)
         1. Energy Labs
         2. Engineered Air
         3. Greenheck
         4. Carrier
         5. York/JCI

C. Minimum Performance Requirements
   1. Cabinet enclosure construction, R-13 minimum
   2. Access Doors: Lockable hinged access doors with continuous hinges and latching hardware.
   3. Fans Sections: Direct drive motor driven fans (wherever applicable)
   4. Compressors: Variable Capacity Scroll Compressors (where applicable)
      a. Capable of modulation from 15-100% of the rated capacity.
   5. Exhaust Fans: Integral to unit
   6. Controls:
      a. Full factory package controls are typically not desired by the District. Review controls requirements with the SVVSD Project Manager and HVAC Maintenance Department to define the extent, BAS integration, and installation responsibilities of field controls.
   7. Other requirements Coordinate with SVVSD PM and Maintenance Staff.
   8. Air Flow Monitoring Sensors: Coordinate with SVVSD PM and Maintenance Staff.
   9. Warranties:
      a. 1-year parts & labor

D. Extra Stock
   1. Complete set of filters
   2. One spare set of belts

23 80 00 Decentralized HVAC Equipment

A. TERMINAL UNITS
    1. Locate baseboard radiation on outside walls as indicated, run cover continuous wall-to-wall unless otherwise indicated. Install end caps where units butt against walls. Install access panels centered in front of each shutoff valve, balancing cock, or temperature control valve.

B. BASEBOARD RADIATION (finned tube): Provide hot water baseboard radiation of lengths, wall to wall enclosure.
    1. Architectural radiator: Provide steel panel radiator elements of lengths and in locations as indicated and of capacities, style and having accessories as scheduled. The wall hung heating panel radiation shall be of one-piece all welded steel construction, consisting of flattened water tubes welded to headers at each end. The radiator shall include an integral heavy gauge (0.09" minimum) all welded perforated top grille (for curved radiators the grille is omitted). RF models to have steel corrugated fins welded to the rear side of the water tubes to increase the convective output of the unit.
    2. The panel radiation shall be capable of being mounted to typical stud wall construction without additional blocking or strapping. Appropriate wall mounting brackets or optional floor post mounting shall be provided with the radiation.
    3. Finishes: The panel radiation shall be cleaned and phosphatized in preparation for the powder coat finish. The radiation is then finish painted with a gloss powder coat finish, for a total paint thickness of 2-3 mils (0.002"-0.003").
    4. Manufacturer: Subject to compliance with requirements, provide products by one of the following:
        a. Baseboard Radiation:
        b. Slant/Fin Corp.
        c. Vulcan
        d. Sterling Radiator; Div. of Reed National Corp.
        e. Runtal (Architectural Radiation)
        f. Rittling (Architectural Radiation)

C. UNIT HEATERS: Provide Horizontal Unit Heaters constructed of steel, phosphatized inside and out, and finished with standard color baked enamel finish casing. Provide motor- mounted panel, minimum of 18-ga steel. Fabricate casing to enclose coil, louvers, and fan blades. Provide louvers for 4-way air diffusion. Fans shall be aluminum, and factory-balance. Provide fan inlet orifice, smooth, and drawn into casing back panel. Coils shall be constructed of plate-type aluminum fins, mechanically bonded to copper tubes. Design coil for use in steam/hot water applications. Provide totally enclosed motors, with built-in overload protection, having electrical characteristics as scheduled. Aquastats are not to be included.
    1. Manufacturer: Subject to compliance with requirements, provide products by one of the following:
        a. Airtherm Mfg. Co.
        b. Vulcan
        c. McQuay Inc.
        d. Modine Mfg. Co.
        e. Trane

D. CABINET UNIT HEATERS: Provide hot water cabinet heaters having cabinet sizes and in locations as indicated, and of capacities, style, and having accessories as scheduled. Include in basic unit chassis, coil, fanboard, fan wheels, housings, motor, and insulation.
    1. Cabinet: 16-ga removable front panel, 18-ga top and side panels. Insulate front panel over entire coil section. Provide access door on coil connection side. Clean cabinet parts, bonderize, phosphatize, and flow-coat with baked-on primer. Custom color as selected by Architect in baked enamel finish.
    2. Water Coils: Construct of 5/8" seamless copper tubes mechanically bonded to configurated aluminum fins. Design for 300 psi and leak test at 300 psi under water. Provide same end connections for supply and return.
    3. Fans: Provide centrifugal, forward curved double width fan wheels constructed of non-corrosive, molded, fiberglass- reinforced thermo-plastic material. Construct fan scrolls of galvanized steel.
    4. Motors: Provide shaded pole motors with integral thermal over-load protection, and motor cords for plug-in to junction box in unit.

5. Filters: Provide 1" thick throwaway type filters in fiberboard frames.
6. Manufacturer: Subject to compliance with requirements, provide products by one of the following:
    a. Airtherm Mfg. Co.
    b. Dunham-Bush, Inc.
    c. McQuay Inc.
    d. Trane
    e. Vulcan

E. FAN-COIL UNITS: Provide fan-coil units included are basic unit chassis, coils, fanboard, drain pan assembly, fans, housing, motor, filter, and insulation.
1. Chassis: Construct chassis of galvanized steel with flanged edges.
2. Insulation: Faced, heavy density glass fiber.
3. Cabinet: Construct of 18-ga steel removable panels, 16-ga front. Provide insulation over entire coil section. Clean cabinet parts, bonderize, phosphatize, and flow-coat with standard factory color selected baked enamel finish.
4. Heating Coils: Construct of 1/2" seamless copper tubes mechanically bonded to configurated aluminum fins. Design for 250 psi working pressure.
5. Drain Pans: Construct of galvanized steel. Insulate with polystyrene or polyurethane insulation. Provide drain connection.
6. Fans: Provide centrifugal forward curved double width wheels of reinforced fiberglass, in galvanized steel fan scrolls.
7. Motors: Provide motors with integral thermal overload protection. Run test motors at factory in assembled unit prior to shipping. Provide quickly detachable motor cords.
8. Filters: Provide 1" thick throwaway type filters in fiberboard frames.
9. Manufacturer: Subject to compliance with requirements, provide products by one of the following:
    a. Airtherm Mfg. Co.
    b. Dunham-Bush, Inc.
    c. McQuay Inc.
    d. Trane
    e. York; Div. of York International
    f. International Fan Coil

F. ELECTRIC UNIT HEATERS: Provide Vertical Unit Heaters constructed of steel, phosphatized inside and out, and finished with standard color baked enamel finish casing. Design casing to enclose fan, motor, and coil, design fan orifice formed into discharge panel.
1. Fans: Construct of aluminum and factory-balance. Design so motor and fan assembly is removable through fan outlet panel.
2. Elements: Unit shall include electric resistance element with manual-reset thermal overload protection, unit mounted thermostat.
3. Motors: Provide totally enclosed motors, with built-in overload protection, having electrical characteristics as scheduled.
4. Surface enclosure: surface mounting plate designed for rough-in on frame ceiling construction. Louvered surface enclosure shall be contoured aluminum extrusion and 20 gauge sheet metal combination with rounded corners; baked enamel finish.
5. Manufacturer: Subject to compliance with requirements, provide products by one of the following:
    a. Q Mark
    b. Singer
    c. Trane
    d. Brasch
    e. Indeeco
    f. Berko
    g. Markel
    h. Modine
    i. Raywall

G. ELECTRIC CONVECTORS: Provide electric convectors having cabinet constructed of 12-ga steel front and top panels, 12-ga side panels, and 14-ga back panels. Phosphatize and finish with standard color, baked enamel finish on enclosure. Secure fronts in place with spanner head tamper resistant screws as provided by heater manufacturer. Heating element shall have full length auto-reset thermal overload disconnect switch, line voltage safety disconnect switch, internal, unit mounted thermostat.

1. Manufacturer: Subject to compliance with requirements, provide products by one of the following:
    a. Q Mark
    b. Singer
    c. Trane
    d. Berko
    e. Markel
    f. Raywall

DIVISION 25 – LIGHTING AUTOMATION

25 00 00 Primary District Requirements

A. DESIGN PREFERENCES
1. Exterior Lighting
   a. All exterior lighting shall be included on the lighting automation and controls system. The incorporation of interior lighting on the lighting automation and controls systems shall be coordinated on a project by project basis.
      1. Typical System: RAB Lightcloud
2. Interior Lighting
   a. Lighting automation and controls shall be a separate and stand-alone system. And shall not be integrated into the HVAC or other building control system(s).
   b. Lighting automation and control systems shall be password protected and should coordinate with the District Technology Services (DTS) user verification system where possible.
   c. All devices shall be provided with a home run to primary equipment, the use or installation of network switches without direct coordination with DTS is prohibited.
   d. Lighting automation and control systems shall avoid use of WIFI data whenever possible. The use of WIFI diminishes reliability and congests WIFI systems.
   e. Lighting automation and controls shall provide "true status" of exterior lights. The system shall verify that current is being provided to the lights rather than simply indicating that the light in set to "on".
   f. Upon project completion the Contractor shall furnish the District with a physical copy of the system program, i.e. thumb drive, CD, etc.
   g. Cable Color: See section 4.03 of Appendix C to this document "Communications, Security, and Technology Guidelines" for District cable coloring scheme.
   h. Products
      1. Preferred
         a. Acuity, nLight, District preference is to standardize on this product.
         b. Cooper, WaveLinx
      2. Acceptable
         a. Crestron

DIVISION 26 – ELECTRICAL

26 00 00 Primary District Requirements

A. POWER
   1. Provide an additional 30A spare circuit (3-5 wire) in all MDF rooms.
   2. Provide surge protection for all 20A outlets for all MDF and IDF room receptacles.
   3. Aluminum conduit is not allowed by the District.

B. LIGHTING
   1. Interior Lighting
      a. Lamp Type: LED, unless otherwise approved.
      b. Lighting Color: 4000+/- Kelvin
      c. Auditorium fixtures must be accessible by catwalk or from floor with approval by District.
      d. Occupancy controls per IECC requirements.
   2. Exterior Lighting
      a. Lamp Type LED, unless otherwise approved.

C. ENERGY MANAGEMENT
   1. Provide electrical energy monitoring unit in the main electrical room.
   2. Manufacturer: eGauge, no substitutions
   3. Coordinate data connection for eGauge monitoring equipment.
   4. Coordinate requirement with the District's Energy Manager.

26 01 00 Common Work Results for Electrical

B. BASIC ELECTRICAL REQUIREMENTS
1. Before any work is installed, determine that equipment will properly fit the space; Coordination drawings for the work with other building systems are a requirement.
2. Ductwork and piping shall not be installed above electrical panelboards, switchboards, motor control centers, and transformers.
3. Sequence, coordinate, and integrate installations of electrical materials and equipment for efficient flow of the work. Give particular attention to large equipment requiring positioning prior to closing-in the building.
4. Where mounting heights are not detailed or dimensioned, install electrical services and overhead equipment in a manner to allow maximum headroom possible.
5. Verify final locations for rough-ins with field measurements and with the requirements of the actual equipment to be connected.
6. All material used on this project shall be UL listed and labeled and be acceptable to the authority having jurisdiction as suitable for the use intended.
7. Install electrical equipment to facilitate maintenance and repair or replacement of equipment components. As much as practical, connect equipment for ease of disconnecting and removal with minimum of interference with other installations.

C. DESIGN CRITERIA:
1. All light fixtures shall be located in a manner for accessibility to maintenance staff without means of rental equipment. Architect is required to review all locations for fixtures with District personal prior to completing bidding documents. Submit light fixtures for O&M review during DD phase of the design
2. Auditorium fixtures must be accessible by catwalk or from floor with approval by District.
3. Install equipment and materials to provide required code clearances and access for servicing and maintenance. Coordinate the final location with piping, ducts, and equipment of other trades to insure proper access for all trades. Coordinate locations of concealed equipment, disconnects, and boxes with access panels and doors. Allow ample space for removal of parts, fuses, lamps, etc. that require replacement or servicing.
4. Extend all conduits so that junction and pull boxes are in accessible locations.
5. Provide access panel or doors where equipment or boxes are concealed behind finished surfaces.
6. Camera boxes to use conduit not MC cable.
7. Outdoor fixtures to have LED lamps.

D. DISTRICT SPECIFIC DESIGN CRITERIA
1. Provide an additional 30A spare circuit (3-5 wire) in all HUB rooms.
2. Provide surge protection for all 20A outlets for all HUB/MDF room receptacles.
3. Backboard lift motor switches shall be 3 position momentary toggles in lockable box; no keyed switches allowed.
4. Provide pull string (Tru-tape) in all empty conduits.
5. All rigid underground conduit including ells will be taped or PVC coated.
6. Provide 30” min. depth below finish grade for all underground conduits.
7. Coordinate required Owner stock of materials with District.

E. PORTABLE / MODULAR CLASSROOM REQUIREMENTS
1. Electrical feed to future potable locations shall be engineered such that if all voltage drops to problematic need shall be served by a 480V switch gear to feed transformer at portables. Conduits will be stubbed up to the switch gears as to safely pull fee wires without shutting down poser to the switch gear.
2. Provide 3’ separation between electrical and low voltage conduit stubs.
3. Provide 6’x10”x10” gutter for Electrical
4. Provide 6’x10”x10” gutter for Communications
5. Provide 3each 2” conduits for fire alarm, fiber and data.
6. Provide 3” ridged 90 degree sweeps only; no PVC 90’s, approved exceptions allowed.

7. Provide expansion joint on all PVC risers.
8. All racks to be welded or bolted on ridged conduit and uni-strut frame.

F. REQUIREMENTS OF LOCAL UTILITY COMPANIES:
   1. Coordinate connection of electrical systems with exterior underground and overhead utilities and services. Comply with requirements of governing regulations, franchised service companies, and controlling agencies. Provide required connection for each service.

G. PERMITS AND FEES:
   1. Contractor shall pay all fees required for connection to municipal and public utility facilities.
   2. Contractor shall arrange for and pay for all inspections, licenses and certificates required in connection with the work.

H. ELECTRICAL COORDINATION DRAWINGS:
   1. Prepare and submit a set of coordination drawings showing major elements, components, and systems of electrical equipment and materials in relationship with other building components. Prepare 11”x17” or 24”x36” drawings to an accurate scale of 1/4"=1'-0" or larger. Indicate the locations of all equipment and materials, including clearances for servicing and maintaining equipment including (but not necessarily limited to) the following:
      a. Electric equipment room layouts
      b. Mechanical equipment room layouts
   2. Prepare coordination drawings for specific equipment installations, including, but not limited to the following:
      a. Generators and automatic transfer switches
      b. Pad mounted and/or dry type transformers
      c. Switchboards and panelboards
      d. Equipment connections
      e. Control panels
      f. Circuit and motor disconnects
      g. Feeder conduits
   3. Wiring Diagrams: Provide wiring diagrams indicating field installed electrical power and control wiring and cabling layouts, overcurrent protective devices, equipment, and equipment connections.

I. EQUIPMENT HOUSEKEEPING PADS:
   1. Provide 4" concrete housekeeping pad for all floor mounted equipment including, but not limited to: switchboards, floor mounted distribution panelboards, floor mounted branch panelboards, and floor mounted dry type transformers.

J. TEMPORARY FACILITIES:
   1. Light, Heat, Power, Etc. - Responsibility for providing temporary electricity, heat and other facilities shall be as identified in Division 01.
   2. If AC power systems or their backup systems serving telecommunications, computer equipment, or their associated HVAC equipment and controls are taken out of service, for any reason, the Contractor shall be responsible for providing temporary systems during the period when the AC power systems or their backup systems are out of service. The Contractor shall be responsible for providing temporary power to all loads being interrupted.

K. WARRANTIES:
   1. Compile and assemble the warranties specified in Division 26, into a separated set of vinyl covered, three ring binders, tabulated and indexed for easy reference.

2. Provide complete warranty information for each item. Information to include product or equipment description, date of beginning of warranty or bond; duration of warranty or bond; and names, addresses, and telephone numbers and procedures for filing a claim and obtaining warranty services.

26 05 00 Low Voltage Electrical Power Conductors and Cables

A. SUMMARY:
   1. This section includes wires, cables, and connectors for power, lighting, signal, control, and related systems rated 600 volts and less.

B. WIRES AND CABLES:
   1. Provide solid conductors for power, control, and lighting circuits 12 AWG and smaller. Provide stranded conductors for 10 AWG and larger. Provide copper for all wires and cables.
   2. Metal Clad Cable - Type MC: Sizes 14 AWG through 10 AWG, copper conductors with 600 volt thermoplastic insulation rated 90 degrees C, aluminum interlocked metal type covering.
   3. Conductors feeding exterior walkway and parking lot lights shall be stranded.

C. CONNECTORS:
   1. Provide UL-type factory-fabricated, solderless metal connectors of sizes, ampacity ratings, materials, types and classes for applications and for services indicated. Use connectors with temperatures equal to or greater than those of the wires upon which used.
   2. Provide long barrel 2-hole compression lugs for all power feeder, neutral, and grounding connections. (Including phase, neutral and
   3. Provide connectors that are designed to accept stranded conductors where stranded conductors are used.

D. PRODUCTS
   1. Wire and Cable:
      a. American Insulated Wire
      b. Anaconda-Ericsson Inc; Wire and Cable Div.
      c. Belden Div; Cooper Industries.
      d. Brand-Rex Div; Pyle National Co.
      e. General Cable Corporation.
      f. Hitemp Wires, Inc.
      g. Phelps Dodge Cable and Wire Co.
      h. Rome Cable Corp.
      i. Southwire Company
      j. Okonite
      k. Superior Essex:
   2. Connectors:
      a. O-Z/Gedney Co.
      b. AMP, Inc.
      c. Burndy Corporation.
      d. Ideal Industries, Inc.
      e. 3M Company
      f. Thomas and Betts Corp.

E. INSTALLATION OF WIRES AND CABLES:
   1. General: Install electrical cables, wires and connectors in compliance with applicable requirements of NEC, NEMA, UL, and NECA's "Standard of Installation", and in accordance with recognized industry practices.
   2. Use pulling means including, fish tape, cable, rope and basket weave wire/cable grips which will not damage cables or raceway. Do not use rope hitches for pulling attachment to wire or cable.
   3. Keep conductor splices to minimum. Splice only in accessible junction boxes. No splices are allowed in feeder, control or fire alarm wiring. Connect unspliced wire to numbered terminal strips at each end.
   4. Install splices and taps which possess equivalent or better mechanical strength and insulation ratings than conductors being spliced.

5. Support cables above accessible ceilings, do not rest on ceiling tiles. Use spring clips and hanger rods independent from the ceiling suspension system to support cables from structure.
6. Provide adequate length of conductors within electrical enclosures and train the conductors to terminal points with no excess. Bundle multiple conductors, with conductors larger than No. 10 AWG cabled to individual circuits. Make terminations so there is no bare conductor at the terminal.
7. Use solderless pressure connectors with insulating covers for copper wire splices and taps, 8 AWG and larger. For 10 AWG and smaller, use insulated screw on type spring wire connectors with plastic caps, push on type are not acceptable.
8. Use copper compression connectors for copper wire splices and taps, 1/0 AWG and larger. Tape uninsulated conductors and connectors with electrical tape to 150 percent of the insulation value of the conductor.
9. Make splices, taps and terminations to carry full ampacity of conductors without perceptible temperature rise.
10. Install exposed cable, parallel and perpendicular to surfaces, or exposed structural members, and follow surface contours, where possible.
11. Make all ground, neutral and line connections to receptacle and wiring device terminals as recommended by manufacturer. Provide ground jumper from outlet box to individual ground terminal of devices.
12. Branch circuits whose length from panel to first outlet exceeds 100 feet for 120 volt circuits or 175 feet for 277 volt circuit shall be #10 or larger, as required to comply with the National Electrical Code.
13. Where conduit is not required for low voltage cable, plenum rated cable shall be provided in ceiling, floor or other air plenum spaces.
14. Tap-cons shall be.25” or larger; no plastic anchors shall be allowed.

26 05 10 Grounding and Bonding for Electrical Systems

A. SUMMARY:
   1. This Section includes solid grounding of electrical systems and equipment. It includes basic requirements for grounding for protection of life, equipment, circuits, and systems. Grounding requirements specified in this Section may be supplemented in other sections of these Specifications.

B. GROUNDING AND BONDING PRODUCTS:
   1. Of types indicated and of sizes and ratings to comply with NEC. Where types, sizes, ratings, and quantities indicated are in excess of NEC requirements, the more stringent requirements and the greater size, rating, and quantity indications govern.
   2. Conductor Materials: Copper.

C. WIRE AND CABLE CONDUCTORS:
   1. Conform to NEC, except as otherwise indicated, for conductor properties, including stranding.
      a. Equipment Grounding Conductor: Green insulated.
      b. Grounding Electrode Conductor: Stranded cable.

D. MISCELLANEOUS CONDUCTORS:
   1. Ground Bus: Bare annealed copper bars of rectangular cross section.
   2. Braided Bonding Jumpers: Copper tape, braided No. 30 gage bare copper wire, terminated with copper ferrules.
   3. Bonding Strap Conductor/Connectors: Soft copper, 0.05 inch thick and 2 inches wide, except as indicated.

E. CONNECTOR PRODUCTS:
   1. Pressure Connectors: High-conductivity-plated units.
   2. Bolted Clamps: Heavy-duty units listed for the application.
   3. Exothermic Welded Connections: Provided in kit form and selected for the specific types, sizes, and combinations of conductors and other items to be connected.

F. GROUNDING ELECTRODES:
   1. Ground Rods: Copper-clad steel with high-strength steel core and electrolytic-grade copper outer sheath, molten welded to core.
      a. Size: 3/4 inch by 10 feet.

G. MANUFACTURERS:
   1. Subject to compliance with requirements, provide products by the following:
      a. Burndy Corporation
      b. Cadweld Div.; Erico Products Inc.
      c. Ideal Industries
      d. OZ Gedney Div.
      e. Thermoweld
      f. Thomas and Betts Corp.
      g. A.B. Chance Co.

H. APPLICATION:
   1. Install separate insulated equipment grounding conductors with circuit conductors for the following in addition to those locations where required by Code:
      a. Feeders and branch circuits: Provide individual grounding and neutral conductors for each isolated ground receptacle. When individual or groups of isolated ground receptacles are on dedicated circuits, individual ground and neutral conductors for each circuit is acceptable.
      b. Nonmetallic Raceways: Install an insulated equipment ground conductor in nonmetallic raceways unless they are designated for telephone or data cables.

c. Air Duct Equipment Circuits: Install an insulated equipment grounding conductor to duct-mounted electrical devices operating at 120-V and above including air cleaners and heaters. Bond the conductor to each such unit.
d. Water Heater, Heat Tracing, and Anti-Frost Heater Circuits: Install separate insulated equipment ground conductor to each electric water heater, heat tracing, and surface anti-frost heating cable. Bond this conductor to heater units, piping, and connected equipment and components.
e. Underground Conductors: Bare, tinned, stranded copper
f. Signal and Communications: For telephone, alarm, and communication systems, provide a #6 AWG minimum green insulated copper conductor in raceway from the grounding electrode system to each terminal cabinet or central equipment location.

2. Electrical Room Ground Bus: Size, location, and arrangement as indicated. Space 1 inch from wall and support from wall 6 inches above finished floor, except as otherwise indicated.
3. Ground Rods: Locate a minimum of two-rod lengths from each other and at least the same distance from any other grounding electrode. Interconnect ground rods with bare conductors buried at least 24 inches below grade. Connect bare-cable ground conductors to ground rods by means of exothermic welds except as otherwise indicated. Make these connections without damaging the copper coating or exposing the steel. Drive rods until tops are 6 inches below finished floor or final grade except as otherwise indicated.
4. Metallic Water Service Pipe: Connect ground conductors to the main metallic water service pipes by means of ground clamps. Where a dielectric main water fitting is installed, connect the ground conductor to the street side of the fitting. Do not install a grounding jumper around dielectric fittings. Provide grounded bushing at conduit ends and bond the ground conductor conduit to the ground bars at each end.
5. UFER Ground: Fabricate grounding electrode conductor by installing lengthwise in form for foundation or footings. Install so conductor is within 2 inches of the bottom of the concrete. Extend conductor below grade and connect to building grounding grid, grounding electrode, or ground bar.
6. Underground distribution System Grounding: For Pad Mounted Gear, install a ¾ inch by 10 feet. driven ground rod inside the cable access blockout of the pad and set the rod depth such that 4 inches will extend above the finished pad. Where necessary, install ground rod before the equipment is placed. Protect ground rods passing through concrete with a double wrapping of pressure-sensitive tape or heat-shrunk insulating sleeve from 2 inches above to 6 inches below the concrete.

26 05 20 Hangars and Supports for Electrical Systems

A. SUMMARY:
   1. This Section includes secure support from the building structure for electrical items by means of hangers, supports, anchors, sleeves, inserts, seals, and associated fastenings.

B. MANUFACTURERS:
   1. Subject to compliance with requirements, provide products by the following:
      1. Slotted Metal Angle and U-Channel Systems:
         a. Allied Tube & Conduit
         b. B-Line Systems, Inc.
         c. GS Metals Corp.
         d. Unistrut Diversified Products
      2. Conduit Sealing Bushings:
         a. O-Z/Gedney
         b. Cooper Industries, Inc.
         c. GS Metals Corp.
         d. Killark Electric Mfg. Co.
         e. Madison Equipment Co.
         f. Raco, Inc.
         g. Spring City Electrical Mgf. Co.
         h. Thomas & Betts Corp.

C. COATINGS:
   1. Supports, support hardware, and fasteners shall be protected with zinc coating or with treatment of equivalent corrosion resistance using approved alternative treatment, finish, or inherent material characteristic. Products for use outdoors shall be hot-dip galvanized.

D. FIRE SEALS:
   1. Firestopping material shall be asbestos free, 100 percent intumescent, have code approval under BOCA, ICBO, SSBC, NFPA 101, NFPA 70, and be capable of maintaining an effective barrier against flame and gases in compliance with the following requirements. Flame Spread: 25 or less, ASTM E84. Subject to compliance with requirements, provide fire seals of the following:
      a. 3M Company
      b. Tremco
   2. Fire Resistance and Hose Stream Tests: Firestopping materials shall be rated "F" and "T" in accordance with ASTM E 814 or UL 1479. Rating periods shall conform to the following:
      a. (F) 3 (T) 3 Time-rated floor or wall assemblies.
      b. (F) 3 (T) 3 Openings between floor slabs and curtain wall.

26 05 30 Raceways and Boxes for Electrical Systems

A. SUMMARY:
   1. Extent of raceway work is indicated by drawings and schedules. Provide complete conduit systems for all conductors unless otherwise specified. Copper wire shall be with conduit only, no UF or Direct-burial ALV allowed.
   2. Types of raceways specified in this section include the following:
      a. Electrical metallic tubing (EMT).
      b. Flexible metal conduit.
      c. Intermediate metal conduit (IMC).
      d. Liquid-tight flexible metal conduit.
      e. Non-metallic Conduit and Ducts.
      f. Rigid metal conduit (RGC).
      g. Wireways.

B. METAL CONDUIT AND TUBING:
   1. Rigid Galvanized Steel Conduit (RGC):
      a. Conduit: Rigid steel, zinc-coated inside and outside, threaded ends.
      b. Fittings: Threaded galvanized steel, bushings shall have nylon insulated throat.
      c. All underground conduit will be taped or PVC-coated to prevent corrosion.
      d. Manufacturers:
         1. Allied
         2. Wheatland
         3. Triangle
         4. Western Tube & Conduit
   2. Intermediate Metal Conduit (IMC):
      a. Conduit: Rigid intermediate grade galvanized inside and outside, threaded ends.
      b. Fittings: Threaded galvanized steel, bushings shall have nylon insulated throat.
      c. Manufacturers:
         1. Allied
         2. Triangle
         3. Western Tube & Conduit
   3. PVC Externally Coated Rigid Steel Conduit:
      a. Conduit: Rigid steel zinc-coated with external coating of PVC.
      b. Fittings: Threaded galvanized steel with external PVC coating, bushings shall have nylon insulated throat.
      c. Manufacturers:
         1. Rob Roy
         2. Perma Cote
         3. OCAL
   4. Electrical Metallic Tubing (EMT):
      a. Conduit: Galvanized steel tubing.
      b. Fittings: Steel compression fittings for rain-tight and concrete-tight applications. Steel set-screw for all other connections. Set-screw quick fit type for 2-1/2 inches and larger may be used. Bushings shall be threaded and have nylon insulated throat or nylon bushing.
      c. Manufacturers:
         1. Allied
         2. Republic
         3. Triangle
         4. LTV
         5. Western Tube & Conduit
   5. Rigid Aluminum Conduit:
      a. Not allowed unless otherwise noted.
   6. Flexible Metal Conduit:

a. Continuous spiral wound, interlocked, zinc-coated steel, approved for grounding.
b. Use of flexible metal conduit shall be held to a maximum of 6'-0" lengths. Other uses shall be specifically approved by the Owner.
c. Fittings: Cadmium plated, malleable iron. Straight connector shall be one-piece body, female end with clamp and deep slotted machine screw for securing conduit, and threaded male end provided with a locknut. Angle connectors shall be two piece body with removable upper section, female end with clamp and deep slotted machine screw for securing conduit, and threaded male end provided with a locknut. All fittings shall be terminated with threaded bushings having nylon insulated throats.
d. Manufacturers:
    1. Carlon
    2. MPF
    3. Can-Tex
    4. PW

7. Liquid-Tight Flexible Metal Conduit:
    a. Conduit: Continuous spiral wound, interlocked zinc-coated steel with polyvinyl chloride (PVC) jacket, approved for grounding.
    b. Fittings: Die-cast with nylon insulated throat or nylon bushings.
    c. Manufacturers:
        1. Carlon
        2. MPF
        3. Can-Tex
        4. PW

C. NON-METALLIC CONDUIT AND DUCTS:
1. Rigid Non-Metallic Conduit (RNC):
    a. Conduit: Schedule 40 polyvinyl chloride (PVC), 90°C for direct burial or concrete encasement.
    b. Fittings: Mate and match conduit type and material. Cement as recommended by manufacturer.

D. CONDUIT BODIES:
1. Types, shapes and sizes, as required to suit individual applications and NEC requirements. Provide matching gasketed covers secured with corrosion-resistant screws.
2. Metallic Conduit and Tubing: Use malleable iron conduit bodies. Use bodies with threaded hubs for threaded raceways and in hazardous locations.
3. Nonmetallic Conduit: Use nonmetallic conduit bodies.
4. Manufacturers:
    a. O/Z Gedney
    b. Killark
    c. Regal
    d. Appleton
    e. Crouse Hinds

E. WIREWAYS:
1. Provide electrical wireways of types, grades, sizes, and number of channels for each type of service as indicated. Provide complete assembly of raceway including, but not limited to, couplings, offsets, elbows, expansion joints, adapters, hold-down straps, end caps,
2. and other components and accessories as required for complete system.
3. Lay-In Wireways: Construct lay-in wireways with hinged covers in accordance with UL 870 with components UL listed. Construct units to be capable of sealing cover in closed position with sealing wire. Provide wireways with knockouts.
4. Manufacturers:
    a. Square D. Co.
    b. Circle AW Products

c. Erickson Electric Equipment Co.
d. Hoffman Engineering Co.
e. Wadsworth Electric Mfg. Co., Inc.

F. CONDUIT SIZES:
1. Conduit sizes shall be as shown on the drawings. If the conduit size is not given on the drawings, the conduit shall be sized in accordance with NEC based on the number of conductors enclosed plus a parity sized equipment ground conductor and be subject to the following minimum sizes:
   a. Rigid, Intermediate, and EMT Conduit: 3/4 inch for all runs except lighting switch legs, 277 volt lighting branch circuits, temperature control and fire alarm which may be 1/2inch.
   b. Flexible and Liquid-Tight Flexible Conduit: 1/2inch for all runs.
   c. MC Cable: 3/8 inch to undercounter luminaires, 1/2inch for all other runs.
   d. Underground or Concrete Encased Nonmetallic Conduit: 3/4inch for all runs.
2. Conduits used for home runs shall contain only the conductors for the circuits indicated on the drawings. Combining multiple home runs into a single conduit will not be permitted.

26 05 40 Cabinets, boxes, and Fittings for Electrical Systems

A. SUMMARY:
   1. This section includes cabinets, boxes, and fittings for electrical installations and certain types of electrical fittings not covered in other sections. Types of products specified in this Section include:
      a. Outlet and device boxes.
      b. Pull and junction boxes.
      c. Floor boxes and service fittings.
      d. Cabinets.
      e. Hinged door enclosures.
      f. Power poles.

B. CABINETS, BOXES, AND FITTINGS:
   1. Provide gaskets for units in damp or wet locations.
   2. Sheet Steel: Flat-rolled, code-gage, galvanized steel.
   3. Fasteners for General Use: Corrosion resistant screws and hardware including cadmium and zinc plated items.
   4. Fasteners for Damp or Wet Locations: Stainless steel screws and hardware.
   5. Cast Metal for Boxes, Enclosures, and Covers; Copper-free aluminum except as otherwise specified.
   6. Painted Interior Finish: Where indicated, white baked enamel. Emergency system cabinets and boxes shall be red.
   7. Fittings for Boxes, Cabinets, and Enclosures: Conform to UL 514B. Malleable iron or zinc plated steel for conduit hubs, bushings and box connecters.
   8. No plastic underground rated electrical J-boxes, provide concrete only.

C. METAL OUTLET, DEVICE, AND SMALL WIRING BOXES:
   1. Malleable or Cast-Iron Boxes: Iron alloy, waterproof, with threaded raceway entries and features and accessories suitable for each location, including mounting ears, threaded screw holes for devices and closure plugs.
   2. Steel Floor Boxes: Sheet steel, concrete tight, fully adjustable, with stamped knockouts, adjusting rings, and brass floor plates. Where indicated, provide multi-section boxes with concealed individual section covers under a common flush floor plate. Provide for power, data, and communication outlets as indicated on the drawings.
   3. Service Fittings for Floor Outlet Boxes: Surface mounted horizontal, cast aluminum type 3-inches high, suitable for finished spaces and finished in satin aluminum, except as otherwise indicated. Provide duplex receptacle or 1-inch bushed opening for telephone or other communications service as indicated. Equip fitting for attaching flat to floor box cover.
   4. Manufacturers - Floor Boxes:
      a. American Electric.
      b. Butler Mfg. Co.
      c. Cooper Industries, Inc.
      d. Raco, Inc.
      e. Thomas & Betts Corp.

D. PULL AND JUNCTION BOXES:
   1. Steel Boxes: Sheet steel with welded seams. Where necessary to provide a rigid assembly, construct with internal structural steel racing.

E. CABINETS:
   1. Comply with UL 50, "Electrical Cabinets and Boxes."
   2. Construction: Sheet steel, NEMA 1 class except as otherwise indicated. Cabinet shall consist of a box and a front consisting of a one piece frame and a hinged door. Arrange door to close against a rabbet placed all around the inside edge of the frame, with a uniformly close fit between door and frame. Provide concealed fasteners, not over 24 inches apart, to hold fronts to cabinet boxes and provide for adjustment. Provide flush or concealed door hinges not over 24 inches apart and not over 6 inches from top and bottom of door. For flush cabinets, make the

front approximately 3/4 inch larger than the box all around. For surface mounted cabinets make front same height and width as box.

3. Double doors for cabinets wider than 24 inches. Telephone cabinets wider than 48 inches may have sliding or removable doors.
4. Locks: Combination spring catch and key lock, with all locks for cabinets of the same system keyed alike. Locks may be omitted on signal, power, and lighting cabinets located within wire closets and mechanical-electrical rooms. Locks shall be of a type to permit doors to latch closed without locking.
5. Manufacturers - Cabinets:
    a. Hoffman Engineering Co.
    b. Erickson Electrical Equipment Co.
    c. Electric Panelboard, Inc.
    d. Parker Electrical Mfg. Co.
    e. Spring City Electrical Mfg. Co.
    f. Square D Co.
    g. Circle AW

F. POWER POLES
    1. Manufacturer: Wiremold Steel Tele-Power Poles 30TP-4V

26 05 50 Electrical Connections for Equipment

A. GENERAL:
   1. Overcurrent Protective Devices (OCPDs): Provide type, rating, adapted to equipment connection installation. Tandem circuit breakers shall not be used. Multiple breakers shall have common trip.
   2. Contacts shall open each ungrounded connection to the motor. Contacts shall be NEMA rated, 75oC.
   3. Overload relays shall be ambient-compensated type with inverse-time-current characteristic. Provide with heaters or sensors in each phase matched to nameplate full load current of the specific motor to which connected with appropriate adjustment for duty cycle and power factor correction supplied with the motor.

B. MANUAL MOTOR STARTERS:
   1. Manual starters shall be flush-mounting type except where conduits are run exposed or as otherwise noted. Manual starters shall be complete with properly sized overload protection and neon pilot light. Manual starters shall be Square D Class 2510 or Allen-Bradley Bulletin 600 with stainless steel plates.
   2. Motor starter shall have a spare set of auxiliary N.O. and N.C. contacts supplied with starter.
   3. Heater units in all manual motor starters shall be sized for approximately 115 percent of full load motor current. Check and coordinate all thermal protective devices with the equipment they protect.

C. CIRCUIT AND MOTOR DISCONNECT SWITCHES:
   1. Provide circuit and motor disconnect switches in types, sizes, duties, features, ratings, and enclosures as indicated. All equipment with maximum fuse size listed in nameplate shall have fusible disconnect switch provided. Provide NEMA 1 enclosure. For outdoor switches and switches indicated as weatherproof, provide NEMA 3R enclosures with raintight hubs. For motor and motor starter disconnects, provide units with horsepower ratings suitable to the loads.
   2. Fusible Switches: Heavy duty switches, with fuses of classes and current ratings indicated. Where current limiting fuses are indicated, provide switches with non-interchangeable feature suitable only for current limiting type fuses. All disconnect switches shall be fusible unless otherwise noted. Provide UL type "HD" 100 percent duty rated switches.
   3. Provide UL type "HD" 100 percent duty rated switches.
   4. Manufacturer: Subject to compliance with requirements, provide circuit and motor disconnects by one of the following:
      a. Square D Company
      b. Cutler-Hammer Inc.
      c. General Electric Co.
      d. Siemens Energy & Automation, Inc.
      e. ITE

D. AUXILIARY CONTROL DEVICES:
   1. Pushbutton Stations, Pilot Lights, and Selector Switches: Heavy-duty type. HANDOFF- AUTOMATIC (H-O-A) switches shall be provided on all magnetic starters. Provide magnetic two-speed starters with a HIGH-LOW selector with H-O-A switch wired to allow manual speed selection in HAND position and remote speed selection in AUTO.
      a. Stop and Lockout Pushbutton Station: Momentary-break pushbutton station with a factory-applied hasp arranged so a padlock can be used to lock the pushbutton in the depressed position with the control circuit open.
      b. Control Relays: Auxiliary and adjustable time-delay relays as required to coordinate with Division 23 Controls Section.
      c. Elapsed Time Meters: Heavy duty with digital readout in hours. Provide on all magnetic starters for motors.
      d. Ammeters, Voltmeters, and Frequency Meters: Panel type, 2-1/2 inch minimum size with 90 degree or 120 degree scale and plus or minus 2 percent accuracy. Where indicated, provide transfer device with an off position.
      e. Current Sensors: Rated to suit application.

f. Phase-Failure Relay: Provide relay for each motor 2 HP and larger. Relay shall have solid-state sensing circuit with isolated contacts for hard-wired connection to socket. Provide matching pin type relay and hold down clamps. Relay shall be arranged to operate on phase failure, phase reversal, current unbalance of from 30 to 40 percent, or loss of supply voltage. Provide adjustable response delay, and adjustable undervoltage setting.

E. EQUIPMENT CONNECTION SCHEDULES:

1. Mechanical Equipment: Unless otherwise indicated, it is suggested that all equipment motors and control shall be furnished, set in place, and wired in accordance with the schedule contained herein. The exact furnishing and installation of the equipment is left to the Contractors involved. Contractor should note that the intent of this schedule is to have the Contractor responsible for coordinating all wiring as outlined, whether or not specifically called for by the Division 23 or Division 26 drawings and specifications.

26 05 60 Identification for Electrical Systems

A. SUMMARY:
1. This Section includes identification of electrical materials, equipment, and installations. It includes requirements for electrical identification components including but not limited to the following:
    a. Buried electrical line warnings.
    b. Identification labeling for raceways, cables, and conductors.
    c. Operational instruction signs.
    d. Warning and caution signs.
    e. Equipment labels and signs.
    f. PPE requirements shall be labeled.
    g. Label panels etc. to keep clear 3' in front
    h. Label receptacles to panel and breaker location
    i. Label all remodeled circuits

A. Manufacturers: Subject to compliance with requirements, provide products by the following:
1. Ideal Industries, Inc.
2. LEM Products, Inc.
3. Markal Corp.
4. Panduit Corp.
5. W.H.Brady, Co.
6. 3M Company

B. ELECTRICAL IDENTIFICATION PRODUCTS:
1. Identify System Raceways with Painted Couplings: Provide painted couplings for all concealed raceways. Install painted couplings at all conduit connecting couplings including end couplings at stub outs. Apply the following colors:
    a. Normal Power: Black
    b. Life Safety Power: Yellow
    c. Standby System Raceways: OSHA Orange
    d. Fire Alarm: OSHA Red
    e. BMS/Temperature Control: Dark Blue
    f. Ground: Safety Green
    g. Data and Telephone System: Light Blue
    h. Security: White
    i. Paging and Sound: Grey
2. Adhesive Marking Labels for Exposed Raceway: Pre-printed, flexible, self-adhesive labels with legend indicating voltage and service (Emergency, Lighting, Power, Power d.c., HVAC, Communications, Control, Fire).
3. Label Size for Raceways and Busway: Kroy or Brother labels 1 inch high by 12 inches long (minimum) with 5/8 inch minimum height letters.
    a. 600 Volt and Below Normal: White letters on black background indicating source equipment designation, circuit number(s), and voltage.
    b. 600 Volt and Below Emergency: Indicate source equipment designation, circuit number(s), and voltage for each branch as follows.
        1. Life safety: Black letters on yellow
        2. Standby: Black letters on OSHA Orange
4. Fire Alarm: White letters on red background indicating "FIRE ALARM".
5. Temperature Control: White or black letters on blue background indicating "TEMP. CONTROL."
6. Ground: White or black letters on green background indicating "GROUND" and equipment and designation.
7. Building Alarms: Orange letters on white background indicating "BUILDING ALARMS."
8. Network Fiber: Black letters on white background indicating "NETWORK FIBER."
9. Where conduits enter or exit a panelboard, pull or junction box, switchboard, or other distribution equipment, conduit labels shall include circuit number in addition to feeder identification and voltage.

26 22 00 Low Voltage Transformers

A. SUMMARY:
   A. This section includes general purpose and specialty dry type transformers and voltage regulators with windings rated 600 V or less.

B. TRANSFORMERS:
   1. Factory assembled and tested, air cooled units of types specified, having characteristics and ratings as indicated. Units shall be designed for 60-Hz service.
   2. Cores: Grain oriented, nonaging silicon steel.
   3. Coils: Continuous windings without splices except for taps.
   4. Internal Coil Connections: Brazed or pressure type.

C. GENERAL PURPOSE, DRY-TYPE TRANSFORMERS:
   1. Comply with NEMA Standard ST 20 "Dry-Type Transformers for General Applications." Transformers shall have the following features and ratings:
      a. Insulation Class: 185 deg C class for transformers 15 kVA or smaller; 220 deg C class for transformers larger than 15 kVA.
      b. Insulation Temperature Rise: 115 deg C maximum rise above 40 deg C.
      c. Windings:
         1. 2-winding type, three phase transformers shall use one coil per phase in primary and secondary. Conductors shall be individually insulated, as small in size as possible, and transposed when necessary to minimize eddy current losses. The primary winding shall be of sufficient size to limit temperature rise to its rated value even with circulating 3rd harmonic current.
         2. Provide aluminum windings.
         3. Windings shall be delta-wye with 30o lagging phase shift to match ANSI standard, unless noted otherwise.
   2. Sound Level: Minimum of 3 dB less than NEMA ST 20 standard sound levels for transformer type and size indicated when factory tested in accordance with that standard.
   3. Taps: For transformers 3 kVA and larger, full capacity taps in high-voltage winding as follows:
      a. 3 kVA through and including 15 kVA: Two 5 percent taps below rated high voltage.
      b. Greater than 15 kVA through and including 500 kVA: Six 2-1/2 percent taps, 2 above and 4 below rated high-voltage.
   4. Core Flux Density: Maintained below saturation point to prevent core saturation caused by harmonics even with a 10% primary overvoltage.

D. CONTROL AND SIGNAL TRANSFORMERS:
   1. Ratings: As indicated and for continuous duty. Where ratings are not indicated, provide capacity in excess of load.
      a. Type: Self-cooled, two-winding dry type.
      b. Enclosure: Indoor, except as indicated.

E. MANUFACTURERS: Subject to compliance with requirements, provide products by the following:
   1. Square D Co.
   2. General Electric Co.
   3. Eaton/Cutler-Hammer
   4. Hevi-Duty Electric.
   5. ITE Siemens Energy & Automation, Inc.

26 24 00 Switchboards

A. SUMMARY:
   1. This Section includes low-voltage power service and distribution switchboards and associated auxiliary equipment rated 600 V or less.
   2. Allow at least (2) 200A, 3-phase spaces in switch gear to accommodate future portables classrooms. If distance is such that voltage drop is not a problem for 120/208v, provide (2) 200A, 3-phase spaces in the low voltage switch gear. The switch gear is to be engineered to accommodate as many portable classrooms as space will allow.

B. EXTRA MATERIALS:
   1. Spare Fuses: Verify with Owner, or furnish spares of each type and rating of fuse for fusible devices amounting to one set of 3 fuses for each 9 fuses installed but not less than 3 fuses of each type. Include spares for:
      a. Fuses for fused circuit breakers.
      b. Fuses for fusible switches.
      c. Fuses for fused power circuit devices.
      d. Control power fuses.
      e. Potential transformer fuses.
   2. Spare Indicating Lamps: Furnish 6 of each type and color installed.

C. SWITCHBOARDS, GENERAL:
   1. Front-connected, front-accessible, with fixed, individually mounted main device, panel-mounted branches, and sections rear aligned. Dead front, metal enclosed, self supporting and conforming to NEMA PB2.

D. MANUFACTURERS:
   1. Subject to compliance with requirements, provide products by the following:
      a. Square D Co.
      b. Siemens Energy & Automation, Inc.
      c. General Electric Co.
      d. ABB Power Distribution, Inc.
      e. Cutler-Hammer, Inc.
   2. Metering Equipment:
      a. Square D Co.
      b. Siemens Energy & Automation, Inc.
      c. General Electric
      d. ABB Power Distribution, Inc.
      e. Westinghouse
      f. Sangamo
      g. TESCO

E. FABRICATION AND FEATURES:
   1. Enclosure: Steel. NEMA 1.
   2. Utility Metering Compartment: Fabricated compartment and section meeting utility company requirements. Where separate vertical section is required for utility metering, match and align with basic switchboard.
      a. Provide integral meter base in accordance with utility standards.
   3. Bus Transition and Incoming Line Pull Sections: Where required, match and align with basic switchboard. Line terminations shall be accessible from the front of the switchboard.
   4. Hinged Front Panels: Provide to allow access to breaker, metering, accessory, and blank compartments.
   5. Pull Box on Top of Switchboard: Provide where indicated or where required by installation conditions, and include the following features:
      a. Adequate ventilation to maintain air temperature in pull box within same limits as switchboard.
   6. Buses and Connections: Three-phase, four-wire except as otherwise indicated. Features as follows:

a. Phase and Neutral Bus Material: Hard-drawn copper, 98 percent conductivity with feeder circuit-breaker line connections. Horizontal cross busses throughout shall be non-tapered – 100 percent rated. Size bus in accordance with NEMA PB2.
b. Contact Surfaces of Buses: Silver plated copper.
c. Main Phase Buses, Neutral Bus, and Equipment Ground Bus: Uniform capacity the entire length of the switchboard main and distribution sections. Provide for future extensions from either end by means of bolt holes or other approved method and connecting links. Provide for any outgoing or incoming bus or cabling as required for each breaker space (i.e. all spares, spaces, and utilized).
d. Neutral Buses: 100 percent of the ampacity of the phase buses except as indicated and equipped with approved pressure connector terminations for outgoing circuit neutral cables.
e. Provide long barrel 2-bolt copper compression lugs for all incoming and outgoing
f. feeders including neutral and ground connections.

7. Ground Bus: 1/4 inch by 2 inch minimum size, hard-drawn copper of 98 percent conductivity, and equipped with pressure connector terminations for feeder- and branch-circuit ground conductors.
8. Supports and Bracing for Buses: Adequate strength for indicated short-circuit currents. Busses shall be bolted with access for future torque maintenance.
9. Provide four (4) spare two-hole lugs for #2 AWG through #4/0 AWG conductors on load side of main circuit breakers.
10. Future Devices: Where provision for future overcurrent protective devices or space is indicated, equip compartments with mounting brackets, supports, bus connections, and necessary appurtenances, designed for the OCPD types and ampere ratings indicated for future installation of devices.

F. OTHER CIRCUIT CONTROL AND PROTECTIVE DEVICES:
1. Control Power: Where electrically operated/ shunt tripped circuit breakers/or other control power functions are required, provide 120 volt control circuits supplied through secondary disconnect devices from a control power transformer. Include the following features:
    a. Control Power Transformers: Dry type. Separate compartments for units larger than 3 KVA and their fuses.
    b. Provide two control power transformers in separate compartments with necessary interlocking relays. Connect the primary of each control power transformer at the line side of the associated main circuit breaker. Connect the 120 volt secondaries through a relay or relays as a control bus.
    c. Control Power Fuses: Include primary and secondary fuses for current-limiting and overload protection.
    d. Provide control power disconnecting means on line side of transformer.
    e. Provide control fuse status indication on front and contacts for remote alarm.

G. IDENTIFICATION:
1. General: Refer to Division 26 Section on Electrical Identification. Identify units, devices, controls, and wiring with factory-applied labels and signs.
2. Compartment Nameplates: Engraved laminated plastic or metal nameplate for each compartment, mounted with corrosion-resistant screws. UL nameplates shall be provided for all switchboards. Information shall include, but not be limited to, manufacturer, model number, serial number, plant or manufacturing location, ampere rating, voltage rating, wire and phase identification and bus short circuit bracing rating.

26 24 15 Panelboards

A. SUMMARY:
   1. Section includes lighting and power panelboards and associated auxiliary equipment rated 600 V or less.

B. PANELBOARDS, GENERAL REQUIREMENTS:
   1. Overcurrent Protective Devices (OCPDs): Tandem circuit breakers shall not be used. Multiple breakers shall have common trip.
   2. Directory Frame: Metal, mounted inside each panel door with card and clear plastic cover. Directory shall match panelboard configuration, i.e. top to bottom, left to right. Provide permanent panelboard labels for each circuit number.
   3. Bus Material: Provide tin plated hard-drawn copper of 98 percent conductivity.
   4. Equipment Ground Bus: Adequate for feeder and branch-circuit equipment ground conductors. Bonded to box.
   5. Provide copper lugs for incoming feeders and grounds.
   6. Provision for Future Devices: Equip with mounting brackets, bus connections, and necessary appurtenances, for the OCPD ampere ratings indicated for future installation of devices.
   7. Special Features: Provide the following features for panelboards as indicated.
      a. Provide 150 percent sized neutral bus and ground bus and double termination lugs for all 120/208 volt panelboards.
      b. Provide two bolt compression lugs for incoming feeders on main lug only (MLO)panelboards, and ground connections.
      c. Subfeed: OCPD or lug provision as indicated.
      d. Feed-Through Lugs: Sized to accommodate feeders indicated.

C. LIGHTING AND APPLIANCE BRANCH CIRCUIT PANELBOARDS:
   1. Branch OCPDs: Bolt-on circuit breakers, replaceable without disturbing adjacent units.
   2. Double-Width Panels: Where more than 42 poles are indicated, provide two panelboards of equal dimensions and with individual fronts on each panelboard.
   3. Panelboards shall be door-in-door trim type and have doors flush with trim, have copper buses.

D. DISTRIBUTION PANELBOARDS:
   1. Branch-Circuit Breakers: Where OCPDs are indicated to be circuit breakers, use bolt-on breakers except circuit breakers 225-ampere frame size and greater may be plug-in type where individual positive locking device requires mechanical release for removal.

E. MANUFACTURERS:
   1. Subject to compliance with requirements, provide products by the following:
      a. Square D Co.
      b. General Electric Co.
      c. Siemens Energy & Automation, Inc. / ITE
      d. Cutler-Hammer, Inc.

26 27 20 Wiring Devices

A. SUMMARY:
1. The extent of wiring device work is indicated by drawings and schedules. Wiring devices are defined as single discrete units of electrical distribution systems which are intended to carry but not utilize electric energy. Types of electrical wiring devices in this section include the following:
    a. Receptacles.
    b. Ground-fault circuit interrupters.
    c. Switches.
    d. Wallplates.
    e. Dimmers.
    f. Plugs and connectors.
    g. Concrete floor boxes
    h. Poke-through assemblies.

B. WIRING DEVICES:
1. Receptacles:
    A. All duplex, single, and special receptacles shall be extra heavy duty type with nylon fronts and backs conforming to Federal Spec #WS-B96K, listed by Underwriter's Laboratories, and have a metal mounting strap with self-grounding and have a hexhead green grounding screw and be side and back wired. Each device shall bear the UL/FS Label. All outlets in public spaces in elementary schools (pre-school included) shall be tamper resistant type.
    B. Convenience Receptacle Configuration: NEMA WD 1; Specification grade type 5-20R, with nylon face. All receptacles connected to emergency circuits shall have a red face. Color selection shall be verified with Architect/Engineer prior to ordering.
    C. Specific-use Receptacle Configuration: NEMA WD 1 straight blade or WD 5 twist-lock; as indicated on drawings, black face.
    D. Isolated Ground Receptacles: Duplex receptacle with metal strap for self-grounding and green hex-head source with dedicated isolated ground and neutral, Hubbel prefix IG. Receptacle shall have an orange triangle located on the face.
    E. Ground-Fault Interrupter Receptacles: Provide "feed-thru" type ground-fault circuit interrupters, with integral heavy-duty NEMA 5-20R duplex receptacles, capable of protecting connected downstream receptacles on same circuit. Provide unit capable of being installed in a 2-3/4" deep outlet box without adapter, grounding type, Class A, Group 1 per UL Standard 943.
    F. Clock with metal hook which permits clock to be flush mounted on wall.
    G. Pendant Cord/Connector Devices: Matching, locking type, plug and plug receptacle body connector, NEMA l5-20P and L5-20R, heavy-duty grade.
    H. Controlled Receptacles
        a. Typical SVVSD preference is for a duplex receptacle where the top outlet is constant power and the lower outlet is controlled. Each project should re-evaluate this approach.

2. Switches:
    a. Wall Switches for Lighting Circuits: extra heavy duty type with nylon fronts and backs conforming to Federal Spec #WS-B96K, listed by Underwriter's Laboratories with toggle handle, rated 20 amperes at 120-277 volts AC, unless noted otherwise. Mounting straps shall be metal and be equipped with a green hex-head ground screw. Each switch shall bear the UL/FS Label.
    b. Handle: Color shall be verified with Architect/Engineer for normal power circuits, red for emergency power circuits.
    c. Pilot Light Type: Lighted handle lit when switch is "on."
    d. Locator Type: Continuously lighted handle.
    e. Manufacturer: Corbin, others acceptable based upon district prior approval.
3. Fluorescent Lamp Dimmers: Provide single-pole, full-wave semi-conductor modular type AC dimmers for fluorescent lamps; wattage and voltage as indicated, and with electromagnetic filters to reduce noise, RF and TV interference to

minimum. Construct with continuously adjustable trim potentiometer for adjustment of low dimming; and with anodized heat sinks. Provide 5-inch wire connecting leads. Dimmers shall be Lutron Nova "T" Star.

C. WIRING DEVICE ACCESSORIES:
   1. Wallplates: Provide wallplates for single and combination wiring devices, of types, sizes, and with ganging and cutouts as indicated. Select plates which mate and match wiring devices to which attached. Construct with metal screws for securing plates to devices; screw heads colored to match finish of plates. Identify all wall plates used for receptacles with branch circuit number per requirements of section on Electrical Identification. Provide blank wall plates for all cable, data, telephone and junction and outlet boxes. Where cables are routed through the wallplate, provide grommets in wallplate openings to protect cables. Provide plates possessing the following additional construction features:
      a. Material and Finish: Steel plate, galvanized for use in unfinished areas, mechanical, and electrical rooms.
      b. Material and Finish: Painted metal plates, in all other areas. Color to be verified with Architect/Engineer/Owner.
   2. Concrete Floor Box: Provide modular floor service outlets and fittings of types and ratings indicated. Construct of die cast aluminum, satin finish. Use design compatible with floor type and outlet wiring methods indicated. Provide 20-amperes, 125-volts, back-to-back duplex receptacles, NEMA configuration 5-20R. Provide with 1" NPT, 1" long, locking nipple for installation.
   3. Poke-Through Assembly Devices: Factory-fabricated poke-through assembly devices with modular service outlets, multi-channeled thru-floor raceway/fire stop assembly and below-floor junction box assembly. Construct service fitting of die cast, satin finished aluminum with 20- ampere 120-volts, gray duplex NEMA 5-20R receptacle and modular communication/ data service outlet with separation barrier between power and low-tension section. Provide integral assembly UL listed as a total unit, with fire rating consistent with that of floor penetrated.
         a. Limit use and location. Coordinate with Owner.

D. ACCEPTABLE MANUFACTURERS:
   1. Manufacturers: Subject to compliance with requirements, provide wiring devices of one of the following:
      a. Devices:
         1. Hubbell Inc.
         2. Leviton Mfg Co.
         3. Pass and Seymour Inc.
         4. Cooper Crouse-Hinds Co.
      b. Concrete Floor Boxes:
         1. Wiremold
         2. Hubbell
         3. Steel City
      c. Poke-Through and Floor Service Outlets:
         1. Wiremold
         2. Hubbell, Inc.

26 28 10 Enclosed Switches and Circuit Breakers

A. SUMMARY:
   1. This Section includes overcurrent protective devices (OCPD's) rated 600 V and below and switching devices commonly used with them.

B. OVERCURRENT PROTECTIVE DEVICES (OCPDS), GENERAL:
   1. Where OCPD's are to be installed in existing panelboards, switchboards, and motor control centers, they shall be of the same manufacture and type as those existing in the equipment. If this is not possible, provide devices which are compatible with the existing equipment and when installed will not void the U.L. label or reduce the short circuit rating of the equipment.
   2. Provide 100 percent rated equipment and feeder breakers unless otherwise noted.
   3. Provide standard rated branch circuit breakers unless otherwise noted.
   4. All overcurrent devices shall be individually rated for the available fault current unless otherwise noted. Series ratings of equipment will only be allowed where specifically called out.

C. CARTRIDGE FUSES:
   1. All fuses used for main, feeder, or branch-circuit protection shall be Underwriters Laboratories listed, current-limiting fuses with 200,000 ampere interrupting rating and shall be so labeled. Fuses used for supplementary protection (other than branch circuit protection) shall be as specified above or shall be U.L. approved or component recognized for such purposes. All fuses provided shall be furnished by the same manufacturer. Should equipment provided require a different U.L. Class or size of fuse, the engineer shall be furnished sufficient data to ascertain that system function will not be adversely affected.
   2. In order to simplify fuse replacement, reduce spare fuse inventory and insure adequate thermal protection, all fuses 600 amperes and below shall be true dual-element time-delay fuses with separate spring-loaded thermal overload elements in all ampere ratings. All ampere ratings shall be designed to open at 400 degrees F or less when subjected to a nonload oven test.
   3. Class L Fuses: UL 198C, "High-Interrupting Capacity Fuses, Current-Limiting Type."
   4. Class RK1 and RK5 Dual Element Time-Delay Fuses: UL 198E, "Class R Fuses."
   5. Class J Low-Peak dual Element Fuse: UL 198C

D. NONFUSIBLE SWITCHES:
   1. Rating: Load-breaking capacity in excess of the normal horsepower rating for the switch.
   2. Operation: By means of external handle.
   3. Interlock: Prevents access to switch interior except when in "off" position.
   4. Enclosure for Independent Mounting: NEMA Type 1 enclosure except as otherwise indicated or required to suit environment where located.
   5. Contacts shall be NEMA rated 75 degrees C.
   6. Provide auxiliary contacts for disconnects supplied from variable frequency drives.

E. FUSIBLE SWITCHES:
   1. Rating: Load-breaking capacity in excess of the normal horsepower rating for the switch. Withstand Capability: In excess of the let-through current permitted by its fuse when subject to faults up to 100,000 RMS symmetrical amperes.
   2. Operation: By means of external handle.
   3. Interlock: Prevents access to switch interior except when in "off" position.
   4. Enclosure for Independent Mounting: Provide NEMA Type 1 enclosure except as otherwise
   5. indicated or required to suit environment where located.
   6. Contacts shall be NEMA rated 75 degrees C.
   7. Provide auxiliary contacts for disconnects supplied from variable frequency drives.

F. MOLDED-CASE CIRCUIT BREAKERS:

1. Construction: Provide bolt-in type, except breakers 225-ampere frame size and larger which may be plug-in type if held in place by positive locking device requiring mechanical release for removal.
2. Tripping Device: Quick-make, quick-break toggle mechanism with inverse-time delay and instantaneous overcurrent trip protection for each pole. Trip unit to be interchangeable within frame sizes for breakers 200 amperes or larger. Breakers 250 amperes and above shall have adjustable trip selection for trip units. All 120/208 volt rated breakers shall be rated and labeled "High Magnetic".
3. Adjustable Instantaneous Trip Devices: Factory adjusted to low-trip-setting current values. Provide adjustable instantaneous trip devices for each circuit breaker supplying individual motor loads and where indicated.

G. COMBINATION CIRCUIT BREAKERS AND GROUND FAULT CIRCUIT INTERRUPTERS:
1. Trip Setting for Ground Fault: 4 to 6 milliamperes, listed and labeled as a class A, type 1device.

H. CIRCUIT BREAKERS WITH SOLID-STATE TRIP DEVICES:
1. Ambient Compensation: Trip device insensitive to temperature changes between minus 20 degrees C and plus 55 degrees C.
2. Ground-Fault Tripping: Adjustable for pick-up and time-delay values. Provide for indicated units.

I. SPARE FUSE CABINET:
1. Cabinet: Wall-mounted, 18-gauge minimum steel unit with full-length, recessed piano-hinged door with key-coded cam lock and pull. Size to be adequate for orderly storage of spare fuses specified with 15 percent spare capacity minimum. Cabinet finish to be gray baked enamel. Stencil legend "SPARE PARTS" in 1-1/2 inches (40-mm) letters on door.

J. MANUFACTURERS:
1. Cartridge Fuses:
   a. Bussmann Div., Cooper Industries, Inc.
   b. Littelfuse Inc.
2. Fusible Switches:
   a. Square D Co.
   b. Allen-Bradley Co.
   c. General Electric Co.
   d. Siemens Energy & Automation, Inc.
   e. Cutler-Hammer, Inc.
3. Molded-Case Circuit Breakers:
   a. Square D Co.
   b. General Electric Co.
   c. Siemens Energy & Automation, Inc.
   d. Cutler-Hammer, Inc.
4. Combination Circuit Breaker and Ground Fault Circuit Interrupters:
   a. Square D Co.
   b. General Electric Co.
   c. Siemens Energy & Automation, Inc.
   d. Cutler-Hammer, Inc.

26 30 10 Generators (Current Version)

A. SYSTEM DESCRIPTION: The generator system shall be factory assembled, liquid cooled unit mounted radiator, natural gas engine-driven generator to provide source of stand-by power. Diesel systems are not allowed. Noise level shall be a maximum of 75 db at 23 feet.(7m.). Design for standby loads of IT closets, freezer, cooler in addition to life safety requirements. Provide minimum two transfer switches, one emergency and one standby. Provide coordination study for emergency panels.

B.

C. ENVIRONMENTAL CONDITIONS:
   1. The equipment specified is intended for application in an environment as described below:
      a. Altitude: 5500 feet above sea level.
      b. Maximum Temperature: 120 degrees F
      c. Minimum Temperature: -20 degrees F
      d. Location: Outdoor

D. ENGINE:
   1. Type: natural gas, liquid cooled with unit mounted radiator and engine driven water pump, multi-cylinder inline or v-type, internal combustion engine. Natural gas engine shall comply with the requirements of SAE 1349, ISO 8528 (Part 2), EGSA 101P and IEEE Standard 446 as they apply to standby application.
   2. Fuel System: Natural gas
   3. Governor: Isochronous type to maintain governed speed at rated frequency regardless of the kW load and shall meet the following requirements:
      a. Stability: 0.25 percent maximum frequency variation at any constant load from no load to full load.
      b. Regulation; 0.25 percent maximum frequency deviation at steady state. Provide a separate overspeed device, independent of the governor, to prevent runaway in the event of any failure which may render the governor inoperable.
   4. Engine Accessories: Provide the following engine accessories.
      a. Pressure regulators
      b. Lube oil filter.
      c. Intake air filter.
      d. Lube oil cooler.
      e. Strainers
      f. Control valves
      g. Gear-driven water pump.
   5. The engine shall be designed to develop rated speed and torque capable of developing full load within 8 seconds from the time the engine starts. Engine cranking period shall not exceed 30 seconds.
   6. The lubrication system shall incorporate a threaded, spin-on or canister type, full flow, duplex filtration system with a spring loaded, pressure calibrated, bypass valve to allow lubrication to continue in the event of unusually high filter restriction. The bypass valve must be an integral part of the engine filters or filter housings. Locate filters for easy servicing. A local alarm shall be generated if bypass occurs.
   7. Provide a valved oil drain extended past the engine frame with braided stainless steel hose and reusable fittings.
   8. The engine shall be equipped with protective devices to meet control requirements specified elsewhere in this specification. The control devices shall be tied into the engine generator control. All protective devices shall be wired into a master engine terminal box with terminal strips for remote connections and connections to annunciators or controls as required.
   9. All equipment shall operate with minimum vibration throughout the operating range as assembled units. The equipment supplier shall assume all additional shop, shipping and transportation expense, and/or cost at the job site necessary to reduce excessive vibration attributable to unbalance of the equipment.
   10. Provide braided stainless steel, flexible fuel lines.
   11. Provide spare fuses for coordination panel.

E. GENERATOR CHARACTERISTICS:

1. The generator shall be of the single-bearing type, dripproof construction, externally regulated, directly connected to the engine flywheel through a semi-flexible coupling.
2. The generator shall be complete with inboard brushless exciter. The generator rotor shall be dynamically balanced to withstand over speeds up to 125 percent rated speed.
3. The generator shall be capable of continuously delivering its rated output at any power factor from 1.0 to 0.8 lagging at ±5 percent of rated voltage and at 60Hz, in the environmental conditions as specified.
4. The generator voltage regulator shall be solid state, with external voltage adjust rheostat. Provide a setscrew stability adjustment. Voltage regulation shall be +1.0 percent. Steady state voltage stability shall remain within a 0.5 percent band of rated voltage.
5. The engine alternator shall be capable of accepting a one-step application of 100% of nameplate kW load at 0.8 power factor and recover to steady state conditions without disruption of power to the load.
6. The generator shall be capable of providing 300 percent of rated current for ten seconds during a short circuit condition.
7. The alternator shall be capable of operation with reverse kVAR of 0.15 per unit.

F. STARTING EQUIPMENT:

1. The engine shall be equipped with a 24 volt electric starting system of sufficient capacity to crank the engine at a speed which will allow starting of the engine. Starting system shall be capable of starting the engine either manually or automatically.
2. Batteries: The batteries shall be dedicated to the engine. Provide battery blanket for 120 volt a.c. supply. Provide a battery rack and necessary cables and clamps.
3. A suitable battery charging alternator shall be provided with sufficient capacity to recharge the batteries back to normal starting requirements within 90 minutes.
4. Battery Charger: Provide a battery charger specifically designed for engine starting batteries which shall maintain the starting batteries at full charge. The charger shall be capable of restoring the batteries to a fully-charged state within eight hours after a complete duty cycle discharge.
5. The charging system shall permit charging from either a normal or emergency 120 volt AC power source.
6. The charger shall be a three stage, temperature stabilized type with automatic and manual charge control.

G. MANUFACTURERS: Subject to compliance with requirements, provide natural gas generator sets of one of the following:

1. Preferred Manufacturer(s)
    a. Cummins/Onan Engine Co.
2. Acceptable Manufacturer(s)
    a. Generac

H. MOUNTING:

1. Provide a suitable full length sub-base for mounting the engine generator unit on a concrete foundation. Provide assembly with spring type vibration isolators for use between the sub-base and the concrete foundation. A pin shall be installed through the spring isolators, sub-base and bolted into concrete inserts for lateral movement limitation.

I. WARRANTY:

1. Provide manufacturer's warranty for a duration of not less than two (2) years from the date of substantial completion of the project. The warranty shall include, but not be limited to, the replacement of materials and equipment used in natural gas generator systems.

26 30 10 Generators (Proposed Version)

A. PRIMARY REQUIREMENTS
    a. Fuel: Natural Gas preferred
    b. Maximum noise level: 75 dB at 23 feet.
    c. Provide annunciator or other approved means of communicating status with District maintenance and the fire alarm panel.

B. COMPONENTS TO BE SUPPORTED BY GENERATOR
    a. Life safety components as required by Code.
    b. All network equipment in Main Distribution Frame (MDF) room
        i. Discuss Intermediate Distribution Frame (IDF) rooms with project team
    c. Walk-in freezers and coolers
    d. Door access control systems
    e. Intercom systems

C. MANUFACTURERS: Subject to compliance with requirements, provide natural gas generator sets of one of the following:
    3. Preferred Manufacturer(s)
        a. Cummins/Onan Engine Co.
    4. Acceptable Manufacturer(s)
        a. Generac

D. WARRANTY:
    2. Provide manufacturer's warranty for a duration of not less than two (2) years from the date of substantial completion of the project. The warranty shall include, but not be limited to, the replacement of materials and equipment used in natural gas generator systems.

26 36 00 Transfer Switches

A. SUMMARY:
   1. This Section includes transfer switches rated 600 V and less. It includes the following items:
      a. Automatic transfer switch (ATS).
      b. Remote annunciation system.

B. MANUFACTURERS:
      a. Same as generator

C. TRANSFER SWITCH PRODUCTS, GENERAL:
   1. Units smaller than 400 amperes shall not have different current ratings for different classes or mixtures of loads, including 100 percent tungsten filament lamp or 100 percent inductive load.
   2. Tested Fault-Current Ratings: Closing and withstand ratings shall exceed the indicated available rms symmetrical fault current at the equipment terminals based on testing according to UL Standard 1008, conducted at full-rated system voltage and 20 percent power factor. Rate each product for withstand duration time as follows when tested for rated short-circuit current correlated with the actual type of circuit protective device indicated for transfer switches for this Project.
      a. 150 Amperes or Smaller: 1.5 closing and withstand duration cycles.
      b. Larger than 150 Amperes: 3 closing and withstand duration cycles.
   3. Annunciation and Control Interface Components: Provide devices at transfer switches for communicating with remote annunciators or annunciator/control panels which have communications capability matched with the remote device.

D. AUTOMATIC TRANSFER SWITCHES (ATSS):
   1. Comply with Level 1 equipment according to NFPA 110, "Standard for Emergency and Standby Power Systems."
   2. Switching Arrangement: Double-throw type, incapable of pauses or intermediate position stops during normal functioning [except for neutral position, to enable programmed transition.]
   3. Manual Switch Operation: The switch shall have provision for manual operation under load with the door closed with either or both sources energized. Transfer time shall be the same as for electrical operation. Control circuit shall automatically disconnect from electrical operator during manual operation.
   4. Digital Communications Interface: Provide full-duplex RS - 232 type, matched to capability of remote annunciator.
   5. Programmed Transition: Switch Operator shall have a programmed neutral position arranged to provide a midpoint between the 2 working switch positions with an intentional, controlled, timed pause during transfer at the midpoint. The midpoint pause shall be adjustable from 0 to 60 seconds minimum, and factory set at 5 seconds, except as indicated. Time delay shall occur for both transfer directions.
   6. Provide dry contacts for connection to Building Automation System (BAS) head-end. Provide separate contacts to indicate:
      a. Transfer to Emergency Power Source
      b. Generator Start/Run.

E. AUTOMATIC TRANSFER SWITCH FEATURES:
   1. Provide for normal source voltage sensing of each phase of normal source. Pick-up voltage shall be adjustable from 85 percent to 100 percent nominal, and drop-out voltage is adjustable from 75 percent to 98 percent pick-up value. Factory set for pick-up at 95 percent and dropout at 85 percent.
   2. Provide for emergency source voltage sensing to prevent premature transfer. Voltage pick-up shall be adjustable from 85 percent to 100 percent of nominal. Factory set to pick-up at 90 percent. Pick-up frequency shall be adjustable from 90 percent to 100 percent of nominal and factory set to pick-up at 98 percent.
   3. Provide a transfer switch signal time delay to override normal source voltage-sensing, delay transfer signal and engine start signal. Delay shall be adjustable from 0 to 6 seconds, and factory set at 5 sec.
   4. Provide a transfer to emergency time delay to delay transfer switch changeover after transfer signal. Delay shall be adjustable from 0 to 5 minutes and factory set at 0 minutes.

5. Provide a retransfer time delay to provide for automatic defeat of the delay upon loss of voltage or sustained undervoltage of the emergency source, provided the normal supply has been restored. Delay shall be adjustable from 0 to 60 minutes and factory set at 15 minutes.
6. Provide an engine shut-down time delay adjustable from 0 to 15 minutes and factory set at 5 minutes.
7. Provide source available indicating lights to supervise sources via the transfer switch normal and emergency source-sensing circuits.
8. Provide a transfer override switch to override automatic retransfer control so the ATS will remain connected to the emergency power source regardless of the condition of the normal source. Provide a pilot light to indicate the override status.
9. Provide engine starting contacts, one isolated normally closed and one isolated normally open. Contacts shall be gold flashed or gold plated and rated 10 amperes at 32 V d.c. minimum.
10. Provide two normally open and two normally closed spare SPDT contacts for each switch position, rated 10 amps at 240 VAC.

F. REMOTE ANNUNCIATION SYSTEM FEATURES:
1. Provide a remote annunciator panel which shall annunciate conditions at indicated transfer switches including:
    a. Sources available (as defined by actual pick-up and drop-out settings of transfer switch controls).
    b. Switch position.
    c. Switch in test mode.
    d. Failure of digital communications link.
    e. Malfunction of the annunciator unit or communication link shall not affect functions of the ATS. In the event of a failure of the communication link, the ATS automatically reverts to stand-alone, self-contained operation. No ATS sensing, controlling, or operating function shall depend on the remote panel for proper operation.
    f. Transfer switch shall be designed, constructed, and mounted to comply with applicable seismic zone requirements.

26 40 20 Electrical for Press Box Sound Systems

A. SUMMARY
1. The sound reinforcement system, audio signal, and control wiring, and components will be furnished, installed, and terminated by the SS System Contractor.
2. Prior to installation of raceway, verify conduit sizes and routing, and coordinate locations of all outlets with the SS System Contractor.
3. Provide 120 VAC power for the sound system equipment.
4. All sound system conduit shall be metallic conduit only. Conduit installed in concrete slabs shall be metallic type conduit rated for such use. No PVC conduit shall be used for sound reinforcement systems raceways unless specifically noted otherwise.
5. Furnish and install conduit, j-boxes, wireways, and floor boxes for audio, and control wiring as shown on the SS drawings. Conduit shall have pull strings left in place.
6. Provide non-metallic conduit connections from the wireways to sound system equipment racks to isolate equipment ground from conduit ground.
7. Provide 20A, 120VAC, single-phase, dedicated circuits with independent third wire grounds as shown on the electrical drawings. Terminate circuits as noted on the SS drawings.
8. Provide a separate insulated ground conductor run in conduit from the sound system equipment racks to the ground at the electrical panel feeding 120VAC power to the equipment. Size of conductor to provide resistance of 0.1 Ohm or less to earth ground based on length of run.
9. Provide emergency power PA system and egress lighting.

26 40 30 Transient Voltage Suppression for Low-Voltage Electrical Power

A. SUMMARY:
   1. Types of TVSS equipment required may include the following:
      a. Category b or c for distribution panelboards.
   2. The specified system shall be factory-tested before shipment. Testing of each system shall include but shall not be limited to quality control checks, "hi-pot" tests at twice rated voltage plus 1000 volts per UL requirements, IEEE c62.41 category b surge tests, UL ground leakage test, and operation and calibration tests.
   3. The project shall be life cycle tested following suggested wait times as defined by ANSI/IEEE C62.45 (1987) and shall be capable of surviving 1000 sequential category C surges of 10,000 amps without failure.
   4. The system shall be UL listed as a complete system under UL 1449 (rev 7/2/87) standard for transient voltage surge suppressors (TVSS) and the rating shall be permanently affixed to the TVSS.

B. TVSS EQUIPMENT:
   1. Components: The system shall be a symmetrically balanced, metal oxide varistor (MOV) array system, constructed using surge current diversion modules. Each module shall be capable of withstanding over 1000 pulses of the 10 kAmps IEEE C62.41 Category C surge current without degradation of clamping voltage. The module shall consist of multiple gap-less metal oxide varistors, with each MOV individually fused. The modules shall be designed and constructed in a manner which ensures reasonable MOV surge current sharing. No gas tubes or silicon avalanche diodes shall be used. The status of each varistor shall be monitored and green LED shall be illuminated if the module is in full working order. When module performance is degraded, such as if one or more fuses or varistors have failed, the LED shall indicate a failed module.

   2. High Frequency Tracking Filter: The unit shall include a UL 1238 high-frequency extended range tracking filter. The filter shall provide for high frequency transient filtering of up to 5 dB attenuation (per MIL-STD E220A) for the band width extending from 100 KHZ to 100 MHZ. This filtering must remove low level surges and sharp wavefronts associated with fast rise-time transients, thus eliminating disturbances which may lead to "system upset".
   3. Connections: Terminals shall be provided for all of the necessary power and ground connections. The terminals shall accommodate wire sizes of #14 to #2/0 AWG for two conductors per required connection. The units shall use standard parallel wiring techniques.
   4. Internal Connections: All surge current diversion module intra-unit connections shall be by way of low impedance busbars or wiring. Surge current diversion modules shall use low impedance connections. All module mounting hardware and power wiring shall be captive or remain in place when a module is removed or replaced.
   5. Enclosure: The specified system shall be provided in a heavy duty NEMA 12 dustight, driptight enclosure with no ventilation openings.
   6. .Integral Test Point: The unit shall incorporate an integral test point allowing easy off-line diagnostic testing which verifies the operational integrity of the unit's suppression/filter system.

B. ACCESSORIES:
   1. LED indicators shall be provided on the hinged front cover to redundantly indicate unit module status. Additionally, a Form C (one N.O. and one N.C.) summary alarm contact rated for at least 120 VAC and 1 ampere shall be provided for remote annunciation of unit status. The summary alarm contact shall change state if any one or more of the surge current diversion modules has failed.
   2. Integral Fused Disconnect Switch: The unit shall include an integral fused and safety interlocked disconnect switch located in the unit enclosure with an externally mounted manual operator. The switch shall disconnect all ungrounded circuit conductors from the distribution system to enable testing and maintenance without interruption of power to the facility's distribution system. The switch shall be rated for 600 Vac. Each current carrying ungrounded circuit conductor connected to the facility's distribution system shall be individually fused with 200,000 AIC rated fuses in order to provide maximum fault current protection. The unit shall be UL 1449 Listed with the integral fused disconnect switch

and the UL 1449 Suppression Rating for this configuration shall be provided. (1.4.6 Performance Ratings.) Specify Integral Fused Disconnect Switch by adding a "DF" suffix to the model number.

C. MANUFACTURERS:
   1. Manufacturer: subject to compliance with requirements, provide TVSS products of the following:
      a. Current technology
      b. GE/Liebert

D. WARRANTY:
   1. The manufacturer shall provide a full five year warranty from date of shipment against any part failure when installed in compliance with manufacturer's written instructions, UL listing requirements, and any applicable national or local electrical Codes.

26 50 00 Lighting

A. District Design Criteria:
   1. LED is the District Standard for all luminaires
      a. Where LED products are not available, appropriate, or cost effective review exceptions with Project Manager.
   2. HID is not allowed in Auditoriums; the basis of lighting for Auditoriums space should be dimmable LED system.
   3. A/E shall consider LED lighting systems in Auditoriums both for house lighting and stage lighting. Review final recommendations with District prior to completing bidding documents.
   4. Fixture accessibility must be considered when locating lights during the design phase. A/E shall pay specific attention to insuring that fixtures are accessible in sloped floor conditions, high volume spaces and stairs.
   5. Provide a spare circuit above each classroom and hallway to accommodate 120V spare circuit.
   6. Conduit from all lighting controls to HVAC controllers shall be ¾" minimum
   7. Lighting control wire shall be to DMS, 24 volt.
   8. Provide controls that allow adjustment of individual fixtures or devices without interrupting other fixtures or devices in the building or area.

B. EXTRA MATERIALS:
   1. Lamps: Furnish stock or replacement lamps amounting to 15%, but not less than 4 lamps in each case, of each type and size lamp used in each type luminaire.
   2. Ballasts: Furnish stock or replacement ballasts amounting to 5%, but not less than 2 of each type used in each type luminaire.
   3. Lenses: Furnish stock or replacement lenses amounting to 3%, but not less than one, of each type and size used in each type luminaire.

C. MANUFACTURERS:
   1. Coordinate manufacturer and rebate opportunities with the District.
   2. Fixtures shall be either:
      a. On the Design Lights Consortium (DLC) Qualified Products List
      b. Energy Star Rated

D. EQUIPMENT:
   1. Provide lighting of sizes, types and ratings indicated; complete with, but not limited to, housings, energy-efficient lamps, lamp holders, reflectors, energy efficient ballasts, starters and wiring. Ship luminaires factory-assembled, with those components required for a complete installation. Design luminaire with concealed hinges and catches, with metal parts grounded as common unit, and so constructed as to dampen ballast generated noise and as to disconnect ballast when door is opened for HQI lamps.
   2. Wiring: Provide electrical wiring within luminaire suitable for connecting to branch circuit wiring as follows:
      a. NEC Type TFN for 120 and 277 volt and shall be minimum No. 18 AWG.
   3. Lamps:
      a. Fluorescent: Provide fluorescent lamp types as specified in the luminaire schedule and types compatible with luminaires. Lamps shall be low mercury type. Provide 5000°K color temperature lamps unless otherwise noted in schedules or on drawings. Provide fluorescent fixtures for display cases.
      b. LED Lamps: Provide LED lighting systems as specified on drawings. System shall include LED lamps and electronic driver(s) as required for a complete and operable installation. LED lighting for site and parking lot fixtures is acceptable.
   4. Interior Fluorescent Ballasts:
      a. Provide high frequency programmed rapid start electronic ballasts for all fluorescent lamps with voltage as indicated on the plans and luminaire schedule.
      b. The ballast total harmonic distortion shall be less than 20% with the number of lamps as identified per luminaire type or as indicated in the luminaire schedule.

c. The ballast shall have a crest factor of less than 1.5 and shall have transient protection which meets IEEE 587, Category A (ANSI C62.41) requirements.
d. The ballast shall have a power factor of 0.98 or higher, and shall have a ballast efficiency of 90% or higher.
e. The ballast shall be UL listed Class P and shall have a sound rating better than A.
f. The ballast electromagnetic interference and radio frequency interference shall comply with FCC CFR 47, Part 18 requirements for both conducted and radiated EMI/RFI.
g. The ballast shall contain no PCB's.
h. Minimum lamp starting temperature of 0 degrees F.

5. Exterior wall packs need to be 120V LED-type lamps.

26 50 20 Sports Field Lighting

A. GENERAL SYSTEM REQUIREMENTS
   1. LED luminaires
   2. Drivers shall be located not more than 20'-0" above the ground.

B. LIGHTING PERFORMANCE
   1. Illumination level: 50 footcandles, unless otherwise directed by the District.
   2. Calculation grid spacing: 30'-0" x 30'-0"
   3. Uniformity Ratio: 2-1/2 : 1

C. ENVIRONMENTAL LIGHT CONTROL
   1. Manufacturer shall provide a submittal of design data to confirm compliance with the requirements below.
   2. Spill Light Control
      a. Not more than 0.5 footcandles at a distance of 150'-0" from the edge of the field.
   3. Glare Control
      a. Not more than 12,000 candelas at a distance of 150'-0" from the edge of the field.

D. CONTROLS
   1. On-Off manual control on-site, as well as web-based remote control capability shall be provided.
   2. Web-based controls shall utilize a cellular connection maintained by the luminaire manufacturer.

E. STRUCTURAL
   1. Structural elements shall be designed in accordance with the edition of the International Building Code currently adopted by the local jurisdiction.
   2. Foundations and Poles
      a. Preferred: Concrete foundations with steel poles affixed to concrete above grade.
      b. Prohibited: Concrete poles, direct bury steel poles, wood poles

F. WARRANTY AND GUARANTEE
   1. 25-Year Manufacturer's Warranty
      a. Luminaire manufacturer shall guarantee light levels; lamp replacements; system energy consumption; monitoring, maintenance and control services, spill light control, and structural integrity.
      b. Shall include all parts, labor, service, equipment, etc. to assess and correct warranty claims.
         1. Warranties that require the District to remove lamps or other equipment and ship to the manufacturer are not sufficient.
         2. The District will be responsible for checking fuses in the event of a luminaire outage.

26 20 00 Electrical Vehicle Charging Stations

A. Coordination
    1. Coordinate Electric Vehicle Charging Stations with the District's Energy Manager.

B. Charging Station Scope
    1. District intent is typically to provide at minimum one charging station with new construction and major renovation projects, however specific project budget and requirements may modify this scope, verify with District Project Manager
    2. Where projects do not provide charging stations consider providing infrastructure to allow future installation.

C. Standard products
    1. Charge Point 4000 Series

D. Validation
    1. Coordinate product installation and validation with the District.

E. Signage
    1. EV station signage will be District provided and District Installed.
    2. The Contractor shall provide and install the sign post.

F. General Requirement
    1. EV charging stalls should be located along an ADA accessible route, but does not need to provide preferential parking for EV users.
    2. Provide bollards where necessary to protect EV charging equipment and infrastructure.

DIVISION 27 – COMMUNICATIONS

27 00 00 Communications See Appendix C

A. See: **Appendix C – Communications, Technology, and Security Guideline** for detailed requirements, including District Standards for:
   1. Cabling
   2. Pathway
   3. Communication Equipment Rooms
   4. Public Address / Intercom / Bell
   5. Wireless Clock System
   6. Door Access Control
   7. Intrusion Detection / Bugler Alarm Systems
   8. Radio re-amplification
   9. Digital signage

27 00 01 Energy Management

A. Provide data connection in MDF room for “eGauge” electrical energy monitoring equipment.
B. Provide dedicated telephone line for gas meter.
C. Coordinate requirements with the District’s Energy Manager:

27 20 00 Energy and Environmental Monitoring

A. Electrical Usage Monitoring
    1. The District monitors electrical usage with “eGauge” devices and platform.
    2. Review sub-metering and monitoring with the District

B. Indoor Air Quality (IAQ) Monitoring
    1. Scope
        a. The District monitors IAQ at select existing facilities and intends to expand this program at new construction and renovation projects.
    2. Platform/Headend
        a. Attune cloud based
    3. Devices
        a. UL-2905 Compliant
        b. Preferred product: Attune IAQ Monitoring Devices
    4. Monitoring Metrics
        a. Temperature
        b. Relative Humidity
        c. Volatile Organic Compounds (VOCs)
        d. Carbon Dioxide
        e. Airborn Particulate Matter
            1. PM10, particulate matter greater than 10 microns
            2. PM2.5, particulate matter greater than 2.5 microns
    5. Locations and Quantities
        a. Avoid space which may have uses which may skew results due to activities such as:
            1. Science rooms
            2. Art rooms
            3. Gymnasiums
            4. Cafeterias
            5. Auditoriums
        b. Provide devices distributed throughout the building in a variety of spaces with the following general quantities:
            1. Elementary: 4 units minimum
            2. PK8: 8-10 units
            3. Middle: 6-8 units
            4. High: 12-15 units

27 50 00 Bi-Directional Amplification (BDA) Systems

A. Preferred Manufacturers
    1. Honeywell
        a. This manufacturer coordinates with District standard fire alarm panels and other devices.
B. Discouraged Manufacturers
    1. Comba

DIVISION 28 – ELECTRONIC SAFETY & SECURITY

See: **Appendix C – Communications, Technology, and Security Guideline** for detailed requirements, including District Standards for:

A. ~~Access Control~~
B. Classroom Audio/Visual Technology
C. Data and data cabling
D. ~~Video Doorbell~~
E. ~~Video Surveillance~~

28 00 00 Automated Emergency Systems

A. Automated Emergency Systems refers to the integration of various security and low voltage systems to align with the Standard Response Protocol and allow simple activation of select actions. See the **Automated Emergency Systems Design Bulletin**, which is posted independently on the SVVSD Bluebeam project as well as at the end of the Technical Guidelines document.
    1. Actions or systems required may include, but are not limited to:
        a. Intercom / mass notification
        b. Door access control
        c. Fire alarm / magnetic door hold-opens

28 31 01 Intercom, Clocks, & Electronic/Digital Signage

A. Intercom
    1. System Platform
        a. Singlewire / InformaCast
    2. Devices
        a. Typical Intercom Devices
            1. IP based speakers
            2. Integrated multi-color visual displays
            3. Microphone/talkback function
            4. Preferred Product
                a. Atlas IED, IPX series, or current equivalent
        b. Other Speakers
            1. Preferred Manufacturers
                a. Washega
                b. Atlas Sound
            2. Acceptable Manufacturers, with SVVSD approval
                a. Valcom
            3. Zone Controllers
                a. Shall coordinate with other system components.
    3. General Requirements
        a. Provide ceiling-mounted speakers throughout the building for public address and bell notification. Wall-mounted speakers may be considered where ceiling-mounted speakers are not feasible. The speaker

system shall be 25v or 70v distribution, and integrate into the IP communications platform via amplifiers and zone controllers.
b. Two-way communications shall be accomplished through the building telephone system.
c. In high noise areas, provide visual notification option. This includes music rooms, gymnasiums, weight rooms, locker rooms, and kitchens.
d. During a zoned page, local Multi-Media systems are to be overridden/auto-muted to facilitate the broadcast of the message. Vendor shall provide tie-in cable from speaker to multi-media system for the page override muting connection. Paging connection leads should be capped until the system is tapped. Integrate with the Wireless Clock System to provide synchronized bell operations.
e. The public address system will integrate with the premises IP voice communication system.
f. Head-end server shall be located in the building Primary TR (MDF).
g. Auditoriums, closets, common areas, coordinate with SVVSD Construction and Maintenance Department

B. Clocks
1. Classroom clocks may be integrated int the intercom devices. Where not integrated with intercom devices or otherwise required:
a. Type: Radio Controlled Stand Alone Clocks
b. General Requirements
1. 1-year warranty min.
2. AA Battery
3. Automatic synchronization to WWVB time signal
4. Time change for ST/DST fully automatic
5. Integrated antenna
6. Manual time setting backup feature
c. Preferred manufacturers
1. American Time, or similar

C. Electronic Signage
1. Interior Electronic Signage and Video Displays
a. Performance Requirements
1. Display shall be a commercial-grade LED-type video display, 42-inch nominal size or larger, with slim bezel. Inputs shall include Ethernet/Network, HDMI, VGA, and Control/RS232.
2. Mount display with the bottom of the monitor at a minimum of 84” above finished floor, unless location requires a lower height.
3. Display mount shall be of solid steel or aluminum construction, with tilting (locked in place) articulation. Mount shall allow for security locking of display.
b. Power Requirements
1. Provide (1) 20-amp, 120-volt duplex (NEMA 5-20R) receptacle minimum. Coordinate with selected mount.
2. Provide (1) power outlet for sign control computer.
c. Control Computer
1. Provided by District
2. Coordinate data requirements
d. Preferred Manufacturers
1. Displays
a. LG
b. NEC
c. Samsung
d. Other products may be acceptable, but require District approval.
2. Mounts
a. Chief Manufacturing
b. Other products may be acceptable, but require District approval.

2. Outdoor Electronic Message Center (EMC) Signs
    a. Performance Criteria
        1. Type: Color LED
        2. Resolution: 19mm min.
        3. Automatic Dimming
        4. Controls: Web Based
        5. Contractor shall provide support and equipment (e.g. computers) as required to deliver a fully functioning system.
    b. Preferred Manufacturers
        1. Daktronics
        2. Watchfire
        3. Other products may be acceptable, but require District approval.
    c. Infrastructure
        1. Fiber network cabling required for all outdoor signs.
        2. Provide separate conduit runs for power and communications. Power and communications may not share pull boxes unless appropriate division plates are provided.
        3. Locate in-grade communications pull boxes along the conduit pathway. One (1) box shall be required at a minimum if conduit length exceeds 100 feet or if more than 180 degrees of bends exists in the entire length. For longer runs, pull boxes shall be installed every 200 feet.
        4. Default preference is for the Contractor to provide all necessary fiber transceivers. Coordinate scope and responsibilities for fiber transceiver at each end per project.
    d. Network Connection
        1. Network connection to the sign will be provided with 50 micron, 6 strand, OM4, outdoor rated fiber in conduit. The fiber must be terminated with LC connectors in a fiber patch panel or block at both ends. Fiber installation must be performed by a District approved contractor. Installation with termination in the MDF is preferred; coordinate with owner for alternates.
        2. Fiber transceivers for both ends of the fiber connection to the sign must be provided by the installer. Fiber patch cables will be provided by the owner.
        3. Proper surge/lightning protection shall be installed on both ends of cable.

28 13 00 Access Control

A. Platform and System Components
    1. Access Control Platform and Head End
        a. Mercury, with Avigilon ACM Software
    2. Proximity Readers
        a. Allegion aptiQ (MTB 11 & MTB 15)
    3. Video Doorbell
        a. Avigilon Video Intercom with surface mount and safety relay
B. General Requirements
    1. Approx. 4’ x 8’ of wall space in the Main Distribution Frame (MDF) room.
    2. Dedicated electrical circuit
    3. Electrical emergency generator backup.
    4. Provide separate raceway and conduit from other cabling systems as required by other sections.
C. Typical system scoping
    1. The following outlines a typical system scoping approach; however, every project is unique and close coordination with SVVSD Construction and Maintenance Department will be necesarry.
        a. Card Reader Locations, exterior
            1. Primary access points only. Additional exterior door to be added as program requires.

b. Card Reader Location, interior
    1. Card readers are typically not provided for interior doors.
        a. Exceptions:
            i. Interior doors associated with the secure vestibule and visitor entry sequence.
            ii. Administration suite
            iii. Main Distribution Frame (MDF) Room.
            iv. Others as required by project programming and approved by SVVSD Operation Department.
c. Monitoring
    1. Coordinate extents and infrastructure for door position monitoring on a case-by-case basis with SVVSD Operations Department.

28 16 00 Intrusion Detection (Burglar Alarm)

A. Manufacturer / System
    1. Head End: Honeywell Ademco Vista-128BPT
    2. Sensor: Honeywell
    3. IP Communicator: Honeywell 7847i
    4. Control Panel: Honeywell Coded Keypad
    5. Motion sensors
        a. Infrared
        b. Ceiling mounted under 12', wall mounted where ceilings are above 12'
B. General Requirements
    1. Head end located in MDF with sufficient wall space for working clearances.
    2. Provided dedicated electrical with backup.
    3. Coordinate locations of keypads, horns, and motion sensors with SVVSD Operations.
C. Areas of Coverage
    1. Interior corridors and administration areas
    2. Coordinate additional coverage with SVVSD Operations.
D. Other Requirements
    1. Freezer/Cooler Alarm
        a. Provide a zone to receive alarm from walk-in freezers and coolers when the temperature limits are exceeded. This system is intended to facilitate telephone notification to technicians. This requirement is separate from Building Automation System monitoring.
        b. Provide low voltage pathway or conduit from freezer or cooler head end to burglar alarm head end.

28 23 00 Video Surveillance (CCTV) & Video Doorbell

A. System Format
    1. IP-based distribution
B. Camera Software
    1. Verkada
C. Camera Devices
    1. Verkada. Confirm product with SVVSD Operations Department on a case-by-case basis.
D. Computer
    1. The computer shall be a Windows based system that exceed the recommended requirements for the operating system of the software.
E. Areas of Coverage

1. Cameras will generally cover the areas listed below; however final locations shall be coordinated with SVVSD Operations department.
    a. Outdoor spaces including (typically from building-mounted cameras):
        1. Parking lots
        2. Plazas
    b. Interior side of exterior doors.
    c. Interior corridors
    d. Cameras shall NOT be located in instructional spaces.

F. General Requirements
1. Server and Video Storage Quality
    a. A CCTV server will provide for recording and viewing of cameral video.
        1. Video storage, minimum two weeks.
        2. Video recorded at 7-8 frames per second minimum.
        3. Resolution: 2048 x 1536 (3 megapixel)
        4. Capacity: 128 cameras
        5. Compression: H.264
2. Cabling
    a. Reference **Appendix C** for general cabling requirements, including cable color coding.
    b. CCTV systems may use fibers from general data network backbones for distribution.
    c. Video from the CCTV system cannot be sent over the District's general data network without coordination with SVVSD DTS.
    d. Any camera that exceeds 290 feet of cabling must be installed with fiber optic cable to the MDF.
3. Infrastructure and Support
    a. Provide 4-post server frame in MDF
    b. Provide electrical backup
    c. Poles supporting exterior cameras shall be reinforced to reduce shake.
    d. Mount roof located cameras to parapet walls, where possible.
    e. Exterior cameras will be specified with appropriate environmental housing. Consideration shall be given to utilizing pan-tilt-zoom (PTZ) cameras in justifying circumstances.

28 31 00 Fire Alarm

A. Quality Assurance
1. Installer shall be certified by NICET as fire-alarm Level IV technician.

B. Products
1. Manufacturer
    a. Preferred: NOTIFIER, a Honeywell Company
    b. Other Acceptable Manufacturers: None

C. Fire Alarm Monitoring
1. System: AES Corporation
2. Vendor/Monitoring: Trident Fire & Security
3. Programming to be by certified fire alarm contractor.

28 31 76 Emergency Responder Radio Coverage Systems

A. Radio re-amplification systems are used for enhancing signals such as public safety radio frequencies or cellular coverage and are not required by the District, but may be required by the Authority Having Jurisdiction (AHJ). Determine if this system is required by the AHJ. Provide as required.

DIVISION 32 – EXTERIOR IMPROVEMENTS

32 00 00 General District Preferences

A. Mowing Considerations
    1. Provide concrete apron or other buffer around mowing obstruction in turf areas to prevent damage to both mowing equipment and objects such as signs, light poles, valve boxes, etc.
    2. Provide radius corners at turf areas and drivable pavement to prevent pavement damage and assist with mowing.
B. Unless other preferences are indicated, defer pavement and other site specifications to project engineer or to City of Longmont "Public Improvement Design Standards and Construction Specifications".

32 10 00 Asphalt and Concrete Paving and Markings

A. Asphalt
    1. Asphalt paving is acceptable except where concrete is required, see below.
B. Concrete
    1. General
        a. Provide fiber mesh or other reinforcement at all exterior flatwork.
    2. ADA ramps
        a. Detectable surface: Provide cast-in, steel truncated dome
            1. Paver, plastic, fiber glass, or mechanically fastened products are prohibited.
    3. Required concrete locations (where asphalt is not accepted)
        a. Curb & Gutter
        b. Sidewalk
        c. Trash pickup and dumpster areas
        d. Service and loading dock areas
        e. Bus pickup and drop off zones
    4. Preferred concrete locations (these locations may be paved with asphalt if budget dictates)
        a. Bus traffic areas (bus loops), see above for drop off zones
        b. Traffic and parking areas
    5. Snow removal
        a. Provide sidewalks designed to accommodate snow removal equipment.
            1. 6'-0" min. width
            2. 6" min. depth
            3. 4,000 psi mix design, or as engineered to accommodate snow removal equipment.
        b. Provide rolled curbs where necessary to accommodate snow removal.
    6. Preferred sidewalk chase
        a. SVVSD Typical Detail # 32-2
C. Markings
    1. All line markings shall be painted by machine.
    2. Consult with project manager regarding painting of curbs at Fire Lanes.
    3. Typical Specs:
        a. 14 mils minimum thickness
        b. Latex acrylic fast dry
        c. 4" lines, typ for parking stalls
        d. 50-degree Fahrenheit minimum ambient outdoor temperature for application.
        e. Handicap symbol, blue with white border, 42" x 42" (to coordinate with District stencil and future re-painting).

f. Yellow paint for hatched areas.
g. Cross walk, 2’-0” x 6’-0” (to match District stencil)
h. Stop Bar, 9’-4” x 2’-0” (to match District stencil)

32 10 20 Athletic Surfacing

A. ATHLETIC EQUIPMENT
    1. See Sections 11 65 00 and 11 65 10

B. SYNTHETIC TURF
    1. Review the following features with the District for project specific design:
        a. Heat reducing infill
        b. Concussion pad
        c. Infill composition and depth
        d. Fiber type and depth
    2. Striping, provide striping for the following sports in listed colors
        a. Football, white
        b. Soccer, yellow
        c. Lacrosse, black (verify)

C. TENNIS
    1. Substrate
        a. Preferred: Post Tensioned Concrete Tennis Courts
        b. Discouraged: Asphalt, with District approval only.
    2. Surfacing
        a. Acrylic

D. TRACK AND FIELD
    1. Track surface, high school
        a. Type: Synthetic
        b. Performance Criteria
            1. Impermeable/Encapsulated (non-porous) Polyurethane systems
                a. Protects asphalt for longer lifespan.
            2. Warranty, 10 year min.
            3. Event markings: per IAAF / NCAA standards
        c. Basis of Design Product
            1. Beynon BSS 300, or similar
            2. Consider future resurfacing with BSS 2000 RE, or similar

    2. Track surface, middle school
        a. Type: Natural
            1. District standard for middle school or K8 applications.
            2. Basis of Design Products:
                a. Cinder Track Surfacing mixed with clay compliant with Colorado High School Activities Association requirements.
                    i. Alternate: Standard warning track material may be a more readily available and typical product for this application.
                b. “StaLok” or similar natural paving surfaces are discouraged. These products don’t allow management and repair.
        b. Type: Synthetic
            1. District may elect to install synthetic track surfaces at select location, coordinate with SVVSD Construction and Maintenance Department.
            2. Performance Criteria
                a. Impermeable/Encapsulated (non-porous) Polyurethane systems
                    i. Protects asphalt for longer lifespan.
                b. Warranty, 5 year min.
            3. Basis of Design Product
                a. Beynon BSS 200, or similar

b. Consider future resurfacing with BSS 2000 RE, or similar, to achieve a BSS 300 level surface.

c. General Requirements
   1. Curbs:
      a. Provide 12” min curb depth to 7” min. below grade.
      b. Provide 5” min. curb width.
   2. Event markings
      a. Provide permanent brass markers in concrete curb.
      b. Coordinate event markings with District staff.

3. Shot Put
   a. Provide raised concrete boarder.

E. BASEBALL
1. Infield Material
   a. Preferred product:
      1. Golf & Sport Solutions, Razum Red
      2. No substitutions, District Standard & stocked material
2. Warning Track Material
   a. Preferred product:
      1. Golf & Sport Solution, Pro Warning Track with Stabilizer
      2. No substitutions, District Standard & stocked material

32 30 00 Fences and Gates

A. Fences
1. Typical fence type: Chain-link
   a. 6’-0” typical height
   b. Posts: Schedule 40 Galvanized
      1. Line posts: 2-3/8”
      2. Terminal, corner, and pull posts: 2-7/8”
      3. Single Gate: 2-1/2”
      4. Double Gate: 3”
      5. Top Rail: 1-5/8”
   c. Fabric: 9 gauge with crimped ends
      1. If vinyl coating is selected specify similar metal thickness to non-coated gauge specified above.
   d. Hinges: 180 degree
   e. Latches: Strong-arm
   f. Over 12’-0” provide cantilever style gates
   g. Fence ties: Steel
2. Non-typical fencing
   a. Where fencing may improve student safety, but chain-link is not appropriate, the following fencing options may be discussed with the project manager:
      1. Preferred:
         a. Concrete split rail
         b. Architectural Metal Fencing
      2. Unacceptable: Wood fencing
3. Set property-line fencing 6” inside of property lines
4. Mow strip: provide 12” x 4” continuous concrete mow strip at all fencing

B. Gates
1. Where mower access is required provide 12’-0” min. double gates.

32 40 00 Site Furnishings & Site Signage

A. Furnishings
    1. Preferred:
        a. Bike Racks: Madrax, Winder model bike racks, single or multi-wave.
        b. Picnic tables: Concrete or Powder Coated Steel
        c. Benches: Concrete or Powder Coated Steel
        d. Also consider natural stone furnishings, i.e. Arkins Park Stone, Loveland, CO
    2. Unacceptable:
        a. Picnic table and Bench materials
            1. Wood, consult District regarding possible exceptions.
            2. Recycled Plastic
            3. Thermoplastic coated steel
B. Site Signage
    1. Set all signage clear of sidewalks and provide concrete apron around sign post for mowing.

32 80 00 Irrigation

A. General
1. Submit irrigation design at Design Development level for District review. Drawings shall indicate all turf and pavement areas, buildings, piping, and sprinkler heads, valves, controllers, and sprinkler coverage areas. Indicate type, number and manufacturer of all products.
2. Provide domestic water supply backup whenever non-potable water is being used.
3. When using ditch water, use of the equalizing pond with a minimum capacity of two water cycles is preferred.
4. When using pump systems, constant pressure systems are preferred.
    1. Preferred Products: SyncroFlo Pump Stations
5. When only sump wells are used, install a low maintenance filtration system with an additional debris collecting well.
6. Valve sleeves shall not be deeper than 5'-0", and valve head extensions used on valves if deeper than 5'-0".
7. System controllers shall be installed in a protective area and preferably within view of zones they control.
8. There shall be an extra zone valve wire for every 4 zones, each wire to have 6'-0" excess loop inside each of the valve boxes. If installing a two-wire system, and extra set of two-wire shall be installed.
9. All wire splices to be located in Ametek 10" diameter valve boxes.
10. Isolation valves shall be installed in such a way that in the event of a main line break the area of the break can be isolated, allowing the rest of the system to operate.
11. A Quick Coupler should be installed on the mainline between an isolation valve and endpoint, or another isolation valve.
12. A drain shall be installed at the end of each main line.
13. Each zone valve shall have a shut off valve upstream to do repairs on zone valves while not having to drain the mainline. This valve should have proper clearance inside an approved valve box to allow operation.
14. Only one zone valve with shut-off per valve box, unless complete system design allows for manifolds.
15. No more than 10'-0" elevation change between a zone valve and the last head of that zone.
16. Zone valves should be installed in inconspicuous locations. No valves shall be installed in playing fields.
17. Pop-up impact and rotor sprinklers shall be used wherever possible. Sprinkler system shall be designed for triangular head to head coverage in all areas. Mixing of head types (pop-ups/impact) in the same zone is prohibited.
18. Mixing of full circle, half circle, or quarter circle within a zone is discouraged.
19. Mixing of nozzle sizes within a zone is discouraged.
20. As-build irrigation drawings are required from the contractor.
21. District approved valve assembly:
    a. SVVSD Typical Detail # 32-1

B. Products
1. Pipe:
    a. Schedule 40 for all mains and laterals 6" or less diameter.
    b. C900 at 8"
2. Joints: Solvent weld on 3" or less pipe.
    a. Mainlines 4" and above to have ductile iron fittings with knuckle joint restraints.
        1. Preferred Manufacturers:
            a. Harco or Leemco
3. Fittings:
    a. Manufactured N.S.F./DW PVC type 1 Schedule 40D 2466.
    b. Pressure rating shall be same as piping system that the fittings are used in.
    c. Ductile iron fittings on all pipe 6" or greater with joint restraints.
4. Sprinkler Heads:
    a. Pop Up Impact Sprinklers
        1. Manufacturer: Rain Bird, no substitutions, Model Numbers: 41-51A, 27A, 37A, 47A.

b. Pop Up Spray Sprinklers:
    1. Rain Bird, no substitutions, Model Numbers: 1804, 1806.
    2. Model 1812 and above pop-up sprays are discouraged.
c. Pop Up Rotors:
    1. Rain Bird, Model Numbers: Rain Bird 5000, 6500 Falcon, 7005, 8000 series or better.
    2. No MP rotors.

5. Quick Coupler Valve:
    a. Rain Bird, one piece valve, bronze casting
6. Control Valves:
    a. Rain Bird, model PESB, Self-flushing nylon screen, Pressure Regulation Module on drip zones, Grit Filter, Flow Control Stem, Manual Open/Close Control, Normally closed, reverse flow position.
7. Isolation Valves:
    a. Manufacturer: Matco, Secure to mainline with megalugs, Construction Resilient wedge with square nut.
8. Drain Valves:
    a. Manufacturer: Cambridge, Type: Ball, quarter turn operation, Construction: Bronze, Ball: Hank chrome plate brass ball, Seals and Packing: Teflon
9. Valve Boxes:
    a. Manufacturer: Armor with drop in lid and snap locks, no substitution.
    b. Type: Heavy duty.
    c. Material: Thermo-plastic.
    d. Size as required by valve size for proper valve clearance and valve operation.
    e. All valve boxes shall be branded to indicate zone number.
10. Hydrometers
    a. Products
        1. Baseline Irrigation Solutions, BHM Series
            a. Netafim Hydrometer
    b. Other Requirements
        1. Electronic Digital Register (ER).
        2. Minimum output of 10 pulses per Gallon.
11. Automatic Irrigation Controllers
        1. Type
            a. Two-wire
            b. One-way communication
                i. Two-way communication is not acceptable per SVVSD network requirements.
        2. Products
            a. Baseline Irrigation Solutions, Base Station 3200
        3. Other Requirements
            a. Grounding Rods
                i. Per manufacturer's recommendations.
            b. Grounding Plates
                i. Per manufacturer's recommendations.
            c. Two-Wire Grounding
                i. Provide surge arrestors at frequency and locations recommended by controller manufacturer.
12. Soil Moisture Sensors
    a. Products
        1. Baseline Irrigation Solutions, BL5315B
    b. Other Requirements
        1. Provide at least one sensor per head type.
        2. Coordinate with SVVSD Irrigation department for placement on-site.
13. Backflow Preventer
    a. Type: As required by local jurisdiction.

b. Location: On all sprinkler main take-offs from domestic water.

14. Booster Pump
    a. Type: End suction centrifugal
    b. Manufacturer: Syncroflo Pump Station
    c. Accessories: Discharge line size – CLA-VAL Model 90-01, combination pressure reducing and check valve
        1. Housing: Above grade weatherproof, vandal resistant enclosure vented housing.

C. EXECUTION

1. Pipe Installation
    a. Trenches shall be straight with uniform depth and shall be excavated to sufficient depth and width to permit proper handling and installation of the pipe and fittings.
    b. Interior of pipe shall be kept free from dirt and debris. When pipe laying is not in progress, open ends of pipe shall be closed by an approved means.
    c. Multiple pipes in same trench is prohibited.
    d. Thrust blocking or joint restrains are required on 3" or larger pipes
    e. Mainlines shall be a minimum 24" and a maximum of 36" depth.
    f. Laterals shall be a minimum of 12" and a maximum of 18" depth.
    g. Trench backfill shall be free of rocks over 1" in diameter
    h. The backfill shall be water settled and compacted. Trench shall be evened off with the adjacent soil level and finished in a fine graded condition.
    i. Construction debris is not to be buried in the trenches.
    j. Location of valves: Design with a minimum of 6'-0" between valves on mainlines of 4" or smaller. Design with a minimum of 8'-0" between valves on the mainline of 6" or larger.
    k. Gluing of pipe shall not be done when temperatures are below product specifications.
    l. It is preferred that all piping shall be properly graded so that the entire system of piping may be completely drained.
    m. Drain valves shall be installed at the end of each mainline, with approved gravel drainage sump, valve box and locations marked on the "as-built" drawing.
2. Pipe Under Concrete or Asphalt
    a. Pipe sleeves shall not be installed deeper than 36"
    b. Piping under concrete or asphalt shall be installed before the start of construction or be installed by jacking, boring or hydraulic driving.
    c. Where any cutting or breaking of sidewalks, concrete work and/or asphalt is necessary, it shall be removed and replaced by the Contractor at his expense. Permission to cut or break sidewalks, concrete and/or asphalt shall be obtained by SVVSD grounds leads.
    d. Only one pipe will be allowed per sleeve, all sleeves shall be at least one nominal size larger than the pipe running through it.
    e. Control wires shall be installed in a separate sleeve than mainline/lateral piping

D. TESTING

1. Irrigation System Quality Assurance: A Distribution Uniformity Test shall be performed on each irrigation zone with the acceptable results being 60% or better. A net precipitation rate shall be furnished to SVVSD before acceptance of system.
2. Hydrostatic Test: Test mainline piping and valves, before backfilling trenches, to a hydrostatic pressure of not less than 100 psi for a period of 24 hours. Piping may be tested in sections to expedite work. Remove and repair piping, connections or valves which do not pass hydrostatic testing. No pipe joints will be accepted within sleeve pipe.
3. Demonstrate to Architect that system meets coverage requirements and the automatic controls function properly.
4. Flush and adjust all irrigation heads for optimum performance and to prevent overspray onto sidewalks, roadways and buildings.

32 90 00 Landscaping

A. Edging
    1. Preferred: 6” to 8” Concrete, with reinforcement.
    2. Acceptable: Coordinate options with District.
    3. Unacceptable:
        a. Trex/Composite
        b. Steel
B. Planting restrictions
    1. 0’-0” to 5’-0” from building
        a. Drip irrigation only, minimize plant material in this zone. Few low water plants such as ornamental grasses may be acceptable.
    2. 5’-0” to 15’-0” from building
        a. Limit to plants will full growth height of less than 10’-0”.
        b. Preferred exclusion of sucralose and other shallow rooted plants that may pose maintenance problems
    3. Entire site
        a. Limit or eliminate any plants with thorns or berries.
C. Xeric Design
    1. Xeric landscape design is encouraged. The use of native and adaptive species rather than excessive rock and cactus is preferred.

32 93 00 Trees

A. Acceptable tree list:

Deciduous Trees	Height	Width	Flowers	Water needs
*Cottonwood, Lanceleaf	40-60'	30-40	no	Moderate
*Cottonwood, Narrowleaf	40-60'	30-40'	no	Moderate
*Cottonwood, Plains Straight	40-50'	40-50'	no	Moderate
Aspen, Mountain Sentinel Quaking	30-40'	5-10'	no	Moderate
Aspen, Swedish Columnar	30-35'	5-10'	no	Moderate
Birch, Heritage	45-50'	30-35'	no	Mod/High
Birch, Rocky Mountain	15-25'	15-20'	no	Mod/High
Catalpa, Chinese	20-30'	20-30'	no	Low/Mod
Catalpa, Northern	40-50'	25-30'	no	Moderate
Catalpa, Western	35-45'	25-30'	no	Moderate
Coffeetree, Kentucky	40-50'	30-40'	no	Moderate
Elm, Emerald Sunshine	30-40'	20-30'	no	Moderate
Elm, New Horizon	35-40'	25'	no	Moderate
Elm, Patmore Brandon	50-60'	20-30'	no	Low
Hackberry	50-60'	40-50'	no	Moderate
Hackberry, Prairie Sentinel	40-50'	10-15'	no	Moderate
Honeylocust, Shademaster	40-50'	30-40'	no	Moderate
Honeylocust, Skyline	35-45'	30-40'	no	Moderate
Honeylocust, Sunburst	30-40'	30-40'	no	Moderate
Horsechestnut, Bauman's	40-50'	35-45'	White	Moderate
Linden, American Sentry	30-40'	20-25'	no	Moderate
Linden, Dropmore	50'	35'	no	Moderate
Linden, Glenleven	40-50'	30-40'	no	Moderate
Linden, Greenspire	30-40'	25-30'	no	Moderate
Linden, Summer Sprite	15-20'	12-15'	no	Moderate
Maple, Autumn Blaze	45-50'	35-40'	no	Moderate
Maple, Autumn Fantasy	40-50'	30-40'	no	Moderate
Maple, Fall Fiesta Sugar	30-50'	30-40'	no	Moderate
Maple, Norway Deborah	40-50'	40-50'	no	Moderate
Maple, Norway Emerald Queen	40-50'	30-40'	no	Moderate
Maple, Norwegian Sunset	30-35'	20-25'	no	Moderate
Oak, Columnar English	40-60'	15-20'	no	Moderate

* - Do not plant around building or playgrounds

Evergreens				
Fir, Blue Cloak Concolor	20-30'	8-12'	no	Moderate
Fir, Rocky Mountain Douglas	50-60'	20-25'	no	Low/Mod
Fir, Select Concolor	40-50'	20-30'	no	Moderate
Pine, Austrian	35-50'	25-35'	no	Moderate
Pine, Bosnian	20-35'	7-10'	no	Moderate
Pine, Bristlecone	10-12'	4-6'	no	Low
Pine, French Blue Scotch	30-40'	20-25'	no	Moderate
Pine, Ponderosa	50-80'	30-40'	no	Moderate
Pine, Vanderwolf Pyramid Limber	30-40'	10-15'	no	Moderate
Spruce, Black Hills	40-60'	10-20'	no	Moderate
Spruce, Colorado Blue	30-40'	25-30'	no	Moderate
Spruce, Fat Albert	25-40'	20-30'	no	Moderate
Spruce, Hoopsi	30-40'	10-20'	no	Moderate
Spruce, Norway	40-60'	25-30'	no	Moderate
Spruce, Serbian	50-60'	20-25'	no	Moderate

Ornamental Trees				
Maple, Ruby Slippers Ginnala	15-20'	15-20'	Red seed pod	Moderate
Pear, Capital Flowering	25-35'	8-10'	White	Moderate
Pear, Chanticleer	25-35'	12-15'	White	Moderate
Pear, Cleveland Select	25-35'	12-15'	White	Moderate
Pear, New Bradford Flowering	25-30'	20-25'	White	Moderate
Pear, Redspire Flowering	25-35'	18-20'	White	Low/Mod
Plum, Crimson Pointe Flowering	25-30'	8-10'	Pink	Moderate
Plum, Mount Saint Helens	20'	20'	Pink	Moderate
Plum, Newport Flowering	15-20'	15-20'	Pink	Moderate
Redbud, Eastern	20-35'	20-25'	Pink	Moderate

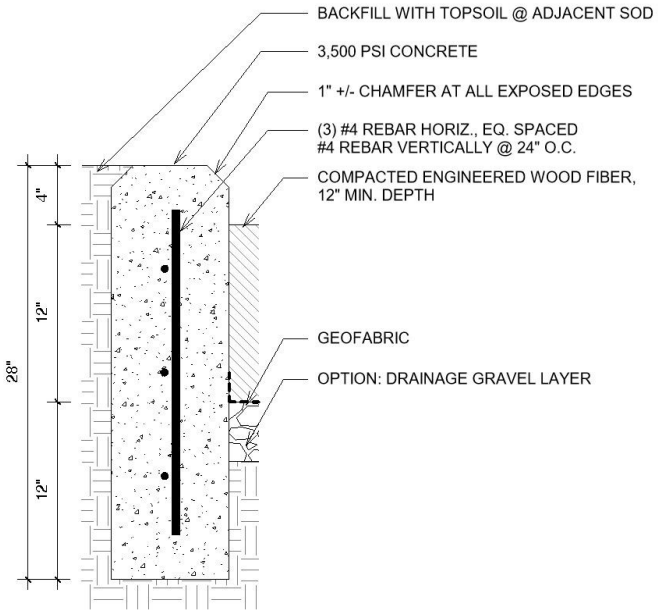
32 91 00 Playgrounds

A. PLAYGROUND DESIGN INPUT
   1. In addition to the DAG, the designer shall coordinate playground design with other District user groups which may include: Early Childhood, Physical Therapy, Operations & Maintenance Grounds Department.

B. PLAYGROUND EQUIPMENT MANUFACTURERS
   1. Preferred
      a. Miracle, Miracle is a **strong preference**. The District is working to transition to Miracle wherever possible to create consistency in parts, maintenance, and support.
   2. Acceptable
      a. Burke
      b. Little Tikes
      c. Playworld
   3. Unacceptable
      a. Game Time

C. SURFACING
   1. Preferred:
      a. Engineered Wood Fiber, 12" Depth min.
      b. Poured-in-place
      c. Synthetic turf
   2. Unacceptable:
      a. Pea Gravel

D. PLAY PIT EDGING
   1. Preferred
      a. Cast-in-place concrete edging only.
   2. Unacceptable
      a. Timbers
      b. Composite
   3. Typical Detail:
      a. ***Note: Coordinate play equipment installation height with top of EWF, not top of concrete.***



DIVISION 33 – SITE UTILITIES

33 00 00 Primary District Requirements

A. All underground utilities shall be provided with **tracer wire** per section 33 50 00 of this guideline.
B. Where allowable by Code PVC piping is preferred to ductile iron or other.

33 40 00 Storm Drainage

A. DESIGN CRITERIA
   1. Storm water shall not discharge over pedestrian walks, pipe underground where possible.
   2. Design grades around buildings to achieve positive drainage away from building (prefer minimum of 2% slope around building.)
   3. Protect inlet and outlet structures with tamper resistant screen to prevent passage of objects larger than 6 inches.
   4. All day lighted pipes shall have concrete headwalls.
   5. Permanent pond water will not be allowed.
   6. Fencing around detention basins is not necessary except as required around outlet structures.
   7. Storage and reuse of storm water for irrigation systems is not allowed by the State of Colorado except by special permit.

33 40 10 Sub-drainage System

A. DESIGN CRITERIA

1. Include subdrain system when recommended by Geotechnical Engineer.
2. Geotechnical Engineer, employed by School District, will observe layout, excavation and installation of subdrain system.
3. Include grade cleanouts at each major change of direction of pipe. In planting areas pour 12" x 12" concrete pad around cleanout.
4. Slope of drain and location of pipe to foundation per recommendation of Geotechnical Engineer. Use perforated pipe adjacent to foundations and unperforated elsewhere. All perforated pipe to be enclosed with ¾" washed gravel and gravel to be isolated from adjacent soil with one layer of Mirafi filter fabric.
5. Changes in direction of drain shall be made with 1/8 bends and Y fittings shall be used at all intersections.

B. PRODUCTS
1. Unperforated Pipe: Rigid polyvinyl chloride sewer pipe and solvent welded fittings.
2. Perforated Pipe: Perforated rigid polyvinyl chloride leaching system piping and solvent welded fittings. Provide solid end caps at terminations.
3. Cleanout pipe and fittings shall be of cast-iron soil pipe. Each cleanout shall have a brass ferrule and cast-brass screw jointed plug with socket for wrench.
4. Bedding around drain pipe: Washed 3/4" to No. 4 aggregate with less than 3% passing the No. 200 sieve.
5. Foundation drainage mat:
6. Mirafi Miradrain 6000.
7. American Wick Drain, Amerdrain 200

33 50 00 Utility Trace Wire

A. GENERAL
1. All trace wire shall comply with applicable Codes and Standers, which shall override the requirements of this section where conflicts exist.
2. All trace wire shall have HDPE insulation intended for direct bury, color coated per APWA standard for the specific utility being marked.
3. Trace wire installation shall be performed in a such a manner that allows proper access for connection of line tracing equipment, proper locating of wire without loss or deterioration of low frequency (512Hz) signal for distances in excess of 1,000 linear feet, and without distortion of signal caused by multiple wires being installed in close proximity to one another.

B. TRACE WIRE
1. Open Trench – Trace wire shall be #12 AWG Copper Clad Steel, High Strength with minimum 450 lb. break load, with minimum 30 mil. HDPE insulation thickness.
2. Directional Boring – Trace wire shall be #12 AWG Copper Clad Steel, Extra High Strength with minimum 1,150 lb. breaking load, with minimum 30 mil. HDPE insulation thickness.
3. Pipe Bursting/Slip Lining – Trace wire shall be 7x7 Stranded Copper Clad Steel, Extreme Strength with 4,700 lb. break load, with minimum 50 mil. HDPE insulation thickness.

C. CONNECTORS
1. All mainline trace wires must be interconnected in intersections, at mainline tees and mainline crossed. At tees, the three wires shall be joined using a single 3-way lockable connector. At crosses, the four wires shall be joined using a 4-way connector. Use of two 3-way connectors with a short jumper wire between them is an acceptable alternative.
2. Direct bury wire connectors – shall include 3-way lockable connectors and mainline to lateral lug connectors specifically manufactured for use in underground trace wire installations. Connectors shall be dielectric silicon

filled to seal out moisture and corrosion, and shall be installed in a manner so as to prevent any uninsulated wire exposure.

3. Non-locking friction fit, twist on or taped connectors are prohibited.

D. TERMINATION/ACCESS

1. All trace wire termination points must utilize an approved trace wire access box (above ground access box or grade level/in-ground access box as applicable), specifically manufactured for this purpose.
2. All grade level/in-ground access boxes shall be appropriately identified with “sewer” or “waste” cast into the cap and be color coded.
3. A minimum of 2 ft. of excess/slack wire is required in all trace wire access boxes after meeting final elevation.
4. All trace wire access boxes must include a manually interruptible conductive/connective link between the terminals for the trace wire connection and the terminal for the grounding anode wire connection.
5. Grounding anode wire shall be connected to the identified or bottom terminal on all access boxes.

E. GROUNDING

1. Trace wire must be properly grounded at all dead ends/stubs.

F. INSTALLATION

1. Trace wire systems must be installed as a single continuous wire, except where using approved connectors. Lay mainline trace wire continuously, bypassing manholes, valves, fittings, etc.
2. No looping or coiling of wire is allowed.
3. Trace wire shall be secured to pipe at 5 ft. minimum intervals.
4. Mainline trace wire shall not be connected to existing conductive pipes.

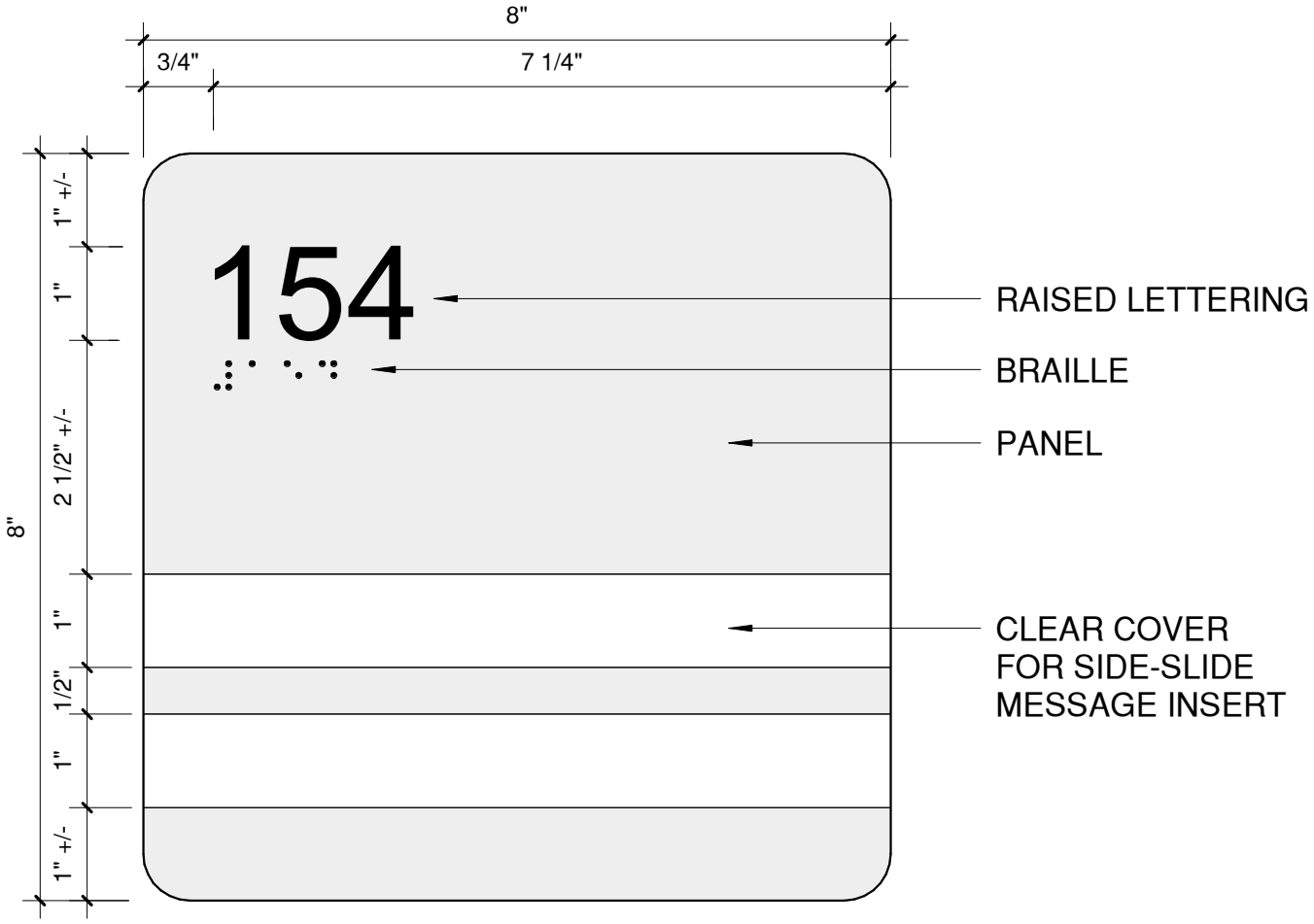
G. TESTING

1. All new trace wire installations shall be tested using typical low frequency (512 Hz) line tracing equipment.

SVVSD TYPICAL DETAILS

10-1 Typical Room Signage
32-1 Typical Irrigation Control Box
32-2 Preferred Sidewalk Chase

ST. VRAIN VALLEY SCHOOLS
academic excellence by design



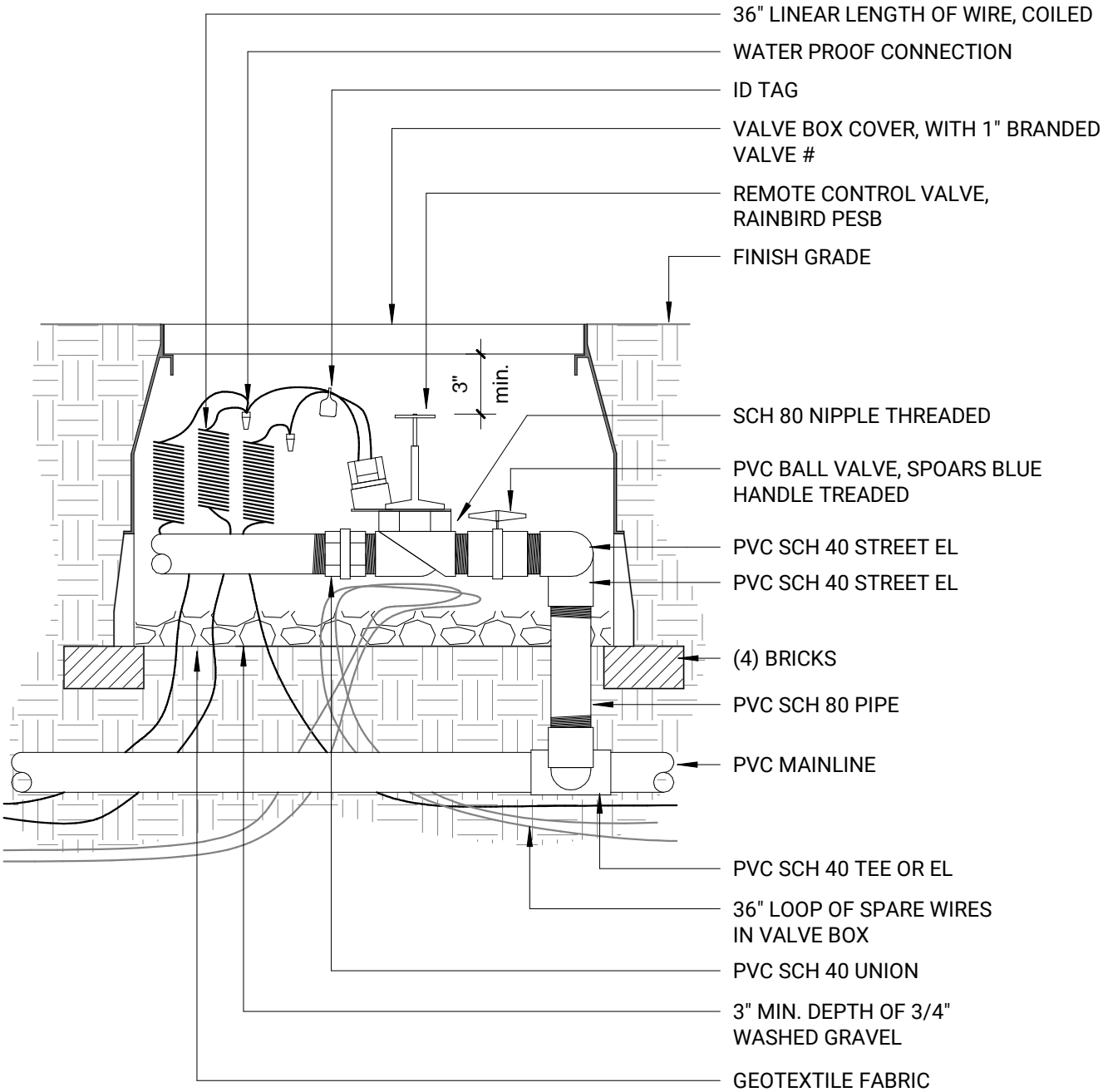
Typical Room Signage
scale: 6" = 1'-0"

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SVVSD Typical Details

Detail # 10-1

ST. VRAIN VALLEY SCHOOLS
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REMOTE CONTROL VALVE ASSEMBLY

Irrigation Control Box

scale: 1 1/2" = 1'-0"

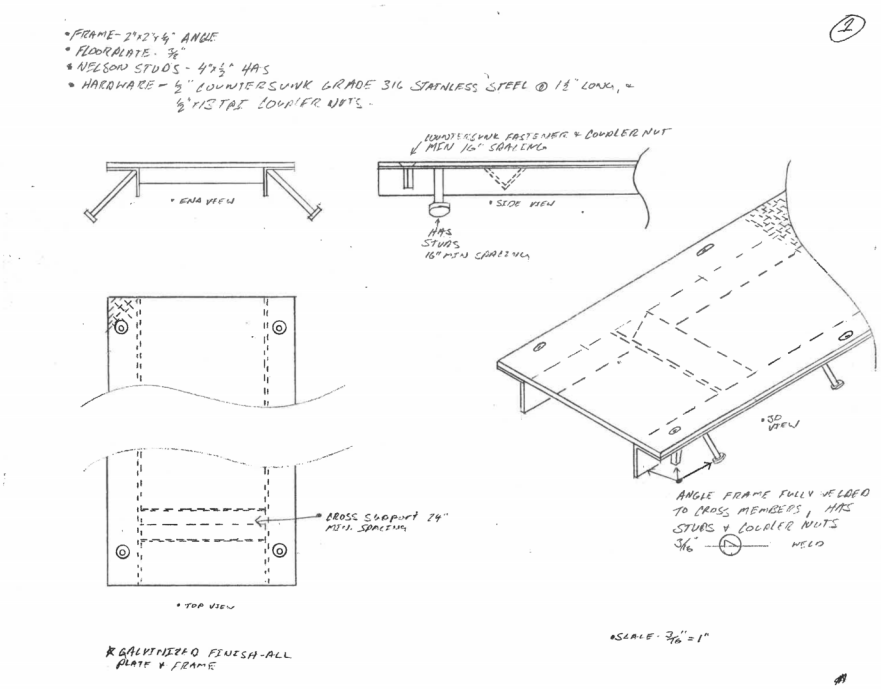
1/23/2026 12:44:45 PM

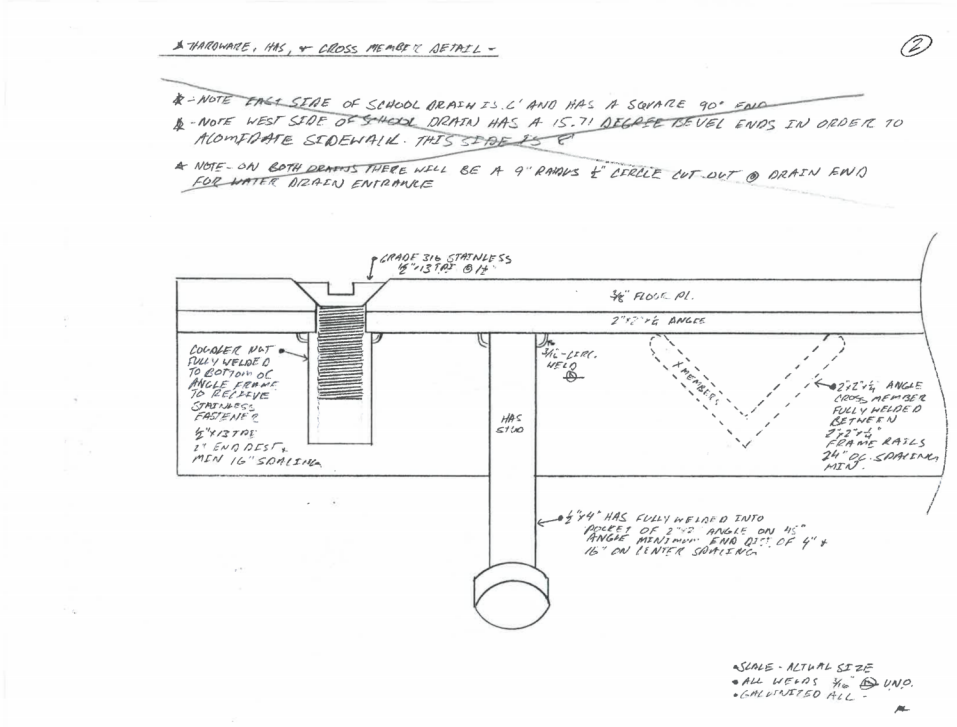
SVVSD Typical Details

Detail # 32-1

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Drafted detail in develpment





Sidewalk Chase
scale: 12" = 1'-0"

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SVVSD Typical Details

Detail # 32-2